



**GREEN
CLIMATE
FUND**

Meeting of the Board
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Provisional agenda item 12

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26 June 2023

Consideration of funding proposals – Addendum II

Funding proposal package for FP207

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

Table of Contents

Funding proposal submitted by the accredited entity	3
No-objection letter issued by the national designated authority(ies) or focal point(s)	160
Environmental and social report(s) disclosure	161
Secretariat's assessment	165
Independent Technical Advisory Panel's assessment	186
Response from the accredited entity to the independent Technical Advisory Panel's assessment	194
Gender documentation	196

Funding Proposal

Project/Programme title:	Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management
Country(ies):	Pakistan
Accredited Entity:	World Wildlife Fund, Inc. (WWF)
Date of first submission:	2022/12/03
Date of current submission	2023/06/23
Version number	V.11



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Contents

Section A	<u>PROJECT / PROGRAMME SUMMARY</u>
Section B	<u>PROJECT / PROGRAMME INFORMATION</u>
Section C	<u>FINANCING INFORMATION</u>
Section D	<u>EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA</u>
Section E	<u>LOGICAL FRAMEWORK</u>
Section F	<u>RISK ASSESSMENT AND MANAGEMENT</u>
Section G	<u>GCF POLICIES AND STANDARDS</u>
Section H	<u>ANNEXES</u>

NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	<u>Not applicable</u>			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF Contribution	Co-financers' contribution ¹	
	Mitigation total	%	%	
	☐ Energy generation and access	%	%	
	☐ Low-emission transport	%	%	
	☐ Buildings, cities, industries and appliances	%	%	
	☐ Forestry and land use	%	%	
	Adaptation total	%	%	
	☒ Most vulnerable people and communities	16%	42%	
	☐ Health and well-being, and food and water security	%	%	
	☐ Infrastructure and built environment	%	%	
☒ Ecosystems and ecosystem services	84%	58%		
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	<u>Co-benefit:</u> 26,450 M-Tons annually 1,037,928 tons of CO₂ (over ~40 years)	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Indicate total number of direct and indirect beneficiaries	
			687,336 people (357,414 men and 329,922 women)	7,024,361 people (3,652,668 men and 3,371,693 women)
			0.3% of the total population of ~200 million people.	3.5% of the total population of ~200 million people.
A.7. Total financing (GCF + co-finance ²)	77.8 million USD	A.9. Project size	Medium (Up to USD 250 million)	
A.8. Total GCF funding requested	66 million USD <i>For multi-country proposals, please fill out annex 17.</i>			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the [Policy of Co-financing](#) of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant ☐ Equity ☐ Loan ☐ Results-based payment ☐ Guarantee		
A.11. Implementation period	7 years	A.12. Total lifespan	30 years
A.13. Expected date of AE internal approval	June 12 th , 1 week ahead of the FP publication date for B.36	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	WWF-Pakistan will be the Executing Entity (EE) for the proposed project. Details of its respective roles and responsibilities during project execution are described in Section B.4 of this Funding Proposal.		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate Change Problem

- Pakistan is facing a period of unprecedented climate-related pressures, having recently experienced a disastrous flood in August 2022 that displaced an estimated 33 million people. The country is the sixth most populous in the world and lies in a mostly arid and semi-arid geographic area prone to recurring large-scale floods, droughts, and landslides. Increasingly severe and frequent floods and droughts caused by climate change are already impacting Pakistan's population, key ecosystems, and economy. With an average per capita water availability of less than 1,000 m³ per annum³ Pakistan is one of the most water-stressed in the world. The country is also strongly dependent on water to support its predominantly agriculture-based economy, which accounts for 19% of GDP, 42% of jobs⁴, and 91% of the annual freshwater usage⁵.

³ National Water Policy, 2018.

⁴ Government of Pakistan, 2017.

⁵ UNDP, 2016.

2. Observed trends and projected changes in climate under RCP 4.5 show increasing temperatures and greater variability in rainfall patterns. These changes lead to increased instances of droughts, flooding and landslides, which negatively affect ecosystems nationwide and the communities that depend upon their proper functioning for water regulation. The recent 2022 flooding in Pakistan in August/September destroyed 556,000 houses, damaged 1.17 million homes, damaged 2 million acres of crops, resulted in the loss of 755,000 livestock, injured 12,700 people and caused the death of 1,400 people^{6,7}. In 2010, severe floods impacted more than 18 million people and caused an estimated US\$10 billion in damages resulting in the design of Recharge Pakistan as a large-scale project response to future climate events. These floods were closely followed by heavy monsoon rains in 2011 that affected 9.7 million people⁸. The impacts of such floods are compounded by longer dry spells and increasingly severe droughts experienced between flood events which make water availability less predictable in Pakistan and exacerbate the country's risk to climate change.
3. The Indus Basin serves as Pakistan's agricultural lifeline, supporting 90% of agricultural production, and is particularly vulnerable to the impacts of climate change. Under future climate change projections, the Indus Basin will experience an increase in rainfall in the higher volume months of July and August, with small decreases in the early parts of the year. In the projected future, these changes in rainfall patterns are expected to increase flood peak intensity, volume, depth and overall velocity, particularly in the upper catchments of the project sites. These changes are further expected to shorten the peak lag time and result in an earlier arrival of the flood peak.
4. Vulnerable communities living in the basin depend on it for reliable water supply and recharge, agricultural production, climate regulation, and other ecosystem benefits. The assessments conducted as part of the Project's Feasibility Study (Annex 2) indicated that the catchments in the Indus Basin of DI Khan and Chakar Lehri present both high levels of ecosystem and community vulnerability to climate change, and opportunity for adaptation benefits through Ecosystem-based Adaptation (EbA) and green infrastructure interventions. For these reasons, four sites in these catchments were selected as the target sites for the project. The project sites of DI Khan and Ramak fall within the DI Khan catchment and the project sites of Chakar Lehri and Manchar fall within the Chakar Lehri catchment.
5. Despite expected increases in annual rainfall, it is projected that rainfall will occur over shorter timeframes, therefore increasing the number of dry spells annually. In comparison to the current climate, the long-term SPEI index as a proxy for drought shows a clear tendency towards drier years over the next century. See Annex 2: Feasibility Study for further details on the site-specific climate and hydrological analyses conducted for the proposed project⁹.
6. Current efforts in Pakistan to manage severe floods and droughts caused by climate change are largely reactive and rely on grey infrastructure to address the impacts of these events on the country's population, economy and ecosystems. This is due to the country's resource and economic constraints, high level of indebtedness, and reliance on intermittent donor financing for disaster recovery, which prevent the Government of Pakistan (GoP) from making proactive investments in flood and drought risk reduction interventions that provide sustainable benefits.
7. In addition to these constraints, Pakistan's current policy and regulatory instruments for water resource management — namely the National Water Policy (2018) and National Adaptation Plan (under development) (NAP), and Provincial Adaptation Plans¹⁰ — do not consider EbA and green infrastructure interventions to help reduce Pakistan's climate vulnerabilities. They also do not account for the critical role of healthy freshwater ecosystems in increasing affected communities' resilience to climate change. There is an urgent need to shift away

⁶ OCHA. Pakistan: 2022 Monsoon Floods. Situation Report No. 03. 26 August 2022.

⁷ OCHA. Pakistan: 2022 Monsoon Floods. Situation Report No. 05. 9 September 2022.

⁸ World Bank Group, Climate Change Knowledge Portal. Available at: <https://climateknowledgeportal.worldbank.org/country/pakistan/vulnerability>

⁹ The observational and projected analysis/data is referenced in the relevant sections of Annex 2. These data sources include: i) Station data from the Pakistan Meteorological Department (PMD) for selected stations near project sites. Available at: <https://www.pmd.gov.pk/en/> obtained 25 May 2022, NASA earth exchange global daily downscaled projections (NEX - GDDP)-; ii) <https://www.nasa.gov/nex/gddp>, Global SPEI database -; and iii) <https://spei.csic.es/index.html>. The projected data is based on CORDEX (Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, Version 20/7/2012) and WorldClim version 2.1. Available at: <https://worldclim.org/>

¹⁰ Provincial Adaptation Plans are in the early ideation stages. Evidence generated by the proposed project's EbA and green infrastructure will establish proof of concept for the integration of these interventions into the Provincial Adaptation Plans, which will be done under the project.

from an over-reliance on grey infrastructure interventions, many of which are financially unsustainable because of their high capital and operational costs. They also do not optimize the potential environmental and social benefits of EbA and green infrastructure such as improved infiltration, groundwater recharge and water availability for vulnerable communities. Public sector reform around flood and water resources management is needed which includes EbA and green infrastructure investments and formal procedures which prioritize and diversify public sector solutions to managing floods and droughts in Pakistan. The absence of such a shift in Pakistan's business-as-usual flooding and water resources management approach will continue to exacerbate communities' vulnerability to extreme events caused by climate change such as the recent 2022 floods.

Proposed Approach

8. Recharge Pakistan is a cornerstone of the Government of Pakistan's vision — as highlighted in its updated 2021 Nationally Determined Contributions (NDCs) — to leverage natural systems to enhance Pakistan's resilience to climate change with the goal of reducing flood and drought risk across the Indus Basin. The project's approach is strongly aligned with the GCF's goal of funding initiatives that catalyse climate change benefits through the creation of enabling environments for climate action, accelerating climate innovation, and funding scalable and replicable adaptation interventions.
9. The project will catalyse transformational change in Pakistan by investing in EbA and green infrastructure interventions at the four project sites in the Indus Basin to reduce the impacts of increasingly severe floods and droughts on vulnerable communities and ecosystems. As the largest investment at the national level to date in an ecosystems-based approach to flood and water resources management, this project will accelerate climate innovation in Pakistan by establishing proofs of concept for EbA and green infrastructure interventions that provide evidence of their cost-effectiveness and efficiency, flood and drought risk reduction benefits, and sustainability. EbA interventions will include the restoration of 14,215 ha of degraded watersheds in DI Khan, restoring 34 km of flow paths, as well as desilting and restoring channels in Ramak Watershed and Manchar Lake. An additional 127 green infrastructure interventions (including recharge basins and retention areas) will be implemented at specific locations in DI Khan, Ramak, Manchar, and Chakar Lehri to maximise the flood reduction benefits to vulnerable communities from these interventions.
10. The implementation of these tangible adaptation interventions will be used as evidence to create an enabling environment for climate action in Pakistan that removes barriers to novel climate solutions through the promotion of integrated strategies, planning and policymaking. Economic valuations of the project's EbA and green infrastructure interventions will demonstrate their cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan. Using this evidence, a National Working Group will be established under the project, chaired by the Ministry of Climate Change (MoCC) and comprised of regulators within the MoWR and its FFC, as well as the relevant Provincial Departments. This National Working Group will lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, and Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. This will lead to the creation of well documented procedures backed by a rigorous scientific base that enables the GoP to channel future investments into EbA and green infrastructure interventions instead of grey infrastructure only to respond to increasingly severe floods and droughts caused by climate change.
11. Together, project interventions will directly benefit a total of 687,336 people, including 357,414 men (52% of the total population) and 329,922 women (48% of the total population), and will indirectly benefit a total of 7,024,361 people, including 3,652,668 men (52% of total population) and 3,371,693 women (48% of total population) in Pakistan's Indus Basin.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Local context

12. Pakistan is characterized by diverse topography, ecosystems, and climate zones. Rich in natural resources, including fertile agricultural lands, natural gas reserves, and mineral deposits, Pakistan faces challenges in balancing competing objectives between economic development and environmental protection. A semi-industrialized country, Pakistan has grown from a primarily agriculture-based to a mostly service-based economy

(with services constituting 49.4% of the Gross Domestic Product — GDP — in 2019). As of 2019 agriculture was still the largest employer, occupying 42.6% of the workforce. By 2013 approximately 29.5% of the population still lived below the national poverty line and by 2018, 12.3% of the population remained undernourished. Pakistan is situated at the western end of the great Indo-Gangetic Plain. Of the total area of the country, about three-fifths consists of rough mountainous terrain and plateaus, and the remaining two-fifths constitutes a wide expanse of level plain¹¹. The land can be divided into five major regions: the Himalayan and Karakoram ranges and their subranges; the Hindu Kush and western mountains; the Balochistan plateau; the submontane plateau; and the Indus River plain¹². The majority of Pakistan's 225.1 million people (2021)¹³ live in the Indus River plain, an area prone to severe flooding in July and August. Major earthquakes are also frequent in the mountainous northern and western regions of the country.

13. Climatically, Pakistan lies in a temperate zone and its climate is as varied as the country's topography —generally dry and hot near the coast and along the lowland plains of the Indus River and becoming progressively cooler in the northern uplands and Himalayas (Köppen-Geiger classifications include BWh, BWk, BSh, BSk, Cwa, Cwb, Cfa, Dfa, Dfb, Dsa, Dsb and Dfd)¹⁴. Four seasons are recognized: i) a cool, dry winter from December to February; ii) a hot, dry spring from March through May; iii) the summer rainy season, also known as the southwest monsoon period, occurring from June to September; and iv) the retreating monsoons from October to November. Average monthly rainfall and temperature range from 6.6 mm in October to 53.4 mm in August, and 9.3 °C in January to 29.7 °C in June, respectively. Most of the country receives very little rainfall, with the exception of the Northern regions, where monsoons can bring upwards of 200 mm a month from July to September (1991–2020 trends)¹⁵. Inter-annual rainfall varies significantly, often leading to successive patterns of floods and drought. El Niño is a significant influence on climate variability in Pakistan, with anomalies in both temperature¹⁶ and flood frequency and impact¹⁷ correlated with the El Niño cycle¹⁸.
14. The proposed project will be implemented in areas of the Indus River basin including Dera Ismail (D.I.) Khan in the Khyber Pakhtunkhwa province, and Chakar Lehri in the Balochistan and Sindh provinces (Figure 1). These areas were identified using a three-phase screening mechanism during the development of the Concept Note. The three phases include: i) assessments based on biophysical, socio-economic, and ecological data and information; ii) risk and opportunity assessments based on spatial maps of hazard, vulnerability, exposure and technical feasibility; and iii) suitability assessments based on respective ecosystems. Ecosystem-based adaptation (EbA) and green infrastructure intervention sites within the project sites were selected based on EbA/NbS archetypes (developed in consultation with stakeholders with clear climate objectives, co-benefits and components on preferred intervention types) that were tailored and applied to each selected site. Additional detail on specific sites and their selection is presented in Section 3 of Annex 2.
15. See Annex 2: Feasibility Study for further details on the site-specific climate and hydrological analyses conducted for the proposed project, the results of which are presented in the relevant sections below¹⁹.

¹¹ Climate Risk Country Profile: Pakistan (2021): The World Bank Group and the Asian Development Bank. Available at: https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf

¹² Britannica: Pakistan <https://www.britannica.com/place/Pakistan>

¹³ World Bank Data <https://data.worldbank.org/indicator/SP.POP.TOTL?locations=PK>

¹⁴ BWh: arid desert – hot and arid; BWk: arid desert – cold and arid; BSh: arid steppe – hot and arid; Cwa: warm temperate – winter dry and hot summer; Cwb: warm temperate – winter dry and warm summer; Cfa: warm temperate – fully humid and hot summer; Dfa: Snow – fully humid and hot summer; Dfb: Snow – fully humid and warm summer; Dsa: Snow – summer dry and hot; Dsb: Snow – dry and warm summer; Dfd: Snow – fully humid and extremely continental. Source: <http://koeppen-geiger.vu-wien.ac.at/present.htm>

¹⁵ World Bank Group, Climate Knowledge Portal <https://climateknowledgeportal.worldbank.org/country/pakistan>

¹⁶ del Río, S., Iqbal, M.A., Cano-Ortiz, A., Herrero, L., Hassan, A. and Penas, A. (2013). Recent mean temperature trends in Pakistan and links with teleconnection patterns. *International Journal of Climatology*, 33, 277–290. URL: <https://rmets.onlinelibrary.wiley.com/doi/full/10.1002/joc.3423>

¹⁷ Ward, P.J., Jongman, B., Kumm, M., Dettinger, M.D., Weiland, F.C.S. and Winsemius, H.C. (2013). Strong influence of El Niño Southern Oscillation on flood risk around the world. *Proceedings of the National Academy of Sciences*, 111, 15659–15664. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4226082/>

¹⁸ Climate Risk Country Profile: Pakistan (2021): The World Bank Group and the Asian Development Bank. Available at: https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf

¹⁹ The observational and projected analysis/data is referenced in the relevant sections of Annex 2. These data sources include: i) Station data from the Pakistan Meteorological Department (PMD) for selected stations near project sites. Available at: <https://www.pmd.gov.pk/en/> obtained 25 May 2022, NASA earth exchange global daily downscaled projections (NEX - GDDP); ii) <https://www.nasa.gov/nex/gddp>, Global SPEI database -; and iii) <https://spei.csic.es/index.html>. The projected data is based on CORDEX (Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, Version 20/7/2012) and WorldClim version 2.1. Available at: <https://worldclim.org/>

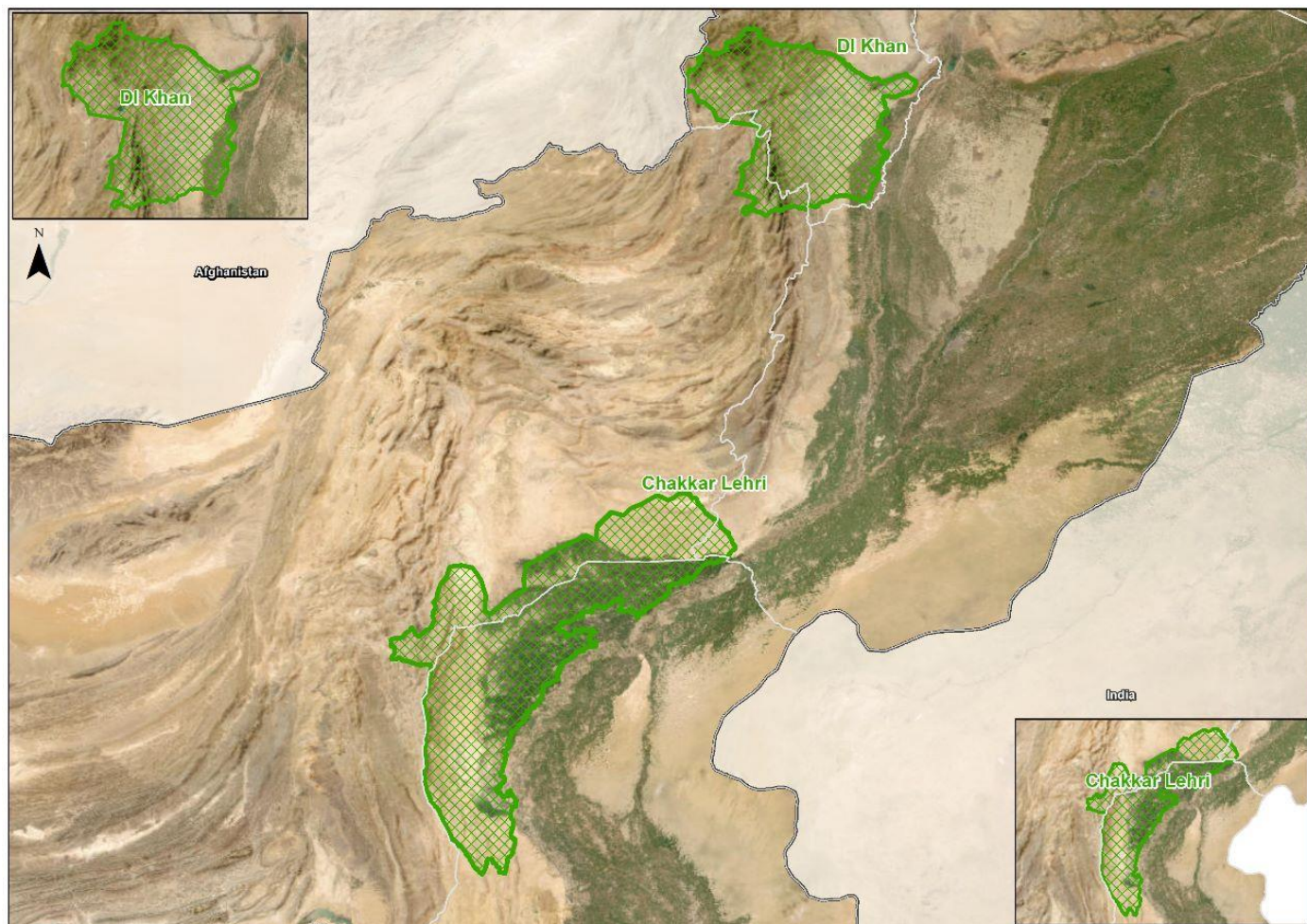


Figure 1. Map of the two catchments in which the project's four sites are located. The project sites of DI Khan and Ramak fall within the DI Khan catchment and the project sites of Chakar Lehri and Manchar fall within the Chakkar Lehri catchment.

Climate change problem²⁰

16. Pakistan, the sixth most populous country in the world, lies in a mostly arid and semi-arid geographic area prone to recurring large-scale floods, earthquakes, droughts, and landslides. Much of the economy relies on natural resources, such as river flows from rain and glacial melt for irrigation and hydroelectricity. Over the last century, there have been long-term increases in day and night-time temperatures as well as more varied rainfall totals. Further, among these baselines' changes are an increase in the number of severe extreme events. There has been an increase in the number and severity of flooding events as well as crop and livestock exposure to severe drought events and extreme temperatures. Additionally, there has been a shift away from the "normal" conditions and years tending to be on the upper or lower ranges of the baseline climate ranges making seasonal and medium-term forecasting more challenging. These current climate changes are anticipated to continue and be exacerbated into the future with increased rainfall intensity projected over most months, a concentration of rainfall into a shorter duration, potentially longer dry spells, and more severe drought. The challenges currently faced in Pakistan are therefore expected to only increase under future climate scenarios.^{21,22}

17. The projected changes in climate include:

- Temperature increases of 1.4°-3.7°C by the 2060s and 6.0°C by the 2090s, with projected increases higher during winter and in the north²³.
- Uncertainty about changes to mean annual precipitation, with projections for changes in monthly rainfall ranging from a 20% decrease to a 41% increase by the 2090s²⁴.
- Decreased glacier volume and snow cover.
- Increased frequency and intensity of extreme climate events, including floods and droughts.

Local vulnerability

18. Pakistan holds considerable social vulnerability to disaster. High poverty and malnutrition rates prevail, and many communities and minority groups are marginalized by socio-economic status, location, and political circumstances. Pakistan sits 125th out of 169 countries on the Human Development Index. Pakistan's high exposure to multiple natural hazards and its likely exposure to above average climate changes should be seen in the context of its vulnerability. Over decadal timeframes mortality attributable to natural hazards is dominated by Pakistan's exposure to earthquakes. However, in recent years flooding has also had a very significant impact²⁵. This was highlighted in recent years when unusually heavy rainfall during the 2010 monsoon caused the worst flood in Pakistan's history, affecting one-fifth of the country. Approximately 2,000 people were killed, with 12 million homes and 2.2 million hectares of crops damaged or destroyed. This event highlighted the country's vulnerability to flooding in both physical and economic terms. More recently, severe flooding in Pakistan in August/September 2022 destroyed 218,000 houses, damaged 452,000 houses, negatively impacted 2 million acres of crops, resulted in the loss of 794,000 livestock, injured 1,343 people and resulted in the death of 937 people²⁶. See additional details on climate change and its impacts in Section B.1. and Annex 2. According to UNISDR, of an estimated US\$ 1.3 billion of average annual losses to natural hazards, around 75% are attributable to floods²⁷. GermanWatch put a higher estimate on Pakistan's average annual losses, at US\$3.8 billion²⁸. In contrast, the probability of severe meteorological drought in any given year in Pakistan has historically been around 3%, and the consecutive droughts of 1999 and 2000 caused crop failure and mass starvation. Furthermore, during Pakistan's 2015 heatwave, over 65,000 people were hospitalized with heat stroke. Under all climate scenarios, all of these climate impacts are expected to worsen in the future. These effects will be experienced in terms of physical, livelihood, ecosystem, and economic exposure²⁹.

- **Agriculture** employs 45% of the workforce in Pakistan, and nearly all arable land is cultivated. The irrigation network is heavily reliant on river systems to feed production. Climate change, such as rising temperatures and extreme weather events, has resulted in devastating crop losses, altered growing seasons, increased pest incidence, and soil erosion. Salinization of land and aquifers as a result of saline intrusion from reduced river flows and rising seas endangers the fertile Indus River plain, which has lost nearly 810,000 ha of arable land and experienced a 40–60% reduction in major crop production in recent years. Livestock production is also highly vulnerable to rising temperatures and drought, which cause heat stress and decreased productivity, as well as decreased animal fodder production and stressed rangelands/water sources³⁰.
- In Pakistan's current climate, **water resources** are both a benefit and a threat. Pakistan's water surface resources are heavily reliant on the Indus River as its primary freshwater source. However, the majority of river flows are caused by snow and glacier melt, with the remainder being caused by monsoon rains. Temperature increases are rapidly melting glaciers and reducing winter snowfall, increasing stream base flows and glacial lake outburst floods, as seen in the Passu village (Gilgit-Baltistan region) in 2022. This has happened at least 16 times in 2022 due to heat waves, compared to an average of 5–6 times per year. While this poses a current flood threat and increases baseline flow in the short term, these water resources are not sustainable under current melt conditions, and significantly reduced water levels are expected in the coming decades as glaciers melt. Furthermore, while uncertainty exists, projections indicate that severe drought conditions are becoming more common in already arid and semi-arid areas, with the frequency of severe drought increasing in Pakistan's wetter northern areas. Water scarcity in Pakistan is exacerbated by the country's limited 30-day storage capacity³¹.
- Pakistan's **energy** supply is augmented by hydropower, which accounts for 31% of total generation. Reduced water supplies caused by unreliable or variable rainfall and rising temperatures (and subsequent evaporation), particularly during the summer when cooling demand is highest, have resulted in ongoing power outages and

²⁰ Climate information and analysis taken from Appendix 2 chapter 1/2/4/7.

²¹ ClimateLinks - Climate Risk Profile: Pakistan: <https://www.climatelinks.org/countries/pakistan>

²² World Bank Group, Climate Change Knowledge Portal: <https://climateknowledgeportal.worldbank.org/country/pakistan>

²³ ClimateLinks - Climate Risk Profile: Pakistan: <https://www.climatelinks.org/countries/pakistan>

²⁴ Ibid.

²⁵ Climate Risk Country Profile: Pakistan (2021): The World Bank Group and the Asian Development Bank. Available at: https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf

²⁶ UN Office for the Coordination of Humanitarian Affairs (OCHA). Pakistan: 2022 Monsoon Floods. Situation Report No. 03. 26 August 2022. Available at: Pakistan: 2022 Monsoon Floods - Situation Report No. 03: As of 26 August 2022 - Pakistan | ReliefWeb.

²⁷ UNISDR (2014). PreventionWeb: Basic country statistics and indicators. Available at: <https://www.preventionweb.net/countries>

²⁸ GermanWatch (2017). Global Climate Risk Index 2017. GermanWatch, Bonn. Available at: <https://germanwatch.org/sites/germanwatch.org/files/publication/16411.pdf>

²⁹ World Bank Group, Climate Change Knowledge Portal: <https://climateknowledgeportal.worldbank.org/country/pakistan/vulnerability>

³⁰ Ibid.

³¹ Ibid.

crippled the economy, with an estimated annual loss of 2% of GDP. It is anticipated that this will be further exacerbated in the future³².

- **Population health** is directly linked to observed extreme events and indirectly linked to water and food security vulnerability. Most of Pakistan's population lives along the flood-prone Indus River, and diarrheal disease caused by contaminated water is a leading cause of morbidity and mortality. In 2010, unprecedented monsoon rainfall flooded 20% of the country, affecting 21 million people and causing infectious disease outbreaks and diarrheal illness due to contaminated drinking water and unsanitary conditions in makeshift camps. Heatstroke is more likely in hot weather, as was the case in 2015 when an unprecedented heat wave with temperatures as high as 49°C killed over 1,200 people in Karachi. Climate risks to food security, such as crop loss/failure and water stress, have implications for malnutrition, which is already severe in Pakistan, with nearly half of all children suffering from malnutrition³³.
- **Poverty and inequality.** Many of the climate changes projected are likely to disproportionately affect the poorest groups in society. Heavy manual labour jobs are commonly among the lowest paid whilst also being most at risk of productivity losses due to heat stress³⁴. Poorer businesses are least able to afford air conditioning, an increasing need given the projected increase in cooling days, and poorer farmers and communities are least able to afford local water storage, irrigation infrastructure, and technologies for adaptation. The high temperature increases projected for Pakistan present particular challenges in a nation where the agricultural sector remains the largest employer. Only 17.5% of the people below the poverty line in Pakistan are in urban areas³⁵. Rural groupings include those who are owners of small farms (20%), those who are landless farmers (10%) and agricultural laborers (12%) all of whom are likely to be among the most affected by the above pressures³⁶.
- The level of climate change impacts and coping strategies of populations depends heavily on their socio-economic status, socio-cultural norms, access to resources, poverty as well as **gender**. Research has also provided more evidence that the effects are not gender neutral, as women and children are among the highest risk groups. Key factors that account for the differences between women's and men's vulnerability to climate change risks include: gender-based differences in time use; access to assets and credit, treatment by formal institutions, which can constrain women's opportunities, limited access to policy discussions and decision making, and a lack of sex-disaggregated data for policy change³⁷. See Annex 8 Gender Assessment and Action Plan for an analysis of the current conditions for women and vulnerable groups in the Indus Basin and the proposed approach to reduce their climate vulnerability.
- Countries in South Asia, including Pakistan, will experience an estimated 17–36 million internal climate migrants by 2050 as a result of slow-onset climate changes. Under all scenarios, the poorest and most climate-vulnerable communities are likely to be hardest hit. Without significant adaptation action, beyond 2050 the climate-induced **migration** rate is likely to accelerate considerably. It is expected that 'hotspots' of in and out-migration are likely to form. The large majority of migrants are expected to come from communities dependent on drought- and flood-vulnerable rain-fed croplands for their livelihoods³⁸.

Observed climate change and noted impacts

19. Climate change is already occurring in Pakistan. Temperatures are estimated to have risen by 0.57°C over the twentieth century, with recent warming accelerating by 0.47°C between 1961 and 2007. These increases are most noticeable in the winter and post-monsoon months of November to February. The southern regions of the country are experiencing the most warming. As maximum temperatures have risen, there has also been an increase in the frequency of heat wave days.

³² Ibid.

³³ Ibid.

³⁴ Kjellstrom, T., Briggs, D., Freyberg, C., Lemke, B., Otto, M., Hyatt, O. (2016). Heat, human performance, and occupational health: A key issue for the assessment of global climate change impacts. *Annual Review of Public Health*: 37: 97–112. Available at: <https://www.annualreviews.org/doi/abs/10.1146/annurev-publhealth-032315-021740>

³⁵ Ahmed, S., Gautam, M. (2013) Agriculture and water policy: Towards sustainable inclusive growth. World Bank Policy Paper Series. Available at: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/57029146808332554/agriculture-and-water-policy-toward-sustainable-inclusive-growth>

³⁶ Climate Risk Country Profile: Pakistan (2021): The World Bank Group and the Asian Development Bank. Available at: https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf

³⁷ World Bank Group (2016). Gender Equality, Poverty Reduction, and Inclusive Growth. Available at: <http://documents1.worldbank.org/curated/en/820851467992505410/pdf/102114-REVISED-PUBLIC-WBG-Gender-Strategy.pdf>

³⁸ Rigaud, K., de Sherbinin, A., Jones, B., Bergmann, J., Clement, V., Ober, K., Schewe, J., Adamo, S., McClusker, B., Heuser, S., Midgley, A. (2018). Groundswell: Preparing for internal climate migration. World Bank Group, Washington DC. URL: <https://openknowledge.worldbank.org/handle/10986/29461>

20. Precipitation in Pakistan's historical profile is significantly more complex. Both the annual volumes and the SPEI drought indexes show long term oscillations. The early twentieth century saw a prolonged decline in annual rainfall, but since 1960, there has been a slight increase. This overall trend conceals significant sub-national variation. Since 1960, mean rainfall in Pakistan's arid plains and coastal belt has decreased by 10% to 15%. The majority of other regions have seen a slight increase, both during the monsoon and dry seasons. Since 1960, the number of heavy rainfall events has increased, with the nine heaviest rains recorded in 24 hours occurring in 2010.
21. Rainfall trends dating back to the 1980s show general increases in recent years (since 2000) compared to the lower volume years of the 1980s and 1990s (Figure 2). While there is interannual variability, there are clusters of dry years in the 1990s, some of which are consecutive, as well as clusters of wetter years in the 2000s, especially in the 2010s. These trends of +30 mm per decade are statistically significant over this time frame, but longer timeframes indicate an oscillation of rainfall characteristics.

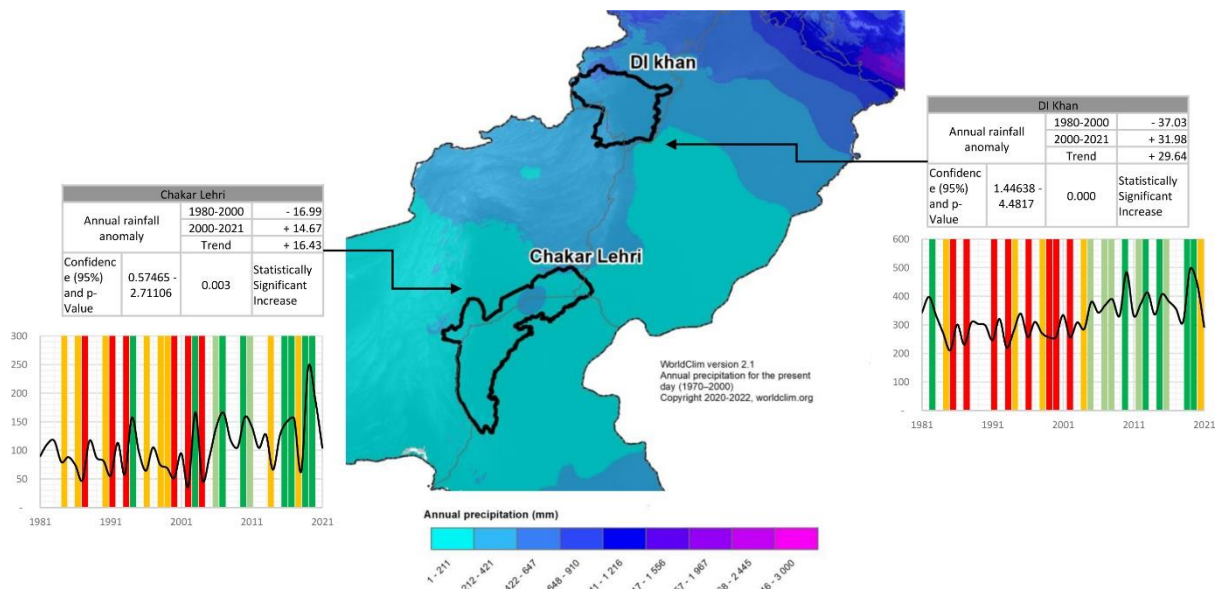


Figure 2. Observational annual rainfall climate 1985-2015 (map), longitudinal rainfall (line) and rainfall classification (bars) – dry year <1.0 standard deviation (red), dry year <0.5 standard deviation (orange), wet year >0.5 standard deviation (light green), wet year >1.0 standard deviation (green) for DI Khan and Chakar Lehri.^{39,40}

22. There are more extreme rainfall events in the northern and eastern parts of the country, and fewer in the southern and western parts of the country (Figure 3). This corresponds to the spatial patterns of the observed annual rainfall volumes. The observational record shows that the number of days exceeding their local 95th percentile daily volume per year has increased over time. There is also an increase in the number of days with more than 10 mm of rain in 24 hours. Both are indicators of larger-scale events that could result in flooding.

³⁹ Station data from the Pakistan Meteorological Department (PMD) for selected stations near project site, <https://www.pmd.gov.pk/en/>, obtained 25 May 2022

⁴⁰ Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>

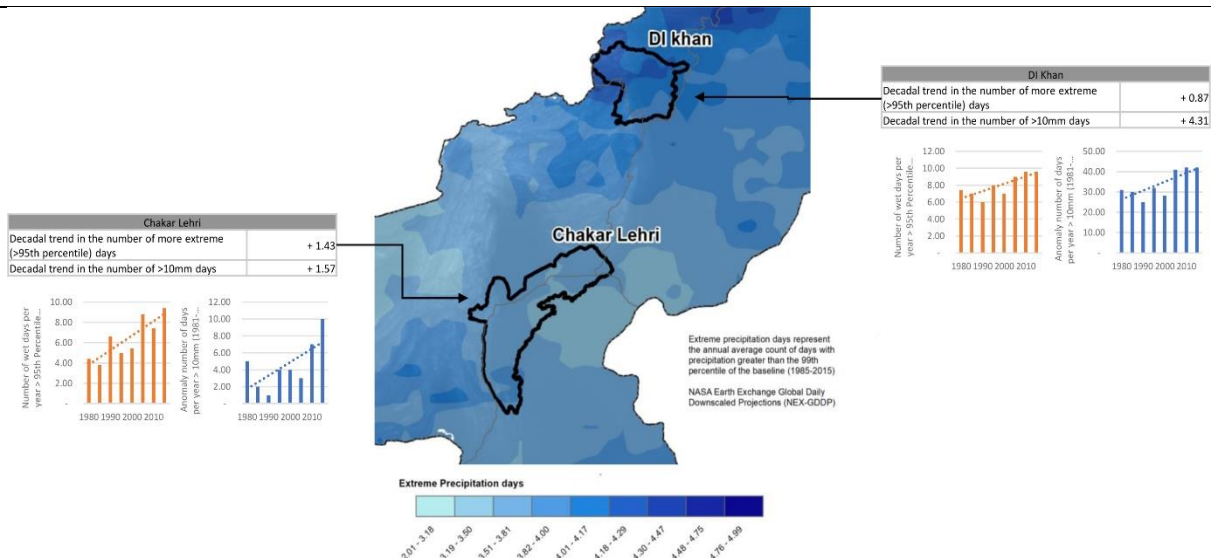
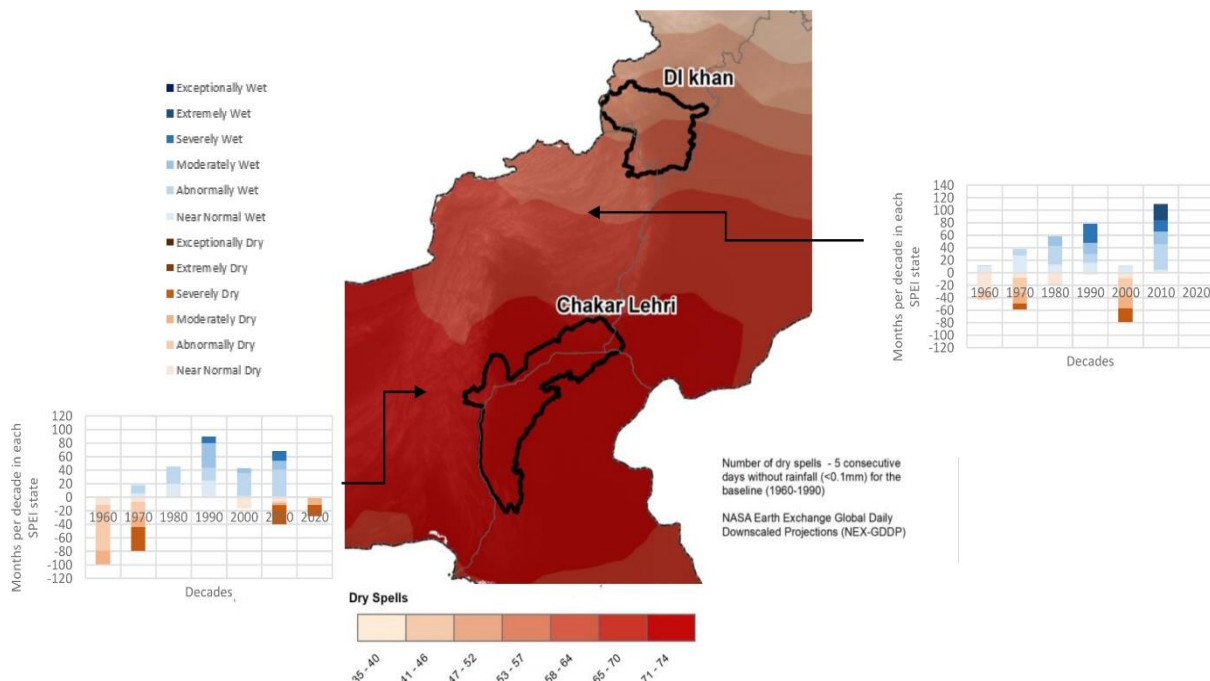


Figure 3. Observational annual extreme rainfall days 1985-2015 (map), change in the number of days exceeding 95th percentile rainfall (orange column graph) and change in the number of days exceeding 10mm (blue column graph) for DI Khan and Chakar Lehri.^{41,42}

23. There are numerous dry spells (5 consecutive dry days) across the country, with the majority concentrated in the south and fewer in the north (Figure 4). This is reflected in lower annual rainfall volumes in the south and higher volumes in the north. The 48-month SPEI classification shows a time-dependent oscillation in the leaning of wet and dry years. The earlier parts of the record have a more even distribution of wet and dry months in each decade. When we look at recent decades, we see that there are strong leanings one way or the other, and the decade is less balanced than previously noted. There is also a decrease in the number of months within the normal SPEI range, indicating that the index is becoming more extreme in one direction or the other. The more recent decades have also shown a trend toward wet years, though there are still some dry months.



⁴¹ Station data from the Pakistan Meteorological Department (PMD) for selected stations near project site: <https://www.pmd.gov.pk/en/> obtained 25 May 2022

⁴² Historical Raster data obtained from WorldClim version 2.1: <https://worldclim.org/>

Figure 4. Observational annual dry spells 1985-2015 (map), change in monthly SPEI classification representing drought index for DI Khan and Chakar Lehri.^{43,44}

24. On the ground, these climate changes have had a significant impact and these natural disasters, which already disrupt people's lives and the economy, are expected to become more frequent and intense (Figure 5). Flooding is the most serious threat to most people and is rated at 8.9 out of 10 in the INFORM 2019 Index for Risk Management for Pakistan of risk. In 2010, floods affected 20 million people and caused an estimated US\$10 billion in damage. They were followed in 2011 by heavy monsoon rains that affected 9.7 million people. In recent years, this trend has affected an increasing number of people. A significant number of people are also affected by droughts, ranked at 5.1 out of 10, though the distinction between being affected by drought and living in arid conditions is difficult to draw. Storm events' impacts have also grown over time, with more people affected in recent years. There is no clear trend in the changes associated with landslides; however, as seen in the 2010 events, there is a correlation between landslides and flooding events. It is reasonable to expect increased flooding activity to result in increased rainfall-induced landslides.

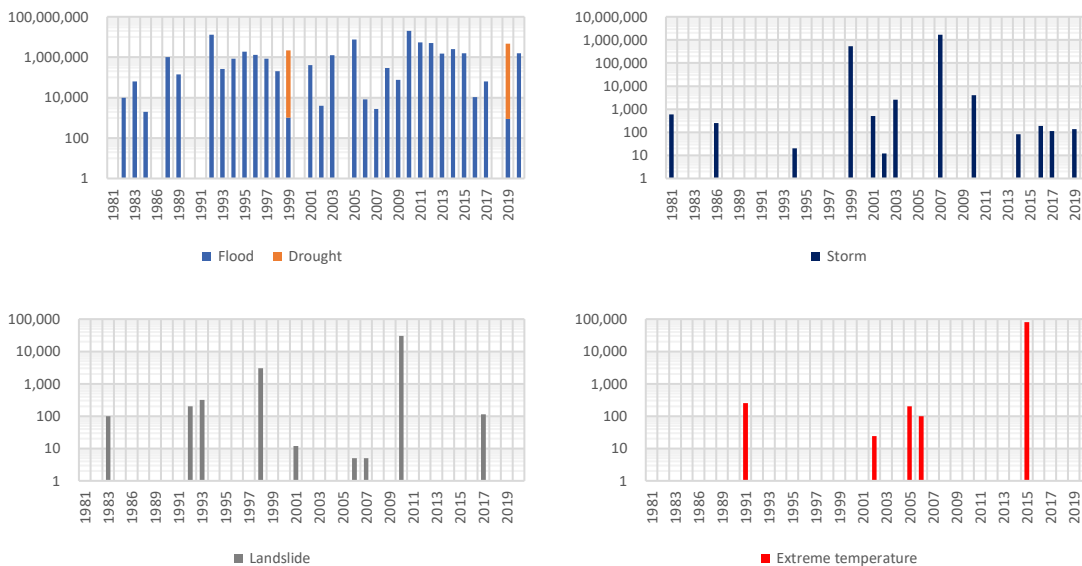


Figure 5. Number of people affected from hydro-meteorological hazards in Pakistan (1980-2020) for flood and drought (top left), storms (top right), landslides (bottom left), and extreme temperatures (bottom right).

25. The 2010 “super-floods” mentioned above inundated large swaths of Pakistan, resulting in the displacement of more than 20 million people. This flood event was called one of the worst humanitarian disasters ever experienced by the country. Twelve years later, in 2022, an even more significant massive flooding event caused by monsoon rains, resulted in such devastation that United Nations Secretary-General Antonio Guterres referred to the inundation as being at an “epochal level”⁴⁵. The 2010 floods began in late July, resulting from heavy monsoon rains in Khyber Pakhtunkhwa, Sindh, and parts of Balochistan. Approximately 20% of the land area of Pakistan and over 20 million people were affected. Total loss of lives stood at 1,985 with Sindh the worst hit, followed by Punjab. The 2022 floods, by comparison, have affected a larger proportion of Pakistan with over 33 million people affected (Figure 6). About 81 districts in Sindh, Punjab, Balochistan, and Khyber Pakhtunkhwa have been officially notified as ‘calamity hit’ with close to 1,400 dead since June 2022, according to data released by the National Disaster Management Authority (NDMA)⁴⁶. 2022’s monsoon season and subsequent flooding are unprecedented in comparison to the disasters of 2010. The consequences of the 2022 floods are far-reaching and unprecedented as they have engendered the imminent threat of food insecurity, water-borne diseases, malnutrition, and social unrest in Pakistan.

⁴³ Raster data obtained from NASA earth exchange global daily downscaled projections (NEX - GDDP) - <https://www.nasa.gov/nex/gddp>

⁴⁴ SPEI data obtained from Global SPEI database - <https://spei.csic.es/index.html>

⁴⁵ Aljazeera, Are catastrophic floods Pakistan's new normal? <https://www.aljazeera.com/news/2022/9/1/pakistans-floods-what-caused-the-floods>

⁴⁶ ProPakistani, 2010 vs 2022: Which Flood Wreaked More Havoc in Pakistan <https://propakistani.pk/2022/09/13/in-numbers-how-damages-due-to-2022-floods-compared-to-2010-floods/>

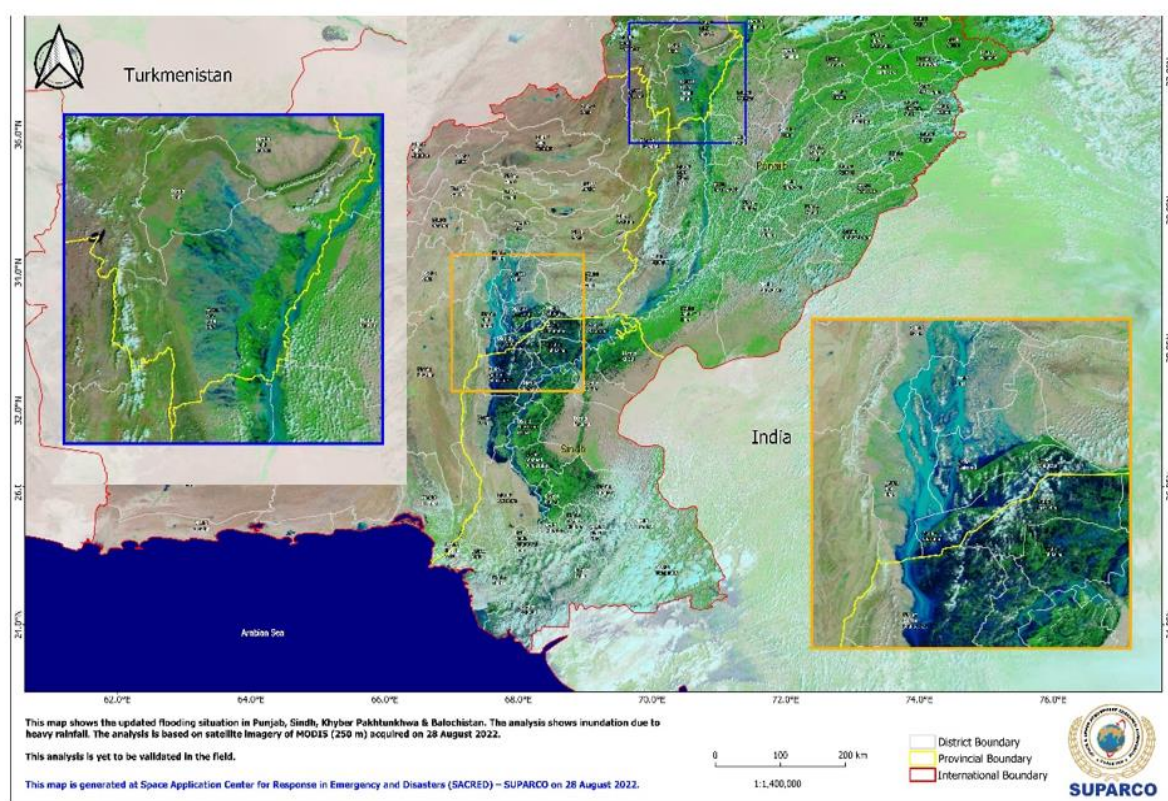


Figure 6. Updated flooding situation in Punjab, Sindh, Khyber Pakhtunkhwa and Balochistan (28 August 2022).

26. The quantification of the damages from extreme hydrometeorological events is difficult to assess, however estimates for the period 1900-2020 are provided from the Emergency Events Database (EM-DAT) in Table 1.

Table 1. Summary of Natural hydro-meteorological Hazards in Pakistan from 1900 to 2020.

	Total Deaths	Total Affected	Total Damage ('000 US\$)
Drought	143	2 200 000	247 000
Heat wave	2 936	80 600	18 000
Flash flood	3 590	22 114 300	10 184 100
Riverine flood	9 229	34 967 400	9 727 000
Other flood	5 286	23 863 300	2 670 000
Landslide	222	29 700	18 000
Mudslide	16	-	-
Convective storm	402	1 900	-
Tropical cyclone	11 555	2 599 900	1 715 000
Other storms	184	3 000	-

Projected climate baseline and extremes

27. Pakistan is already an arid/semi-arid country prone to recurring large-scale floods, droughts, and landslides, and future climate conditions are expected to exacerbate these effects. Over the different scenarios from the 1986-2005 baseline, Pakistan is expected to warm faster than the global average, with increases ranging from 1.3°C to 4.7°C by the 2060s and up to 6.0°C by 2090. There is also spatial uncertainty in Pakistan's rainfall, with models predicting decreases ranging from 20% to 41% by 2090. However, there is an increase in the likelihood of drought, increased evapotranspiration, and higher temperatures. There is also an increase in the frequency and intensity of extreme climate events, such as higher rainfall days, rainfall intensity, flooding, and the duration and intensity of heat waves. The number of people affected by extreme river flooding is expected to rise by 5 million between 2035 and 2044.

By 2070-2100, sea levels are expected to rise by 30-80 cm, exposing an additional 1 million people to coastal flooding⁴⁷.

28. The project focuses on the two catchments of the Indus River basin: DI Khan, and Chakar Lehri. Climate assessments were conducted in these areas to assess future climate exposure. Because of the size of the Chakar Lehri catchment, different climate change signals may be present. The evaluation was divided into two parts: northern and southern.
29. Rainfall patterns vary spatially, with a decrease in the far west and patches of northern areas and an increase in the southern and central northern areas (Figure 7). While each catchment has a different seasonal character, most locations see an increase in the July/August months, with small decreases in the early parts of the year resulting in a potential focusing of rainfall into these already higher volume months. This is visible in the accumulation anomaly, which shows small decreases in early-onset rainfall volumes before volumes increase mid-year and end with positive accumulation in the studied areas. Total rainfall volume increases less in the years 2020-2050 than in the years 2050-2080. The December-January-February season also sees the greatest increase in rainfall variability, implying that rainfall cessation volumes will be more volatile.

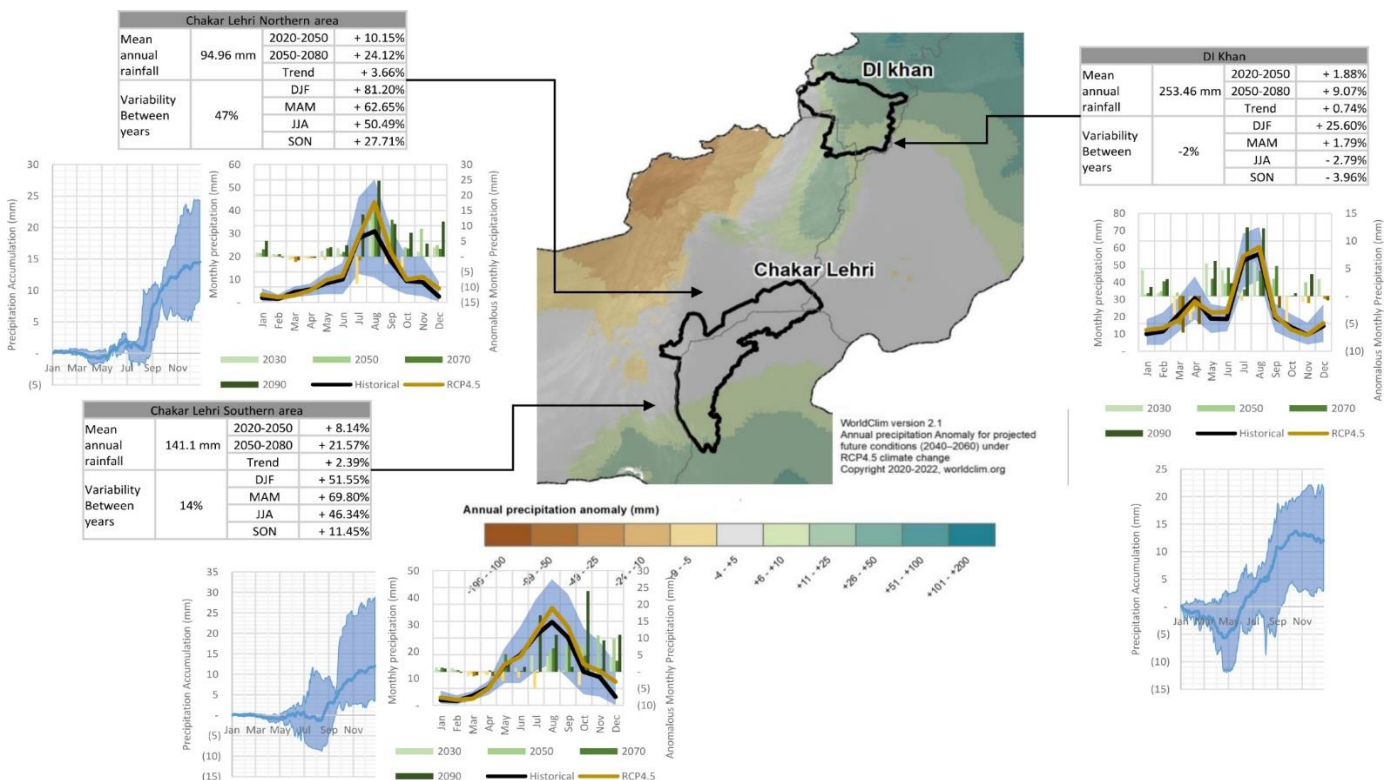


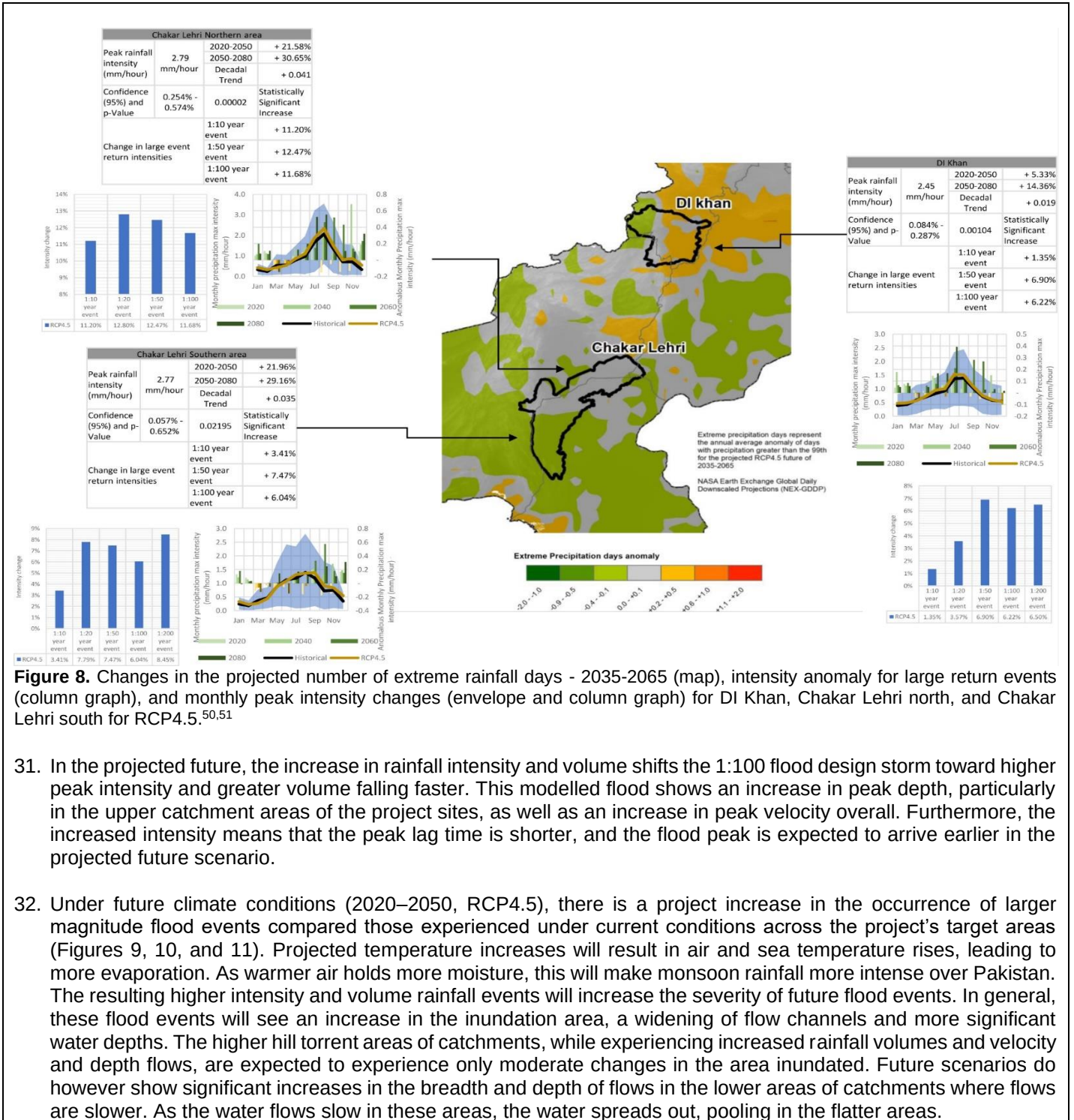
Figure 7. Changes in projected annual rainfall - 2040-2060 (map), monthly rainfall anomaly (envelope and column graph), and daily rainfall accumulation (envelope graph) for DI Khan, Chakar Lehri north, and Chakar Lehri south for RCP4.5.^{48,49}

30. The observed concentration of rainfall in the mid-late part of the year, combined with the projected increase in temperature, results in more atmospheric moisture and increased convective activity (Figure 8). The number of extreme rainfall days is increasing, primarily in northern parts of the country where ocean mitigation is less effective. The number of extreme rainfall days decreases very slightly in the southern areas. However, there is an increase in average peak hourly rainfall intensity in most locations, with the focus once again being on the July/August months. There is also an increase in the early part of the year, but it is milder compared to the later part of the year. The increased intensity of rainfall also increases the magnitude of large flood return events in all locations.

⁴⁷ World Bank Group, Climate Change Knowledge Portal https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf

⁴⁸ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁴⁹ Projected Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>



⁵⁰ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁵¹ Projected Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>

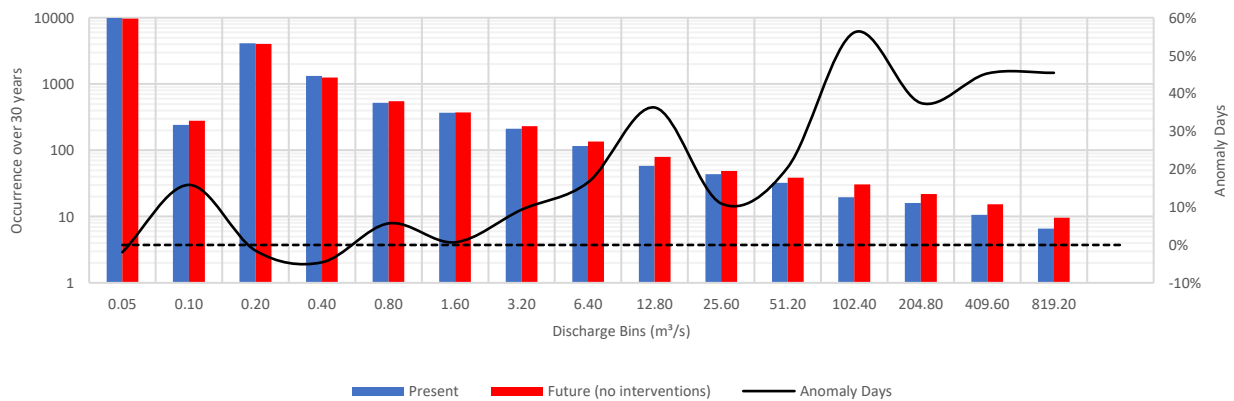
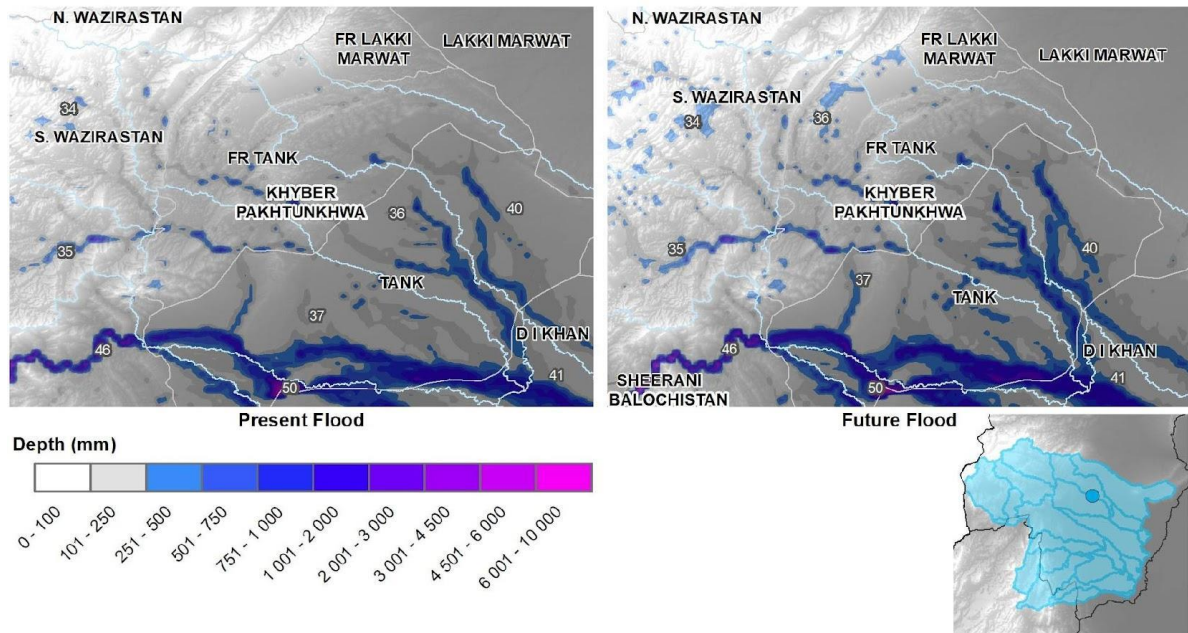
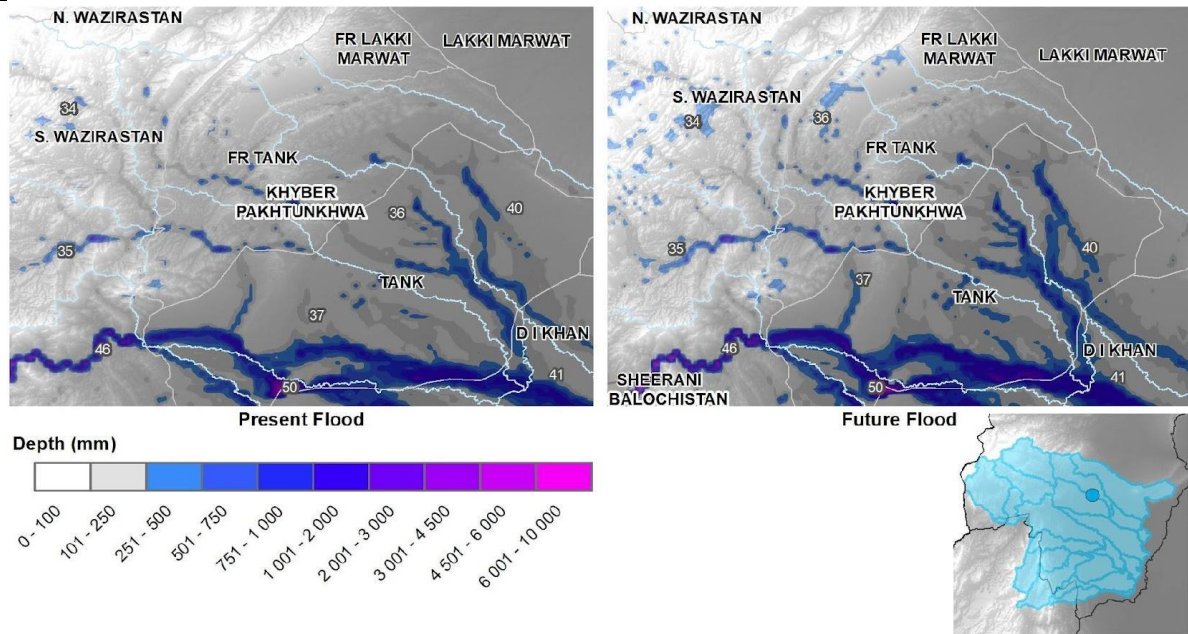
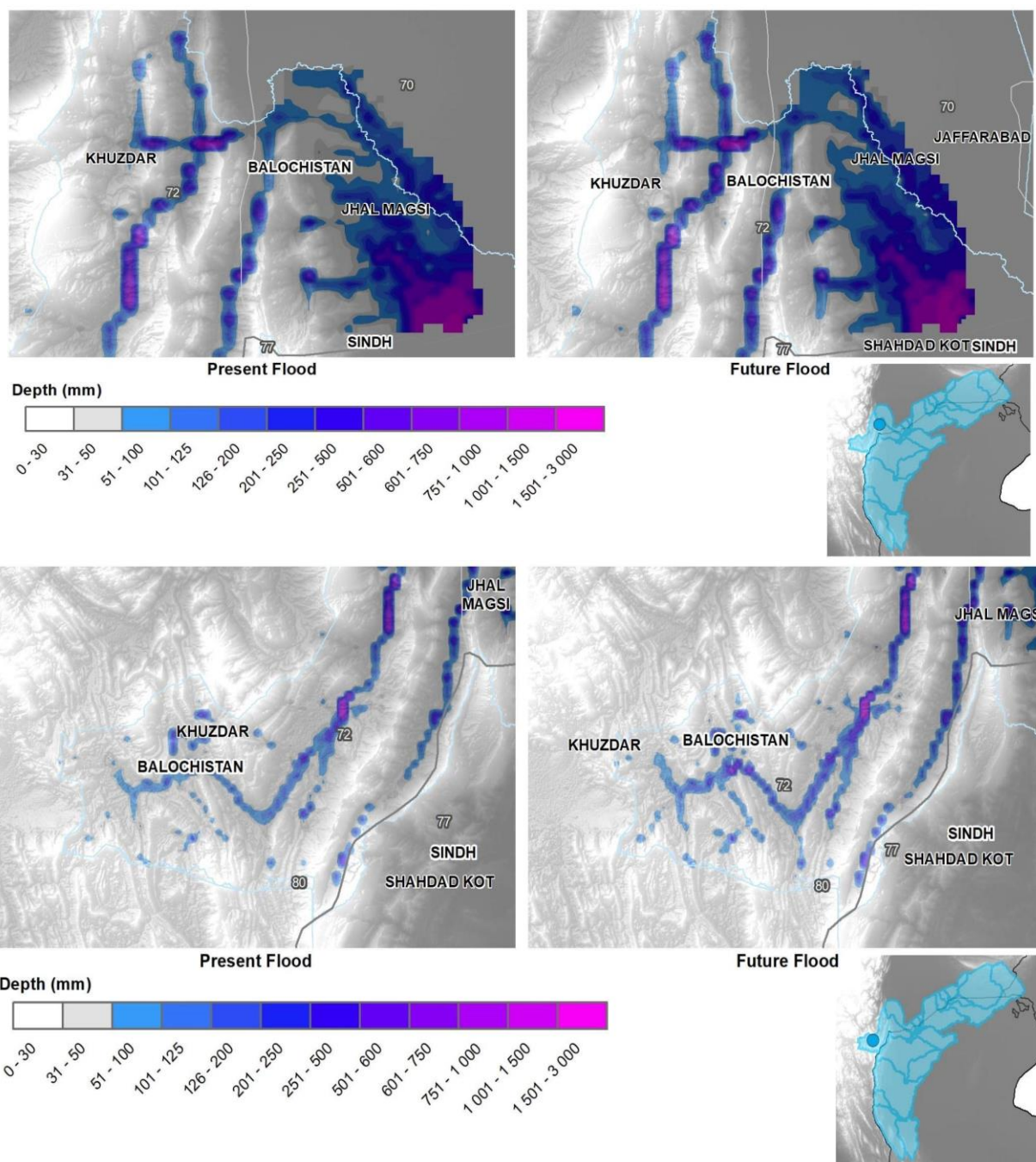


Figure 9. Flood inundation under current climate conditions (left) and future projected (2020-2050 RCP4.5) climate change conditions (right) for the northern areas of DI Khan (top), and southern areas of DI Khan (middle), and average hydrograph histogram of present and future flood discharge events and occurrence anomaly (bottom).⁵²



⁵² Output of RRI modelling

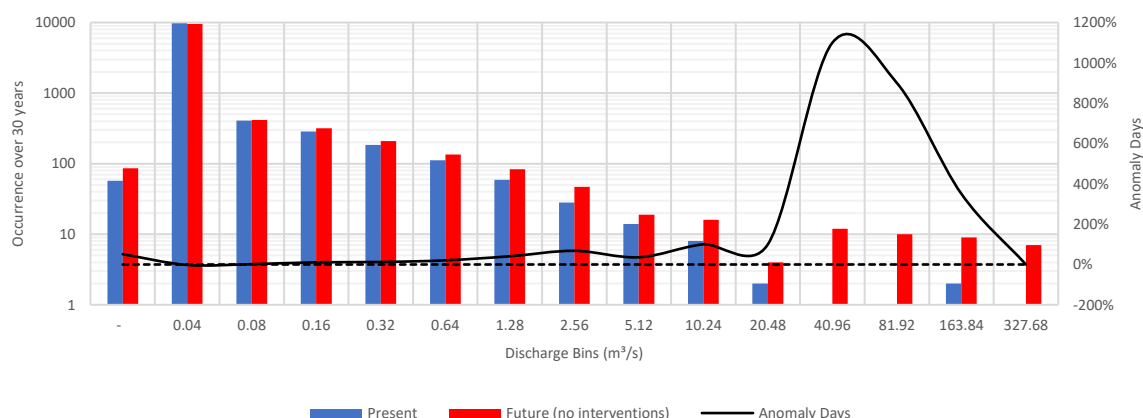


Figure 10. Flood inundation under current climate conditions (left) and future projected (2020-2050 RCP4.5) climate change conditions (right) for Balochistan northern areas (top) and Balochistan southern areas (middle) at the discharge outlet assessment point, and average hydrograph histogram of present and future flood discharge events and occurrence anomaly (bottom).⁵³

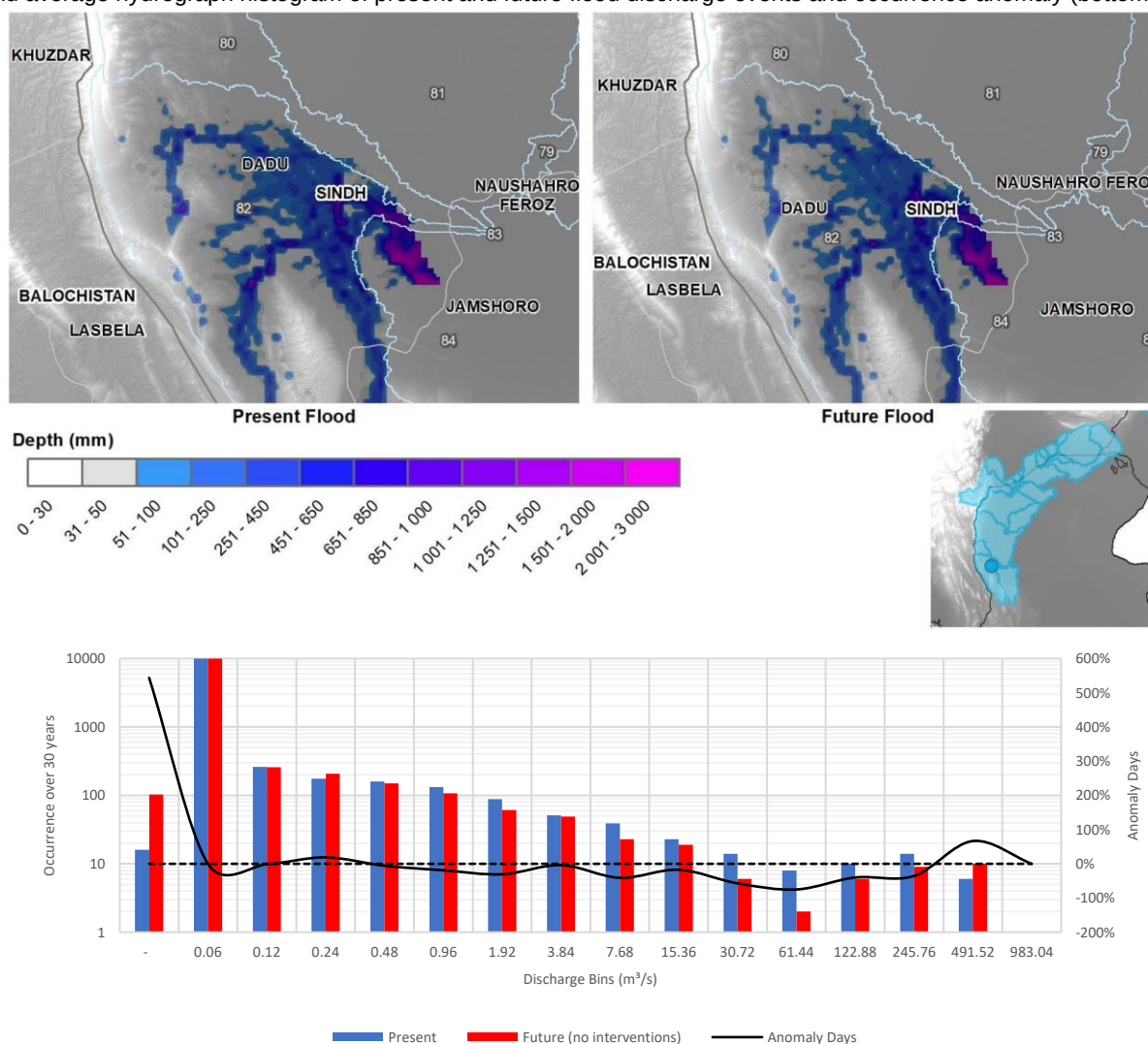


Figure 11. Flood inundation under current climate conditions (left) and future projected (2020-2050 RCP4.5) climate change conditions (right) for Manchar, and average hydrograph histogram of present and future flood discharge events and occurrence anomaly (bottom).⁵⁴

⁵³ Output of RRI modelling

⁵⁴ Output of RRI modelling

33. Despite expected but variable increases in annual rainfall, there is a focus of rainfall in shorter timeframes and thus an increase in the number of dry spells per year (Figure 12). This is most noticeable in the north, but there is a positive anomaly running northeast to the southwest. The number of dry spells in the eastern, far northern, and southern areas decreases slightly each year. However, this is only true in the medium-term projected future; there is significant uncertainty near the end-of-the-century scenario. The locations show an annual change in dry days ranging from -1.15 to +4.03. In comparison to the current climate, the long-term SPEI index as a proxy for drought shows a very clear tendency toward negative (dry) years. Droughts are expected to become more severe and frequent across the globe over the next century.

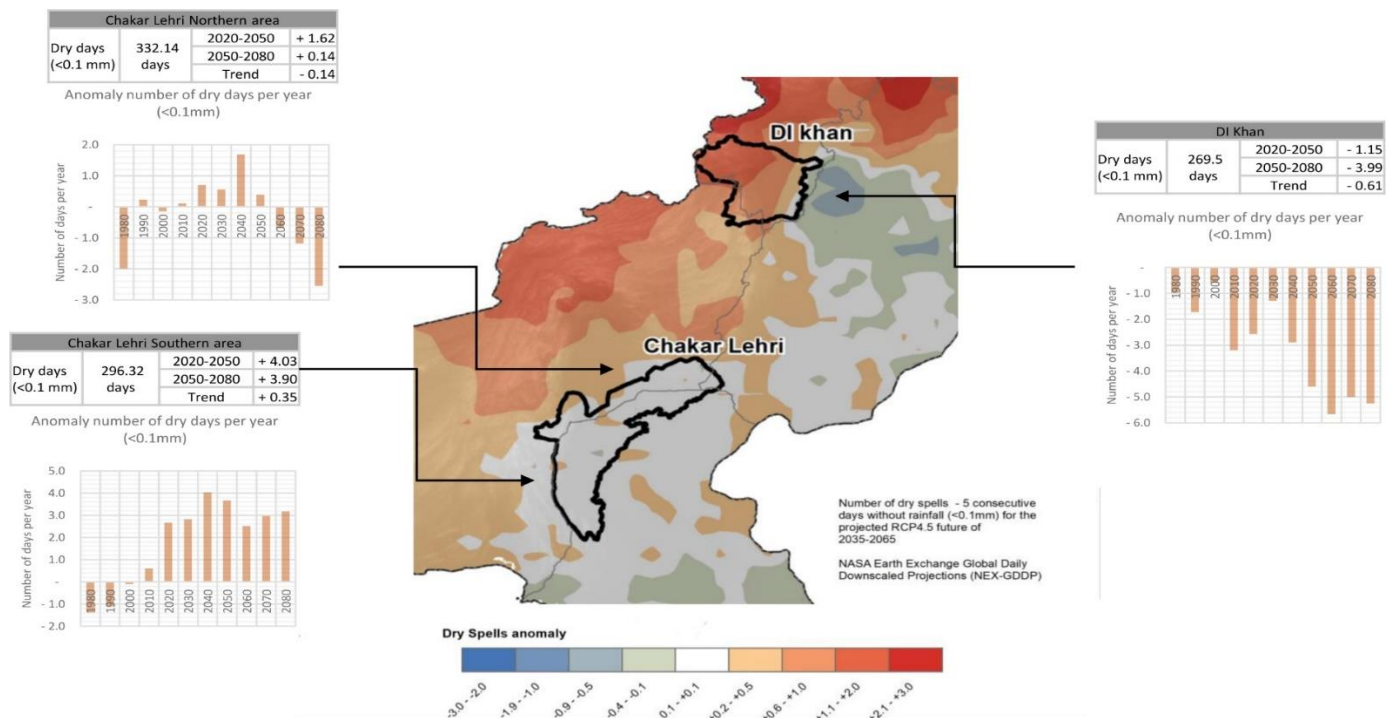
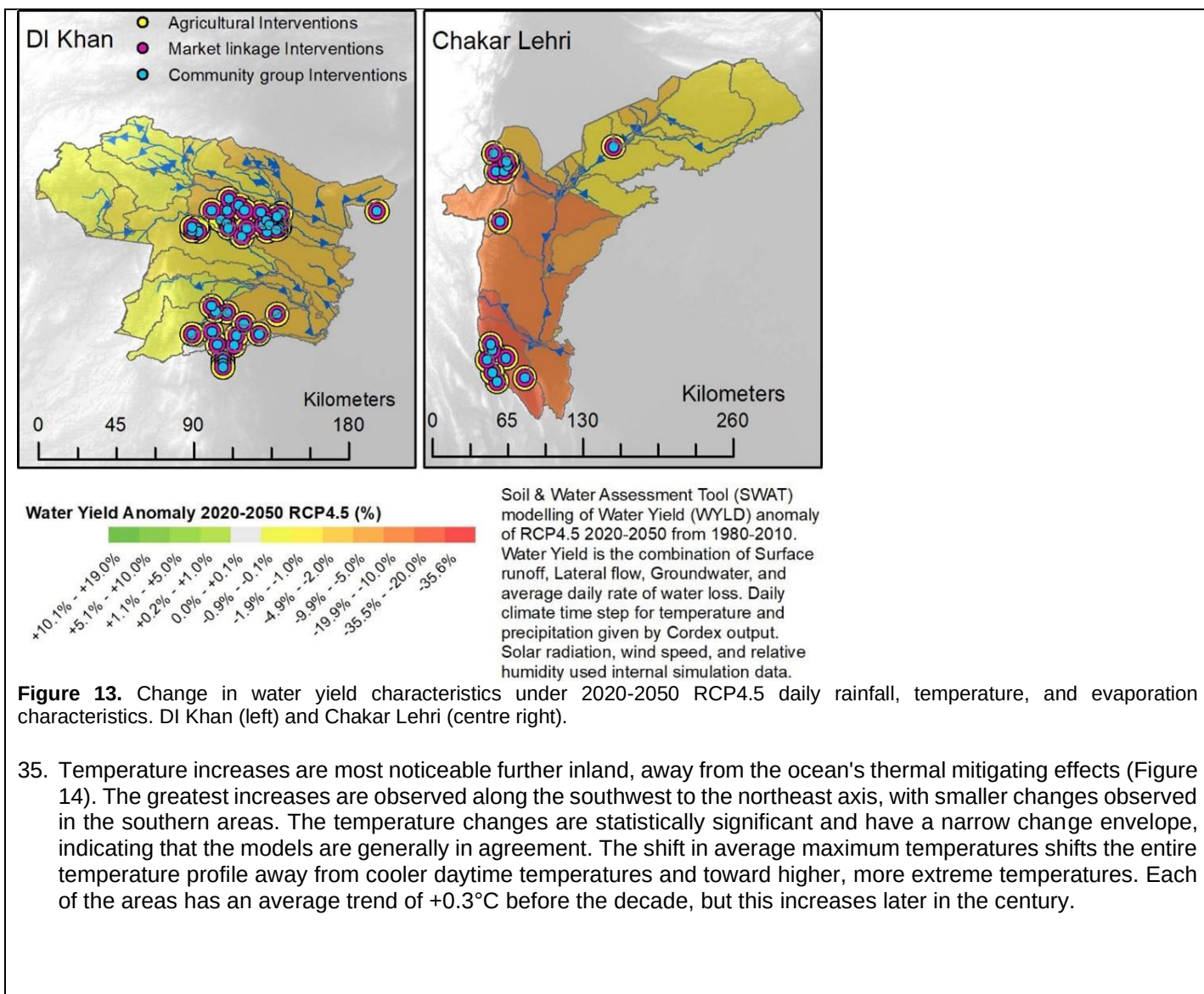


Figure 12. Changes in the projected number of dry spells - 2035-2065 (map), decadal anomaly number of dry days annually (column graph), and long-term SPEI (standardized evapotranspiration index) (line graph) for DI Khan, Chakar Lehri north, and Chakar Lehri south for RCP4.5.^{55,56}

34. The projected future shows variable annual rainfall volume, but more concentrated rainfall with longer dry spells and more dry days per year (Figure 13). Furthermore, an increase in maximum and minimum temperatures, as well as an increase in evaporation and evapotranspiration, affects surface runoff, lateral flow, groundwater, and the average daily rate of water loss. For RCP4.5 2020-2050, there is a general decrease in water yield, particularly in the lower reaches of the catchments, compared to 1980-2010 values.

⁵⁵ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁵⁶ Projected Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>



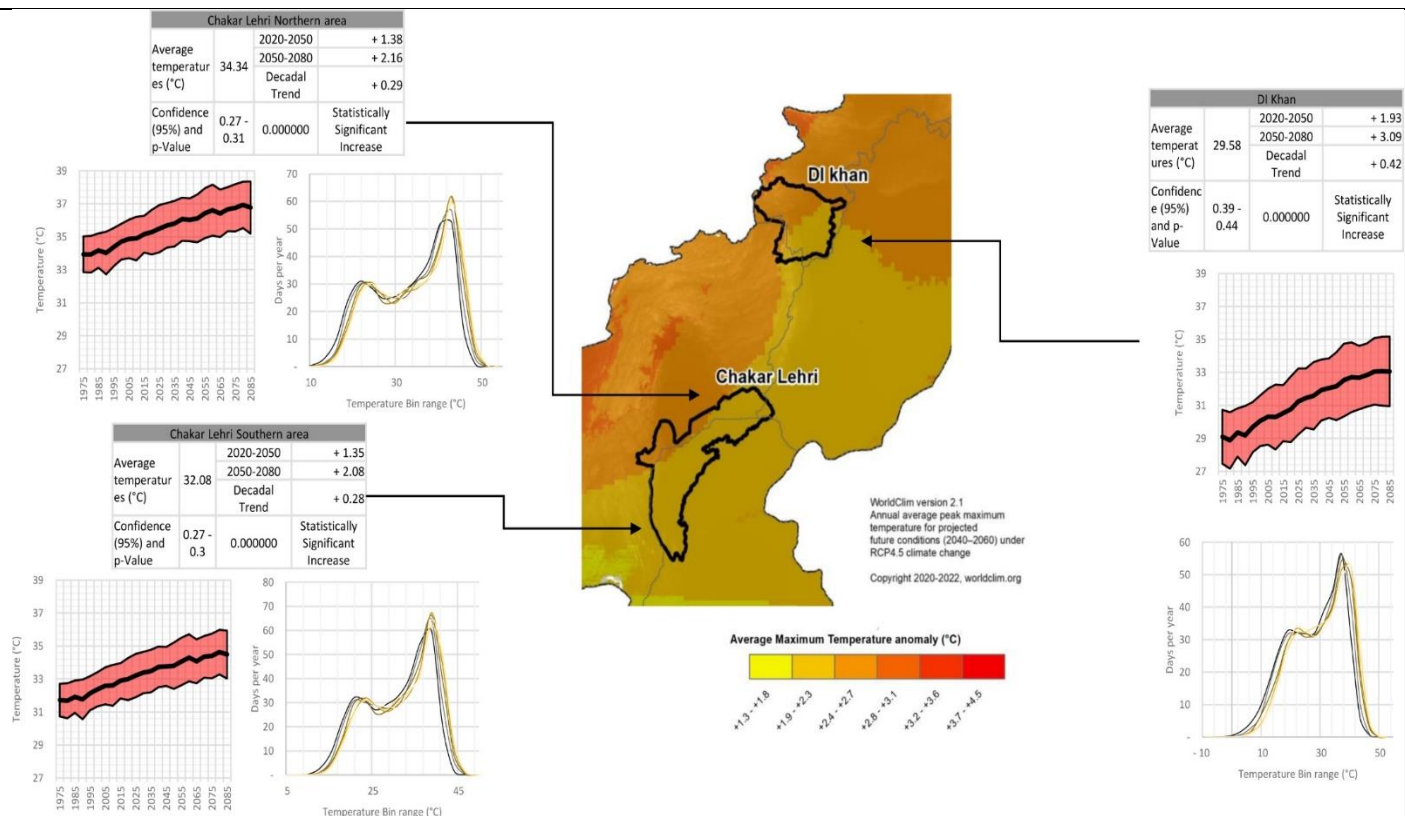


Figure 14. Changes in projected annual average peak maximum temperature - 2040-2060 (map), long-term maximum temperature regression (envelope graph), and maximum temperature occurrence profile (line graph) for DI Khan, Chakar Lehri north, and Chakar Lehri south for RCP4.5.^{57,58}

36. The shift from cooler to hotter and more extreme temperatures results in an increase in the number of extreme temperatures over most of the country each year. The number of extreme temperature days has increased the most in the already hot south-west of the country and the least in the cooler areas to the north. The number of extremely hot days (+40°C) has increased significantly in all locations. By mid-century, the number of heatwaves will nearly double, and these heatwaves will be several degrees warmer than current heatwaves.

⁵⁷ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁵⁸ ProjectedRaster data obtained from WorldClim version 2.1 - <https://worldclim.org/>

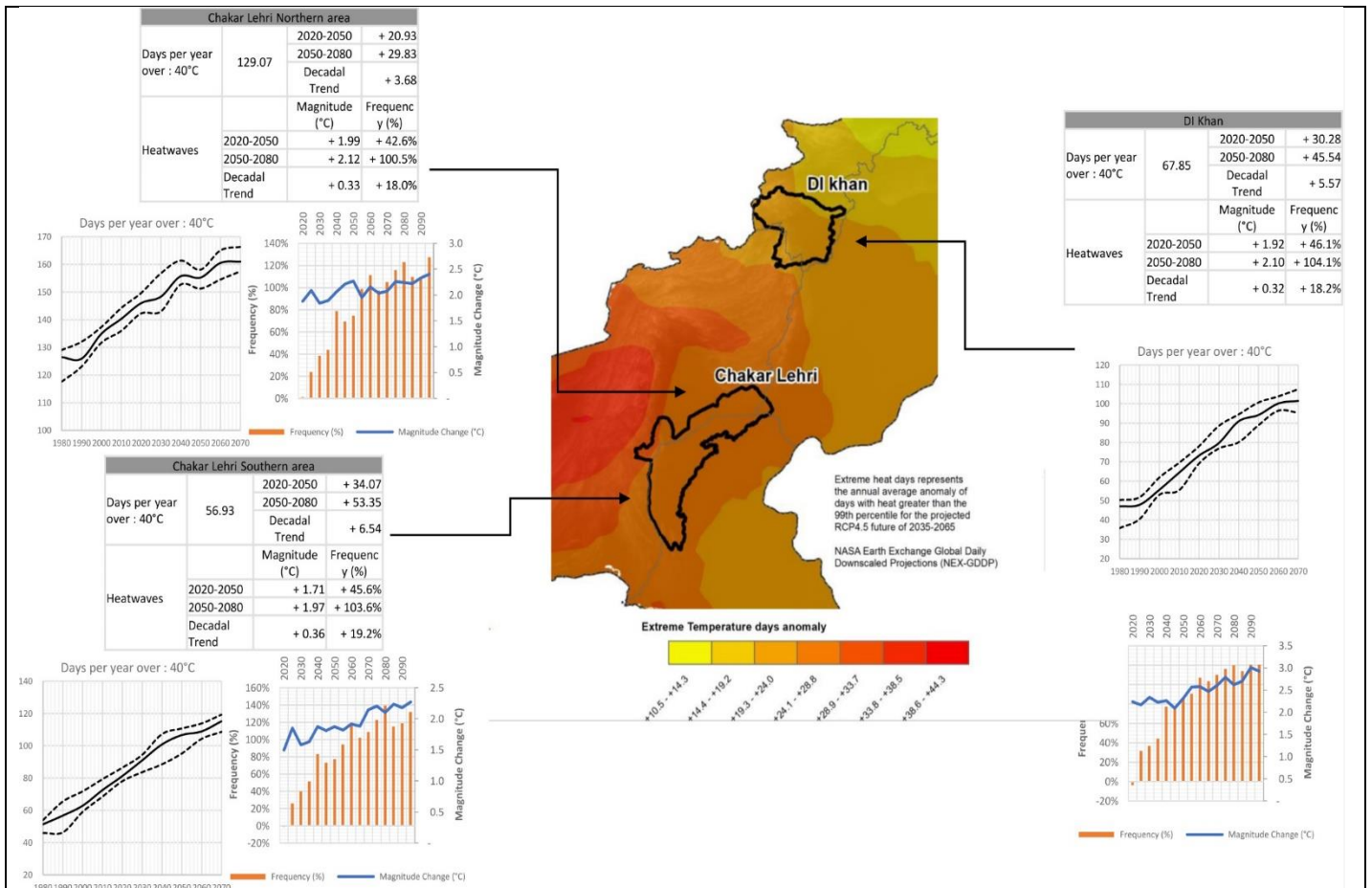


Figure 15. Changes in projected extreme heat days - 2035-2065 (map), number of days over 40°C (line graph), and heatwave frequency and magnitude changes (line and column graph) for DI Khan, Chakar Lehri north, and Chakar Lehri south for RCP4.5.^{59,60}

37. There is a larger increase in minimum temperatures to the northeast, which is the coldest part of the country, and a smaller increase to the southwest (Figure 16). The increases seen in maximum temperatures are also seen in minimum temperatures, with near linear increases over the next century and majority model agreement. The rate of increase of minimum temperatures, on the other hand, is greater than that of maximum temperatures. While both maximum and minimum temperatures are rising, the fact that minimum temperatures are rising faster results in a reduction in the diurnal range. This will change the country's thermal interchange character.

⁵⁹ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁶⁰ Projected Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>

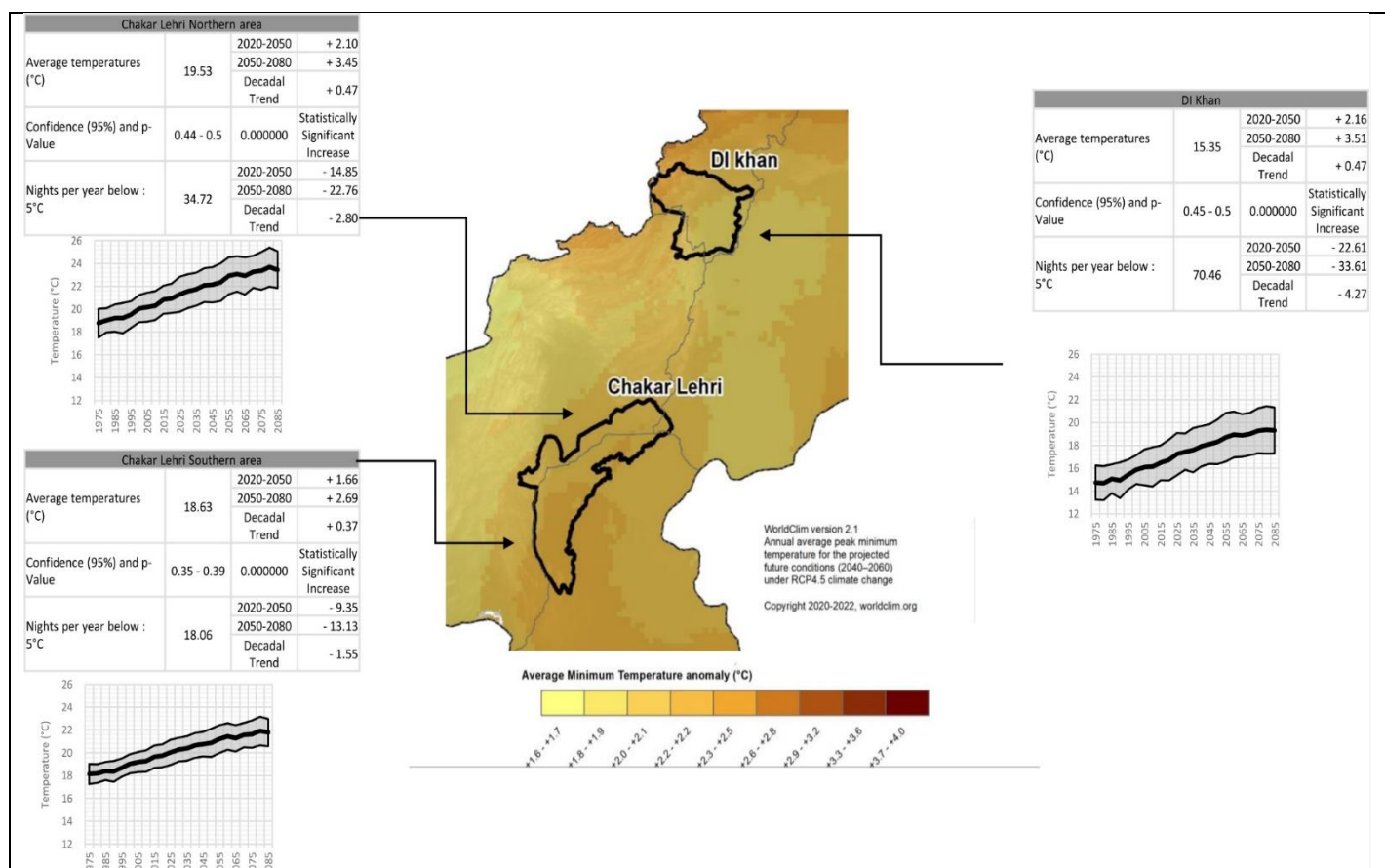


Figure 16. Changes in projected annual average peak minimum temperature - 2040-2060 (map), long-term minimum temperature regression (envelope graph) for DI Khan, Chakar Lehri north, and Chakar Lehri south for RCP4.5.^{61,62}

Most likely scenario without any climate action

38. Current climatic conditions in Pakistan indicate an increase in temperatures, with projected changes indicating that this trend will continue in the future. There will most likely be an increase in the number of severe heat waves, similar to those observed in 2015 and 2022, but more frequently and potentially with greater magnitude. Evaporation and evapotranspiration will most likely increase, further drying out arid and semi-arid areas, especially given the uncertainty of future rainfall trends. This will increase the likelihood of more severe dry months occurring more frequently. This would exacerbate the current drought's impacts and increase the frequency in occurrence and length of extreme droughts in the future.
39. Long-term cyclical annual rainfall trends are visible in Pakistan's current climatic conditions. The rainfall forecast is currently uncertain. While some models predict generally increased rainfall, there is a lot of variability in these. Rainfall is expected to remain highly variable, with the effects of multiple wet and dry years compounding. There is also a trend toward departing from the current "normal" rainfall volumes and SPEI month. This variability is likely to exacerbate drought potential, especially given the likelihood of increased maximum and minimum temperatures and extremely severe heating events.
40. The observed and projected rainfall intensities have both increased, a trend which is expected to continue. Rainfall will most likely concentrate during peak times, increasing the likelihood of large volume and high-intensity rainfall events, as was the case in 2010, resulting in major floods causing wide-reaching environmental, social and economic impacts. Return events of this type, and the floods associated with them, are more likely to occur in the future.
41. The climate impacts mentioned, particularly rainfall changes and the associated droughts and floods, will likely further expose natural resource-based sectors, particularly the most vulnerable poor and minority groups — such as those living in the flood- and drought-vulnerable Indus River Valley. If no cost-effective and sustainable

⁶¹ Christensen OB, Gutowski B & Nikulin G. 2012. CORDEX Archive Design, version 20/7/2012

⁶² Projected Raster data obtained from WorldClim version 2.1 - <https://worldclim.org/>

adaptation solutions, such as EbA, are implemented to strengthen the climate-resilience of vulnerable rural communities and the ecosystems that they depend on for their lives and livelihoods, the costs of the impacts on floods and droughts will continue to increase exponentially. These will include loss of life, infrastructural damages, loss of livestock, damage to and loss of crops, and reduced ecosystem integrity, ultimately worsening socio-economic conditions and increasing the vulnerability of Pakistan's most poor.

Adaptation solution

42. To respond to the impacts of increasingly severe floods and droughts on ecosystems and vulnerable communities in Pakistan, it is necessary to implement an integrated approach to watershed management. This approach should prioritise the use of EbA and green infrastructure interventions above the current paradigm of using grey interventions (such as dams) reactively to address the impacts of floods and droughts on vulnerable communities and ecosystems. Feasible and effective interventions under the proposed adaptation solution that will address past, present and expected future behaviour of the climate indicators described above are presented in Table 2 below. Barriers to the proposed adaptation solution and the project's approach to addressing them are presented in Section B.2.
43. The EbA and green infrastructure interventions selected by the project are inherently 'no-regret' approaches, focusing on maximizing positive and minimizing negative aspects of nature-based strategies and options for adaptation to floods and droughts. Such options will not worsen vulnerabilities to climate change, increase adaptive capacities and will have a positive impact on livelihoods and ecosystems regardless of the how the climate changes. The project has been designed (see Section B.3. and Annex 2 of detailed intervention designs) to avoid environmental, sociocultural, and economic maladaptation. *Environmental maladaptation* is prevented by activities that: i) avoid degradation that causes negative effects *in situ*; ii) avoid displacing pressures onto other environments (neighbouring areas or areas that are connected ecologically or socio-economically); iii) support the protective role of ecosystems against current and future climate-related hazards; iv) integrate uncertainties concerning climate change impacts and the reaction of ecosystem; and v) set the primary purpose as being to promote adaptation to climate-related changes rather than to reduce greenhouse gas emissions. *Sociocultural maladaptation* will be prevented by activities that: i) start from local social characteristics and cultural values that could have an influence on risks and environmental dynamics; ii) consider and develop local skills and knowledge related to climate-related hazards and the environment; and iii) call on new skills that the community is capable of acquiring. *Economic maladaptation* will be prevented by activities that: i) Promote the reduction of socio-economic inequalities; ii) support the relative diversification of economic and/or subsistence activities; and iii) integrate any potential changes in economic and subsistence activities resulting from climate change.

Table 2. Project EbA and green infrastructure approaches that will address droughts and floods (more detail is presented in Annex 2).

Climate change indicator	Adaptation intervention	Adaptation benefits
Flooding	Restoration of riverine vegetation	- Reduced peak stream flow^{63,64} - Reduced flood peaks and related damages to land, agriculture and infrastructure^{65,66,67,68} - Reduced soil erosion⁶⁹

⁶³ Vouk, I., Pilechi, V., Provan, M. and Murphy, E. (2021). Nature-Based Solutions for Coastal and Riverine Flood and Erosion Risk Management. csagroup.org. Available from https://www.researchgate.net/publication/355649746_Nature-Based_Solutions_for_Coastal_and_Riverine_Flood_and_Erosion_Risk_Management

⁶⁴ Evaristo, J. McDonnell, J.J (2019). Global analysis of stream flow response to forest management. *Nature*. 10.1038/s41586-019-1306-0, 570(7762). 455-461

⁶⁵ United Kingdom Eddleston Water (n.d). River restoration for natural flood management in Eddleston Water. Available from <https://casestudies.naturebasedsolutionsinitiative.org/casestudy/river-restoration-for-natural-flood-management-nfm-in-eddleston-water-scotland/pdf/>

⁶⁶ Smith, A. and Chausson, A. (2021) Nature-based Solutions in UK Climate Adaptation Policy. A report prepared by the Nature-based Solutions Initiative at the University of Oxford for WWF-UK and RSPB. Available at: <https://www.rspb.org.uk/globalassets/downloads/policy-briefings/nature-based-solutions-adaption-report.pdf>

⁶⁷ Nisbet, T. (n.d). Case study 27. Investigating the Impacts of Upland Land Use Management on Flood Risk at Pontbren, Wales. Available from <https://www.gov.uk/flood-and-coastal-erosion-risk-management-research-reports/working-with-natural-processes-to-reduce-flood-risk#case-studies>

⁶⁸ Thomas, H. And Nisbet, T. (n.d) .Case study 28. Investigating the impact of floodplain woodland on flood flows in the River Cary catchment, Somerset. Available from: <https://www.gov.uk/flood-and-coastal-erosion-risk-management-research-reports/working-with-natural-processes-to-reduce-flood-risk#case-studies>

⁶⁹ Karlen, D. L., Stott, D. E., Cambardella, C. A., Kremer, R. J., & McCarty, G. W. (2014). Surface soil quality in five Midwestern cropland conservation effects assessment project watersheds. *Journal of Soil and Water Conservation*, 69,393–401. <https://doi.org/10.2489/jswc.69.5.39>

Drought	Flow path restoration	• River height reduction during flood peaks resulting in reduced damages to land, agriculture and infrastructure⁷⁰ • Reduced flood peak magnitude and time⁷¹ • Reduced surface water runoff⁷²
	De-siltation of flow paths and wetlands	• Decrease in rate and volume of stormwater runoff, reducing the threat of damage to land, agriculture and infrastructure during extreme rainfall events⁷³
	Embankments (rock detention structures)	• Reduction in peak flows during rainfall events^{74,75} • Reduced sediment deposition
	Constructed embankments	• Flood peak reduction and flood-risk prevention^{76,77} • Floodwater retention and diversion⁷⁸
	Restoration of riverine vegetation	• Improved wetland water inflows and reduced losses⁷⁹
	Flow path restoration	• Reduction in drought severity by improving basal flows of rivers⁸⁰ • Increase in water table and river flow stability and discharge⁸¹
	Embankments (rock detention structures)	• Increase in total runoff⁸² • Extension of summer base flows and flow volume • Increased water tables and natural restoration of downstream wetlands⁸³

Related projects/interventions

44. Considerable baseline investments are being made through public expenditure and donor-funded initiatives to increase the resilience of Pakistan's vulnerable communities to the impacts of climate change. Lessons learned and best practices from these investments have been incorporated into the design of the proposed project to replicate successful techniques and ensure that there is complementarity between the project and existing actions. The most relevant baseline projects that will be complemented by the proposed project are presented in Table 3 below, with further projects and information provided in Annex 2.

⁷⁰ Climate – Adapt (2021). Room for the River Waal – protecting the city of Nijmegen. Available from https://climate-adapt.eea.europa.eu/metadata/case-studies/room-for-the-river-waal-2013-protecting-the-city-of-nijmegen/#adapt_options_anchor

⁷¹ Addy, S., Holden, M and Sargeant, L (n.d). Case study 1. New Forest LIFE III Project. Available from <https://www.gov.uk/flood-and-coastal-erosion-risk-management-research-reports/working-with-natural-processes-to-reduce-flood-risk#case-studies>

⁷² Vouk, L., Pilechi, V., Provan, M. and Murphy, E. (2021). Nature-Based Solutions for Coastal and Riverine Flood and Erosion Risk Management. csagroup.org. Available from https://www.researchgate.net/publication/355649746_NatureBased_Solutions_for_Coastal_and_Riverine_Flood_and_Erosion_Risk_Management

⁷³ Feibel, S. (2022). Removing Sediment to Restore Wetlands Clears Revenue Paths. Available from <https://www.conservationfinancenetwork.org/2022/01/24/removing-sediment-to-restore-wetlands-clears-revenue-paths>

⁷⁴ Norman, L. M. (2020). Ecosystem Services of Riparian Restoration: A Review of Rock Detention Structures in the Madrean Archipelago Ecoregion. Air Soil 28% more flow volume per area in the treated watershed compared to the untreated watershed. Water Res. 13. 117862212094633. doi: 10.1177/117862212094633

⁷⁵ USGS, (2022). Re-greening a dryland watershed. Available from <https://www.usgs.gov/media/videos/re-greening-dryland-watershed>

⁷⁶ Serra-Llobet, A., Jaehnig, S.C., Geist, J., Kondolf, G.M., Damm, C., Scholz, M., Lund, J., Opperman, J.J., Yarnell, S.M., Pawley, A., Shader, E., Cain, J., Zingraff-Hamed, A., Grantham, T.G., Eisenstein, W. and Schmitt, R. (2022). Restoring Rivers and Floodplains for Habitat and Flood Risk Reduction: Experiences in Multi-Benefit Floodplain Management from California and Germany. Front. Environ. Sci. 16. Available from <https://www.miragenews.com/flood-risk-reduction-with-multiple-benefits-775685>

⁷⁷ McIlwrath, C. (2016). Case study 10. Padgate Brook River Restoration – part of the Warrington FRM Scheme. Available from <https://thefloodhub.co.uk/wp-content/uploads/2018/09/WWNP-Case-Study-10-Padgate-Brook.pdf>.

⁷⁸ Climate Adapt (2021). Lower Danube green corridor: floodplain restoration for flood protection. Available from https://climate-adapt.eea.europa.eu/metadata/case-studies/lower-danube-green-corridor-floodplain-restoration-for-flood-protection/#adapt_options_anchor

⁷⁹ Construction 21 Luxembourg. (2019). Changyuan River Wetland Natural Park (China - Qixian, Shanxi). Available from <https://www.construction21.org/luxembourg/data/exports/pdf/changyuan-river-wetland-natural-park-china-qixian-shanxi.pdf>

⁸⁰ Wen, L., Rogers, K. Ling, J. (2011). The impacts of river regulation and water diversion on the hydrological drought characteristics in the Lower Murrumbidgee River, Australia. Journal of Hydrology 405(4-5). 1099-1100

⁸¹ Hejran, H. Mitusov, A. and Azami, A. (2021). Syako Karez recharge and flood control dam in Logar province, Afghanistan. Earth Sciences. 15 (191). 61-63

⁸² Norman, L.M., Brinkerhoff, F., Gwilliam, E., Guertin, D.P., Callegary, J., Goodrich, D.C., Nagler, P.L., Gray, F., (2016). Hydrologic Response of Streams Restored with Check Dams in the Chiricahua Mountains, Arizona. River Res. Applic. 32, 519–527. <https://doi.org/10.1002/rra.2895>

⁸³ Borderlands Restoration Network (2022). Cuenca Los Ojos Foundation. Available from <https://www.borderlandsrestoration.org/cuenca-los-ojos.html>

Table 3. Summary of the most relevant past and ongoing projects and investments that will be complemented by the proposed project.

Project	Description	Status
<u>Scaling-up of glacial lake outburst flood (GLOF) risk reduction in Northern Pakistan.</u> 2015—2022. USD 37.5m. Category B.	The objective of the project is to empower communities to identify and manage the risks associated with GLOFs and other related impacts of climate change and to strengthen the public service systems to lower the risks of GLOF-related climate disasters. Focused geographical boundaries are Gilgit-Baltistan (GB) and Khyber Pakhtunkhwa. The project is building 250 engineering structures including dams, ponds, spillways, tree plantations, and drainage to reduce the risks of GLOF from 33 glacial lakes. The project also includes the development of disaster management policies and introducing weather monitoring stations, flood gauges, hydrological modelling, and early-warning systems.	It is a GCF-funded adaptation project and is being implemented by UNDP. This project is benefiting 696,342 direct and 28.5m indirect beneficiaries.
<u>Transforming the Indus basin with climate resilience agriculture and water management.</u> 2019—2026. USD 47.7m. Category B.	The major objective is to transform agriculture in the Indus basin by increasing resilience among the most vulnerable farmers and strengthening the government's capacity to support their communities to adapt to climate change. This project is building farmers' climate resilience through skills development, and knowledge, and supporting them on technological fronts for creating a wider enabling environment for continuous adaptation. It will see the impact of climate change on agriculture and water management in selected basins through the modelling of climate change scenarios for Pakistan. While there is no overlap in geographies and participating stakeholders between the proposed project and this existing GCF-funded project, lessons from this ongoing project such as efficient and effective ways to drive public sector reform and sustainable and cost-effective climate-resilient agricultural techniques will be of value and used to inform the implementation of relevant activities under Components 2 and 3 of the proposed project. Additionally, the proposed project's PSC will support the PMU to, among others, maintain complementarity between the proposed project and this ongoing GCF-funded project, including through coordinating with the FAO-led project's PMU to ensure lessons learned from the impacts of the 2022 floods on their project's implementation are transferred to the proposed project's PMU from the first year of project implementation.	It is a GCF-funded adaptation project and is being implemented by FAO. This project is benefiting 1.3m direct and 16m indirect beneficiaries in 5 districts of Punjab (Muzaffargarh, DG Khan, Khanewal, Lodhran, Multan) and 3 districts of Sindh (Sanghar, Badin, Umerkot) ⁸⁴ .
Living Indus Initiative	Under the Living Indus Initiative, 25 high-impact projects/programmes have been identified through a series of extensive consultations with senior policy makers and experts; academic, civil society, business and technical specialists. The overall objective of the initiative is to revitalize and restore the ecosystem functioning of the Indus Basin under future climate change conditions using primarily EbA and nature-based solutions. The proposed project is identified as a main contributor to the Living Indus Initiative's objective of improving the resilience of the Indus Basin through its focus on implementing EbA and green infrastructure interventions to reduce the increasing impacts of floods and droughts on the basin and most vulnerable groups that depend on it. The close link between the proposed project and the Living Indus Initiative will enhance country ownership, facilitate the	This initiative is ongoing, developed jointly by the Ministry of Climate Change (MoCC) and United Nations in Pakistan under a directive of the Prime Minister's Committee on Climate Change, and led by the MoCC (NDA). The initiative is intended to serve as a strategic framework for Pakistan, coordinating all climate change adaptation work focused on enhancing the resilience of the country's Indus Basin.

	NDA and provincial government agencies in ensuring inter-alignment of related initiatives, and reduce potential for overlaps or redundancies in future proposals to the GCF or other sources of climate finance.	
<u>Balochistan small-scale irrigation project (BSSIP).</u> 2008—2014. USD 25m.	In 2008, the World Bank funded a project in Balochistan to increase groundwater availability for farmers, restore water storage, develop small-scale irrigation schemes, and strengthen the capacities of farmers. BSSIP, implemented by the Government of Balochistan, restored around 15 groundwater aquifers (Karez systems), raising productivity levels by 40% and crop intensity by 25% and allowing farmers to produce crops throughout the year. BSSIP did not only focus on improving the existing Karez and channels but also provided flood protection and associated structures such as small storage tanks and flow diversions.	It is a World Bank-funded adaptation project and is being implemented by Balochistan Irrigation Department. This small-scale project brought significant large-scale benefits to Balochistan and built the resilience of farmers and water users against climate-related disasters.
<u>Economic transformation initiative – Gilgit-Baltistan</u> ⁸⁵ . 2015—2022. USD 120.15m.	The objectives of the IFAD initiative are to increase the agricultural production of 100,000 farmers ⁸⁶ by introducing high-value crops and linking them to local markets, developing irrigation schemes, organizing smallholder farmers into producers' groups, and formulating value chain development.	The International Fund of Agricultural Development (IFAD) is supporting an initiative in Gilgit-Baltistan for the economic growth of the region. Total beneficiaries are 100,000.
<u>The US—Pakistan partnership for agricultural market development (AMD).</u> 2014—2019.	It is focused to support the development of Pakistan's commercial agriculture. The programme aimed to improve the quality and efficiency of production for agricultural/livestock products (mango, citrus, meat, high-value/off-season vegetables) in Pakistan by providing training and matching grants to encourage innovation and private-sector investment. The programme also facilitated the increased demand for Pakistani agricultural products and fostered supply-demand synergies between producers and buyers.	It is a USAID-funded programme and was implemented by CNFA. This project created 25,477 new full-time employment opportunities and contributed to strong GDP growth.
<u>Water for livelihoods (W4L).</u> 2011—2017. USD 11.8m.	It was implemented in Khyber Pakhtunkhwa and the tribal area (FATA) ⁸⁷ of Pakistan. The project aimed to achieve institutional and legal reforms to eliminate constraints of inefficient water management and introduce better protection of regional water resources, mainly for agriculture and human consumption. The focus of the project was to promote dialogue among stakeholders to achieve an effective, equitable, and efficient use of water resources on a local level based on the Integrated Water Resource Management (IWRM) approach. The project was not supposed to implement the activities, but it supported many actors (local authorities, water user' associations, and governmental line departments) by building their capacities so that they were able to implement several activities/interventions which increased their access to water and protected them from climate-related disasters.	The Swiss Agency for Development and Cooperation (SDC) funded this project and was implemented by Inter cooperation (IC). Total beneficiaries were 22,484.
<u>Watershed rehabilitation and irrigation improvement: Demonstrating in Pakistan and disseminating the best</u>	The objective of this project was to promote better management of land and water resources. The project led to the dissemination of information on efficient water consumption which enabled farmers to adopt sustainable agricultural production techniques to store and use water resources effectively in the context of climate change.	It was implemented in Sindh supported by the International Center for Agricultural Research in Dry Areas (ICARDA) and the United States Department of Agriculture (USDA) and implemented by the Institute of Water Resources Engineering and Management (IWREM).

⁸⁴ FAO, Inception workshop held to launch the "Transforming the Indus Basin with Climate Resilient Agriculture and Water Management" Project in Sindh and Punjab, <https://www.fao.org/pakistan/news/detail-events/en/c/1304449/>

⁸⁵ IFAD, <https://www.ifad.org/en/web/operations/-/economic-transformation-initiative-gilgit-baltistan-eti-gb->

⁸⁶ Economic Transformation Initiative, <https://www.etiqb.com.pk>

⁸⁷ Karak, D.I. Khan, Chitral, Tank, Lower Dir in Khyber Pakhtunkhwa; FR D.I. Khan in the FATA.

<u>technologies to help rural farmers.</u> 2011—2016. USD3.3m.		
<u>Australian Water Programme for Pakistan.</u> 2016—2021. AUD15m.	A 15-million-Australian-dollars programme on water management was launched by the Australian Government to improve water management and boost agricultural productivity in Pakistan ⁸⁸ . In this regard, the Australian Government's Centre for International Agricultural Research (ACIAR) had worked with the Pakistan Council for Water Research in Water Resources (PCRWR), Pakistan Agricultural Research Council (PARC), provincial agriculture and livestock departments, universities, and research institutes to address technical, social, economic and policy constraints, which hinder the growth of the agriculture sector. Moreover, the Australian scientists from the Commonwealth Scientific and Industrial Research Organization (CSIRO) also worked with the Ministry of Water and Power to build their capacity for efficient water management. The programme also included a component on groundwater management ⁸⁹ , which is a step toward more effective groundwater use for social, economic, and environmental benefits. This component aims to build the capacity of researchers, farmers, farming communities, and relevant government and non-government agencies to improve groundwater management in ways that enhance farming family livelihoods in Pakistan. It is designed to provide an enabling environment for communities to participate in the research process and to develop socially acceptable solutions. It is implemented in three provinces of Pakistan (Balochistan, Sindh, and Punjab).	It was implemented by many public-sector Pakistani actors.
<u>Disaster and climate resilience improvement project (DCRiP).</u> 2015—2019. USD125m.	DCRiP funded by the World Bank, aimed to build resilience through the restoration of [critical] flood protection infrastructure and strengthen government capacity to manage disasters and climate variability, for instance, upgrading hydro-meteorological observation network and associated flood early warning systems; restoring flood protection infrastructure; enhancing physical resilience and upgrading flood management systems; emergency response.	It was implemented by Punjab Agriculture and Irrigation Departments.
<u>Sindh resilience project (SRP).</u> 2016—2024. USD100m.	It aims to mitigate flood and drought risks in selected vulnerable regions of Sindh while strengthening the capacity of provincial institutes to manage natural disasters. The project plans to improve resilience by improving embankments at vulnerable sites along the Indus River for protection against riverine floods, as well as the construction of small recharge dams and gravity weirs for addressing drought and flash-flood risks. The project would also support the improvement and upgrading of modelling facilities at the Irrigation Research Institute (IRI), river morphology studies, floodplain mapping, groundwater assessment studies, etc. Currently, several civil works, institutional capacity, and hydrological modelling are under process.	Funded by the World Bank. The implementing agencies are Sindh Irrigation Department (SID) and Provincial Disaster Management Authority (PDMA), Sindh.
<u>Development of national flood</u>	It focuses on integrated water management with equal emphasis on measures other than flood protection works, for instance, improved rules of reservoir operation, flood	Funded by the Pakistani government (Ministry of Water Resources) and

⁸⁸ Daily Pakistan, Australia launches \$15 million fund for water management in Pakistan [Press release](#) dated 31.01.2017.

⁸⁹ Improving ground water management to enhance agriculture and farming livelihoods in Pakistan, <https://www.pcrwr.gov.pk/ACIAR/watermanagement/index.php?readmore=read>

<u>protection plan-IV (NFPP-IV).</u>	forecasting and early warning system, flood risk zoning and proofing for disaster- and climate-related insurance, water management at catchment level, disaster risk reduction and management, etc.	NDRMF and being implemented by the Federal Flood Commission of Pakistan.
<u>Ten billion trees tsunami programme: up-scaling of green Pakistan programme.</u> 2019—2023. ~USD700m.	The overall objective of this programme is to revive forest and wildlife resources in Pakistan, conserve protected areas, and encourage eco-tourism, community engagement, and job creation. The project has created 84,609 green jobs to date. A consortium including WWF-Pakistan, IUCN and FAO has been established to conduct third-party monitoring of this baseline initiative. As part of its monitoring work, the consortium has developed Silvo-Ecological Zones for the entire country, including a list of best suited native species for each zone. This list will be used to directly inform the selection of species for plantation under the proposed project. The consortium also developed a detailed monitoring and evaluation methodology, which will be used by the proposed project's M&E staff for monitoring the proposed project's achievements of its afforestation targets (including how these targets contribute to the targets under this baseline initiative) and measuring the adaptation impact of its interventions.	It is being primarily implemented by the Ministry of Climate Change along with provincial forest and wildlife departments. UNEP is supporting the implementation of this initiative.
<u>Mainstreaming Climate Change Adaptation through Water Resource Management in Leather Industrial Zone Development.</u> 2015—2019. USD3.3 million GEF grant and USD14.7 million in co-financing.	This project focuses on reducing vulnerability and building resilience through the integration of climate change adaptation into urban development in Pakistan. The main components of the project include: i) mainstreaming adaptation into urban and rural development planning; ii) capacity building of targeted communities and leather business owners; iii) supporting sustainable water use by leather industries in the Sialkot Tannery Zone and demonstrating improved wastewater treatment. The proposed project will benefit from lessons generated by this GEF-funded project around effective implementation of water resources management solutions implemented in the Sialkot Tannery Zone.	This GEF-funded project was approved for implementation in 2015. The Implementing Agency is UNIDO and the Executing Agencies included Sialkot Tannery Association Guarantee Ltd (STAGL) (Lead Executing Partner) District Government Sialkot, Irrigation and Environment Departments, Provincial Government Punjab.
<u>Enhance community, local and national-level urban climate change resilience to water scarcity, caused by floods and droughts in Rawalpindi and Nowshera, Pakistan.</u> 2020—2023. USD 6 million.	This baseline project focuses on addressing the impacts of floods and droughts on people living in urban areas in the cities of Rawalpindi and Nowshera. The project comprises three components, namely: i) enhance community- and household-level flood resilient water harvesting facilities; ii) enhance city- and district-level water harvesting facilities in public buildings and on water storages in public gardens, as well as develop district/city-level tools for improved planning and response to the impacts of climate change; and iii) strengthen national and provincial-level capacity to guide/direct city-level development considering climate change and disaster risks and impacts. Lessons learned from this baseline project, particularly related to efficient and effective use of water storage systems, will be used to inform the implementation of the proposed project as the relevant information becomes available.	This is an AF-funded project currently under implementation by MoCC, and supported by UN-Habitat, relevant provincial and municipal government actors, as well as NGOs and CBOs.
<u>Reducing Risks and Vulnerabilities from Glacier Lake Outburst Floods in Northern Pakistan.</u>	This baseline project focuses on reducing the risks and vulnerabilities from glacial lake outburst floods (GLOFs) and snow-melt flash floods in Northern Pakistan. Its main objectives are to develop the human and technical capacity of public institutions and vulnerable communities in Northern Pakistan to understand and address immediate	This project was funded by the Adaptation Fund (AF) and completed in 2015. It was led by Pakistan's Ministry of Environment.

2011—2015. USD3.9 million.	GLOF risks, including through policy, strengthening knowledge and information sharing, and implementing community-based GLOF risk management in vulnerable mountain valleys of Northern Pakistan. While the focus of this baseline project is different to the proposed project, lessons learned on effective strategies for community-based adaptation work will be drawn from during the implementation of the proposed project.	
<u>Water resource accountability in Pakistan (WRAP)</u> . 2021—2026. £15m.	To improve water governance and management issues in Pakistan to be able to adapt to changing climate while ensuring environmental sustainability with a focus on collaboration and engagement with provinces, research, and academia.	Funded by FCDO. £5m component on water governance is being implemented by IWMI. £5m component on nature-based solutions is being implemented by WWF.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Impact: Paradigm shift

45. **IF:** EbA and green infrastructure interventions are implemented in the Indus basin and are supported by corresponding changes in the National Water Policy, the National Adaption Plan and Provincial Adaptation Plans, this will operationalize EbA and green infrastructure integration in future adaptation interventions by the Government of Pakistan. Furthermore, if targeted communities are engaged in ecosystem restoration and management practices that align with these interventions, planning and regulations, **THEN:** vulnerable communities in the Indus Basin will benefit from additional protection to floods and droughts through additional EbA and green infrastructure interventions, national and provincial governments will have the necessary regulations and planning documents in place to sustain these interventions and operationalise future EbA and green infrastructure measures, and community businesses and livelihood practices will be better suited to withstand future climate threats **BECAUSE:** the project will address the above mentioned barriers by convincing key stakeholders of the benefits and cost-effectiveness of EbA and green infrastructure, by fostering an enabling environment for public sector reform for flood and drought prevention, and by strengthening government and community capacity to enhance resilience of key ecosystems and vulnerable communities to climate change impacts like frequent and severe floods and droughts.

Target communities

46. The proposed project will target local communities, including small-scale farmers, living in four project sites in the Indus Basin, namely the DI Khan Watershed, Ramak Watershed, Chakar Lehri Watershed, and Manchar Lake and Watershed. The people living in these areas are particularly vulnerable to the impacts of climate change-induced floods and droughts, which are already negatively affecting their water and livelihood security.

Outcomes and co-benefits

47. The project will achieve the following outcomes:

48. Outcome 1: EbA and green infrastructure interventions with improved community-led management reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin will be achieved by i) implementing EbA interventions to reduce flood impacts on populations close to the intervention sites and people living downstream through enhanced ecosystem services such as slowing of flood waters delays in flood peak onset, and increased access to water during drought periods, while downstream communities will benefit from the slowing of upstream velocity and delayed peak flood; ii) implementing green infrastructure interventions to reduce flood impacts on nearby communities and ecosystems by reducing soil erosion and river sedimentation, diverting and reducing the velocity of intense flows in the hill torrents away from nearby villages located in the floodplain, and increasing infiltration and groundwater recharge; and iii) providing organization and capacity development to Community-Based Organisations (CBOs) and creating Water User Groups and Community Watch and Ward systems to support community-based natural resource management and water supply and flood attenuation services provided by EbA interventions implemented floodplains and watersheds, empower communities to responsibly manage the ecosystems they rely on.

49. Outcome 2: Enabling environment for climate action that has new procedures for implementing EbA and green infrastructure interventions for flood and water resources management in Pakistan will be achieved by using the

evidence of the effectiveness and efficiency of EbA and green infrastructure interventions in enhancing flood and water resources management generated under Outcome 1 to expand Pakistan's regulations and policies for flood and water resources management through updated procedures that prioritise EbA and green infrastructure interventions. This will be supported by strengthening the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated procedures. The outcome of this work will enable the GoP to channel future investments into EbA and green infrastructure interventions and replicate similar adaptation strategies in other flood- and drought prone areas in Pakistan to those demonstrated under the proposed project.

50. Outcome 3: Enhanced resilience of community livelihoods in the Indus Basin will be achieved by training communities and smallholder farmers on how to enhance existing agricultural practices, prevalent across the Indus Basin, to better withstand current and future climate change scenarios through improved irrigation practices to conserve water, soil management to enhance yields and reduce agricultural expansion, and introducing new crop varieties which require less water and can withstand more extreme temperatures caused by climate change. Additionally, new opportunities for sustainable enterprises will be supported to provide alternative revenue generating opportunities for communities in the Indus Basin and reduce ecosystem degradation from traditional livelihoods directly dependent on natural resources. Project pipeline sourcing, incubation, and technical assistance to environmentally sustainable businesses will help reduce ecosystem degradation and increase revenue generation for businesses and families.

51. The project will achieve the following co-benefits, with additional co-benefits described in Section D.3 of this Funding Proposal:

- Co-benefit 1: GHG emission reduction from ecosystem restoration and improved ecosystem management. This co-benefit will be achieved by restoring 14,215 ha of degraded agro-ecological landscapes in watersheds (under Activity 1.1.1). These mitigation co-benefits are described in further detail in Section D.1 of this Funding Proposal.
- Co-benefit 2: Improved biodiversity as a result of the project's EbA and green infrastructure interventions. This will include: i) increases in migratory birds' population at Manchar Lake; increases in migratory birds' population at Indus Game Reserve (D.I Khan); ii) increases in fish species richness and abundance/Macro-invertebrates species richness; iii) increases in population abundance of the Indus River Dolphin (EN); iv) increases in population abundance of freshwater turtles (hard-shelled and soft-shelled turtles); and v) the contribution of the project's EbA and green infrastructure interventions towards the designation and nomination of restoration sites as sites of global ecological important such as Man and Biosphere Reserve and Ramsar Sites.
- Co-benefit 3: New jobs created and related new skills developed through the implementation of the project's EbA and green infrastructure interventions.

Outputs and activities

52. The project will deliver the following outputs through the associated activities:

53. Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems will be achieved by restoring 14,215 ha of degraded agro-ecological landscapes in watersheds in DI Khan (Activity 1.1.1) and rehabilitating degraded wetlands, flow paths, and channels in Ramak Watershed and Manchar Watershed/Lake (Activity 1.1.2). Improving the management of the Indus Basin's existing ecosystems and restoring their function through EbA and green infrastructure interventions is critical to enhance the resilience of these ecosystems and secure the flood and drought risk reduction benefits they provide to communities living in the basin. The interventions under Output 1.1 will enhance ecosystem functioning and the delivery of goods and services to vulnerable communities living in the Indus Basin and secure their livelihoods.

54. Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin will be achieved by designing and implementing 127 green infrastructure interventions to reduce the impacts of flooding on communities living in the four project sites in the Indus Basin, namely the DI Khan Watershed, Ramak Watershed, Chakar Lehri Watershed, and Manchar Lake and Watershed (Activity 1.2.1). Green infrastructure interventions such as storage ponds, embankments, and gabion bunds will be developed to complement the EbA interventions (under Output 1.1) to create new water diversion and storage capacities in the Indus Basin to enhance existing ecosystems' capacities to regulate water and reduce flooding and drought risk.

55. Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds will be achieved by: i) strengthening the capacity of CBOs to adopt EbA and green infrastructure interventions and undertake climate-resilient community-based natural resource management (Activity 1.3.1); and ii) strengthening communities'

knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions (Activity 1.3.2).

56. Outputs 1.1, 1.2 and 1.3 contribute directly to achieving Outcome 1 described above.
57. Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan will be achieved by conducting economic valuations of the project's EbA and green infrastructure interventions (under Outputs 1.1 and 1.2) to demonstrate their cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan (Activity 2.1.1). This evidence-base will then be used to inform the development and dissemination of an evidence report to national and provincial government for implementing EbA and green infrastructure interventions in response to flooding and droughts in Pakistan (Activity 2.1.2).
58. Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan will be achieved by using the evidence generated under Output 2.1 to develop and adopt updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, and Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. This will remove barriers to novel climate solutions through the promotion of integrated strategies, planning and policymaking, and lead to the creation of well documented procedures backed by a rigorous scientific base that enables the GoP to channel future investments into EbA and green infrastructure interventions instead of grey infrastructure only to respond to increasingly severe floods and droughts caused by climate change. A National Working Group established under the project will lead this process (Activity 2.2.1). The National Working Group will be chaired by the Ministry of Climate Change (MoCC) and comprised of regulators within the MoWR and its FFC, as well as the relevant Provincial Departments.
59. Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management will be achieved by strengthening the capacity of national and sub-national staff, as well as community leaders/representatives (under Activity 2.3.1), to apply the updated procedures under Activity 2.2.1 and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management. This will ensure national and subnational policy makers and community leaders have sufficient capacity to effectively implement and manage EbA and green infrastructure interventions in future government-led initiatives.
60. Outputs 2.1, 2.2 and 2.3 contribute directly to achieving Outcome 2 described above.
61. Output 3.1: Pipeline of feasible climate-resilient businesses will be achieved through WWF-Pakistan leveraging its experience developing pipeline businesses for the Dutch Fund for Climate and Development (DFCD) to provide incubation services to 7 small local businesses already identified to help them access debt and equity finance from the DFCD and other investors. This will contribute to the project's efforts to enhance community resilience by providing important case studies for business models which create positive environmental impacts while creating economic resilience for vulnerable communities and business owners (Activity 3.1.1).
62. Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities will be achieved by implementing community-based adaptation interventions to increase the climate resilience of the most vulnerable smallholder farmers in the project sites, including through the adoption of climate-resilient agricultural inputs (e.g., drought-resilient crop varieties), enhanced capacity and regular access to improved extension services for implementing resilient and sustainable irrigation and soil management techniques, and improved access to 20,250 m³ of additional water from new water storage tanks. This will optimise on-farm water efficiency and support agricultural production under increasingly severe drought conditions.
63. Outputs 3.1, 3.2 contribute directly to achieving Outcome 3 described above.

Barriers to the adaptation solution

64. Barrier 1: Lack of proof of concepts for EbA and green infrastructure as viable adaptation interventions in Pakistan.
The GoP's current regulations and policies for flood and water resources management do not consider EbA and green infrastructure interventions and the country has limited capacity to finance the establishment of these interventions through domestic public sector financing. There is therefore little evidence of EbA and green infrastructure interventions being used as cost-effective climate change adaptation interventions that also provide

a wide variety of environmental and social benefits (e.g., GHG emissions avoidance/reduction benefits and the protection of agricultural land houses from floods and droughts). This leads to the GoP being trapped in the reactive use of grey infrastructure, which often needs to be financed through international donors and foreign aid, to assist in post-disaster situations, including for flood and drought recovery.

65. Barrier 2: EbA and green infrastructure interventions are not included in Pakistan's National Water Policy, NAP, and Provincial Adaptation Plans. There are currently no planning arrangements in the Indus Basin for dealing with increasing instances of flooding and drought brought on by climate change. Following the passage of the National Water Policy in 2018, there has been no operationalization of the policy's objective in Pakistan's four Provinces. The Implementation Framework of the National Water Policy, NAP, and Provincial Adaptation Plans for Sindh, Punjab, Balochistan, and Khyber do not consider EbA and green infrastructure and prioritize the use of grey infrastructure interventions. As a result, the country remains locked into managing floods and droughts reactively with grey infrastructure interventions, which do not optimize potential environmental and social benefits, and are financially unsustainable because of their high capital and operational costs and low adaptive capacity over the long term. To effectively reduce climate change risk under future scenarios, the Government of Pakistan must formulate integrated water management plans which combine grey infrastructure solutions with improved freshwater ecosystem management to maximize risk reduction benefits for communities and conserve the Indus Basin so critical to Pakistan's overall productivity.
66. Barrier 3: Inadequate technical capacity across government and communities for implementing EbA and green infrastructure interventions in Pakistan. Government staff in the MoWR and its Executive Department the FFC, the MoCC, relevant Provincial Departments such as the Provincial Departments of Irrigation, Forestry and Agriculture and Provincial Disaster Management Authority (PDMA) do not have sufficient technical and institutional capacity to implement, monitor, operate and maintain EbA and green infrastructure interventions. These capacity constraints also extend to communities who are often involved in the operations and maintenance of infrastructure initiatives through the relevant provincial governments. Because of these limitations, there is a limited knowledge and understanding of the benefits of EbA and green infrastructure for flood and water resources management and how to shift away from the current paradigm of using grey infrastructure reactively to address climate change impacts such as floods and droughts.
67. Barrier 4: Limited access to finance among the most vulnerable communities and micro-, small- and medium-enterprises (MSMEs) for climate-resilient, sustainable businesses. There are few institutions and resources available in Pakistan to incentivise and assist the development of enterprises and bankable projects which help enhance ecosystem functions and contribute to Pakistan's economic development. Sustainable businesses currently need to overcome challenging regulatory requirements and processes with limited access to resources and technical support. While there is strong interest among development banks and lending institutions to invest in bankable projects, they are often unwilling to provide the necessary upfront financing to support earlier stage business development needs. There have been past efforts to try and promote community based sustainable enterprise in the Indus Basin, but these efforts are challenged to design the necessary financial instruments to package these businesses to help reach a scale sufficient to attract development banks or institutional investors. Assistance projects have also typically focused on market assessments and business identification efforts and less on the actual technical assistance, incubation services, and access to finance.
68. Barrier 5: Current land use practices and livelihoods do not consider climate risk. Communities in the Indus Basin currently rely on traditional land use practices to sustain their livelihoods, including farming and cattle raising, and often fail to account for the changing water dynamics with increased instances of flooding and drought brought on by climate change. These traditional practices have often relied on the conversion of ecosystems or the extraction of natural resources to support agrarian livelihoods. Communities in the Indus Basin are often unaware of the climate change impacts (e.g., increasingly severe floods and droughts) they will likely face in the future and opportunities for reducing these impacts through improved ecosystem management and stewardship. Communities also often lack the extension services and support needed to successfully adopt alternative livelihood practices, such as drought resistant crop cultivation and do not yet consider the benefits of these alternative practices as means to reduce household and community vulnerability to climate change.

Assumptions

69. Key assumptions are listed below, with more detailed notes and assumptions presented as required in Section E: Logical Framework.
- Communities are committed and have sufficient incentives to reduce their increasing vulnerability to threats from flooding and drought to engage with and sustain project activities.

- The Government of Pakistan maintains its commitment to mainstream EbA and green infrastructure solutions to climate change within its national climate change strategy.
- Climate change projections remain stable and/or predictable and climate change impacts are adequately addressed by interventions of this project.

Market failure and absence of financing in respect to CSA

70. Regarding Output 3.2, 84% of the total investment of US\$1,369,605 for the output is directed at the provision of TA and training which is in the scope of grant financing. A detailed explanation on the need for grant funding for the remaining US\$222,600 (related to the provision of CSA inputs) of the total investment for Output 3.2 can be found below.

Crops and market entry

71. Smallholder farmers in Pakistan cultivate a diverse range of crops that vary depending on the region and climate. Commonly grown crops include wheat, rice, cotton, sugarcane, maize, and vegetables such as tomatoes, onions, and potatoes⁹⁰. The smallholder farmers targeted under the project primarily grow crops such as wheat, rice and vegetables primarily for subsistence and household use. In the years where there is a surplus yield, this surplus is sold in the open local markets, however, this is not representative of a bankable income stream because in some years meeting subsistence needs is challenging. Additionally, due to limited access to resources such as transportation and storage facilities, many smallholder farmers in Pakistan face challenges in effectively marketing their produce and often receive low prices for their crops⁹¹. As a result, their ability to engage in commercial agriculture is limited and producing primarily for subsistence and household consumption to support household food security.

Financial challenges for small holders

72. In Pakistan, banks do not provide loans to smallholders for climate-smart agriculture related interventions to demonstrate or pilot on the ground. Traditional loaning facilities primarily focus on financial products that cover procurement of equipment and farm machinery such as tractors and harvesters. Smallholders struggle to avail these traditional financial products due to pledging requirements in the form of land, which often times is estimated to be less than half of its existing market value. Financial literacy is also a challenge that smallholder farmers face, as filling loaning applications is a task that requires significant understanding of financial jargon and being able to demonstrate positive cashflows.

73. According to an analysis conducted by FAO of nine countries, including countries from South Asia, limited financial literacy, collateral, credit history, and few alternative sources of income are common obstacles faced by smallholders when attempting to secure financing for improving productivity and adopting sustainable farming practices⁹². A farm level analysis that studies the demand for credit by smallholder farmers in Sindh, Pakistan found that small-scale farmers have low access to agricultural credit as compared to landlord farmers due to lack of collateral, low-income level and their less landholding size⁹³. GCF FP 108 states that “some 63–85 percent of farmers are smallholders, predominantly tenant farmers. Around 70 percent of these individuals are women, and about 43 percent are managing landholdings of less than one hectare”.

74. **Limited Access to Credit and Finance:** Smallholder farmers in Pakistan often face difficulties in accessing credit and finance to invest in climate-smart agriculture practices. They may not have collateral to secure loans or may not meet the requirements of lending institutions. Even when credit is available, high interest rates and short repayment periods make it difficult for farmers to invest in long-term projects.

Additional Financial Challenges for CSA

75. Smallholders do not have sufficient funds to adopt new climate resilient agriculture practices. These smallholders already struggle to cover their daily expenses and lack the necessary capital to purchase relevant inputs for planting and harvesting climate resilient crops. There are several studies that highlight the lack of collateral as a significant challenge for farmers in accessing loans for climate-smart agriculture. According to a report by FAO, smallholder farmers encounter considerable challenges when seeking financial support for sustainable farming practices due to their insufficient credit history and collateral. The report highlighted that lack of access to finance can limit the

⁹⁰ [Climate-Smart Agriculture in Pakistan](#)

⁹¹ [Smallholder access to markets in Pakistan](#)

⁹² [The economic lives of smallholder farmers: An analysis based on household data from nine countries \(FAO, 2015\)](#)

⁹³ [Determinants of demand for credit by smallholder farmers: a farm level analysis based on survey in Sindh, Pakistan](#)

adoption of climate-smart agriculture practices, which can in turn limit the ability of smallholders to adapt to the impact of climate change⁹⁴.

76. For smallholders to adopt climate-smart agriculture practices, a strong business case that demonstrates cost savings and profitability at farm level needs to be demonstrated. For this, pilot projects need to be funded by grant financing to raise awareness around the financial benefits of climate-smart agriculture, allowing farmers to see first-hand the positive impacts. A study conducted by the International Fund for Agricultural Development (IFAD) found that grants are the predominant financial instrument used for small-scale agriculture climate finance, providing 50% of finance committed. This is because small-scale producers face significant obstacles in accessing loans due to a lack of collateral and limited land tenure rights. In addition, grants for capacity building allow to bridge the knowledge gap for the adoption of climate smart agricultural practices by small-scale producers and can contribute as a de-risking or first-loss tranches to blended finance instruments to catalyse private investments⁹⁵. Thus, grant-funded pilot projects allow for de-risking the adoption of climate-smart agriculture practices and provides the technical support and handholding for farmers to transitions towards this new approach to farming.
77. Lack of support from financial institutions are a major concern for smallholder farmers who need access to credit and finance to invest in climate-smart agriculture practices. Without affordable loans or other financial incentives, farmers may not have the means to purchase the inputs and technologies required for climate-smart agriculture, such as drought-resistant seeds and efficient irrigation systems. It is important for financial institutions to recognize the potential of climate-smart agriculture practices and offer financial products and services that incentivize and support smallholder farmers to adopt these practices. This can include offering preferential loans, grants, or other financial support to farmers who implement climate-smart agriculture practices. By doing so, financial institutions can contribute to building a more climate-resilient and sustainable agricultural sector in Pakistan.
78. In addition, financial institutions may not fully understand the benefits of climate-smart agriculture practices, or the risks associated with climate change. They may not see these practices as profitable or may view them as too risky, particularly in the short term. This lack of understanding can result in a lack of investment in climate-smart agriculture practices, further hindering their adoption among smallholder farmers in Pakistan. There is a need for more research and data on the costs and benefits of climate-smart agriculture practices. In sum, smallholder farmers in Pakistan face the following financial challenges to climate-smart agriculture:
 1. **High Upfront Costs:** The upfront costs of inputs and technologies required for climate-smart agriculture, such as drought-resistant seeds, efficient irrigation systems, and weather monitoring tools, can be prohibitively expensive for smallholder farmers in Pakistan. These costs can make it difficult for farmers to invest in climate-smart agriculture practices, especially when they are already struggling to make ends meet.
 2. **Lack of Market Opportunities:** Even if smallholder farmers in Pakistan invest in climate-smart agriculture practices, they may struggle to find markets for their products and services. Climate-smart agriculture products, such as organic and sustainable produce, can be niche and may not be in high demand. Farmers may also lack the knowledge and skills to access and navigate markets, limiting their ability to make a profit.
79. There are several technical challenges that inhibit smallholder farmers in Pakistan's agricultural sector from adopting and implementing a shift toward more climate-resilient and sustainable agricultural practices. These conditions are highlighted in GCF FP 108 (*Transforming the Indus Basin with Climate-resilient Agriculture and Water Management*), which confirms that smallholders are most vulnerable to the impacts of climate change due to their lack of assets to buffer shocks and their limited access to the information, new technologies, finance and government services needed to undertake adaptive actions.
80. Specifically, smallholder farmers in Pakistan face the following challenges to adopt climate-smart agriculture:
 1. **Limited Technical Knowledge and Skills:** Many smallholder farmers in Pakistan lack the technical knowledge and skills required to implement climate-smart agriculture practices. They may not know how to select and plant drought-resistant crops, how to use efficient irrigation systems, or how to use weather monitoring tools. This lack of technical knowledge and skills limits their ability to implement effective climate-smart agriculture practices.
 2. **Inadequate Access to Information and Extension Services:** Smallholder farmers in Pakistan may not have access to information and extension services on climate-smart agriculture practices and technologies. This limits their ability to learn about new practices and technologies that could help them adapt to climate change.

⁹⁴ [The State of Food and Agriculture: Climate Change, Agriculture And Food Security \(FAO, 2016\)](#)

⁹⁵ [Examining the Climate Finance Gap for Small-Scale Agriculture \(IFAD, 2020\)](#)

Extension services that are available may not be accessible to all farmers or may not be tailored to the specific needs of smallholder farmers.

3. **Access to Inputs:** Small holders do not have access to key inputs that they could deploy on their lands to target the impacts of increasing heat and rainfall extremes on crop productivity in existing farmlands, including micro-irrigation, rainwater harvesting and climate resilient seeds.

81. Therefore, shifting agricultural practices among smallholder farmers in Pakistan that better equips these vulnerable populations to cope with the challenges of climate change requires:

1. **Building the capacity of farmers by providing training and technical assistance** on climate-smart agriculture practices. This helps farmers improve their technical knowledge and skills and increases their openness to and confidence in adopting new practices.
2. **Demonstrating climate-smart agriculture practices** through on-farm trials and demonstrations. On-farm demonstrations move climate smart agriculture from knowledge into practice. The live setting facilitates adoption by small holder farmers because it shows how the practices are operationalized and produced in the same setting in which the small holder will be applying the practices. In this way, on-farm demonstrations “derisk” adoption by the farmer as they can see the concrete benefits of these practices firsthand and encourage them to adopt these practices on their own farms.
3. **Linking farmers to bio-inputs, such as drought-resistant seeds, bio-fertilizers, and bio-pesticides.** This helps farmers access the inputs they need to implement climate-smart agriculture practices.

82. Under the proposed project, a total of US\$1,528,910 will be channeled towards improving climate-resilient livelihoods among vulnerable communities in Pakistan (Output 3.2). Regarding points 1 and 2 above, promoting climate-smart agriculture practices among smallholder farmers, building the capacity of farmers by providing training and technical assistance on climate-smart agriculture practices, and demonstrating climate-smart agriculture practices through on-farm trials and demonstrations are interventions frequently mainstreamed across many donor-funded and development projects to address the technical and financial challenges face to adopt these climate-resilient practices on their own and without grant support.

83. These traditional grant funded activities total US\$1,306,310, or 85% of the activity total, distributed to staff, consultant, travel, workshop communications and convening costs as detailed in Annex 4.

84. GCF grant funds will be used to directly address the technical challenges faced by smallholder farmers in the project’s sites in Pakistan’s Indus Basin by strengthening the capacity of small-scale farmers and people dependent on agriculture on climate-resilient agriculture and sustainable practices through a Farmer Field School (FFS) approach whereby training sessions are conducted over an entire crop season. This will involve: i) the development of province-specific curriculum, training modules, and workplans for Climate-Smart FFSs by WWF-Pakistan, supported by Technical Assistance (TA) provided by international and local experts; ii) providing training through the FFS on improved climate resilience of soil management practices and introducing on-farm best management practices for improved soil conservation and productivity; and iii) mobilize women participation in the implementation and adoption of project activities under Output 3.2, as specified in Annex 8: Gender Assessment and Action Plan. Specifically, WWF-Pakistan will work with relevant local government departments to provide training and technical capacity building of local women to adopt gender-responsive climate-resilient/alternative livelihoods under Output 3.2 that will enhance rural income generation opportunities, agricultural productivity, livelihood security, and sustainable rural development.

85. Grant funding is required to enable the implementation of these activities focused on delivering technical assistance and extension services to the most vulnerable agricultural communities and smallholder farmers in the project’s sites.

86. The remaining US\$222,600 (15% of the total investment) will be used to: i) improve the water efficiency of farming through improved farm management and irrigation systems by providing farmers the required components and inputs — e.g., drip irrigation, sub surface irrigation systems, and tunnel farming and greenhouses; ii) introduce drought-resistant crop varieties, including the provision of seed varieties to participating farmers; and iii) provide 15 water storage tanks on communal land for public access for vulnerable agricultural communities across the three provinces (Sindh, Balochistan and Khyber Pakhtunkhwa).



Overcoming Barriers to Achieve Impact

87. The project will achieve these benefits through three interconnected components, described in detail in Section B.3. Table 4 articulates which project activities and sub-activities have been designed to address the described barriers and achieve the Theory of Change presented in Figure 17 below.

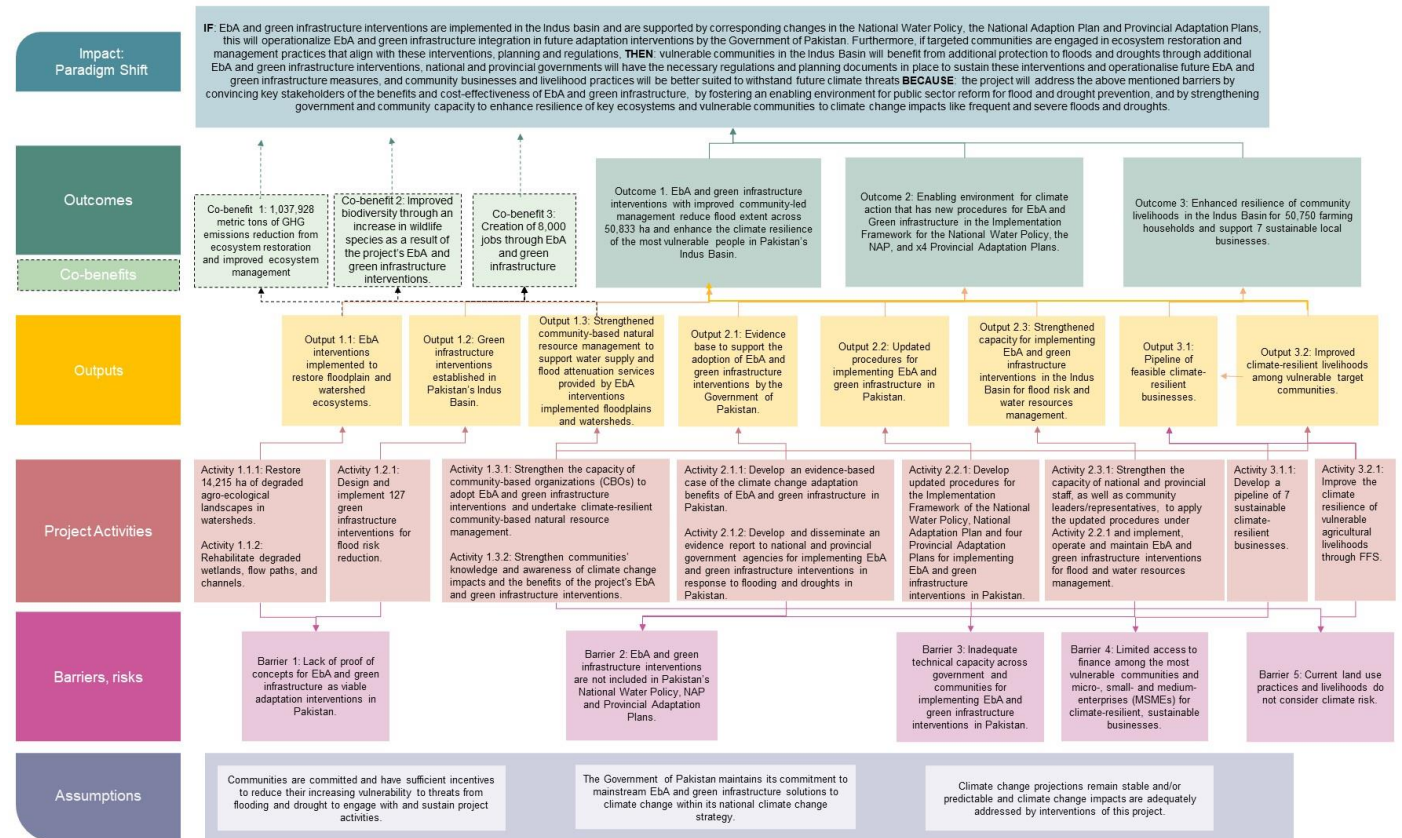


Figure 17. Theory of Change diagram.

Overcoming Barriers to Achieve Impact

88. To address Barrier 1, the project will invest US\$64 million in implementing EbA and green infrastructure interventions on the ground in four sites in the Indus Basin to reduce the impacts of increasingly severe floods and droughts on vulnerable communities and ecosystems. This investment will accelerate climate innovation in Pakistan by establishing proofs of concept for EbA and green infrastructure interventions that provide evidence of their cost-effectiveness and efficiency, flood and drought risk reduction benefits, and sustainability. Overall, these interventions will enhance the resilience of a total of 7,711,697 people (687,337 direct beneficiaries and 7,024,361 indirect beneficiaries) to floods and droughts, while simultaneously delivering the co-benefits of reduced/avoided GHG emissions, improved biodiversity through safeguarding habitats for a variety of species, particularly migratory birds, and creating job opportunities for 8,000 people through local community involvement on a permanent, part-time or temporary (e.g., daily) basis in the implementation, operations and maintenance of the EbA and green infrastructure interventions.

89. Barrier 2 will be addressed by expanding Pakistan's regulations and policies for flood and water resources management from relying only on grey infrastructure to address the threats of floods and droughts to prioritising EbA and green infrastructure interventions. This will be achieved through a step-wise approach. An economic valuation of the project's EbA and green infrastructure interventions will be undertaken which will include monitoring and analysis of these interventions' cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan, and developing an evidence report of their adaptation benefits to initiate public sector reform in flood and water resources management at the national and provincial levels in Pakistan. Using this evidence, a National Working Group will be established under the project, chaired by the MoCC and comprised of regulators within the MoWR (including FFC) and the relevant Provincial Departments. This working group will lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation

Framework for the National Water Policy, the NAP, and the Provincial Adaptation Plans of Sindh, Khyber, and Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.

90. The project will address Barrier 3 by securing the uptake and application of these updated procedures by national and provincial government involved in flood and water resources management in Pakistan. This will be achieved by training project planners and technical staff within the MoCC, MoWR (including FFC), and Provincial Departments of Irrigation, Agriculture, and Forestry on the requirements of the updated procedures and how to apply these procedures in the design and implementation of EbA and green infrastructure interventions in future country-driven projects. Technical staff and community representatives will also be trained on effective and efficient procedures for implementing, monitoring, operating and maintaining these interventions to enable them to maximise their flood and drought risk reduction potential over the long term.
91. To address Barrier 4, WWF-Pakistan will leverage its experience developing pipeline businesses for the Dutch Fund for Climate and Development (DFCD) to will provide incubation services to 7 small local businesses already identified to help them access debt and equity finance from the DFCD and other investors. Supporting sustainable enterprises provides a signal that business models can contribute to ecosystem and community resilience and generate revenues. Assistance to these businesses is intended to incentivize additional sustainable business development and help launch the creation of an investment market in sustainable enterprises.
92. To address Barrier 5, the project will support the adoption of EbA and green infrastructure interventions and implementation of climate-resilient land use practices at the basin level. This will be achieved by establishing operational agreements between 8 CBOs and the Provincial Departments of Forestry for the provinces of Punjab, Sindh, Khyber Pakhtunkhwa and Balochistan which meet the requirements of the updated procedures for the Implementation Framework of the National Water Policy, the NAP, and the four Provincial Adaptation Plans, as well as assign specific operation, maintenance, and stewardship roles among communities and Provincial Departments of Forestry for implementing and maintaining the EbA and green infrastructure interventions. Technical assistance will also be provided to CBOs so that they are adequately equipped to undertake sustainable ecosystem management practices, including maintenance and monitoring of wetlands, floodplains, flood bypasses and washlands, watersheds and community plantations. Operating agreements will also be established between farmer groups and the Provincial Departments of Agriculture in the same four provinces which define extension services for the use of improved irrigation systems, alternance crop varieties, and farm management practices to encourage participation and sustainability.

Table 4. Project outputs, activities and sub-activities mapped against the barriers they are designed to address.⁹⁶

Barrier	Output	Project activity	Project sub-activity
Barrier 1: Lack of proof of concepts for EbA and green infrastructure as viable adaptation interventions in Pakistan.	Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.	Activity 1.1.1: Restore 14,215 ha of degraded agro-ecological landscapes in watersheds.	Sub-activity 1.1.1.1: Restore 14,215 ha of degraded riverine ecosystems in D.I. Khan.
		Activity 1.1.2: Rehabilitate degraded wetlands, flow paths, and channels.	Sub-Activity 1.1.2.1: Excavate 264 m of flow paths in Badri Village, Ramak, to restore the natural hydrology of connected wetlands.
			Sub-Activity 1.1.2.2: De-silt the 2 ha Badri Pond in the Ramak wetlands to enhance its water-holding capacity.
			Sub-Activity 1.1.2.3: Enhance and reinforce 410 m of embankments in Badri Village to reduce erosion.
			Sub-Activity 1.1.2.4: Excavate 4,000 m of flow paths in Paniala Village to restore natural hydrology of Karez Systems (surface and subsurface natural channels).

⁹⁶ Further details of interventions are provided in Section B.3 of the Funding Proposal and Section 11.2 of Annex 2.

			Sub-Activity 1.1.2.5: Excavate 30,000 m of flow paths in the existing water channels of Manchar Lake to restore the natural hydrology of the 25,000 ha Manchar Wetland.
	Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.	Activity 1.2.1: Design and implement 127 green infrastructure interventions for flood risk reduction.	Sub-Activity 1.2.1.1: Design the green infrastructure measures. Sub-Activity 1.2.1.2: Build 9 flood protection embankments. Sub-Activity 1.2.1.3: Build 7 flood dispersal embankments. Sub-Activity 1.2.1.4: Build 45 gabion bunds. Sub-Activity 1.2.1.5: Build 42 small retention areas. Sub-Activity 1.2.1.6: Build 9 large retention areas. Sub-Activity 1.2.1.7: Build 15 recharge basins.
Barrier 2: EbA and green infrastructure interventions are not included in Pakistan's National Water Policy, NAP, and Provincial Adaptation Plans.	Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.	Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan.	Sub-Activity 2.1.1.1: Prepare baselines and indicators for the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services. Sub-Activity 2.1.1.2: Monitor the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services using the baselines and indicators prepared under Sub-activity 2.1.1.1. Sub-Activity 2.1.1.3: Conduct an economic valuation of the benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services. Sub-Activity 2.1.1.4: Identify additional opportunities and sites at the national level across Pakistan for scaling up and replicating the project's EbA and green infrastructure interventions for flood risk and water resource management. Sub-Activity 2.1.1.5: Collate and store the data and information generated under 2.1.1.1–2.1.1.4.
		Activity 2.1.2: Develop and disseminate an evidence report to national and provincial government agencies for implementing EbA and green infrastructure interventions in response to flooding and droughts in Pakistan.	Sub-activity 2.1.2.1: Prepare an evidence report on the use and benefits of EbA and green infrastructure interventions in response to flooding and droughts in Pakistan. Sub-activity 2.1.2.2: Disseminate the evidence report developed under Sub-activity 2.1.2.1 to national and provincial planners and regulators involved in flood and water resources management, as well as academic and research institutions.

	Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.	Activity 2.2.1: Develop updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for implementing EbA and green infrastructure interventions in Pakistan.	Sub-Activity 2.2.1.1: Establish a National Working Group to develop and adopt updated procedures for using EbA and green infrastructure interventions in flood and water resources management in Pakistan. Sub-Activity 2.2.1.2: Review the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for the four provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh to identify entry points for EbA and green infrastructure as cost-effective adaptation interventions for addressing floods and droughts. Sub-Activity 2.2.1.3: Draft updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. Sub-Activity 2.2.1.4: Validate the updated procedures within the National Working Group for adoption into the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans.
Barrier 3: Inadequate technical capacity across government and communities for implementing EbA and green infrastructure interventions in Pakistan.	Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.	Activity 2.3.1: Strengthen the capacity of national and provincial staff, as well as community leaders/representatives, to apply the updated procedures under Activity 2.2.1 and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management.	Sub-Activity 2.3.1.1: Train and raise awareness of national and sub-national policymakers and planners on the contents and application of the updated procedures developed under Activity 2.2.1. Sub-Activity 2.3.1.2: Train national and sub-national technical staff on the implementation, monitoring, operations, and maintenance of EbA and green infrastructure interventions. Sub-Activity 2.3.1.3: Train local community leaders/representatives on EbA and green infrastructure interventions and their benefits, as well as on the implementation, monitoring, operations, and maintenance of these interventions.
Barrier 4: Limited access to finance among the most vulnerable communities and micro-, small- and medium- enterprises (MSMEs) for climate-resilient, sustainable businesses.	Output 3.1: Pipeline of feasible climate-resilient businesses.	Activity 3.1.1: Develop a pipeline of 7 sustainable climate-resilient businesses.	Sub-Activity 3.1.1.1: Build on the initial study completed by WWF-Pakistan identifying bankable project opportunities in the Indus Basin in the agriculture, forestry, and water and sanitation sectors. Sub-Activity 3.1.1.2: Formulate technical assistance and business development plans for shortlisted climate-resilient DFCD businesses. Sub-Activity 3.1.1.3: Assist Bankable enterprises with financial close within the DFCD funding window (loaning application, investment pitch preparation, etc.).

Barrier 5: Current land use practices and livelihoods do not consider climate risk.	Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.	Activity 1.3.1: Strengthen the capacity of CBOs to adopt EbA and green infrastructure interventions and undertake climate-resilient community-based natural resource management.	Sub-Activity 1.3.1.1: Strengthen CBOs to adopt EbA and green infrastructure interventions and implement community-based natural resource management. Sub-Activity 1.3.1.2: Organize Water User Groups (WUGs) under CBOs and build their capacity for improved water resources management. Sub-Activity 1.3.1.3: Establish community-based watch and ward systems under CBOs to protect the project's EbA and green infrastructure interventions from degradation.
		Activity 1.3.2: Strengthen communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.	Sub-Activity 1.3.2.1: Consult with tribal elders to support the sensitization of the project's adaptation interventions in beneficiary communities. Sub-Activity 1.3.2.2: Develop and implement awareness-raising campaigns for target communities to increase their knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.
	Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.	Activity 3.2.1: Improve the climate resilience of vulnerable agricultural livelihoods.	Sub-Activity 3.2.1.1: Strengthen the capacity of small-scale farmers and people dependent on agriculture on climate-resilient agriculture and sustainable practices. Sub-Activity 3.2.1.2: Improve the water efficiency of farming through improved farm management and irrigation systems. Sub-Activity 3.2.1.3: Introduce drought-resistant crop varieties. Sub-Activity 3.2.1.4: Improve the climate resilience of soil management practices. Sub-Activity 3.2.1.5: Provide water storage tanks to vulnerable agricultural communities.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

93. Project outcomes:

- Outcome 1: EbA and green infrastructure interventions with improved community-led management reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin.
- Outcome 2: Enabling environment for climate action that has new procedures for implementing EbA and green infrastructure interventions for flood and water resources management in Pakistan.
- Outcome 3: Enhanced resilience of community livelihoods in the Indus Basin.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☐	☐	☒

Outcome 2	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 3	☐	☐	☐	☐	☒	☐	☐	☐

94. Project co-benefits:

- Co-benefit 1: GHG emission reduction from ecosystem restoration and improved ecosystem management
- Co-benefit 2: Improved biodiversity through an increase in the population of migratory birds in Manchar Lake
- Co-benefit 3: New jobs created through the project's EbA and green infrastructure interventions

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☐	☐	☐	☐	☐	☒
Co-benefit 2	☒	☐	☐	☐	☐	☐
Co-benefit 3	☐	☒	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

95. The objective of the proposed project is to transform Pakistan's approach to flood and water resources management, which is underpinned by the reactive use of costly grey infrastructure interventions to address climate change impacts such as floods and droughts. Because these climate change impacts are surpassing the country's existing water infrastructure's capacity to prevent large scale economic damages and loss of human life, Pakistan will become increasingly vulnerable to climate change in the future in the absence of urgent adaptation.

96. The project will achieve transformational change in Pakistan's approach to flood and water resources management by investing US\$64 million in implementing EbA and green infrastructure interventions on the ground in four project sites in the Indus Basin, namely the DI Khan Watershed, Ramak Watershed, Chakar Lehri Watershed, and Manchar (Lake and Watershed). Specific interventions and sites were selected through detailed feasibility assessments, which included climate threat analysis and community needs assessments — conducted via stakeholder consultations and qualitative and quantitative analyses (see Annex 2: Feasibility Study). The assessments identified areas highly vulnerable to flooding and/or drought, as well as interventions with the highest demonstrable capacity to generate climate change adaptation benefits for at-risk communities.

97. EbA interventions will target highly valuable freshwater and terrestrial ecosystems in the Indus Basin that play a critical role in regulating and providing water for downstream communities, and reducing flooding and landslide risks, with critical co-benefits in conserving globally unique biodiversity and capturing GHG emissions. These interventions will include the restoration of 14,215 ha of degraded watersheds in DI Khan, restoring 34km of flow paths, as well as desilting and restoring channels in Ramak Watershed and Manchar Lake. These interventions will reduce flood impacts on people living near the intervention sites by slowing down flood waters, as well as provide increased access to water to communities during drought periods. In addition, 127 green infrastructure interventions (including recharge basins and retention areas) will be implemented at specific locations in DI Khan, Ramak, Manchar, Chakar Lehri to protect nearby communities from flooding by reducing soil erosion and river sedimentation, diverting and reducing the velocity of intense flows in the hill torrents away from nearby villages located in the floodplain, and increasing infiltration and groundwater recharge.

98. The investment into these EbA and green infrastructure interventions will accelerate climate innovation in Pakistan by establishing proofs of concept for EbA and green infrastructure interventions that provide evidence of their cost-effectiveness and efficiency, flood and drought risk reduction benefits, and sustainability. This evidence will be used to create an enabling environment for climate action in Pakistan that removes barriers to novel climate solutions through promoting integrated strategies, planning and policymaking. Specifically, a National Working Group will be established under the project, chaired by the MoCC and comprised of regulators within the MoWR (including FFC) and the relevant Provincial Departments. This working group will lead the development and adoption of updated procedures for implementing EbA and green infrastructure interventions to address the threats of floods and droughts in Pakistan. These procedures will enable the GoP to channel future investments into EbA and green

infrastructure interventions instead of grey infrastructure to respond to increasingly severe floods and droughts caused by climate change.

99. These project interventions align directly with the GCF's paradigm shift objective of increased climate-resilient sustainable development and contribute to GCF Adaptation Result Area (ARA) 1.0 — Increased resilience and enhanced livelihoods of the most vulnerable people, communities, and regions; and ARA4.0 — Improved resilience of ecosystems and ecosystem services.
100. To support the uptake of the EbA and green infrastructure interventions among communities, ensure sufficient application of the updated procedures in the implementation, monitoring, operation, and maintenance of these interventions, and strengthen community resilience, the project will provide technical assistance and capacity-building to technical staff in national and provincial government and representatives from CBOs. This capacity-building will be led by WWF-Pakistan, in close collaboration with MoCC, MoWR (including FFC) and relevant provincial government departments.
101. An overview of the design of the project's components is provided below. Specific details on the technical design of the interventions are presented in Annex 2: Feasibility Study.

Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan.

Outcome 1. EbA and green infrastructure interventions with improved community-led management reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin.

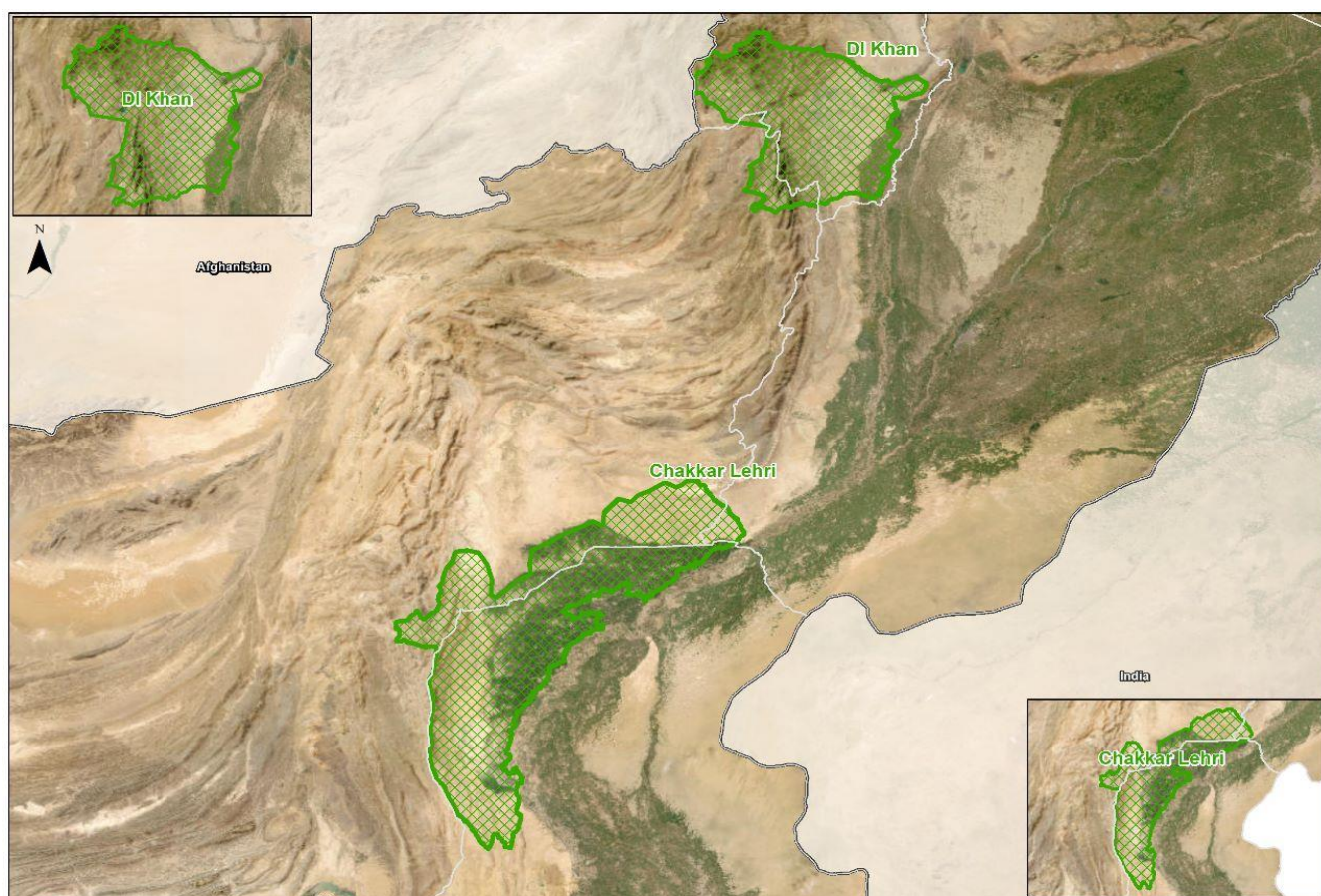


Figure 18. Project intervention areas.

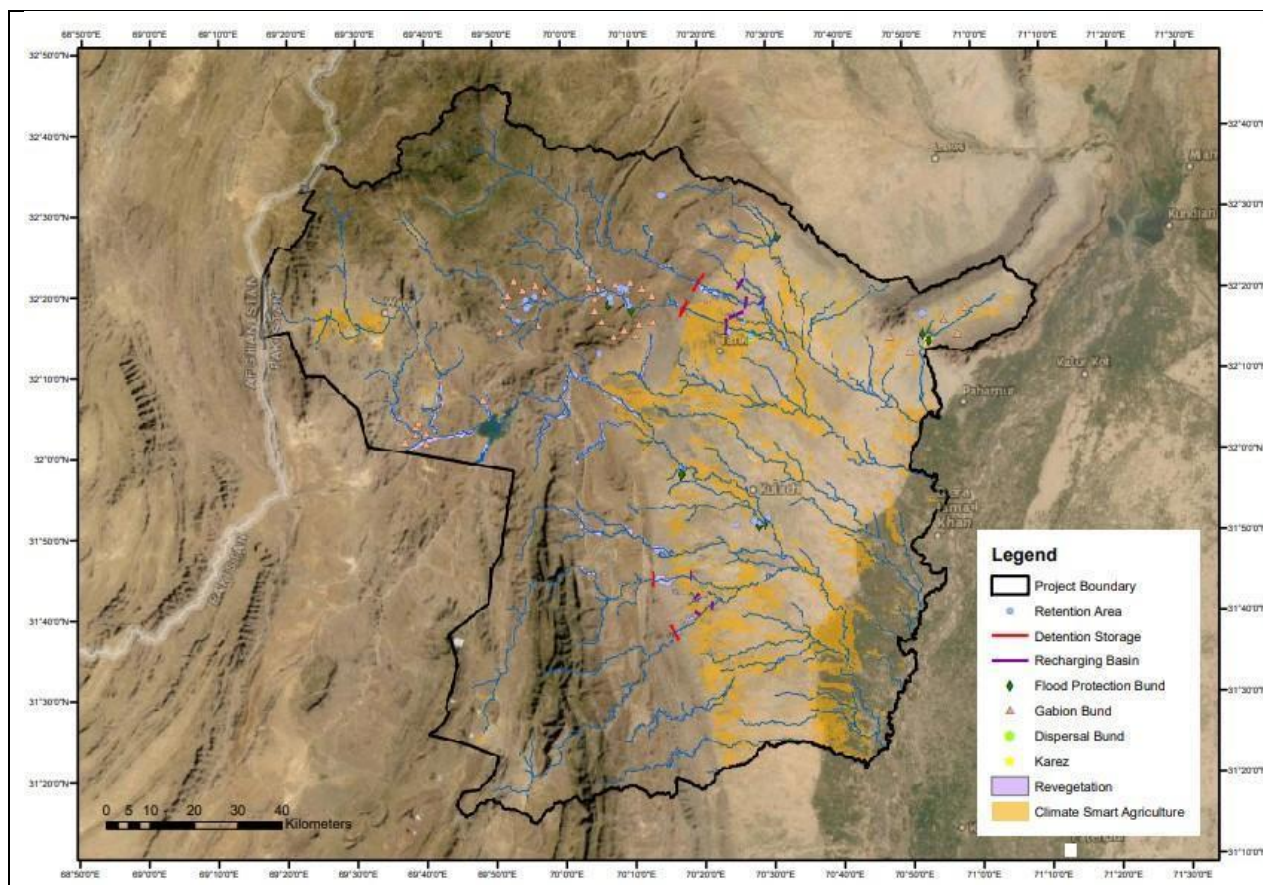


Figure 19. Map showing locations for Eba and Green Infrastructure interventions in DI Khan Watershed.

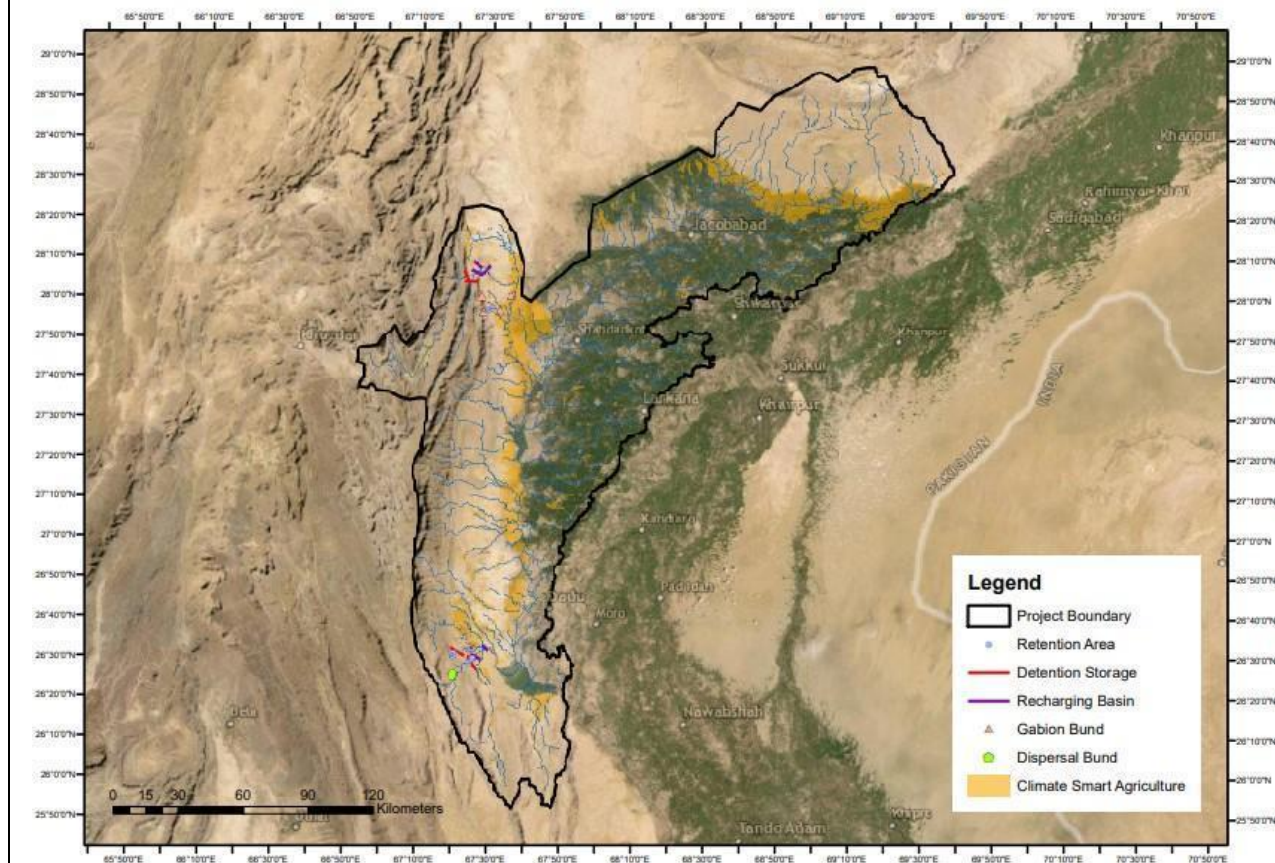


Figure 20. Map showing locations for Eba and Green Infrastructure in Manchar & Chakkar Lehri Watershed.

Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.

102. Under Output 1.1, a suite of EbA interventions will be implemented to restore degraded floodplain and watershed ecosystems in DI Khan, Ramak Watershed and Manchar Lake which have been damaged by past flood events and historically provided critical flood and drought protection to communities. These target ecosystems all fall within Pakistan's Indus Basin and include forests, riverine areas (including riverbanks), wetlands, water flow paths, and channels. The restoration of vegetation in these ecosystems will stabilize currently exposed slopes, slowing run-off and stabilizing soils. This will reduce the risk of flooding, landslides, and erosion, and concurrently enhance groundwater recharge, as well as the basal flows of rivers and springs. Revegetation of degraded ecosystems along riverbanks, springs, and lakes will also help to ensure sustainable water flows and reduce sedimentation. As a result of Output 1.1, 14,215 ha of degraded watersheds and floodplains will be brought under restoration. Furthermore, this output is expected to benefit a total of 427,300 people (310,200 direct beneficiaries and 117,100 indirect beneficiaries) through reduced adverse impacts of flooding, including damages to nearby community assets such as agricultural lands, livestock, and community infrastructure, as well as increased water availability for household use during droughts.

Activity 1.1.1: Restore 14,215 ha of degraded agro-ecological landscapes in watersheds.

103. Block and linear plantations will be established along identified hill torrents and catchment areas in D.I. Khan. This activity will be executed by WWF-Pakistan who will work with the Provincial Forest Department of Khyber Pakhtunkhwa through which planting stock will be sourced and plantation establishment will be executed. The Provincial Forest Departments will also provide the necessary excavation equipment and labour for this activity. For smaller sites and when available, saplings will be sourced from and planted by community owned and managed nurseries. Through this activity vegetation cover will be restored (natural regeneration and afforestation) across 14,215 ha of degraded riverine forests in the DI Khan regions. For natural regeneration activities of existing forested areas, replanting sites have been selected where at least 30% of tree cover is already available. Afforestation activities will include the planting of indigenous tree species which have close association with native biodiversity and adapted to local conditions. Revegetation of native species at this scale will help reduce flooding, erosion, flow velocities, and increase infiltration through enhancing soil porosity, ultimately increasing the climate resilience of surrounding communities. The GCF grant for the Provincial Forestry Departments will be managed by WWF-Pakistan's PIU in each province and the technical staff in the SIUs will coordinate with the Department staff in each province on the project planning, development and reporting.

Sub-activity 1.1.1.1: Restore 14,215 ha of degraded riverine ecosystems in D.I. Khan.

104. Under this sub-activity, 14,215 ha of degraded riverine ecosystems in D.I. Khan will be restored. The area of land that will be restored is located downstream, near existing agricultural areas. Restoring this area will reduce the impacts of flooding on communities and agriculture through improved attenuation of flood flows and provide water regulating functions to enhance agricultural productivity. Considering Pakistan's climate, restoration in DI Khan will involve the use of native trees well acclimated for the dry climate and capable to thrive on environments with limited water availability is essential. The project will largely plant Ghaf trees (*Porsopis cineraria*), Farash (*Tamarix aphylla*), Phulai (*Acacia modesta*), Ber (*Zizyphus Spp.*), Jand (*Porsopis cineraria*), Kikar (*Acacia nilotica*), and Shisham (*Dalbergia sissoo*) commonly used for revegetation in dry climates for areas such as the Indus Basin.

Activity 1.1.2: Rehabilitate degraded wetlands, flow paths, and channels.

105. Existing flow paths, channels, and wetlands in the Indus Basin are often filled with silt and debris preventing the channelling and holding of water, exacerbating peak flows (increasing flood risk), and preventing water retention (adversely impacting local water security and exacerbating the impacts of droughts). Under this activity, 34 km of flow paths/channels will be restored at the Ramak Watershed and Manchar Lake project sites. Activity 1.1.2 will be executed by WWF-Pakistan who will work with the Provincial Irrigation Departments of Khyber and Sindh who will be responsible for providing equipment and resources for restoration efforts in the area. The grant with the Provincial Irrigation Departments will be managed by WWF-Pakistan's PIU in each province and the technical staff in the SIUs will coordinate with the Department staff in each province on the project planning, development and reporting.

106. DI Khan watershed has an area of 2,763 km² and lies in the DI Khan district of Khyber Pakhtoon Khwa (KPK) province. The watershed of DI Khan generates flash floods from its main five hill torrents consisting of small perennial "nallahs"/"zams". The annual rainfall is low, sporadic and ranges between 350—400 mm. Several small streams are entering these Zams downstream of the proposed dam locations. Owing to the perennial nature of these Zams and favourable soil conditions, cultivation, and forestation in the upper reaches of the CRBC command area is a natural feature. The DI Khan and Ramak watersheds are located in Dera Ismail Khan, Pakistan's

southernmost district of KPK. The Ramak project area has been included in the same watershed delineation as the DI Khan project areas because they are influenced by some of the same streams and torrents.

107. The Manchar-Hamal watershed covers an area of 9,057 km² and lies in the Kohistan region. Farmers usually practice irrigation by harvesting stormwater by constructing small diversion dykes across the flowing streams. The Manchar Lake project area comprises two lakes, Manchar and Hamal, and is located in Qambar Shahdadkot District in Sindh, Pakistan. The Manchar Lake is linked to the Hamal Lake via the Main Nara Valley Drain and receives additional water from the Kulachi River, small hill torrents from the Khirthar Mountain range and small canals from the Indus. As the main part of the watershed is connected to the watershed of the Chakkar Leri area, the delineation is the same for both project sites.

Sub-Activity 1.1.2.1: Excavate 264 m of flow paths in Badri Village, Ramak, to restore the natural hydrology of connected wetlands.

108. Restoring the flow paths of the Badri channel, which is currently filled with silt, debris, and other obstructions, will allow better freshwater flows via this natural channel into connected wetlands (including the Badri Pond), reducing the risk of water directly flooding community agricultural lands, assets and infrastructure. Excavation of the flow paths will include: determining proper depth to maximize water flows, reinforcing of channel embankments, and removal of debris materials from the flow paths using machinery and labour. Furthermore, the water-holding capacity of the Badri channel, which supplies the Badri Pond (Sub-activity 1.1.2.2), will be enhanced. These two features of Ramak's natural hydrology are the only sources of freshwater for agriculture and domestic use for Badri village, which consists of 300 households with an estimated population of 2,100 people. Their functioning is therefore critical, not only to flood attenuation, but also water supplies during drought periods.

Sub-Activity 1.1.2.2: De-silt the 2 ha Badri Pond in the Ramak wetlands to enhance its water-holding capacity.

109. The build-up of silt will lead to a reduction in water depth, an increased risk of flooding. This usually occurs when plants die, and trees lose their leaves it is also caused by run off from surrounding land as well as the decomposing of natural materials. In order to extend the life of a lake or body of water, it is necessary to periodically remove the build-up of silt from the lakebed using local labour. This sub-activity will enhance the water holding capacity of the Badri Pond by 20,000 m³. As mentioned above, the pond, together with the Badri channel, is the only freshwater source for Badri Village. The silt from the pond will be used to strengthen the embankments of the pond as described under Sub-activity 1.1.2.3.

Sub-Activity 1.1.2.3: Enhance and reinforce 410 m of embankments in Badri Village to reduce erosion.

110. Flood embankments provide protection for peoples, livestock, and community assets such as crops against water surges. These embankments can be damaged by excessive rains and flooding. This sub-activity will improve bank stability by compacting clay soils and planting of native vegetation (such as *Tamarix Spp*, *Populus euphratica* and indigenous grasses) and stone pitching (placement of loose stone at the toe of embankments) to reduce the erosion of embankments during flooding.

Sub-Activity 1.1.2.4: Excavate 4,000 m of flow paths in Paniala Village to restore natural hydrology of Karez Systems (surface and subsurface natural channels).

111. Rehabilitation of Karez Systems⁹⁷ in the Ramak watershed will be an integral part of addressing the flood risks related to Khisor hill torrents, within which the Karez system is situated. Due to flow debris (such as silt) accumulated over years, the existing capacity of the channels is insufficient to accommodate current and future floodwaters. Consequently, they overflow, resulting in damage to nearby community infrastructure and agricultural land. The protection and rehabilitation of the Karez systems, will not only reduce the risk of flooding, but also enhance water supplies through a traditional conveyance system of waterbeds and banks, and multiple storage ponds, for agricultural and domestic use — increasing resilience to drought. Restoration, including desilting and repairs to natural flow paths of the Karez systems, will improve freshwater flows into three nearby natural ponds, which have a combined water-holding capacity of 10,000 m³. This will effectively capture water during flood events and reduce flood impacts, providing a natural flood buffer, while at the same time increasing dry season water supplies for the approximately 1,000 households (6,000–7,000 people) located in Paniala Village.

⁹⁷ Karez is an indigenous method of irrigation in which groundwater is tapped by a tunnel. After running for some distance, the tunnel comes out in the open and the water is supplied to the area that requires irrigation or for domestic use.

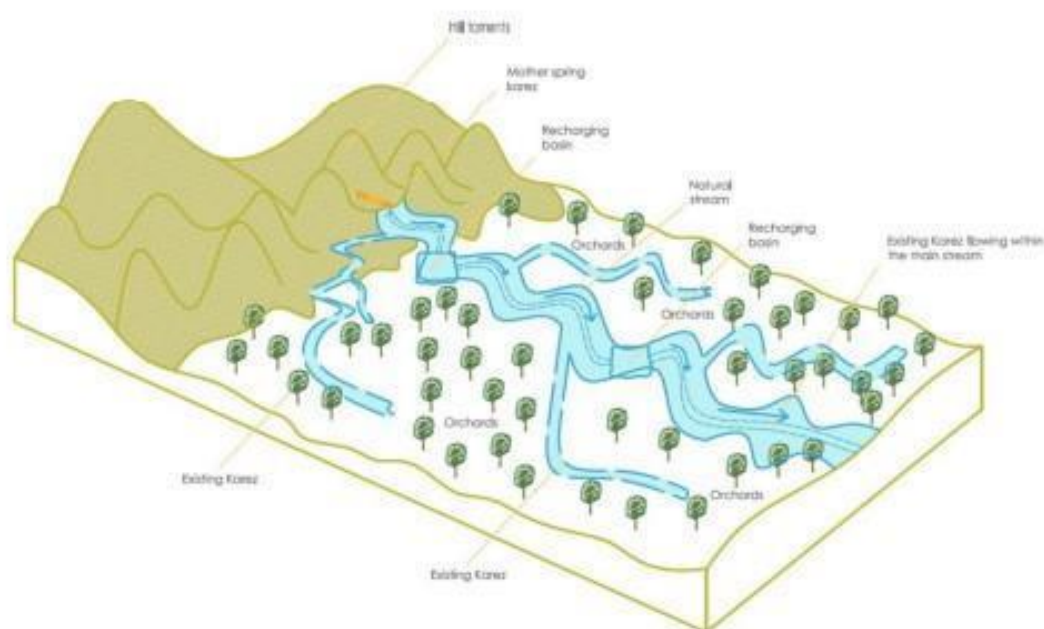


Figure 21. Conceptual Diagram of Karez Channels in Paniala Village, Ramak Watershed.

Sub-Activity 1.1.2.5: Excavate 30,000 m of flow paths in the existing water channels of Manchar Lake to restore the natural hydrology of the 25,000 ha Manchar Wetland.

112. The Indus River flows directly into the Manchar Wetland through three main river channels Aral Wah, Mor lak, and the Danistar Canal. Freshwater flows through these channels are currently obstructed due to blockage from debris and sediment, leaving local communities vulnerable to drought, and the risk of flooding of surrounding areas is increased. The freshwater flows will be restored through excavating the channels to reduce flooding of nearby communities, increase water availability, and restore natural ecosystem functioning essential to the livelihoods of local farmers and fishers. The excavation work will include desilting the channel using small machinery provided by the Provincial Irrigation Departments. Manchar Lake supports the livelihoods of approximately 10,000 households, including 2,000 Indigenous Mohana peoples.

Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.

113. To complement Output 1.1's EbA interventions, green infrastructure interventions will be established to reduce the impacts of floods and droughts on people living in the Indus Basin. In this project, "green infrastructure" is considered as *'interventions which are not naturally occurring or existing in an ecosystem but are designed using natural elements and can contribute to the improved functioning and health of an ecosystem'*. Green infrastructure interventions will include the establishment of flood protection embankments, flood dispersal embankments, gabion bunds, small and large retention areas, and recharge basins.

114. These interventions are expected to benefit a total of 462,600 people (311,300 direct beneficiaries and 151,300 indirect beneficiaries) by reducing flood impacts on nearby communities and ecosystems through reducing soil erosion and river sedimentation, diverting and reducing the velocity of intense flows in the hill torrents away from nearby villages located in the floodplain, and increasing infiltration and groundwater recharge. People living near the green infrastructure interventions, particularly the retention and recharge areas, will benefit from the provision of temporary storage and infiltration of water from these interventions for use during droughts.

Activity 1.2.1: Design and implement 127 green infrastructure interventions for flood risk reduction.

115. The project will commission the design and implementation of green infrastructure measures by a specialized local engineering firm. WWF Pakistan will execute this activity and provide funding to the Provincial Irrigation Departments who will commission and manage the engineering services firms responsible for the design and implementation of green infrastructure measures under the oversight of WWF Pakistan. Green infrastructure interventions will include: i) 9 flood protection embankments spread across the D.I. Khan and Ramak Watersheds; ii) 7 dispersal embankments in the D.I. Khan, Ramak, and Chakar Lehri Watersheds; iii) 45 gabion bunds across the D.I. Khan, Ramak, Manchar, and Chakar Lehri Watersheds; and iv) 42 small and 9 large retention areas, and 15 recharge basins across the D.I. Khan, Manchar, and Chakar Lehri Watersheds. The type, location, quantity and

size of the green infrastructure Projects are included in Table 5 below. More details on the feasibility of the green infrastructure interventions are presented in Annex 2.

Table 5. Details of green infrastructure interventions to be implemented under the project for flood and drought risk management.

#	Location	Village	Quantity	Size
Flood protection embankments				
1	DI Khan Watershed	Jandola, Shahbaz Khel, Kulachi	6	Length: 150 m Width: 150 m Height: 10 m
2	Ramak Watershed	Paniala	3	Length: 150 m Width: 150 m Height: 10 m
Total			9	
Dispersal embankments				
1	DI Khan Watershed	Tank	1	Length: 150 m Width: 150 m Height: 10 m
2	Manchar Watershed	Gaji Shah	1	Length: 150 m Width: 150 m Height: 10 m
3	Chakar Lehri Watershed	New Mitho, Barija, Misar, New Shambani	5	Length: 150 m Width: 150 m Height: 10 m
Total			7	
Gabion bunds				
1	DI Khan Watershed	Jandola, Gomai Zam, Spalipon, Tangi Abbas Khel, Juma Gul Manrai	34	Length: 150 m Height: 6 m
2	Ramak Watershed	Tutti Khel and Abdul Khel	5	Length: 150 m Height: 6 m
3	Manchar Watershed	Wahi Pandi and Qasbo	2	Length: 150 m Height: 6 m
4	Chakar Lehri Watershed	New Mitho, Barija, Misar and New Shambani	4	Length: 150 m Height: 6 m
Total			45	
Small retention areas				
1	DI Khan Watershed	Tank, Kulachi, Jandola, Tangi Abbas Khel, Juma Gul Manrai	34	Length: 275 m Width: 275 m Height: 1 m
2	Manchar Watershed	Tando Rahim Khan	6	Length: 245 m Width: 245 m Height: 1 m
3	Chakar Lehri Watershed	New Mitho	2	Length: 180 m Width: 180 m Height: 1 m
Total			42	
Large retention areas				
1	DI Khan Watershed	Khargai, Kot Nasran, Daraban, Chaudhwan	2	Length: 874 m Width: 874 m Height: 1 m
2	Manchar Watershed	Tando Rahim Khan	2	Length: 1,000 m Width: 1,000 m Height: 1 m
3	Chakar Lehri Watershed	Barija	5	Length: 1,000 m Width: 1,000 m Height: 1 m
Total			9	
Recharge basins				
1	DI Khan Watershed	Tank, Daraban	9	Length: 150 m Width: 14.5 m Height: 2 m
2	Manchar Watershed	Haleli Nadi, Haji Khan Shahani, Chhini	3	Length: 150 m Width: 14.5 m Height: 1 m

3	Chakar Lehri Watershed	Saifabad	3	
Total			15	

Sub-Activity 1.2.1.1: Design the green infrastructure measures.

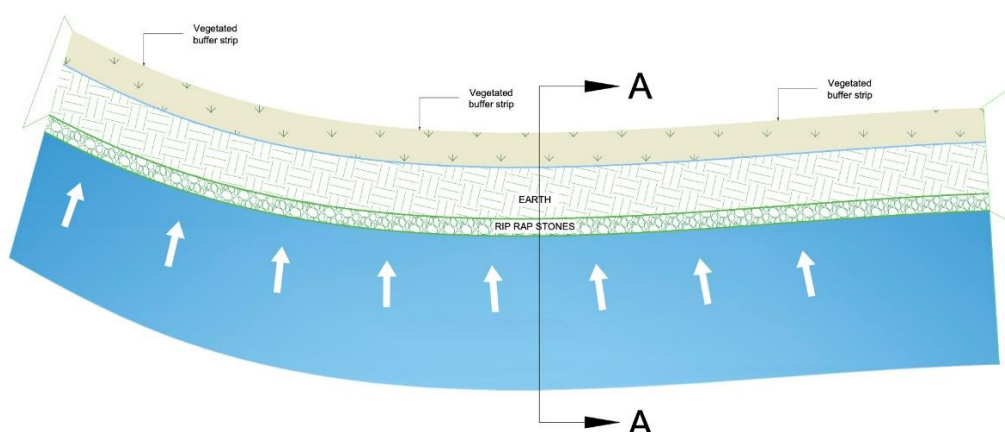
116. Under this sub-activity, the green infrastructure interventions under Sub-activities 1.2.1.2–1.2.1.7 will be designed. Preliminary assessments of the green infrastructure measures have already been completed as part of the Feasibility Study (Annex 2) to inform their design and implementation. Specifically, these assessments determined the type and cost-effectiveness of implementing green infrastructure interventions to reduce flood impacts in the Indus Basin, as well as the areas where such measures will have maximum flood and drought reduction benefits for communities. The proposed locations for these interventions are presented in Section 3 of Annex 2, along with concept drawings showing their preliminary design specifications.

117. The design process will include: i) a site validation visit, and assessment of each proposed location; ii) the preparation of site-specific engineering designs for each proposed structure; and iii) an assessment of the site-specific costs for constructing each structure. The site-specific assessments will start in Year 1 of the project period and be completed by the end of Year 2. A certified local engineering firm will be procured to develop the detailed site-specific designs for all the green infrastructure measures under Activity 1.2.1.

Sub-Activity 1.2.1.2: Build 9 flood protection embankments

118. Sandbags traditionally used to retain and divert floodwaters fail to provide adequate protection for community assets and homes, particularly as flood events continue to intensify under conditions of climate change. Six (6) flood protection embankments will be built in D.I. Khan and three (3) in the Ramak Watershed using locally available materials, including clay-silt, earth, stones, and gravel (Figure 22). The embankments will retain floodwaters moving at fast velocities and control lateral erosion along riverbanks, protecting crops, property, and infrastructure from damage. Grass will be planted, and other supplementary structures established on the embankments to provide additional protection against erosion and overtopping where needed. To further reinforce the grass on the flood embankments, the riverbank slope will be armoured with stone pitching or riprap (loose stone).

PLAN OF FLOOD PROTECTION EMBANKMENT



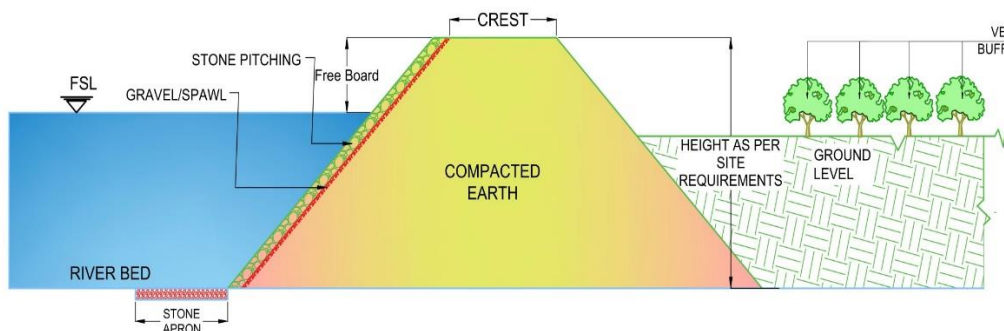
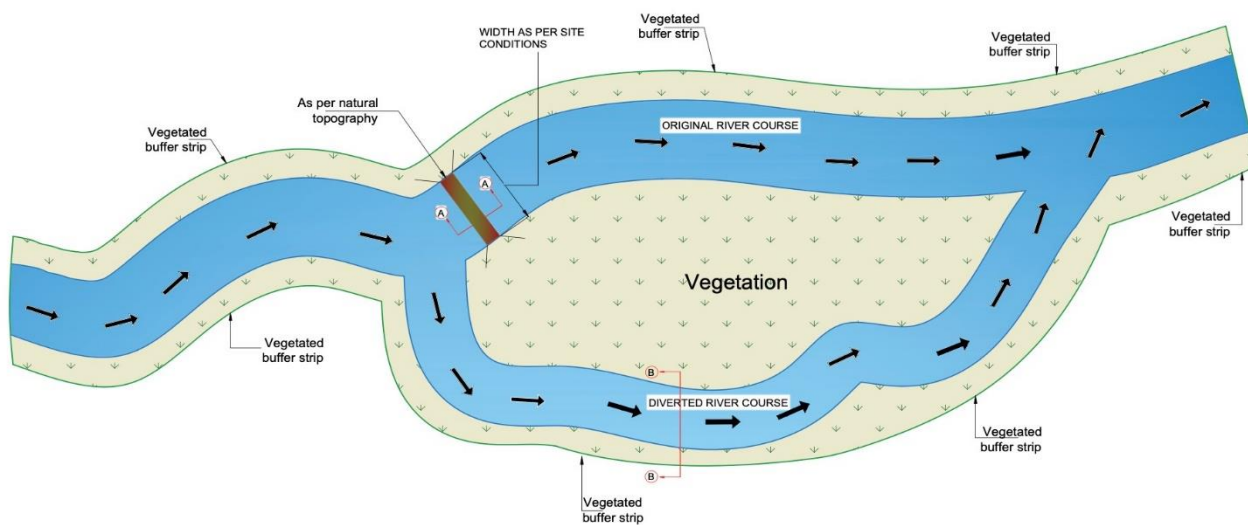


Figure 22. Conceptual drawing of flood protection embankments for D.I. Khan and the Ramak watershed. Top: flood protection embankment. Bottom: Cross-section view of flood protection embankment.

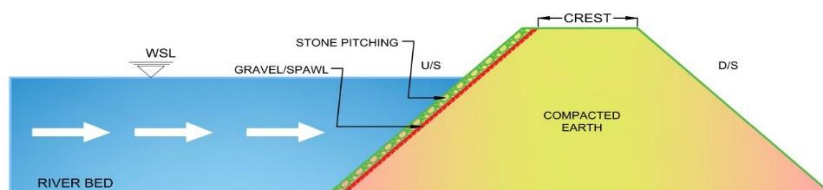
Sub-Activity 1.2.1.3: Build 7 flood dispersal embankments

119. Floodwater dispersal measures include earthen embankments, earth-clay bunds, levees, watercourses, water channels, and other structures that modify the natural flows of a waterway. Water dispersal or diversion consists of multiple measures that intercept clear surface water runoff upstream of an area at risk of flooding, transport it through the natural path, away from at-risk property, infrastructure or land, and discharge it safely downstream. Dispersal embankments will be built with local materials such as earth, clayey-silt, sandy loam stones, and gravel in the catchments of D.I. Khan and Manchar — 1 flood dispersal embankment will be built in each catchment (Figure 23). The embankments will be armoured with stone, placed in the form of riprap or stone pitching. The crest of the flood dispersal embankments will be further strengthened with murram (gravel) on the overflow sections to protect them against extreme flood flows, which are more likely to occur under future climate scenarios. The diverted path will be crossed through existing channels and/or non-perennial streams so local farmers can use the now controlled floodwaters for irrigation purposes.

PLAN INDICATING THE FLOOD FLOW AFTER DISPERSAL EMBANKMENT



SECTION A-A: SECTION OF FLOOD DISPERSAL EMBANKMENT



SECTION B-B: FLOOD DISPERSAL EMBANKMENT

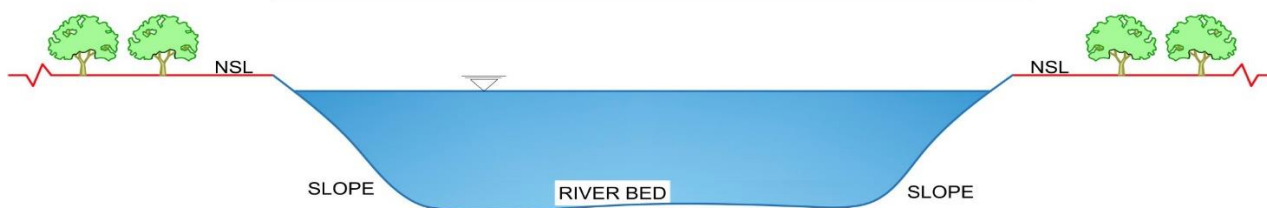
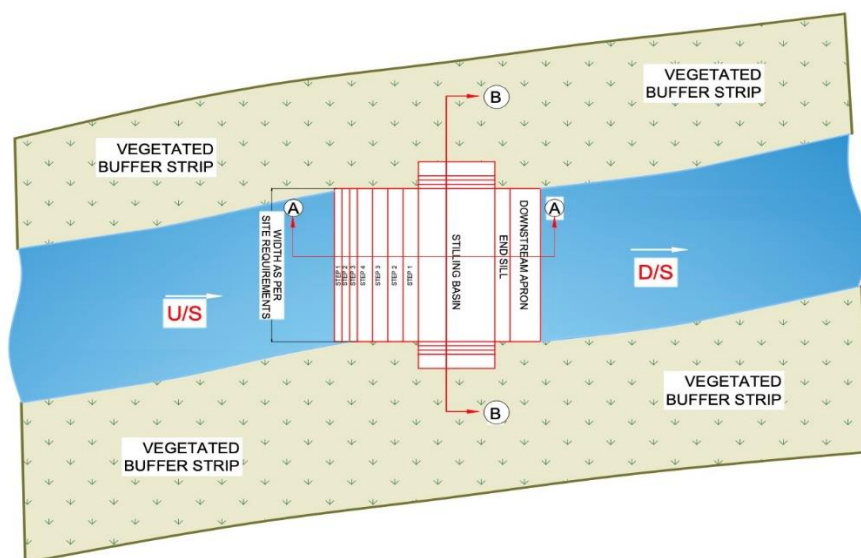


Figure 23. Conceptual drawing of flood dispersal embankment for D.I. Khan and Manchar watershed. Top: Plan of flood dispersal embankment. Bottom: Cross-section view of a flood dispersal embankment.

Sub-Activity 1.2.1.4: Build 45 gabion bunds.

120. A gabion bund is a wire cage filled with stones, boulders, rocks, or sand/soil. It is often used to stabilize and protect riverbeds, riverbanks, or slopes against erosion. Gabion stepped weirs are commonly used for river training and flood control; the stepped design enhances the rate of energy dissipation in the channel. The interlocking of the individual stones and rocks within the wire mesh ensure internal stability, and their mass coupled with vegetation will assist in resisting hydraulic forces. Gabion bunds will be established in the catchments of Manchar (2), DI Khan (34), Ramak (5) and Chakkar Lehri (4) to mitigate the risks of floods through delaying and diverting floodwaters (Figure 24). The gabion bunds will be designed to act as a spillway, safely diverting floodwaters.

PLAN OF GABION BUND



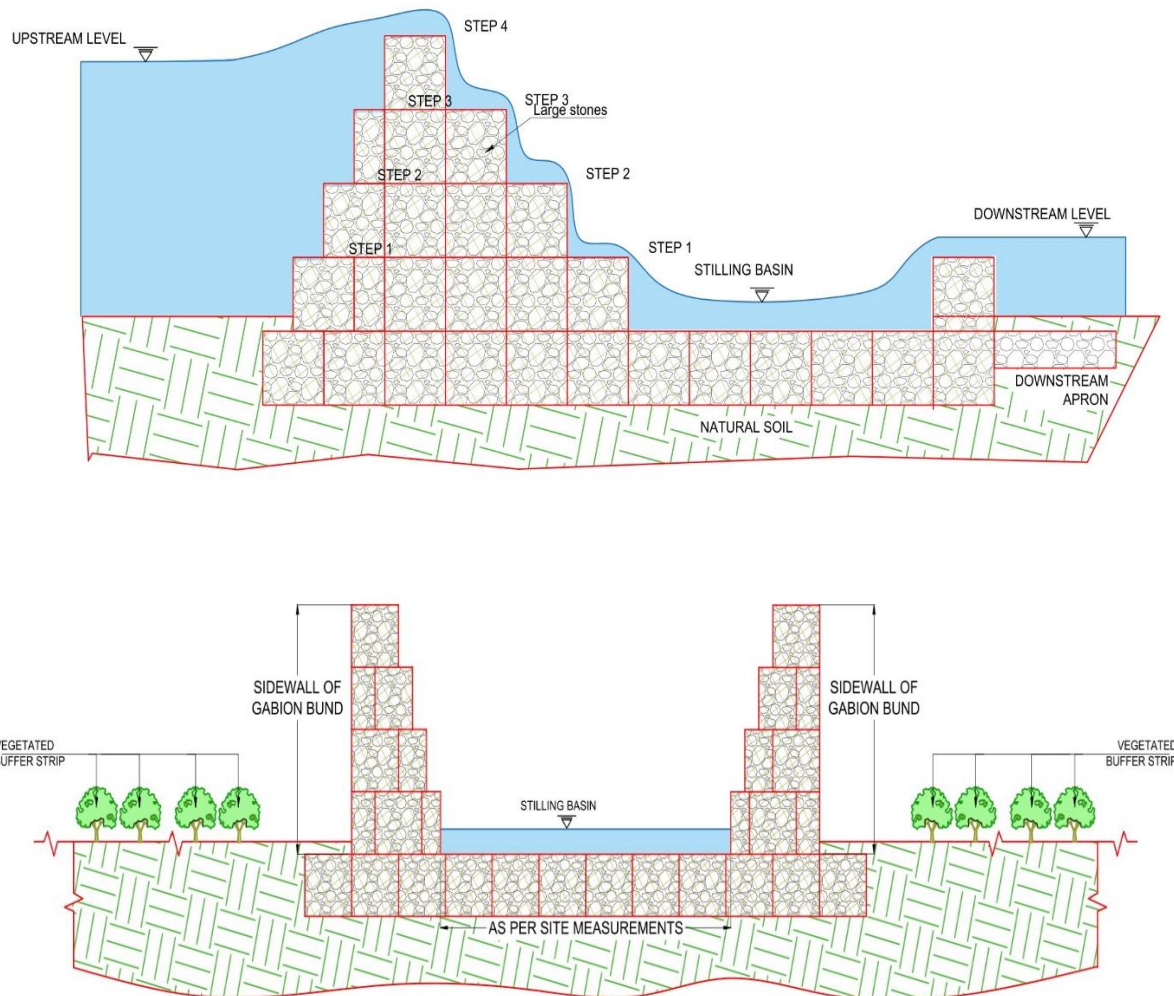


Figure 24. Conceptual drawing of gabion bunds for DI Khan, Ramak, Manchar and Chakar Lehri Watershed. Top: Plan of gabion bund. Middle and bottom: Cross-section view of gabion bund.

Sub-Activity 1.2.1.5: Build 42 small retention areas.

121. Retention areas are ponds where excess river discharge or floodwater is stored (Figure 25). Generally, they are connected to a river via channels so that in periods of high discharge or flooding, the retention areas are filled, which reduces the peak flow, increases infiltration, and stores water for domestic and/or agricultural purposes. During the field survey conducted as part of the project's Feasibility Study (see Annex 2), potential intervention locations were identified and were prioritized if they overlapped with hot-spot areas for water level fluctuation and discharge, i.e., areas with high flood peaks. For maximum impact (water harvesting and flood attenuation), retention areas were mainly identified in upstream areas of the target watersheds. Sites for the establishment of retention ponds were also selected to take advantage of natural depressions and existing contours with some augmentation to ensure water remains trapped for a sufficient amount of time to reduce flood peaks, aid infiltration, and recharge groundwater. The retention areas will be established using local materials (such as stones and gravel), in natural depressions and watercourses as mentioned above. This will limit the transformation of the landscapes and ecosystems — reducing any risk of environmental impact. The detailed design of the retention areas (Sub-activity 1.2.1.1) will ensure that only flood peaks are captured, maintaining downstream water flows. The small retention areas to be established include 34 in D.I. Khan, 6 in Manchar Watershed, and 2 in Chakar Lehri Watershed.

Sub-Activity 1.2.1.6: Build 9 large retention areas.

122. Large retention areas have been proposed in areas of the Indus Basin where a river exits the mountains through a gorge, during the same field surveys described in Sub-activity 1.2.1.5. The flow focus results, and the discharge patterns are theoretically the same as for the small retention areas and served as the basis for the selection of the sites for the large retention areas. The design and functioning of the large retention areas is similar to those of the

smaller retention areas as described above (Sub-activity 1.2.1.5). The large retention areas to be established include 4 in the D.I. Khan watershed, , 2 in the Manchar Watershed, and 1 in the Chakar Lehri Watershed.

RETENTION AREA (CONCEPTUAL DRAWING)

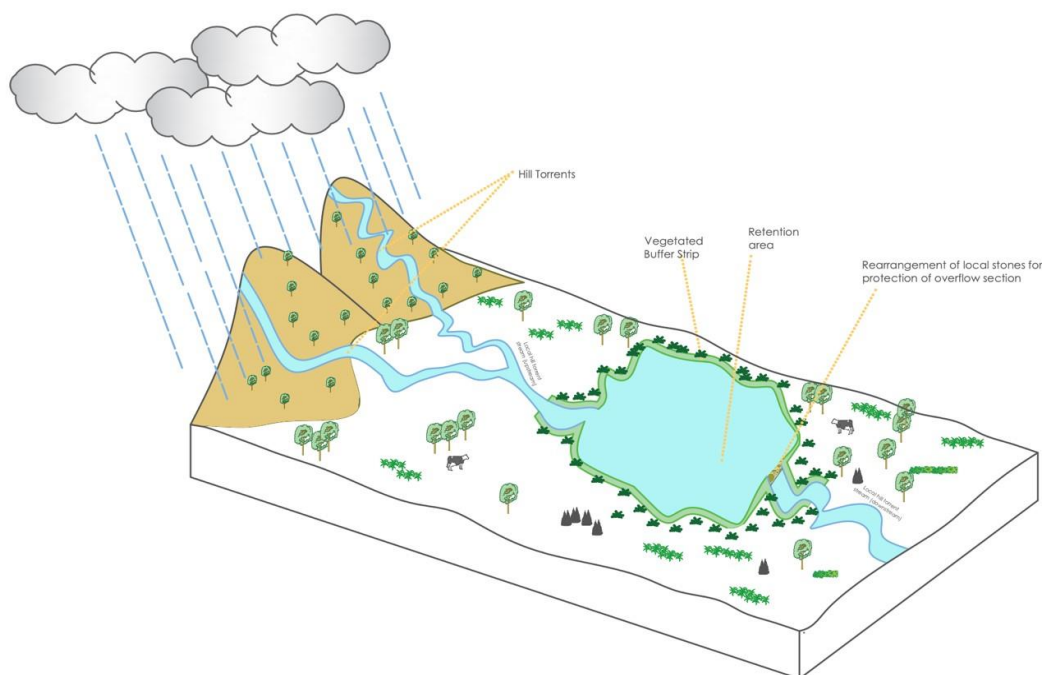
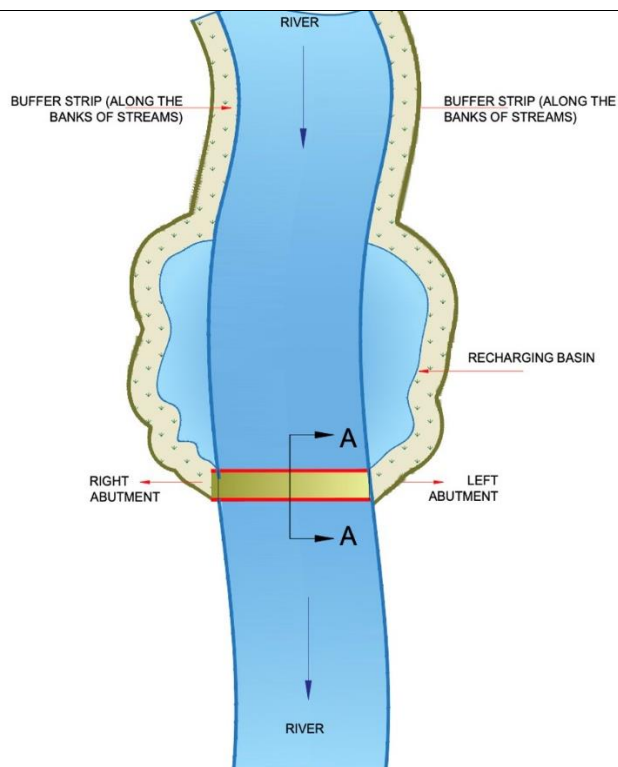


Figure 25. Conceptual drawing of retention areas of DI Khan, Manchar and Chakar Lehri watersheds.

Sub-Activity 1.2.1.7: Build 15 recharge basins.

123. Recharge basins can divert and slow down floodwaters and recharge shallow aquifers (Figure 26). They contribute to adaptation by diverting and slowing down floodwater flows and reducing flood peaks, and the risk of flooding downstream. However, their main goal is to capture water under the sediment layer, providing a constant water supply through recharging of the groundwater table, thereby reducing the impacts of droughts. Due to groundwater increases, biodiversity can benefit as well. This will also lead to an increase in local vegetation which in turn will contribute to a decrease in erosion and flow velocity during heavy rainfall events, further reducing flood risks — complementing the ecosystem restoration interventions to be implemented under Output 1.1. A recharge basin is different from the retention areas in that it has a sandy layer at the bottom to enhance recharge. The recharge basins to be established include 9 in the D.I. Khan Watershed, , and 3 in the Manchar Wetland, and 3 in Chakar Lehri Watershed.



SECTION A-A: SECTION OF RECHARGING BASIN

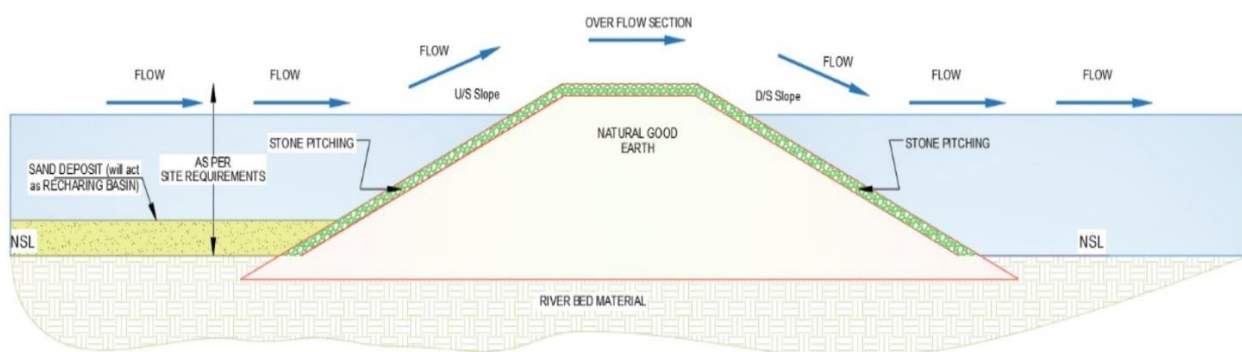


Figure 26. Conceptual drawing of a recharge basin to be established in the DI Khan, Manchar and Chakar Lehri watersheds. Top: Plan view of recharge basin. Bottom: Cross-section view of recharge basin.



Figure 27. Photograph of a recharge basin in Sindh Province captured during a field visit.

Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.

124. Under this output, 18,215 ha of land in the Indus Basin (including the 14,215 ha that will be restored and rehabilitated, as well as an additional 4,000 ha under Component 1) will be placed under improved community-based climate-resilient management to support the adoption, implementation and maintenance of the EbA and green infrastructure interventions by the beneficiary communities, as well as strengthen the health of ecosystems. This will be achieved through directly engaging CBOs⁹⁸ — which are important contributors to water resources management in Pakistan, taking the lead for sustainably managing these resources at the household level — to: i) create community agreements with the CBOs to establish a common set of practices for communities and district-level governments on the use and management of ecosystems (including the EbA and green infrastructure interventions); ii) organize Water User Groups (WUGs) under CBOs and build their capacity for improved water resources management at the local level; iii) establish Community-based Watch and Ward Systems under CBOs to protect the project's EbA and green infrastructure interventions from degradation; and iv) strengthen communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions. These interventions will be supported by capacity-building and stewardship to ensure the project's EbA and green infrastructure interventions continue contributing towards flood and drought risk reduction at the local level during and beyond the project period.

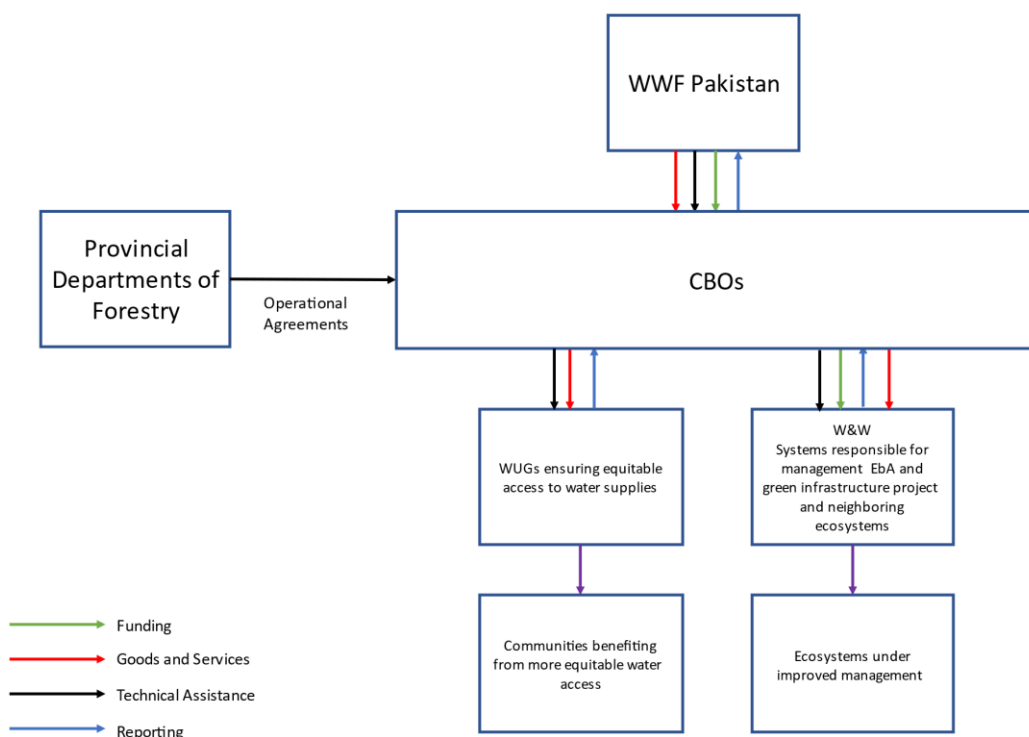
Activity 1.3.1: Strengthen the capacity of CBOs to adopt EbA and green infrastructure interventions and undertake climate-resilient community-based natural resource management.

125. Under this activity, CBOs will be strengthened to support the adoption of EbA and green infrastructure interventions, as well as climate-resilient community-based natural resource management at the local level. WWF-Pakistan's project teams will work closely with at least 8 CBOs and the relevant provincial government departments in the Indus Basin to establish 8 WUGs and 4 Watch and Ward (W&W) Systems under these CBOs (Figure 28).

126. Four Watch and Ward Plans (one plan per project area) will be developed by Watch and Ward Systems under their respective CBOs and the Provincial Forest Departments. These plans will include provisions for ecosystem protection (such as protecting natural forests from illegal cutting and forest fires) and restoration, as well as operations and maintenance of the EbA and green infrastructure interventions. Operational agreements (terms of partnership & FPIC) will be created for the implementation of the Watch and Ward Plans assigning specific operation, maintenance, and stewardship roles among communities and Provincial Departments of Forestry of Sindh, Khyber Pakhtunkhwa and Balochistan. In addition, community agreements will be established with WUGs to define arrangements for accessing and using the additional water made available by the project's EbA and green infrastructure interventions, as well as determine water allocations for agriculture, domestic and disaster risk reduction measures to ensure effective distribution of water resources at the community level.

127. To support these community agreements, the 8 CBOs, 8 WUGs and 4 Watch and Ward Systems will be: i) engaged to implement and support the maintenance of EbA and green infrastructure interventions, incorporating indigenous knowledge and skills; and ii) trained and equipped for climate-resilient land and water resources management to secure the sustainability of the project's EbA and green infrastructure interventions, as well as the health of the ecosystems where they are located, e.g., maintenance of wetlands, floodplains, flood bypasses and washlands, watersheds and plantations (see Outputs 1.1 and 1.2). These capacity-building activities will be supported by field visits for CBO leaders (who sit on executive committees) to the EbA and green infrastructure interventions' sites. This will be done to demonstrate the progress and effectiveness of the project's EbA and green infrastructure interventions, support the uptake of these interventions at the community level, and enable knowledge on the experienced benefits and lessons learned to be shared with additional CBOs and communities across Pakistan's three provinces through the CBO executive committees. Activity 1.3.1 will be executed by WWF-Pakistan who will be providing technical assistance and grant funding to CBOs in the Indus Basin. If the need for new CBOs is determined during execution of the project, WWF Pakistan will take the following steps in keeping with WWF Pakistan's Social Mobilization Strategy:

1. Plan visits to the sites and identify focal points in each village/site and plan focus group discussions;
2. Request that the focal points organise community gatherings to present the project and its objectives in more detail with community members;
3. Request that the focal points organise community gatherings to present the project and its objectives in more detail with community members;
4. Identify and prioritise needs in the communities related to flooding and drought prevention and prepare community action plans. Engage the local people and apply participatory assessment tools for social mapping, wealth ranking, needs identification, prioritisation, SWOT analysis, risk assessment and mitigation planning, seasonal calendar, daily activity chart etc;
5. Organise local communities into CBOs developing participatory decision-making processes to ensure the needs of all constituencies are considered in CBO planning and decision making, ensuring e representation of marginalised and vulnerable households;
6. Once the CBOs are established and have received the necessary capacity development from the project, activities can be outsourced to the CBOs through grant agreements. WWF Pakistan will monitor the execution and progress of those activities.



⁹⁸ which represent Village and Women's Organisations as well as youth committees at the village level

Figure 28. Community-based natural resource management arrangements.

Sub-Activity 1.3.1.1: Strengthen CBOs to adopt EbA and green infrastructure interventions and implement community-based natural resource management.

128. The project will provide capacity development and training to CBOs to: i) strengthen and formalize their organization, governance, and management structures; ii) integrate EbA and green infrastructure interventions for flood and drought risk reduction within their mandates; iii) train and equip CBO members in W&W functions, including the implementation and maintenance of EbA and green infrastructure interventions, ecosystem management and monitoring of wetlands, floodplains, flood bypasses and washlands, watersheds and community plantations; and iv) establish and operate WUGs in communities (Table 6).

129. WWF-Pakistan will carry out detailed assessments for each of the CBOs in Year 1 and develop action plans to enhance participation, including the participation of underrepresented and vulnerable groups. It will design and carry out capacity building and participatory exercises to improve the ability of CBOs to define flooding and drought management priorities and sustainably manage nearby ecosystems to increase community resilience and reduce risks identified in section B.1. Trainings will be delivered to CBOs in each community twice a year in 2-day courses. The trainings will engage CBO managers and future W&W agents. 25 W&W agents will be trained per training (~240 in total). Each W&W agent will be responsible for 40 ha of land to be patrolled weekly. Daily wages for W&W agents during their patrols will be covered by the project grants to the CBOs. Training curriculums for each community will be adapted to meet specific community needs based on their capacities and needs identified in the table below and during further consultations and assessment of community capacity needs in Year 1 of the project.

Table 6. Overview of CBO baseline, gaps and Project approached for strengthening the capacity of CBOs in intervention areas to undertake community-based natural resource management.

Community training and capacity development activities across all provinces		
- Training and capacity development for CBOs in organizational management, including financial management and record keeping, and proposal development to independently secure and execute small grants from local donors for community-based natural resource management Projects. - Working with CBOs to ensure better gender representation in their leadership and decision making. - Establish new W&W systems and/or train existing W&W agents in ecosystem management and monitoring of wetlands, floodplains, flood bypasses and washlands, watersheds and community plantations. - Establish new and/or train existing WUGs to establishing water use allocations for agriculture, domestic and disaster risk reduction uses. - Raise awareness in communities on the impacts of climate change and appropriate adaptation solutions.		
Existing community-based natural resource management arrangements	Gaps	Province specific activities
Khyber Pakhtunkhwa		
The Forest and Wildlife Department of Khyber Pakhtunkhwa established CBOs under the Billion Tree Afforestation Project (BTAP) to engage communities in afforestation activities and community watch and ward systems.	- WWF-Pakistan conducted the third-party monitoring of BTAP, under which it was observed the community W&W agents needed training on the demarcation of enclosures, grazing and fire management, restoration and collecting basic vegetation data (such as survival rate, species composition, record and causes of damage, etc). - No formal WUGs exist to manage community water access disputes.	- Train W&W agents in fire management, illicit grazing management, deforestation prevention, exclusion area establishment and restoration. - Establish WUGs governance mechanism to ensure effective implementation and distribution of water resources.
Balochistan		
CBOs were established in the aftermath of the 2010 and 2012 floods by the UN and other relief agencies to ensure direct involvement of affected communities in relief and reconstruction.	Once the relief Projects were completed, all these CBOs were left without any follow-up or financial support or training in how to access public grants.	- The Project will provide assistance to these CBOs in registering as official organizations with the Provincial Government of Balochistan. - Establish WUGs governance mechanism to ensure effective implementation and distribution of water resources.
Sindh		

The Indus Ecoregion Community Livelihoods Project (IECLP) formed eight CBOs and one CBO Network known as the Manchar Conservation Network in 2018 with the objective to support Project implementation and sustain the Project interventions once Projects are completed.	• Lack of representation from all ethnic groups. • Lack of proper training on organizational management and conflict resolution. • Lack of exposure to development agenda due to the high level of illiteracy and poverty.	• Working with CBOs to ensue better ethnic and gender representation in leadership and decision making. • Train these CBOs in organizational management and proposal development to secure and execute small grants from local donors.
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Sub-Activity 1.3.1.2: Organize WUGs under CBOs and build their capacity for improved water resources management.

130. The GoP has been involved in the transfer of irrigation management to farmers' organizations at different levels of irrigation networks. This sub-activity will build on the GoP's initial work by establishing and strengthening of WUGs (associations of community members responsible for water planning), which will lead to improved water management mainly through reducing water theft and conflicts around water, as well as improving maintenance of waterways. WUGs empower communities to take ownership and get involved in managing and monitoring water-related ecosystems and infrastructure and help fill government capacity gaps in water resource management. The project will organize new and bolster existing WUGs through CBOs and provide technical capacity development to create and/or strengthen WUG governance and management structures, increase WUG membership through community trainings and awareness raising campaigns, and establish WUG operational plans and monitoring frameworks, including for the project's EbA and green infrastructure interventions. If new WUGs are needed, these will be established under the CBOs to ensure equitable access to water resources. CBOs will nominate members of WUGs through a community resolution signed by the executive bodies of the CBOs. The WUGs members will report to the executive body of the CBOs. WWF-Pakistan will work with communities and the Provincial Irrigation departments to ensure alignment of the WUG workplans with provincial adaptation plans. WUGs will operate under the umbrella of local CBOs to improve water governance at the local level and formalize local water resource allocation processes. WUGs will also play a key role in propagating effective water utilization and reduce wasteful consumption at domestic and farm levels.

131. The water users of irrigation water will be the cultivators of lands, owners of private farms, lease-holding farms, owners of garden plots, etc. whereas, the water users of drinking water will primarily be households. The members of the committee will be selected from the CBOs of the respective project sites. The head of the committee will be a chairman who will be elected by the CBO members and a general body. Under the overall supervision of the CBOs and working in close collaboration with local community members, the WUGs will undertake the following specific duties:

- Contribute towards the analysis of general physical conditions of the local villages, including assessment of water availability once the drinking and domestic needs have been met;
- Ensure equitable distribution of drinking and irrigation water among water users;
- Ensure reliable supply of drinking as well as irrigation water to target water users;
- Resolve disputes that may arise on the usage or consumption of water among water users at a local level;
- Ensure the provision of technical resources for the operation and maintenance of irrigation channels and drinking water supply from the members of the water user groups;
- Prevent water theft and take action against those who are involved in water theft;
- Support the Provincial Irrigation Departments with regular Operation and Maintenance (O&M) and facilitate the implementation of water-related interventions in the project village;
- Coordinate with the project team in executing project activities under Output 3.2;
- During project execution, work with the CBOs to explore potential options for fee structures for irrigation and drinking water access to finance the O&M costs of community water infrastructure to be implemented after the GCF project period as part of the exit strategy;
- Assist the local government, CBOs and local communities with planning and scaling of existing systems; and
- Provide progress updates to the CBOs and their executive committees.

132. WUG participation within communities will be incentivized by linking incremental project support with WUG development milestones. To ensure sustainability, the WUGs will also be responsible for establishing a pay per use system in close consultation with local communities and project team to cover operation and maintenance cost once the project ends.

Sub-Activity 1.3.1.3: Establish community-based watch and ward systems under CBOs to protect the project's EbA and green infrastructure interventions from degradation.

133. Community-based W&Ws systems are community managed enforcement networks of individuals responsible for and taking ownership of, in this context, the improved management of ecosystems on which they depend. These W&W systems will be established under the CBOs to facilitate the implementation and protection of the project's EbA and green infrastructure interventions including their required operation and maintenance. This will be operationalized by the development of 4 W&W Plans by the CBOs and Provincial Forest Departments to protect and preserve freshwater ecosystems. These plans will include: i) ecosystem protection and restoration commitments; ii) provisions for the operation and maintenance of EbA and green infrastructure interventions developed under Component 1; iii) a process for operationalizing ecosystem and community impact monitoring; and iv) specific operation, maintenance, and stewardship roles agreed between CBOs and the Provincial Departments of Forestry.

Activity 1.3.2: Strengthen communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.

134. Under this activity, communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions will be strengthened through a tailored communications strategy. The audience will include members of vulnerable communities, particularly women and children, students and academia, policymakers, and the general public. The objectives of the communication strategy include: i) educating communities, students, tourists and other project stakeholders on the impacts of climate change, particularly floods and droughts, on people and ecosystems, as well as on how to respond to these impacts through implementing EbA, green infrastructure and efficient community-based natural resources management; ii) informing relevant stakeholders about the project's EbA and green infrastructure interventions, their benefits and the progress made; and iii) communicating the successes and impacts of the project interventions to build a case for scaling up EbA and green infrastructure interventions. The following communication tools will be utilized: videos and project photography, short documentaries, press releases, news packages (electronic), blogs, signboards, brochures, manuals, briefs, flyers, posters, standees, etc. for stakeholder use, and posters, information sheets and pictorials for community members. This project activity will be executed by WWF-Pakistan.

Sub-Activity 1.3.2.1: Consult with tribal elders to support the sensitization of the project's adaptation interventions in beneficiary communities.

135. Consulting with tribal elders to support the sensitization of the project's adaptation interventions in beneficiary communities is important because many communities still follow tribal leadership. Under this sub-activity, consultations will be held with tribal elders to: i) promote dialogue among communities on the risks and impacts of floods and drought and how these are being intensified by climate change; ii) coordinate with tribal elders on the sensitization of the project's EbA and green infrastructure interventions among local communities; iii) garner support for the on-the-ground implementation, uptake and sustainable management of the project's EbA and green infrastructure interventions at the local level.

Sub-Activity 1.3.2.2: Develop and implement awareness-raising campaigns for target communities to increase their knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.

136. Under this sub-activity awareness-raising campaigns will be developed and implemented for target communities to increase their knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions. To ensure that the information shared via the campaigns is contextually relevant to the different areas across the Indus Basin, it will be tailored to be site- and ecosystem-specific, through engagements with representatives of local communities and institutions who are members of local CBOs. The campaigns will focus on: i) the implementation and benefits of EbA and green infrastructure interventions and improved natural resource management for flood and drought risk reduction; ii) raising awareness about the impacts of climate change (particularly floods and droughts) on ecosystems, communities and economic sectors; iii) the progress of the project's EbA and green infrastructure interventions and how they have helped communities adapt to the impacts of climate change. Several different channels will be used to communicate this information to communities, including signboards, newspaper publications, local radio stations, short videos, photography and other digital media.

Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan

Outcome 2: Enabling environment for climate action that has new procedures for implementing EbA and green infrastructure interventions for flood and water resources management in Pakistan.

Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.

137. To initiate public sector reform in Pakistan's approach to flood and water resources management and ensure EbA and green infrastructure interventions are implemented at scale, decision-makers must have strong evidence of the environmental and economic benefits of these interventions. Under Output 2.1, an evidence base will be established to demonstrate to policy makers and planners at the national and local levels (including the Ministry of Climate Change, Ministry of Water Resources, Federal Flood Commission, Ministry of Planning, Development and Special Initiatives, Pakistan Council for Research in Water Resources, Provincial Planning and Development Departments, Provincial Irrigation Departments, and Provincial Agriculture departments) the effectiveness of EbA and green infrastructure interventions for flood and drought risk reduction, as well as their economic, social and environmental benefits. This evidence will be generated through monitoring and evaluation of the activities under Component 1 of the project.

Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan.

138. The evidence will be generated under the project through an economic valuation of the project's EbA and green infrastructure interventions (described in detail in Component 1) which will include monitoring and analysis of these interventions' cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan. Communities will be engaged throughout this process through their representative CBOs to ensure the analysis of the adaptation benefits of EbA and green infrastructure captures communities' experiences and appropriately addresses communities' needs.
139. The EbA interventions are expected to begin delivering adaptation benefits after 3 years, with the analysis work commencing in Year 4 of the project. The analysis will consider: i) avoided costs and losses — impacts or losses that would otherwise occur in the absence of ecosystem services generated by the project's EbA interventions, such as increased flood extent, damages and/or losses to agricultural land and the subsequent reduced productivity, and reduced water resource quantity and quality; and ii) alternative costs — the implication of implementing 'grey' artificial systems (as opposed to 'green' EbA solutions) that would be required to deliver the same adaptation benefits as ecosystems. The analysis will also consider a wide variety of sustainable development co-benefits, including but not limited to enhanced biodiversity, decreased pollution of water resources and improved access to clean water (see Section D.3 for additional details on the sustainable development potential of the project).
140. To ensure the evidence generated under this output is used to expand Pakistan's regulations and policies for flood and water resources management from only focusing on using grey infrastructure instead of prioritising EbA and green infrastructure interventions to address the impacts of floods and droughts, this evidence will be used for the development of updated procedures under Output 2.2 for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, and Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.

Sub-Activity 2.1.1.1: Prepare baselines and indicators for the biophysical benefits of EbA on ecosystems and their flood attenuation and water resource-related services.

141. Because an economic valuation of the adaptation benefits of EbA and green infrastructure interventions has not yet been conducted for Pakistan, the baseline for the valuation to be undertaken under Activity 2.1.1 will be assessed and established under this sub-activity. The relevant indicators that will be used to monitor and analyse the performance of the EbA and green infrastructure interventions will also be set under this sub-activity, including metrics for the 'Ground Water Atlas' which is included within the monitoring framework of the National Water Policy. The project will help the Provincial Irrigation Departments align their monitoring efforts at the sub-basin for groundwater recharge potential, water quality and water table depth zones for recharging aquifers to the Atlas. Establishing the baseline and setting appropriate indicators is the first step in undertaking the economic valuation, so this will happen in Year 1 before the monitoring and analysis work can begin.

Sub-Activity 2.1.1.2: Monitor the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services using the baselines and indicators prepared under Sub-activity 2.1.1.1.

142. Under this sub-activity, the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services will be monitored annually using the baselines and indicators prepared under Sub-activity 2.1.1.1. This will include measuring these interventions' flood and drought risk reduction benefits, as well as their ecosystem service-related benefits, such as improved infiltration, and enhanced soil and water quality to build a case for integrating EbA and green infrastructure into policy and planning (Output

2.2). The project's M&E staff will undertake this monitoring (supported by community leaders/representatives wherever feasible) and compile the data in an appropriate format for the economic valuation study (Sub-activity 2.1.1.3 below).

Sub-Activity 2.1.1.3: Conduct an economic valuation of the benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services.

143. The economic valuation of ecosystems and their services and benefits will be conducted by WWF-Pakistan under this sub-activity using tools such as Integrated Valuation of Ecosystem Services and Trade-offs (InVEST) to value natural capital and evaluate the contribution of EbA and green infrastructure towards flood and water resources management under future climate change conditions.

Sub-Activity 2.1.1.4: Identify additional opportunities and sites at the national level across Pakistan for scaling up and replicating the project's EbA and green infrastructure interventions for flood risk and water resource management.

144. Once the economic value and benefits of the project's EbA and green infrastructure interventions have been assessed (under Sub-activity 2.1.1.3), site assessments will be conducted to identify opportunities for scaling up and replicating these solutions across other suitable areas in Pakistan that are vulnerable to floods and droughts. The identification of opportunities for scaling up and replicating EbA and green infrastructure interventions across Pakistan under this sub-activity will consider additional sites to the 11 sites identified during the project's Pre-Feasibility Study stage, with the project's EbA and green infrastructure interventions serving as blueprints for addressing increasingly severe floods and droughts caused by climate change across similar contexts nationally and regionally.

Sub-Activity 2.1.1.5: Collate and store the data and information generated under 2.1.1.1–2.1.1.4

145. The data and information generated by sub-activities 2.1.1.1–2.1.1.4 will be collated, managed and stored by WWF-Pakistan for Years 1–5 of the project implementation period. After Year 5, the information will be transferred to the MoCC and the FFC (under MoWR) and the Provincial Irrigation Departments to serve as evidence for future flood and water resources management and planning in Pakistan. While the collated data and information will initially be managed by WWF-Pakistan during Years 1–5 of the project, it will still be made readily available to project partners throughout the project period. The FFC will be responsible for sharing the collated evidence with relevant decision-makers as described in Activity 2.1.2 below.

Activity 2.1.2: Develop and disseminate an evidence report to national and provincial government agencies for implementing EbA and green infrastructure interventions in response to flooding and droughts in Pakistan.

146. Under Activity 2.1.2, the data and information generated under Activity 2.1.1 will be collated and synthesised into an evidence report, highlighting EbA and green infrastructure interventions' flood and drought risk reduction potential, economic value and environmental and social benefits. This report will serve two main purposes. First, it will be used to develop updated procedures under Output 2.2 for Pakistan's key regulations and policies for flood and water resources management on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. Second, this evidence report will stand alone as an important knowledge product and decision-making tool for policymakers and planners in Pakistan to use in other decision making around climate change planning and flood and water resources management. Moreover, because Pakistan has limited data on the cost-effectiveness of EbA and green infrastructure interventions, the economic valuation underpinning this evidence will demonstrate the cost-effectiveness and wide variety of adaptation benefits delivered by these interventions and serve as a key reference for policymakers in future flood and water resources management.

147. WWF-Pakistan will work with FFC to disseminate this evidence report to the MoCC, the Provincial Irrigation Departments, Provincial Agriculture Departments, Provincial Forestry Departments, academic and research institutions, as well as other relevant national and provincial stakeholders. The MoCC, as the NDA and chair of the PSC, will be responsible for the integration of lessons learned and best practices for EbA and green infrastructure from the project into other ongoing and future projects in Pakistan.

Sub-activity 2.1.2.1: Prepare an evidence report on the use and benefits of EbA and green infrastructure interventions in response to flooding and droughts in Pakistan.

148. Under this sub-activity, WWF-Pakistan will lead the preparation of an evidence report on the use and benefits of EbA and green infrastructure interventions in response to flooding and droughts in Pakistan. This report will be informed by the economic valuation undertaken under Activity 2.1.1 and contain step-wise instructions for how to implement EbA and green infrastructure interventions efficiently and cost-effectively in Pakistan. Importantly, this report will be used under Activity 2.2.1 for the development of updated procedures for Pakistan's key regulations

and policies for flood and water resources management, namely the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans. The set of guidelines will be prepared by WWF-Pakistan.

Sub-activity 2.1.2.2: Disseminate the evidence report developed under Sub-activity 2.1.2.1 to national and provincial planners and regulators involved in flood and water resources management, as well as academic and research institutions.

149. The evidence report prepared under Sub-activity 2.1.2.1 will be disseminated by WWF-Pakistan and the FFC to five national-level stakeholders, seven provincial line departments, and at least five academic and/or research organisations through the National Working Group (which will be established under Activity 2.2.1), PSC and existing inter-departmental communication channels. These recipients are presented in the table below.

Table 7. Stakeholders to be involved in capacity-building interventions.

Stakeholder	Organisation type
Ministry of Climate Change (MoCC)	National government
Ministry of Water Resources (MoWR)	
Ministry of Planning, Development and Special Initiatives (MoPSI)	
National Disaster Management Authority (NDMA)	
Provincial Departments of Irrigation, Forestry, Wildlife, and Agriculture	Provincial government
Provincial Department of Environment Protection	
Provincial Disaster Management Authority (PDMA)	
Provincial Department of Planning and Development (PP&D)	
Pakistan Council for Research in Water Resources (PCRWR)	Academia and research
Pakistan Agriculture Research Council (PARC)	
Lahore University of Management Sciences (LUMS)	
National University of Science and Technology (NUST)	
University of Engineering and Technology (UET)	

Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.

150. The Government of Pakistan approved its first National Water Policy in 2018 and MoWR is currently in the process of developing its Implementation Framework. The Implementation Framework is a document that accompanies the policy and contains instructions for how to apply the rules set out in the National Water Policy, including approaches and techniques for addressing flood impacts and droughts on Pakistan's population and key ecosystems. The FFC, an Executive Department under MoWR, is mandated to address the problem of flooding in Pakistan, with the Provincial Irrigation Departments having the authority on-the-ground to enforce the National Water Policy through its Implementation Framework.

151. The Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for the provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh do not currently include EbA and green infrastructure interventions as effective solutions for addressing the impacts of climate change — particularly those related to floods and droughts — on Pakistan's vulnerable population and key ecosystems. Instead, it prioritises hard infrastructure interventions, which have with limited flexibility to changing environmental conditions and high capital expenditure and operational costs.

152. In addition to EbA and green infrastructure interventions not being considered in the Implementation Framework of the National Water Policy, these interventions are not currently prioritised in Pakistan's National Adaptation Plan and Provincial Adaptation Plans for the four provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh. Because the National Adaptation Plan and four Provincial Adaptation Plans are currently under development and not yet finalised, there is a unique opportunity for the project to also integrate the use of EbA and green infrastructure interventions in these planning documents.

153. Output 2.2 will address these gaps through the creation and adoption of updated procedures for the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. The MoWR (through FFC) will require the application of the updated procedures for the Implementation Framework of the National Water Policy, while MoCC in collaboration with each province's Environment Protection Authority (EPA) will require the application of the updated procedures for the National and four Provincial Adaptation Plans. This will lead to the creation of well documented procedures backed by a rigorous scientific base that enables the GoP to channel future

investments into EbA and green infrastructure interventions instead of grey infrastructure to respond to increasingly severe floods and droughts caused by climate change.

Activity 2.2.1: Develop updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for implementing EbA and green infrastructure interventions in Pakistan.

154. Activity 2.2.1 will contribute to transforming Pakistan's approach to flood and water resources management by establishing a National Working Group chaired by the MoCC and comprised of regulators within the MoWR (including FFC) and the relevant Provincial Departments (such as the Provincial Departments of Irrigation, Forestry, Agriculture, Planning and Development) to: i) review the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans to identify entry points for EbA and green infrastructure as cost-effective adaptation interventions for addressing floods and droughts; and ii) lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.

155. The process for developing the updated procedures for adoption by the GoP is presented in Table 8 below, with further details provided in the sub-activity descriptions below. WWF-Pakistan will lead the implementation of this project activity, working with MoCC and MoWR (through FFC).

Table 8. Process for developing and adopting updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for the provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh.

Strategic document being updated	Stage of integration of EbA and green infrastructure into the document	Timeline (i.e., what year of the Project period?)	Responsible entity
Implementation Framework of the National Water Policy	Draft updated procedures for the Implementation Framework of the National Water Policy on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan are produced and presented to the National Working Group.	Q4 of Year 4	MoWR (through FFC)
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture review the draft updated procedures for the Implementation Framework of the National Water Policy.	Q4 of Year 5	
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture receive final updated procedures for the Implementation Framework of the National Water Policy on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan for endorsement and adoption.	Q3 of Year 6	
National Adaptation Plan	Draft updated procedures for the National Adaptation Plan on the proper execution of EbA and green infrastructure interventions	Q4 of Year 4	MoCC

	to reduce the impacts of floods and droughts in Pakistan are produced and presented to the National Working Group.		
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture review the draft updated procedures for the National Adaptation Plan	Q4 of Year 5	
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture receive final updated procedures for the National Adaptation Plan on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan for endorsement and adoption.	Q3 of Year 6	
Four Provincial Adaptation Plans for Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh	Draft updated procedures for the four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan are produced and presented to the National Working Group.	Q4 of Year 4	The Provincial Adaptation Plans will be developed by each province's Environment Protection Authority (EPA), i.e., Khyber Pakhtunkhwa Environment Protection Authority, Punjab Environment Protection Authority, Balochistan Environment Protection Authority and Sindh Environment Protection Authority.
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture review the draft updated procedures for the four Provincial Adaptation Plans.	Q4 of Year 5	
	Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture receive final updated procedures for the four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan for endorsement and adoption.	Q3 of Year 6	

Sub-Activity 2.2.1.1: Establish a National Working Group to develop and adopt updated procedures for using EbA and green infrastructure interventions in flood and water resources management in Pakistan.

156. A National Working Group will be established to lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. This working group will also ensure these procedures align with current national and provincial climate change planning frameworks. The National Working

Group will include representation from the MoCC, MoWR (including FFC), the Provincial Departments of Irrigation, Forestry and Agriculture local NGOs, CSOs, and academic and research organisations⁹⁹. The National Working Group members will be selected based on their subject matter expertise and decision-making capacity or influence within national and subnational government. The National Working Group will lead Sub-activities 2.2.1.2–2.2.1.4 below. At project completion, the National Working Group will be transitioned into specialized working groups within MoCC and MoWR responsible for continuing the uptake and use of EbA and green infrastructure in future flood and water resources management and adaptation initiatives in Pakistan beyond the project.

157. The project Steering Committee (PSC) and Provincial Oversight Committees (POCs) will be critical in facilitating engagements under this sub-activity, securing government and community buy-in, and driving approval/decisions at the national and provincial levels related to adopting the updated procedures. Targeted engagements will be held between the abovementioned national and provincial level water-related stakeholders and programme partners to enhance coordination between these stakeholders, improve knowledge sharing on integrated flood and water resources management strategies, strengthen water governance, and promote evidence-based decision-making for climate change adaptation with an emphasis on EbA and green infrastructure.

Sub-Activity 2.2.1.2: Review the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for the four provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh to identify entry points for EbA and green infrastructure as cost-effective adaptation interventions for addressing floods and droughts.

158. The National Working Group, supported by WWF-Pakistan, will review the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for the four provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh to identify entry points for EbA and green infrastructure as cost-effective adaptation interventions for addressing floods and droughts. This review will consider the evidence generated under Output 2.1.

Sub-Activity 2.2.1.3: Draft updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.

159. Following the review under Sub-activity 2.2.1.2, and using the evidence generated under Output 2.1, the National Working Group, supported by WWF-Pakistan, will draft updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. These draft updated procedures will be presented to the National Working Group for review and finalisation.

Sub-Activity 2.2.1.4: Validate the updated procedures within the National Working Group for adoption into the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans.

160. A series of consultative sessions will be held between the National Working Group and stakeholders mentioned in Table 7 under Sub-activity 2.1.2.2 to review the draft updated procedures produced under Sub-activity 2.2.1.3. Following the review, the required adjustments will be made and the final updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans will be submitted to the National Working Group for endorsement and adoption.

Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.

161. Capacity-building under this output will focus on: i) strengthening the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated regulations under Activity 2.2.1, and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management; and ii) training national and sub-national technical staff, as well as community leaders/representatives, on how to implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management. This capacity development will support the mainstreaming process under Output 2.2 and overall theory of change for the project by ensure that technical staff are aware of the updates to the implementation framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans and are

⁹⁹ Such as the Pakistan Council for Research in Water Resources (PCRWR), Pakistan Agriculture Research Council (PARC), Lahore University of Management Sciences (LUMS), National University of Science and Technology (NUST), and University of Engineering and Technology (UET).

sufficiently equipped to implement EbA and green infrastructure interventions across Pakistan. It will also facilitate a community-driven and participatory approach to flood and water resources management by involving community leaders/representatives in the implementation, monitoring, operation and maintenance of EbA and green infrastructure interventions, and ensuring these interventions directly respond to communities' needs.

Activity 2.3.1: Strengthen the capacity of national and provincial staff, as well as community leaders/representatives, to apply the updated procedures under Activity 2.2.1 and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management.

162. This project activity will strengthen the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated procedures under Activity 2.2.1. Additionally, because wide-scale implementation of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan is not yet institutionalised in Pakistan, there is a need to train national and sub-national technical staff in the necessary skills to design, implement, operate and maintain these interventions to ensure they continue delivering adaptation benefits over the long term. Similar training is also required for community representatives to secure their buy-in to and ownership of these new approaches for Pakistan, fostering the sustainable adoption and use of these approaches to flood and water resources management at the community level.

163. To respond to these capacity needs, Interventions under Activity 2.3.1 will include: i) training and raising awareness of national and sub-national policymakers and planners on the contents and application of the updated procedures of the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan; ii) developing a national pool of experts for scaling up EbA and green infrastructure through investments in curriculum development, academic programmes, training of trainers approaches, and exposure visits; iii) training national and sub-national technical staff involved in flood risk and water resources management on the implementation, monitoring, operations and maintenance of EbA and green infrastructure interventions; and iv) training local communities on the impacts of climate change, benefits of EbA and green infrastructure interventions and how to operate and maintain these interventions; this will build on the awareness raising carried out under Activity 2.3.1 WWF-Pakistan will lead the implementation of this project activity, training 80 national and sub-national regulators, 120 national and sub-national Project planners, and 120 community leaders.

Sub-Activity 2.3.1.1: Train and raise awareness of national and sub-national policymakers and planners on the contents and application of the updated procedures developed under Activity 2.2.1.

164. Through a series of workshops, 80 participants from the following national and provincial governments will be trained on the updated procedures of the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans developed under Output 2.2:

- a. Ministry of Climate Change (MoCC)
- b. Ministry of Water Resources (MoWR)
- c. Ministry of Planning, Development and Special Initiatives (MoPSI)
- d. National Disaster Management Authority (NDMA)
- e. Provincial Departments of Irrigation, Forestry, Wildlife, and Agriculture
- f. Provincial Department of Environment Protection
- g. Provincial Disaster Management Authority (PDMA)
- h. Provincial Department of Planning and Development (PP&D)

165. Participants will be selected from the ministries listed above which have direct influence over flooding and drought regulation and planning. Those departments within these agencies will be sent requests to nominate trainees from their department. The training will focus on: i) the details of the updated procedures and their key considerations for on-the-ground implementation and adoption of EbA and green infrastructure; ii) the relevance of the updated procedures in future climate change adaptation initiatives in Pakistan, including in funding applications prepared by GoP staff to access external (e.g., donor) financing; and iii) how to apply the updated procedures in the design, construction, operation and maintenance of EbA and green infrastructure interventions in future country-driven climate change adaptation Projects. Additionally, WWF-Pakistan will work with the relevant national and sub-national government staff to connect them with international training institutions that provide certified online courses on the design, implementation and benefits of EbA and green infrastructure, such as IHE Delft Institute for Water Education. While IHE Delft Institute for Water Education is an already identified partner, the international training programmes will also include other similar institutes which meet the training needs. Such other institutes will need to be certified training institutes and will be selected based on specific training needs and capacity identified during Year 2 of project execution.

Sub-Activity 2.3.1.2: Train national and sub-national technical staff on the implementation, monitoring, operations, and maintenance of EbA and green infrastructure interventions.

166. Through a series of professional, technical trainings and exposure visits, 120 participants from the following national and provincial governments will be trained on how to implement, operate and maintain EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan:

- i. Ministry of Climate Change (MoCC)
- j. Ministry of Water Resources (MoWR)
- k. Ministry of Planning, Development and Special Initiatives (MoPSI)
- l. National Disaster Management Authority (NDMA)
- m. Provincial Departments of Irrigation, Forestry, Wildlife, and Agriculture
- n. Provincial Department of Environment Protection
- o. Provincial Disaster Management Authority (PDMA)
- p. Provincial Department of Planning and Development (PP&D)

167. Participants will be selected from the ministries listed above which have direct influence over flooding and drought regulation and planning. Those departments within these agencies will be sent requests to nominate trainees from their department. As part of this capacity-building, technical staff from these governments will visit the Component 1 project sites where local stakeholders are engaged in implementing EbA and green infrastructure interventions to enhance their learning and enable scaling up and replication across Pakistan. Furthermore, a pool of experts will be established within MoCC, MoWR (including FFC) and relevant provincial line departments mentioned above to ensure that EbA and green infrastructure interventions are integrated within future country-driven flood and water resources management and climate change planning.

Sub-Activity 2.3.1.3: Train local community leaders/representatives on EbA and green infrastructure interventions and their benefits, as well as on the implementation, monitoring, operations, and maintenance of these interventions.

168. Through a series of workshops, 120 local community leaders/representatives will be trained on the impacts of climate change on their livelihoods, how to implement EbA and green infrastructure interventions at an appropriate scale to reduce climate change impacts, and how to operate and maintain these interventions. The training and capacity building of communities under Sub-Activity 2.3.1.3 will be executed in consultation with the CBOs. The executive body of the CBOs which is responsible for making key decisions on behalf of the CBOs will nominate members from the CBOs and the community who will participate in these training sessions. The CBOs will ensure that the most relevant community members participate, and their capacity is built to ensure the CBO and the local community can effectively implement, monitor and maintain the EbA and Green Infrastructure interventions. Furthermore, to ensure the participation of local communities in decision-making, tools such as Participatory Rural Appraisal (PRA) and Rapid Rural Appraisal (RRA) will be used, and community elders and CBOs will be engaged to ensure the trainings respond directly to the needs of communities and local-level beneficiaries.

Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin.

Outcome 3: Enhanced resilience of community livelihoods in the Indus Basin.

Output 3.1: Pipeline of feasible climate-resilient businesses.

169. Although there is interest among financial institutions in Pakistan to invest in climate-resilient sustainable businesses, they are not willing to invest in the necessary origination and incubation work needed to identify such investments. Important foundational work in this space has been done by WWF-Pakistan in collaboration with the DFCD. In late 2019, WWF-Pakistan submitted a concept for the Indus Ecoregion as a potential target region for DFCD and was subsequently chosen out of 61 submissions to pilot and develop bankable projects. WWF-Pakistan will leverage its experience developing pipeline businesses for the DFCD and provide incubation services to these businesses which achieve adaptation benefits and provide real return opportunities for investors. The DFCD project is still on-going however Phases I and II which cover the period of feasibility assessment, pipeline screening, and grant assistance and TA for the businesses was ended for the DFCD portfolio in Pakistan despite remaining TA needs before being able to access non-grant concessional lending from the DFCD investment window. Through Recharge Pakistan the selected businesses included in Table 9 will receive the additional TA required and when ready will prepare investment pitches to the DFCD investment window. DFCD will not be the only investors to be approached. The Project will also prepare investment packages for other investors including JS Bank and the Acumen Fund in Pakistan.

170. WWF-Pakistan has identified 7 candidate businesses remaining from the DFCD project eligible for incubation assistance to ultimately access debt and equity finance. While these businesses have received some assistance

from DFCD, see table below, they still require additional support to access DFCD's investment window where these can be considered for non-grant debt and/or equity investment. The project will however seek to link business supported under Activity 3.1.1 with other initiatives in Pakistan focused on sustainable investment such as the Acumen Climate Action Pakistan (ACAP) Fund being proposed to the GCF. The pipeline developed through grant financing under Recharge Pakistan will aim to secure investment from ACAP Fund. That primarily focuses on climate resilient agriculture and food systems at large.

Activity 3.1.1: Develop a pipeline of 7 sustainable climate-resilient businesses.

171. The project will conduct due diligence over the existing DFCD Project identified businesses, conduct detailed needs assessments, and prepare business development plans for each business, provide TA, incubation, and business support to each business based on their individual needs (see Table 9) below, and support businesses in closing financial agreements with investors, including with the DFCD investment window. This activity will be executed by WWF-Pakistan who will administer grants to each business. This activity will contribute to the project's efforts to enhance community resilience by providing important case studies for business models which create positive environmental impacts while creating economic resilience for vulnerable communities and business owners.

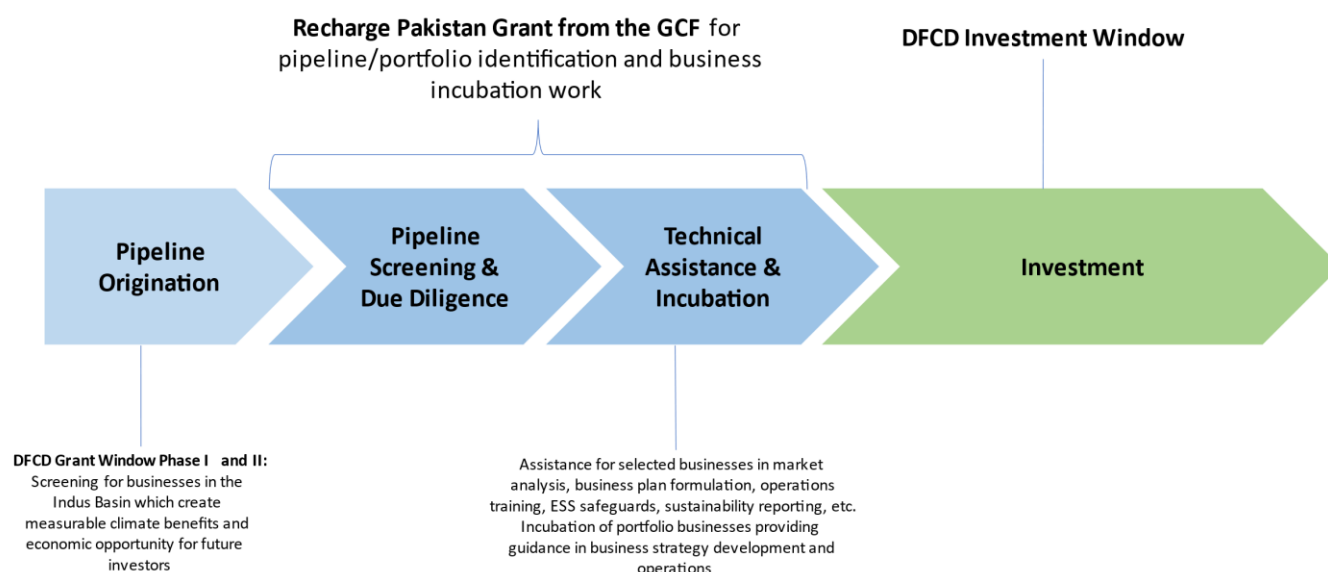


Figure 29. Project support for businesses

Sub-Activity 3.1.1.1: Build on the initial study completed by WWF-Pakistan identifying bankable project opportunities in the Indus Basin in the agriculture, forestry, and water sectors.

172. The project will conduct due diligence over the DFCD pre-screened business to establish the final pipeline for the project. This due diligence will include: i) individual business surveys and visits; ii) identification of technical assistance needs among shortlisted businesses and projects; iii) determining financing needs; iv) business governance and management assessments; v) environmental and social safeguard screenings; and vi) risk identification. Businesses will be selected for support based on the results of these assessments.

Table 9. DFCD businesses needs assessment.

Business Name	Sector - Product	DFCF Phase/ What assistance have they already received if any, including amount of financing.	Estimated additional capital need	General areas of TA to be provided by the proposed GCF project
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Sustainable mushroom farming	Climate Smart Agriculture – Fresh and canned mushrooms	Secured a Euro 50,000 grant from DFCD to conduct pre-feasibility/ feasibility studies in the following areas; market analysis, technical studies, socio-economic and environmental baseline setting, stakeholder engagement & analysis, financial modelling, capital structure, legal & tax advisory, and organizational capacity building.	Pilot development: USD 300K-400K	1. Grant finance to launch pilot Projects; 2. Assistance in integrating sustainability and environmental and social safeguards within their business strategy and process. 3. Improving market linkages and access accessing local and export markets; 4. Technical assistance to access blended finance.
Climate smart horticulture using hydroponics	Climate Smart Agriculture – High value commodities such as Ginger & bell pepper	Secured a Euro 50,000 grant from DFCD to conduct pre-feasibility/ feasibility studies in the following areas; market analysis, technical studies, socio-economic and environmental baseline setting, stakeholder engagement & analysis, financial modelling, capital structure, legal & tax advisory, and organizational capacity building.	Pilot development: USD 300K-400K	Same as above.
Sustainable banana and sugarcane growing using tissue cultures	Banana and Sugarcane	A detailed concept has been developed under the DFCD's scanning of bankable Project study, however the DFCD Project in Pakistan ended before the Project could access any grant finance.	For pre-feasibility/ feasibility studies: USD 15K-20K Pilot development: USD 300K-400K	1. Development of feasibility studies; 2. Grant finance to launch pilot Projects; 3. Assistance in integrating sustainability and environmental and social safeguards within their business strategy and process. 4. Improving market linkages and access accessing local and export markets; 5. Technical assistance to access blended finance.
Sustainable aquaculture of high value species and value chain enhancements	Aquaculture – Freshwater Fishes	A detailed concept has been developed under the DFCD's scanning of bankable Project study, however the DFCD Project in Pakistan ended before the Project could access any grant finance.	For pre-feasibility/ feasibility studies: USD 15K-20K Pilot development: USD 300K-400K	Same as above.
Development of natural sustainable fibre alternatives to cotton	Textile – Alternate fibre from <i>Sisal</i> and <i>Flax</i>	A detailed concept has been developed under the DFCD's scanning of bankable Project study, however the DFCD Project in Pakistan ended before the Project could access any grant finance.	For pre-feasibility/ feasibility studies: USD 15K-20K Pilot development: USD 300K-400K	Same as above.

Horticulture and livestock raising	Climate Smart Agriculture – Dates and other fruit commodities	A detailed concept has been developed under the DFCD's scanning of bankable Project study, however the DFCD Project in Pakistan ended before the Project could access any grant finance.	For pre-feasibility/ feasibility studies: USD 15K-20K Pilot development: USD 300K-400K	Same as above.
Sustainable Medium Density Fiberboard (MDF) manufacturing through agro-forestry	Agroforestry – MDF boards and other raw & processed wood products	A detailed concept has been developed under the DFCD's scanning of bankable Project study, however the DFCD Project in Pakistan ended before the Project could access any grant finance.	For pre-feasibility/ feasibility studies: USD 15K-20K Pilot development: USD 300K-400K	Same as above.

Sub-Activity 3.1.1.2: Formulate technical assistance and business development plans for shortlisted climate-resilient DFCD businesses.

173. The project will provide grant assistance to the selected businesses for: regional market studies; detailed environmental and social impact assessments; financial deal structuring; legal and tax advisory (deal structuring); piloting project activities; institutional capacity building (hiring, operations and financial systems buildout); stakeholder analysis, and community engagement. The aforementioned technical support is vital for bankable enterprises to help them secure the much needed financial and technical assistance to help them grow and reach the investment thresholds of debt and equity investors.

Sub-Activity 3.1.1.3: Assist Bankable enterprises with financial close within the DFCD funding window (loaning application, investment pitch preparation, etc.).

174. For those businesses capable of revenue generation and showing favourable financial forecasts, the project will assist these businesses in the preparation of applications to the DFCD loan and equity window and other investors such as JS Bank in Pakistan (GCF DAE).

Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.

175. Because the agricultural sector is one of the most vulnerable to the impacts of floods and droughts in Pakistan, interventions under Output 3.2 will strengthen the climate resilience of small-sale farmers and vulnerable people in the Indus Basin that depend on agriculture for their livelihoods. These people's vulnerability to climate change is exacerbated by limited technical support, high cost of inputs, lack of access to high-yielding and climate-resilient crop varieties, and the use of unsustainable agricultural practices. The Provincial Departments of Agriculture currently provide extension services to farmers, but these departments have yet to adapt these to be responsive to current and future anticipated climate scenarios.

176. WWF-Pakistan will assist the Provincial Departments of Agriculture to modernize their extension service programs by training extension agents in ecosystem friendly and climate-resilient alternatives to current agricultural practices employed by small-scale farmers. This will allow the Provincial Agriculture Departments in the future to deliver climate-responsive extension services on measures such as: i) improved water-efficient irrigation practices; ii) farming of drought-resistant crop varieties; iii) soil management practices responsive to changing climate conditions; and iv) installation and use of water storage tanks for vulnerable small-scale farming households. Output 3.2 will complement Outputs 1.1 and 1.2 by introducing a community-based climate-resilient approach to farming to enhance agricultural productivity, reduce agricultural expansion, and reduce the degradation of ecosystems that provide critical services that buffer the impacts of climate change. Interventions under this project output will benefit 50,750 small-scale farmers, their families and vulnerable people in the Indus Basin dependent on agriculture for their livelihoods.

Activity 3.2.1: Improve the climate resilience of vulnerable agricultural livelihoods.

177. Under this activity, WWF-Pakistan will partner with the Provincial Departments of Agriculture to enhance agricultural extension services and enable the adoption of climate-resilient agricultural practices in the Indus Basin. This will involve designing and delivering a new climate-responsive extension services curriculum, training the department's existing extension agents in delivering this curriculum, and improving the delivery of climate-responsive extension services to vulnerable farmers to adapt their practices in response to increasingly severe and frequent floods and droughts. The curriculum will include training modules focusing on: i) water-efficient irrigation methods; ii) the use of drought-resistant crop varieties; iii) climate-resilient soil management methods; and iv)

effective use of rainwater tanks, to be provided by the project, to supplement dry season water supplies, providing water for irrigation and livestock.

178. Activity 3.2.1 will use a Farmer Field School (FFS) approach in delivering this new curriculum, whereby training sessions are conducted over an entire crop season. Lessons learned from other initiatives in Pakistan indicate that such an approach leads to higher adoption rates and sustained behaviour change both within government agencies and communities. Through the FFS, the project will introduce new irrigation technologies and methods, facilitate the adoption of drought resistant seed varieties into current crop farming practices as well as improved farming and soil management techniques which are better suited to prolonged periods of drought. Water storage tanks, accompanied by training on their installation, operation and maintenance, will also be provided to capture rainfall and serve as immediate sources of water for irrigation and household use during droughts. The Farmer Field School cohorts will be assembled during execution of the project following calls for participation from local farmers by WWF Pakistan in the target villages and communities. Enrolment in the FFS will be contingent upon farmer willingness to adopt the climate-resilient agriculture practices covered by the curriculum.
179. The Provincial Agriculture Departments, via their extension services, will be responsible for delivering the training to communities in Punjab, Balochistan, Khyber Pakhtunkhwa and Sindh Provinces. The project will provide the necessary materials and equipment for the FFS trainings, including seeds, grow bags, saplings, low-cost material, irrigation systems, etc. The trainers will identify Lead Farmers during their trainings to demonstrate new approaches to others in the area. Lead Farmers within the FFS cohorts will be identified by the participating farmers in each of the communities. These Lead Farmers will be provided additional farming inputs to incentivize their knowledge sharing. Activity 3.2.1 will be executed by WWF-Pakistan who will provide training, capacity development, and grant finance to the provincial departments to provide training and equipment to farmers in the Indus Basin.

Sub-Activity 3.2.1.1: Strengthen the capacity of small-scale farmers and people dependent on agriculture on climate-resilient agriculture and sustainable practices.

180. WWF-Pakistan will develop a curriculum, training modules, and workplans for the FFSs, with Technical Assistance (TA) provided by international and local experts brought on as consultants by the project for the full seven-year implementation period. The curriculum, training modules and workplans for the FFSs will be shared with the Provincial Departments of Agriculture, local NGOs, academic and research institutions such as the Pakistan Agriculture Research Council (PARC), and community representatives to gather input before these are finalized and agreed upon with the Provincial Departments. WWF-Pakistan will organize the necessary logistics for these trainings including venues, training materials, supplies, equipment and materials, and excursions for training exercises on farm. The trainings will be hosted by the Provincial Departments of Agriculture in each province.
181. Eighty-five extension agents and Department of Agriculture staff (15% women) will be trained in each province. The training curriculum will be implemented over 90 days in Year 2 of the Project, with periodic scheduled refresher trainings. Upon completion of the training the trainees (staff and extension agents) will become Master Trainers and collectively responsible for delivering 170 FFS per year (6 trainings per Master Trainer and 25 farmers participating in each FFS) to total of 4,131 participants per year. A total of 24,786 small-scale farmers and people in the Indus Basin will benefit from this training on climate-resilient agricultural practices.
182. Trainers will be responsible for tracking adoption rates, water savings, and changes in yield over time. The PIUs will be responsible for coordinating the training delivery with the Provincial Agriculture Departments in each province and manage all TA consultancies for the development of the training materials. The agriculture activity leads in the SIUs (see Section B.4 for more details) will also be engaged to ensure that the training approaches are used in the project sites for Sub-activities 3.2.1.2, 3.2.1.3, 3.2.1.4, and 3.2.1.5.

Sub-Activity 3.2.1.2: Improve the water efficiency of farming through improved farm management and irrigation systems.

183. The project will organize and administer FFSs on water-efficient farming practices and irrigation technologies. Master Trainers from the Provincial Departments of Agriculture will deliver on farm trainings in best practices for the cultivation of crops and fruit trees more capable of withstanding drought conditions such as orchards, vegetables and horticulture varieties, and fodder cultivation plots. The trainers will distribute and demonstrate the use of water-efficient irrigation technologies including high efficiency irrigation systems (HEIS) and provide farmers the required components and inputs — e.g., drip irrigation, sub surface irrigation systems, and tunnel farming and green houses. Farmers will also be trained on the maintenance of these systems to ensure they can continue to operate

independently. A total of 6,752 small-scale farmers and people in the Indus Basin will benefit from improved irrigation techniques.

Sub-Activity 3.2.1.3: Introduce drought-resistant crop varieties.

184. Farming communities in the Indus Basin have traditionally relied on water-intensive crops such as rice and maize exposing them to an increasing risk of crop failure and lower yields due to climate change-related diminishing seasonal water access for irrigation. Under this sub-activity, 13,277 small-scale farmers will benefit from adopting drought-resistant crop varieties proven to be successful in these areas, including varieties of barley and wheat which require less water, olives, figs, vegetables, and herbs. Further adoption of these climate smart agricultural practices and enhanced seeds is predicated on the economic case to be demonstrated through the farmers plots that the use of these climate smart methods and seeds leads to greater economic efficiency for farmers from reduced irrigation costs and material costs, and increases in crop yields after which farmers would voluntarily purchase farming improved seeds using their own resources because of their economic benefits. This voluntary approach is already in practice in the Provincial Agriculture Departments existing extension programs and is essential to ensure sustainability in the uptake of new farming practices limiting dependence on public or donor funds.

Sub-Activity 3.2.1.4: Improve the climate resilience of soil management practices.

185. Under this sub-activity, climate-resilient management practices for improved soil conservation and productivity will be introduced to farmers in the Indus Basin. Sustainable soil management practices will include the use of bio-fertilizers provided by the project, conservation tillage, efficient use of synthetic fertilizers, improved crop residue management, mulching, and use of compost. These methods will reduce production costs as well as improve soil health and water holding capacity. A total of 4,424 farmers will benefit from the adoption of these improved practices. Agricultural businesses receiving assistance by the Project under Activity 3.1.1 will also receive extension services support to improve their farming practices.

Sub-Activity 3.2.1.5: Provide water storage tanks to vulnerable agricultural communities.

186. Under this sub-activity, rainwater storage tanks will be procured and installed, accompanied by training on their installation, operation and maintenance, to supplement dry season water supplies in agrarian communities in water scarce areas of the Indus Basin. The storage tanks will be used for agricultural purposes and livestock watering. 15 tanks will be installed on communal lands for public access. Specific tank locations will be identified by the communities to ensure these meet local needs and enhance equitable access. The anticipated storage capacity of each tank will be 1,350 m³ (dimensions — 30 m × 30 m × 1.5 m), which will provide 20,250 m³ of additional water to 1,512 smallholder farmers and their families. Communities which receive the water storage tanks will be identified during Year 1 of the project. WWF Pakistan will identify these communities based on the following criteria: presence of a central location allowing for equal access for community members and willingness and ability of communities to assume the O&M costs of the tanks after the GCF financing period. The water collected in the tanks will be provided free of charge for communities and the WUG established under Sub-Activity 1.3.1.2 will be responsible for establishing equitable distribution in each community.

Schematic Diagram of Water Storage Tank

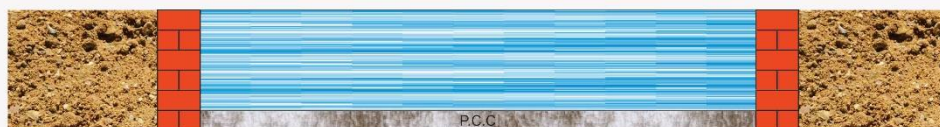
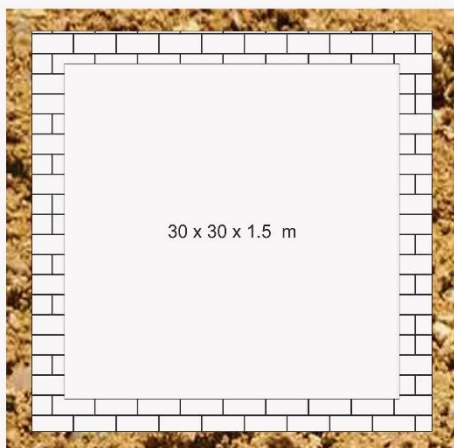


Figure 30. Schematic drawing of a water storage tank.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Accredited Entity

187. **The World Wildlife Fund, Inc. (WWF-US)** will serve as the Accredited Entity (AE) for the project and will provide GCF Proceeds to the Executing Entity (EE), in accordance with the FAA to be entered with the GCF. The AE will be responsible for the overall oversight of this project, including technical, financial, and administrative monitoring and supervision (through reporting, audits, and annual site visits) and review and approval of the EE's annual workplans and budgets. WWF-US will also be responsible for channelling the GCF Proceeds and the relevant co-financing resources from USAID and TCCF to the EE, providing support, guidance and backstopping to the EE, monitoring of the achievement of project results and outputs, reporting to the GCF and the referred co-financiers, and project closure and evaluation. WWF-US will conduct these responsibilities, and disburse GCF Proceeds to the EE, in line with WWF-US' Accreditation Master Agreement (AMA) and anticipated Funded Activity Agreement (FAA) with the GCF. WWF-US also serves as the AE for *FP050 Bhutan for Life*, which was launched in 2018.

Executing Entity (EE)

188. The project will be executed by one Executing Entity (EE), WWF-Pakistan. The EE will be responsible for project execution, among others, including the management of grantees, recipients of goods or services, and procured parties and their activities, and reporting to the AE.

189. **WWF-Pakistan¹⁰⁰**, a private independent organization under the international WWF Network and separate legal entity to WWF-US, will serve as an EE for the project. With its Head Office in Lahore, 6 regional offices, and 30 project site offices, WWF-Pakistan is the leading conservation organization in the country. WWF-Pakistan has significant experience working with the Government of Pakistan (GoP) in policy and regulatory design and reform, including the Environmental Protection Act, the National Water Policy, the National Wetlands Policy, and the Clean Drinking Water Policy. WWF-Pakistan also has a long track record implementing EbA projects in Pakistan's landscapes most vulnerable to climate change including: 1) restoring and protecting, through community

¹⁰⁰ WWF-Pakistan is not controlled by the AE, does not control the AE, and is not under common control with the AE, where "control" means ownership of a majority of the voting power of the either entity. Both organizations are independent legal entities with bilateral contractual and licensing relationships with World Wildlife Fund for Nature, a Swiss foundation that serves as the secretariat for the global network of WWF National Organizations.

management arrangements, mangroves ecosystems in the Indus Delta; 2) improving watershed management and enhancing agricultural livelihoods to better withstand climate change in sub-catchments of Khanpur Dan; 3) the development of pilot projects to serve as replicable models for future wetland conservation projects and their use as an evidence base to support the formulation of national and sub-national wetland regulations and plans under the Pakistan Wetlands Program; and 4), carrying out assessments in the Western Himalayas Ecoregion, Indus Floodplains, and Indus Delta Ecoregion to identify flood damages and formulate plans for ecological restoration as part of the 2010 Flood Early Recovery Program, which helped to restore flood-affected ecosystems through reforestation, natural regeneration through protection, construction of flood protection spurs, and restoration of landslide areas through green infrastructure.

190. WWF-Pakistan will be responsible for the execution of project Activities 1.1.1, 1.1.2, 1.2.1, 1.3.1, 1.3.2, 2.1.1, 2.1.2, 2.2.1, 2.3.1, 3.1.1, 3.2.1, managing all procured parties and their activities, reporting to the AE, and ensuring optimal alignment of the project with the GoP's policies and ministerial contributions to achieve the Project Outcomes and Fund-level impacts, as described in Section B.3. WWF-Pakistan will hold GCF Proceeds, in a new and separate dollar (USD) denominated account. As an EE, WWF-Pakistan will enter into a subsidiary agreement with WWF-US (as the AE) in respect of the GCF Proceeds, grant agreements with each grantee, consulting agreements with private sector service providers, and cooperative agreements with each recipient of goods or services, such as technical assistance for the project activities, retaining responsibility for any delegated authority over financial management and procurement. As part of the AE's due diligence, WWF-US assessed WWF-Pakistan's capacity to execute the project (see Annex 9 for details). This assessment determined WWF-Pakistan to be capable of applying WWF-US and GCF standards and policies in the execution of the project.
191. WWF-Pakistan (EE) will establish and host the **Project Management Unit (PMU)** within WWF-Pakistan's office in the Federal Capital, Islamabad (Figure 29). The PMU (through the project Manager) will report to the AE. The PMU will be led by the Project Director and operate under the guidance of the **Project Steering Committee (PSC)** described below. The Project Director will be a full-time staff member of WWF-Pakistan providing 50% of LOE to the project through high-level support and strategic guidance to the PMU and facilitate engagements with high-level government staff where appropriate and necessary during project execution. The PMU will be responsible for overall project management and planning, providing support to the execution of day-to-day activities, coordinating with the national government and project partners, coordination with the AE, direct supervision of contracted project activities, and coordinating project execution across four provinces. The PMU will also be responsible for reporting on the application of resources and results achieved, preparing management reports including annual reports and any proposals for the adaptive management of the project, promoting inter-institutional linkages and coordination with national initiatives, and disseminating project results.
192. The key staff in the PMU will include a: i) Project Director; ii) National Project Manager (PM); iii) Operations Manager, iv) Manager Administration & Security; v) Coordinator Contract Managements; v) M&E Manager; vi) Environment and Social Safeguards (ESS) Manager; vii) Gender Manager; viii) Senior HR Officer; ix) Officer, Contracts Management; x) 9 Office Attendants to support administration; and xi) Security Support. The PM will be a full-time staff member on the project, recruited competitively, who will be responsible for the daily implementation and management of the project. This will include ensuring that the project achieves the targets set out in the Logical Framework (Section E) to the required quality standards, and within the specified time and budget allocations. In line with this mandate, the PM will: i) report directly to the PSC on project management-related matters; ii) manage the project in accordance with the specified workplans and allocated budget; iii) ensure that all project interventions are implemented according to GCF and WWF rules, policies and standards; iv) work closely with national and local authorities to ensure that the project is managed effectively and that the needs of all beneficiary groups are considered; v) oversee the efficient and effective information- and knowledge-transfer to relevant project partners; vi) oversee the development of annual work plans and budgets; vii) communicate project progress updates to the PSC at biannual PSC meetings, including recommendations from the POCs on project implementation; and viii) develop ToRs for key project staff and implementing partners, including detailed scopes of work and service contracts, in line with the project Procurement Plan (Annex 10). The PM will be required to fulfil the role until project completion and the submission of the final evaluation report — as well as any other required documentation — to WWF US (AE). The Operations Manager will report directly to the PM and be responsible for: i) providing financial support to the PMU; ii) compiling reports on the disbursement procedures for the project and anticipated requirements for project funding during the implementation phase; and iii) ensuring that all project-related tasks are undertaken according to WWF and GCF rules, policies and standards. The Operations Manager will also manage the financial transactions for the project's activities, which will be implemented in accordance with the established project workplan and national priorities.

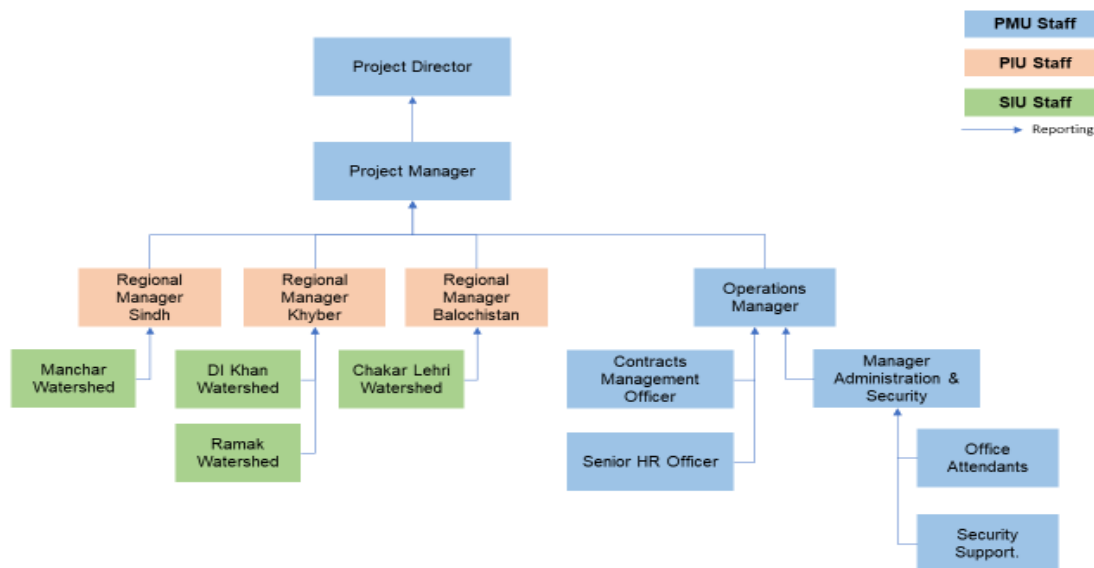


Figure 31. WWF-Pakistan Organigram.

193. There will be four **Provincial Implementation Units (PIUs)** hosted within WWF-Pakistan's regional offices located in the Provincial Capitals (Khyber Pakhtunkhwa, Sindh and Balochistan). These units will coordinate the implementation of project activities under Components 1, 2 and 3 guided by the Provincial Oversight Committees (POCs) described below. The PIUs will be headed by Regional Managers (one for each of the three provinces), that are full time WWF-Pakistan staff working on the project, each reporting to the PM in Islamabad. The Regional Managers will be responsible for: i) managing the Site Implementation Units (SIUs) defined below, coordinating with provincial line agencies, and managing key service providers on the implementation of project activities; ii) providing technical support and oversight to field teams including ensuring compliance with WWF, GCF and national policies and standards, in collaboration with the Provincial Oversight Committees (POCs); iii) undertaking monitoring and evaluation (M&E) of project activities and developing knowledge products to share lessons learned via the PMU and project partners; and iv) development of ToRs for provincial and field level staff and implementing partners in addition to defining scope of service level agreements and procurement plans.

194. **Site Implementation Units (SIUs)** will be based at field offices in the following locations: i) DI Khan Watershed (this office will cover the DI Khan and Ramak Watershed project areas); ii) Chakar Lehri Watershed; and iii) Manchar Lake Watershed. SIUs are necessary for the successful implementation of project activities as the four target provinces under the project cover large areas with difficult terrain and road networks. Some of the proposed sites are as far as six hours away by car from the PIUs requiring smaller site-based offices to manage the daily execution of project activities. These offices will host full-time WWF-Pakistan staff who will report directly to the respective Regional Managers of the PIUs. The field office staff will be responsible for on-the-ground implementation of project activities and supervising the service providers responsible for executing specific project activities. The SIUs will also provide technical support to the CBOs located in the project areas. SIUs will also provide on-the-ground support to project M&E staff, collect data for project reporting and ensure the timely execution of project activities against the project workplan.

Governance Bodies

195. **The Project Steering Committee (PSC)** will be the project's governance body and consist of 10 members that will ensure alignment with Pakistan's national climate objectives and provide technical guidance to the Project Management Unit and Provincial and Site Implementation Units. Specifically, the PSC will: i) review and endorse annual work plans and budgets; ii) provide expert guidance on the project strategy and endorse any changes during implementation; iii) support the PMU with identifying and managing project-related risks; iv) support communication and dissemination of project outcomes to beneficiaries; v) provide strategic and policy guidance to ensure alignment of the project with national climate change regulations and plans; and vi) ensure that EbA and green infrastructure interventions are integrated in the main development agenda of the Government of Pakistan (GoP) at national and sub-national levels.

196. The PSC will support the PMU to implement the project's exit strategy across all levels of government, disseminate key lessons from the project, and maintain complementarity between the proposed project and key planned and ongoing initiatives in the Indus Basin such as the NDRMF-funded project and the two GCF-funded projects in the Indus Basin, FP108 *Transforming the Indus Basin with Climate Resilient Agriculture and Water Management (led by FAO)* and FP018 *Scaling-up of Glacial Lake Outburst Flood (GLOF) risk reduction in Northern Pakistan (led by UNDP)*, respectively. A quorum of at least 50% of the members shall be required with mandatory participation from the Chair and PM. During PSC meetings, all members will receive updates on the status of the project and on the significant project developments from the PM. Members of the PSC will join the AE during annual site supervision visits to monitor project activity progress and engagement with local communities and stakeholders.
197. The PSC for the project will include a representative from each of the following:
- Ministry of Climate Change (MoCC) (Chair & GCF NDA);
 - Ministry of Water Resources (MoWR);
 - National Disaster Risk Management Fund (NDRMF);
 - National Disaster Management Authority (NDMA);
 - Ministry of Planning Development and Special Initiatives (MoPDSI);
 - Provincial Planning and Development Departments (to include one from each province: Sindh, Balochistan and Khyber Pakhtunkhwa); and
 - Environmental Affairs Division (EAD).
 - Community Representatives
 - Academic Institutions
 - NGOs
198. Serving below the PSC and to ensure ownership at the provincial level and by key government line departments, **Provincial Oversight Committees (POCs)** will be established in each of the four provinces. The POCs will be chaired by the Provincial Planning and Development Department who will also represent their Province in the PSC and include representation from appointed Provincial Focal Points from the Provincial Departments of (1) Irrigation, (2) Forest, (3) Wildlife, (4) Fisheries, (5) Environment, and (6) Agriculture.
199. The POCs will: i) assist the PMU with the development of annual work plans and budgets for the project activities for endorsement by the PSC; ii) ensure coordination among all relevant federal and provincial government departments; iii) support the PMU with formalizing service based contracts with service providers who will implement project activities; iv) seek complementarities among on-going government projects to maximize project level impact; v) coordinate with project M&E staff to track project progress at provincial level against the approved work plans; and vii) provide guidance and support to the SIUs during implementation, as needed. The POC will appoint a Chair who will be responsible for consulting with key experts as and when required for knowledge sharing and peer assistance. The POCs shall meet bi-annually before the PSC meetings are convened. This will ensure that POCs recommendations are duly considered and incorporated at the national level.
200. Under the POCs, the Provincial Focal Points will serve as the points of contact at the provincial levels for the implementation of project activities and report to the PM for all project activities. One focal point will be identified from each department during project start-up. The POCs and the respective focal points will provide technical support and guidance to the field teams for the implementation of project activities, including promoting the adoption of the project's adaptation interventions and mainstreaming EbA and green infrastructure interventions into Pakistan's provincial governments' planning for flood and water resources management.
201. Entities under the EE level are characterized as "procured parties". Procured parties will receive grants or contracts from the EEs to provide goods or services to the EEs in support of specific activities. Procured parties include private sector firms, national government agencies, provincial governments, and local communities that are responsible for territorial planning and climate change mitigation and adaptation. Table 10 identifies each procured party and specifies the type of agreement and selection criteria deployed. Grants will be awarded under WWF's accreditation scope of "grant award and/or funding allocation". The table below also indicates which Activities involve grant award and/or funding allocation.

Table 10. Partner entity details: Procured Parties

Executing Entity	Entity Shown in Figure 33 below	Role of that Entity on the Project	If grant award and/or funding allocation,	Form of Agreement between the	Entity pre-selected or will criteria be	Will that entity sub-grant? i.e., will it be an
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			which activities	Entity and the EE	applied during execution	intermediary for cash grants?
WWF-Pakistan	Provincial Departments of Forestry	Procured party	Activity: 1.1.1. Sub-activity: 1.1.1.1, 1.1.1.2.	Grant agreement	Pre-selected	No
WWF-Pakistan	Community managed Nurseries	Procured party	Activity: 1.1.1. Sub-activity: 1.1.1.1, 1.1.1.2.	Grant agreement	Criteria will be applied during execution	No
WWF-Pakistan	Provincial Departments of Irrigation	Procured party	Activity: 1.1.2. Sub-activity: 1.1.2.1, 1.1.2.2, 1.1.2.3, 1.1.2.4, 1.1.2.5	Grant agreement	Pre-selected	No
WWF-Pakistan	Provincial Departments of Irrigation	Procured party	Activity: 1.2.1. Sub-activity: 1.2.1.1, 1.2.1.2, 1.2.1.3, 1.2.1.4, 1.2.1.5, 1.2.1.6, 1.2.1.7.	Grant agreement	Pre-selected	Yes – WWF Pakistan will provide funding to the Provincial Departments of Irrigation to hire the engineering firm(s) (see row directly below). Engineering firms will be selected by the Provincial Departments using Quality cum Cost-Based Selection (QCBS) selection criteria which evaluates cost committed by the bidder and the technical qualification of the bidder.
WWF-Pakistan	Engineering Firm(s)	Procured party	Activity: 1.2.1. Sub-activity: 1.2.1.1, 1.2.1.2, 1.2.1.3, 1.2.1.4, 1.2.1.5, 1.2.1.6, 1.2.1.7.	Consulting agreement	Criteria will be applied during execution	No
WWF-Pakistan	CBOs	Beneficiaries	Activity 1.3.1. Sub-activity: 1.3.1.1, 1.3.1.2, 1.3.1.3 Activity 1.3.2. Sub-activity: 1.3.2.1, 1.3.2.2	Grant agreement	Criteria will be applied during execution	No
WWF-Pakistan	MoCC	Procured party	Activity: 2.2.1. Sub-activity: 2.2.1.1,	Cooperative agreement	Pre-selected	No

			2.2.1.2, 2.2.1.3, 2.2.1.4			
WWF-Pakistan	Provincial Departments of Agriculture	Procured party	Activity: 3.2.1. Sub-activity: 3.2.1.1, 3.2.1.2, 3.2.1.3, 3.2.1.4, 3.2.1.5.	Grant agreement	Pre-selected	No

Table 11. Procurement selection criteria for entities that will not be pre-selected (as specified in the table directly above).

Entity shown in Figure 33 below	Selection criteria applied/to be applied during execution
Engineering Firm(s)	The selection criteria for procurement of this entity will be based on Quality cum Cost-Based Selection (QCBS) criteria and procurement will follow WWF-Pakistan's (EE) and the Procurement & Consultancy Guidelines. The QCBS selection criteria evaluates cost committed by the bidder and the technical qualification of the bidder.
CBOs	8 CBOs have been identified, however, if the need for new CBOs is determined during execution of the project, WWF Pakistan will follow its Social Mobilization Strategy (Appendix 6 of Annex 6: ESMF) to select CBOs, as described under Activity 1.3.1 of Section B.3 in the Funding Proposal.
Community managed Nurseries	Selection criteria will follow WWF Pakistan's (EE) Social Mobilization Strategy (Appendix 6 of Annex 6: ESMF) to select Community Managed Nurseries.

202. **The Ministry of Climate Change** is the **National Designated Authority (NDA)** for Pakistan. It holds the national mandate for climate change, environment, and Multilateral Environmental Agreements, including UNFCCC and NDC commitments. The environmental division of the ministry is led by the Director General (Env/CC), and its responsibilities include: i) Climate Change & Environment; Environmental Legislation and Trans-boundary (inter-provincial, regional and international) issues; and ii) matters relating to sustainable development, water and sanitation, sustainable urbanization, Multilateral Environmental Agreements (MEAs) — including UN Framework Convention on Climate Change (UNFCCC). The MoCC will be a procured party and a recipient of goods and services in the form of technical assistance and logistical support from WWF-Pakistan under Activity 2.2.1. MoCC will lead government efforts to develop and adopt updated procedures for Pakistan's Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans on the execution of EbA and green infrastructure.

203. **The Ministry of Water Resources' (MoWR) Federal Flood Commission (FFC).** The FFC is an Executive Department of the MoWR which was established in January 1977 to address the problem of flooding in Pakistan. FFC provides technical advisory services at national and sub-national levels — particularly to the MoWR and Provincial Departments of Irrigation — on engineering, water and hydropower sectors, including flood control, dams' safety, irrigation, drainage, and hydropower. FFC has over thirty-seven (37) technical officers who have experience in engineering and working in the water sector. The MoWR' FFC will provide technical assistance and coordination support to the Provincial Irrigation Departments on their activities, execution of project Activity 1.2.1.

204. **Provincial governments** are responsible for the administration of Pakistan's four Provinces. Governance over natural resources including forests and waterways is highly decentralized and provincial government Departments are largely responsible for the management of ecosystems and addressing climate change threats within their individual jurisdictions. They are also responsible for ensuring provincial level interventions align with national policies and regulations. Under the project, The Provincial Departments of Forestry will be procured parties for Activities 1.1.1 and 1.1.2, the Provincial Departments of Irrigation will be procured parties for Activity 1.2.1, and the Provincial Departments of Agriculture will be procured parties receiving grant funding and services in the form of technical assistance under Activity 3.2.1. The Provincial Departments are accustomed to managing sizeable projects and budgets considering their direct role in planning and implementation across key sectors. Just as recently as February 2023, the Province of Sindh, Planning and Development Department, Province of Sindh,

Irrigation Department received a 510-million-dollar loan from the World Bank to repair the provinces damaged grey infrastructure network¹⁰¹.

Table 12. AE and EE roles & responsibilities.

WWF US (AE)	WWF-Pakistan (EE)
- Signs the FAA with the GCF detailing all the responsibilities of the AE for the execution of the project. - Signs a Subsidiary Agreement with the EEs to provide the GCF Proceeds and to hold them to the duties of an EE as described in the AMA and FAA. - Signs relevant Co-financing Agreements with USAID and TCCF in respect to the relevant co-financing resources, including the relevant terms and conditions from such donors; and signs separate grant agreements with the EE for the channelling of such co-financing resources and flow down of any donor-specific requirements. - Receives, reviews, and approves the project financial and technical information, and disburses GCF Proceeds and co-financing resources to the EE. - Reports to the GCF on the project technical and financial progress and requests disbursement of grant funding. - Monitors the performance of the EEs and supports their capacity building. - Communicates with the country NDA to facilitate the successful implementation of the project.	- Signs a Subsidiary Agreement with the AE. - Signs separate grant agreements with the AE in respect of the co-financing resources. - Submits requests for disbursement to the AE for GCF-funding in accordance with the timing and conditions of the Subsidiary Agreement. - Receives from WWF-US, the relevant annual instalments of the GCF grant and co-financing resources. - Leads the execution of Activities 1.1.1, 1.1.2, 1.2.1, 1.3.1, 1.3.2, 2.1.1, 2.1.2, 2.2.1, 2.3.1, 3.1.1, 3.2.1 - Acts as financial manager for the project implementation. - Procures all goods and services funded by GCF proceeds in accordance with AE's fiduciary standards. - Prepares and compiles annual work plans and budgets. - Compiles and prepares technical and financial reports for submission to the AE.

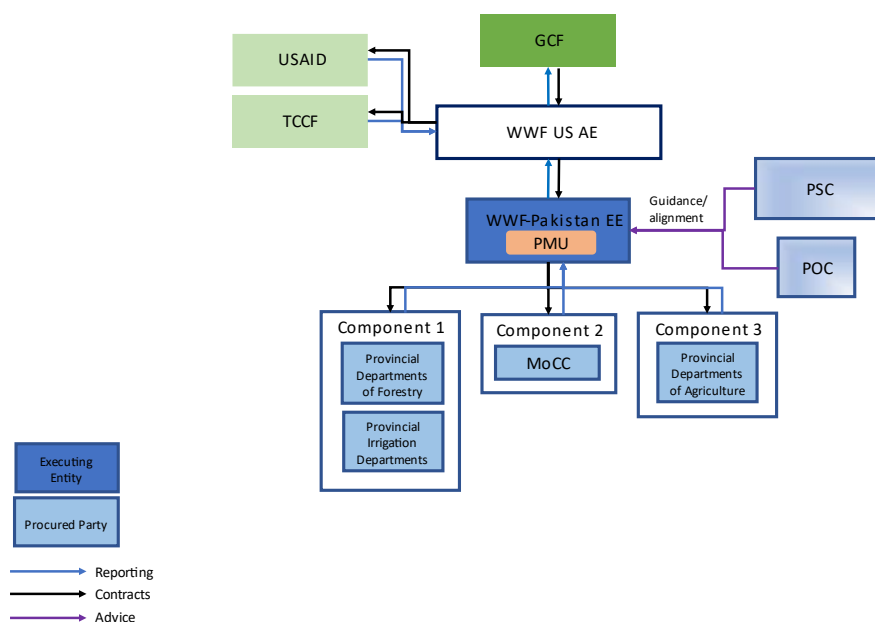


Figure 32. Implementation arrangements for the project.

Funds flow for the project

205. A diagram showing the flow of funds is presented below. WWF-Pakistan will be an EE for the project and receive GCF Proceeds and the co-financing resources channelled through the AE. Under Component 1, WWF-Pakistan will provide cash grants to the Provincial Departments of Forestry and community-managed nurseries as procured parties for the execution of Activity 1.1.1, and to the Provincial Departments of Irrigation for Activity 1.1.2 and 1.2.1,

¹⁰¹ <https://projects.worldbank.org/en/projects-operations/project-detail/P179981>

to competitively procure the services of one or more engineering firms to design and build the green infrastructure measures identified for the project sites. WWF-Pakistan will also provide grant financing to the CBOs to strengthen community-based natural resource management of EbA and green infrastructure interventions implemented in floodplains and watersheds (Activities 1.3.1 and 1.3.2). Under Component 2, WWF-Pakistan will provide goods and services to the MoCC, GCF NDA, to support its role in facilitating the establishment and operationalization of the National Working Group responsible for expanding Pakistan's current flood and water resources management regulations to prioritise EbA and green infrastructure interventions (Activity 2.2.1). Under Component 3, WWF-Pakistan will provide grants to the Provincial Departments of Agriculture as procured parties for the execution of Activity 3.2.1.

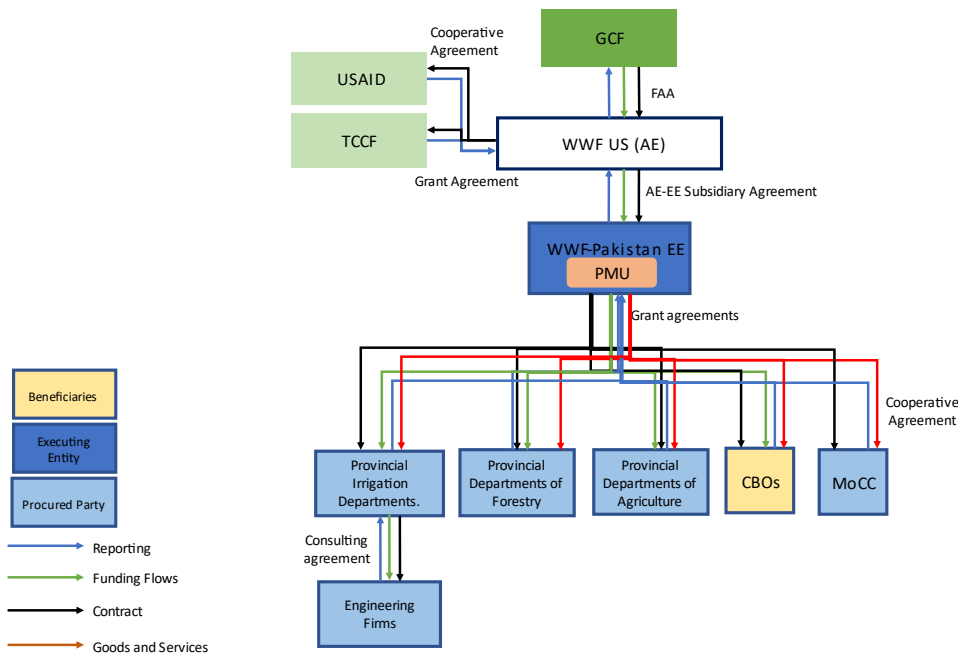


Figure 33. Flow of funds for the project.

Synergy and coordination between the proposed project and FP108

206. There is no geographic overlap and therefore no duplication of investment between the existing FP108 and the proposed project.

207. Given that, broadly, both projects focus on the water and agricultural sectors, there are opportunities for the two projects to coordinate and contribute to each of their respective objectives. The table below specifies which activities will enable these contributions, and how alignment and coordination between the two projects will be ensured during implementation.

Table 13. Synergy and coordination between the proposed project and FAO-led GCF FP 108.

FAO Activity	The proposed GCF project's activities	Coordination area	Coordination process
Activity 1.1.1: Develop and establish a water accounting system, including the equipment needed to operate this system in two canals.	Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan.	Data generated under the proposed project can feed into the FAO project's ACWA Portal to inform future planning, decision-making and development. To ensure the investments into data generation on water under the FAO-led project are maximised, the proposed project will make use of this data through the ACWA Portal in its generation of evidence of EbA and green infrastructure interventions' effectiveness in flood and	The FAO-led project's Water Focal Point for FP108 is based in Islamabad, which is where the proposed project's PMU will be located. The proposed project's PMU will therefore be well-placed to coordinate regularly with the FAO Focal Point on areas of alignment and reciprocal contribution during implementation. Finally, the proposed project's PMU and Site Implementation

		water resources management.	Units will engage with the FAO-led project's Project Facilitators, Coordinators on the ground in Sindh, as well as the Technical Working Group, to ensure coordination on this work. Synergy at the federal level between the proposed project's PSC and the FAO-led project's PSC will also be explored given the similarity between government partners between the two projects.
Activity 2.1.1: Develop and improve training materials to support adoption of CRA and OFWM practices.	Activity 3.2.1: Improve the climate resilience of vulnerable agricultural livelihoods.	- The proposed project will make use of knowledge and best practices generated by the FAO-led project's implementation of its CRA and OFWM activities. - The knowledge and best practices generated by the FAO-led project will be used to inform the implementation of the proposed project's Activity 3.2.1. Specifically, this knowledge will be incorporated into the Farmer Field School approach (and associated curriculum) the proposed project will implement under Activity 3.2.1 through which the project will introduce new irrigation technologies and methods, facilitate the adoption of drought resistant seed varieties into current crop farming practices as well as improved farming and soil management techniques. - This will ensure that investments in the FAO-led project into CRA and OFWM will result in knowledge transfer and replication of climate-resilient techniques under the proposed project, particularly given that both projects intend to work with similar crops (e.g., cotton).	
Activity 2.1.3: Develop a CRA and OFWM practice repository – the Pakistan Climate Agriculture Information System (PCAIS).			
Sub-component 2.2: Training of Trainers on CRA and OFWM.			
Sub-component 2.3: Developing Farmers' Capacity to Transform Agriculture Practices with CRA and OFWM			
Sub-component 3.1: Improving Information and Awareness Raising Campaigns.			

Activity 3.2.1: Increasing Collaboration between Institutions through Knowledge Sharing for Evidence based Policymaking.	Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan. Activity 2.2.1: Develop updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for implementing EbA and green infrastructure interventions in Pakistan. Sub-Activity 2.2.1.1: Establish a National Working Group to develop and adopt updated procedures for using EbA and green infrastructure interventions in flood and water resources management in Pakistan.	The proposed project's National Working Group, as well as the Project Steering Committee (PSC) and Provincial Oversight Committees (POCs), will contribute to the FAO-led project's Activity 3.2.1 by being the vehicle for enhancing coordination between stakeholders in the water sector, improving knowledge sharing on integrated flood and water resources management strategies, strengthening water governance, and promoting evidence-based decision-making for climate change adaptation with an emphasis on EbA and green infrastructure.	
Activity 3.2.2: Mainstreaming Climate Resilient Agriculture and Water Management across All Policy Areas		The proposed project's evidence base of the effectiveness of EbA and green infrastructure for flood and water resources management will build on the evidence generated by the FAO-led project on climate-resilient agriculture and on-farm water management. This evidence will be used to initiate public sector reform in flood and water resources management under the proposed project, as described in Section B.3 of this Funding Proposal.	

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

208. Pakistan already experiences significant economic and human capital losses from climate change and evidence suggests it will only have increasing impacts if no action is taken. According to the Global Climate Risk Index annual report for 2021, Pakistan has lost 0.53%/unit GDP, suffered economic losses worth USD 3.8 billion, and experienced 152 extreme weather events between 1999 and 2018¹⁰². Between 1995 and 2014, climate change cost Pakistan USD 3.9 billion (purchasing power parity) in average economic losses annually. Under future climate scenarios, water availability is expected to be less predictable, and floods and droughts more common. Pakistan is one of the most water-stressed countries of the world with average per capita water availability of less than 1,000 m³ per annum¹⁰³, making it extremely vulnerable to an increasing risk of drought. The 1998–2002 drought, recorded as one of the worst droughts of the last 50 years, affected 1.2 million people and caused the death of 2 million livestock animals and significant losses in crops¹⁰⁴. Major floods that have affected Pakistan in the past decade have shown the vulnerability of homes, workplaces, and infrastructure. For example, the catastrophic 2010 floods

¹⁰² Source: https://www.finance.gov.pk/survey/chapter_20/16_Climate_Change.pdf

¹⁰³ National Water Policy, 2018.

¹⁰⁴ Waseem, M., et al. 2022. Impact of meteorological drought on agriculture production at different scales in Punjab, Pakistan. Water and Climate Change (2022) 13 (1): 113-114. Available at: <https://iwaponline.com/jwcc/article/13/1/113/83806/Impact-of-meteorological-drought-on-agriculture>

in the Indus Basin caused financial damages of an estimated USD 40 billion and affected more than 20 million people; approximately 2,000 lives were lost, 3,000 people were injured, 1.6 million houses and millions of acres of crops were damaged. More recently, severe flooding in Pakistan in August/September 2022 destroyed 218,000 houses, damaged 452,000 houses, negatively impacted 2 million acres of crops, resulted in the loss of 794,000 livestock, injured 1,343 people and resulted in the death of 937 people¹⁰⁵. See additional details on climate change and its impacts in Section B.1. of this Funding Proposal and Annex 2.

209. Combating the impacts of climate change requires action on both mitigation (reduction of GHG emissions) and adaptation (building resilience in natural and human systems). A comparison of the global cost estimates with the current level of adaptation funding indicates that projected global adaptation needs are to be significantly greater than current investment levels, particularly in vulnerable developing countries like Pakistan. The average costs for annual adaptation and mitigation to climate change for Pakistan were estimated to range annually from more than USD 10 billion to up to USD 32 billion by 2050^{106,107}. According to the revised NDCs, climate change adaptation needs in Pakistan range from USD 7 to 14 billion per annum based on current exchange rates¹⁰⁸.
210. Pakistan remains a resource-constrained economy, dependent on foreign financing in the shape of loans, foreign investment, and grant aid. The country relies predominantly on loans from multilateral banks for infrastructure projects and foreign aid to assist in post-disaster scenarios, including for drought and flood recovery. For instance, recent donor funding from, but not limited to, the FAO, WFP, and WHO were channelled through the Central Emergency Response Fund and Pakistan's Humanitarian Fund for targeted post-drought relief across multiple sectors totalling USD 15 million. Post-disaster foreign aid assistance (including for floods and droughts) provided to Pakistan over the last two decades from donors such as USAID, the European Commission, the World Bank and the Asian Development Bank (ADB) have totalled an approximate USD 1.5 billion.
211. As flooding has historically been a major observed and projected climatic hazard in Pakistan, a significant investment is required to meet the country's flood risk management needs. For example, the National Flood Projection Plan (NFPP) IV presents an estimated budget of USD 375 million for the country's flood risk management needs. However, due to insufficient GoP funds, foreign financing is needed to cover these costs, particularly for Ecosystem-based Adaptation interventions, which have historically received less attention and funding than hard infrastructure alternatives. According to the Federal Flood Commission's 2021 Annual Flood Report, Federal and Provincial Government funding needs for the Normal/Emergent Flood Programme, for 2019–20, totalled USD 56 million, but only USD 2.8 million was released for use leaving a funding gap of USD 53.2 million¹⁰⁹. Despite these strong commitments towards climate action from the GoP, economic constraints continue to limit the amount of funding available. Approximately PKR 10 billion (approximately USD 50 million) has been allocated to the Ministry of Climate Change (MoCC) under the 2022-23 national budget, a decrease of PKR 4 billion from the year before likely brought by the economic downturn brought on largely by COVID and only widening the gap between available public climate finance and the growing economic needs brought on by Pakistan's continuous exposure to climate disasters¹¹⁰.
212. In addition, initiatives such as the NDRMF have been established to fund disaster risk reduction projects. With a total fund size of USD 393.56 million (mainly international donor funding), 56% of NDRMF's current grant portfolio targets flood protection, while 2% targets water conservation/drought mitigation. However, most of this finance is used to cover the costs of climate-related disaster relief and reconstruction, with adaptation in the form of disaster preparedness remaining limited. By increasing financial flows to adaptation approaches for climate-related disaster preparedness (such as EbA and green infrastructure), overall spending would be reduced, damages would be avoided and the negative impacts on Pakistan's economy lessened.

¹⁰⁵ UN Office for the Coordination of Humanitarian Affairs (OCHA). Pakistan: 2022 Monsoon Floods. Situation Report No. 03. 26 August 2022. Available at: Pakistan: 2022 Monsoon Floods - Situation Report No. 03: As of 26 August 2022 - Pakistan | ReliefWeb.

¹⁰⁶ UNDP (2015). Pakistan - Climate Public Expenditure and Institutional Review. Government of Pakistan. Source: <http://www.pk.undp.org/content/dam/pakistan/docs/Environment%20&%20Climate%20Change/UNDP%20Climate%20Report%20V1.0.pdf>

¹⁰⁷ Government of Pakistan (2017). Pakistan - Climate Change Financing Framework. United Nations Development Programme, Pakistan. Source: <https://www.climatefinance-developmenteffectiveness.org/sites/default/files/publication/attach/Pakistan-CCFF-Oct-2017.pdf>

¹⁰⁸ Source: https://www.finance.gov.pk/survey/chapter_20/16_Climate_Change.pdf

¹⁰⁹ Government of Pakistan. 2021. Federal Flood Commission Annual Report.

¹¹⁰ United States Institute of Peace. Pakistan's Climate Challenges Pose a National Security Emergency. A whole-of-government approach is needed now before climate change exacerbates conflict in the country. Thursday, July 7, 2022. [https://www.usip.org/publications/2022/07/pakistans-climate-challenges-pose-national-security-emergency#:~:text=Almost%20PKR%2010%20billion%20\(approximately,the%202021%20D22%20national%20budget.](https://www.usip.org/publications/2022/07/pakistans-climate-challenges-pose-national-security-emergency#:~:text=Almost%20PKR%2010%20billion%20(approximately,the%202021%20D22%20national%20budget.)

213. While the GoP's financial commitment to flood and drought risk management is expected to continue increasing in the future, it will still fall far short of the abovementioned funding needs, particularly for initiatives such as the proposed EbA flood risk management project that inherently has high up-front costs. Pakistan debt level and trade imbalance imposes significant limitations on available finance from the GOP on climate adaptation. The country's public debt, including guaranteed debt, reached 70.7% of GDP at the end of December 2021, rising from 72% percent at the end of December 2020. In 2022: i) real GDP growth is expected to slow to 4.3% (and to 4% in 2023); ii) inflation is estimated to rise to 10.7%; iii) the current account deficit is expected to widen to 4.4% of GDP; iv) the fiscal deficit (including grants) is projected to widen slightly to 6.2% of GDP; and v) public debt as a share of GDP is projected to stay high¹¹¹.

214. Along with the country's high level of indebtedness, rural communities in the project areas are too economically vulnerable to make significant financial contributions to climate projects. Local financing institutions also have limited funds and don't have the risk appetite for EbA and green infrastructure investments. In addition, private sector resources, either local or international, for climate change adaptation remain absent in rural areas of Pakistan, as is the case in most low-income developing countries. This is compounded by the lack of evidence on the economic benefits of EbA for flood risk and water resource management in Pakistan, which is a barrier to investment in EbA and green infrastructure approaches particularly in rural areas.

215. Given the funding gap between Pakistan's adaptation priorities and its public finance commitments, a shift is required in national policymaking and planning. Without GCF resources to catalyse this shift, Pakistan will continue to operate under the current paradigm relying on traditional and largely reactive models for flood risk and water resources management that is neither climate resilient nor financially sustainable. As a result, Pakistan's rural communities, such as those in the Indus Basin, will become increasingly vulnerable to the impacts of climate change.

Financial instrument make-up of the proposed project

216. Considering the needs and current adaptation funding constraints presented above, the GoP is requesting a GCF grant of USD 66 million to support the implementation of the proposed project's activities. As a result of WWF-Pakistan's engagements during Concept Note and Funding Proposal development with national (MoCC and MoWR), local (provincial irrigation departments) and international (e.g., World Bank, ADB, KfW, GIZ, etc.), the proposed project was designed to have a total of ~US\$67 million in government co-finance from the NDRMF (~US\$37 million) and MoWR (~US\$29 million) committed to the project to complement the GCF grant. While ~US\$67 million in government co-finance from the NDRMF and MoWR was committed to the project to complement the GCF grant, the recent 2022 floods have required that these funds be repurposed for immediate disaster relief making it very difficult for the Government of Pakistan to commit significant adaptation finance at this time. Despite these constraints, the GoP remains committed to the proposed project but is unable to commit any co-financing for the project due to the economic constraints on public spending due to the high costs for recovery from the 2022 floods.

The AE has engaged private and public sector partners to mobilize resources for this US\$77.8M project. As a result, a commitment letter from The Coca-Cola Foundation (TCCF) for US\$5M and a letter of interest from USAID/Pakistan for up to US\$10M has been secured. USAID requires a 1:1 match for its funding. With these commitments, US\$10M in co-finance will be provided for the project, secured by the AE from TCCF and USAID.

217. The conditions in Pakistan to support a rationale for maximum concessionality to be applied to investing in EbA for flood risk and water resources management are summarised below.

- *Market failures*: markets tend to be skewed toward hard infrastructure as they often do not account for any negative impacts of hard infrastructure and the wider benefits of ecosystem-based approaches and associated ecosystem services are undervalued.
- *Barriers to entry for alternative technologies*: investors may not have the technical knowledge to understand and be able to compare traditional hard infrastructure approaches to EbA alternatives.
- *Transaction costs*: economically viable EbA and green infrastructure interventions are often not considered as economically viable options due to their high transaction costs associated with community and stakeholder engagement and technical assistance needs among local governance.

218. The above market failures and barriers can be overcome by investing in an enabling environment that advances EbA and green infrastructure in Pakistan as a climate change adaptation response and demonstrates

¹¹¹ Source: <https://www.worldbank.org/en/country/pakistan/overview#1>

the potential to further reduce flood risk and improve water resource management but have high upfront costs and limited options for finance. Such activities have the potential to be scaled up once their potential or the ability of investment to generate returns is proven within the country, given expectations of intensified flood and drought risks (resulting from climate change) and consequent increases in the demand for solutions.

219. GCF grant funding will be used to fund and demonstrate the use of EbA and green infrastructure interventions as alternatives or complementary (to hard infrastructure) approaches to flood and drought risk management in Pakistan (Component 1). The implementation of these adaptation interventions will be used as evidence to create an enabling environment for climate action in Pakistan that removes barriers to novel climate solutions through promoting integrated strategies, planning and policymaking. Specifically, this evidence will be used to develop and adopt updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan (Component 2). Additionally, the project will support climate-resilient livelihoods among the most vulnerable communities in the Indus Basin by: i) providing incubation services to 7 small businesses already identified during the DFCD project to incentivize additional sustainable business development and help launch the creation of an investment market in sustainable enterprises; and ii) providing direct support to local communities to improve the sustainability of their land and water resources management practices and strengthen the climate-resilience of their agro-ecological livelihoods (Component 3).

B.6. Exit strategy (max. 500 words, approximately 1 page)

220. The proposed project has been designed in close collaboration with the relevant key government stakeholders and community representatives to ensure it provides sustainable adaptation benefits to the target vulnerable communities in Pakistan during and beyond the lifetime of the project. The project's Exit Strategy is presented in the paragraphs that follow, focusing on institutional, technical and financial sustainability, as well as local-level sustainability.

Strengthened institutional capacity, awareness and ownership

221. The project will be implemented in collaboration with relevant key government stakeholders — including MoCC, MoWR (and FFC), and the Provincial Departments of Irrigation, Forestry, and Agriculture — to ensure ownership for the sustainable adoption, scale up and replication of the project's interventions. This includes addressing institutional and technical needs regarding the ongoing O&M of the project's EbA and green infrastructure interventions identified in consultation with the GoP, as well as under the relevant policies and plans related to climate change planning, flood and water resources management (e.g., National Water Policy and its Implementation Framework, NAP, four Provincial Adaptation Plans, National Flood Protection Plan IV (NFPP IV), National Climate Change Policy, and National Forest Policy).
222. In addition, the project will foster ongoing national ownership by generating new evidence on the benefits and cost-effectiveness of implementing EbA and green infrastructure interventions for flood and water resource management, as well as lessons learned from implementing these interventions. This evidence will serve as a key reference for policymakers in future flood and water resources management. The evidence will also be presented to the Pakistan Climate Change Council (PCCC) — headed by the Prime Minister of Pakistan. In line with its mandate of leading climate change planning in Pakistan, the Council along with the National Working Group that will be established under the project will ensure that the ownership of project interventions is integrated into the relevant regulatory frameworks for ongoing implementation and financial support beyond the lifetime of the project.

Policy and regulatory support

223. The evidence generated under the project will also be used to develop updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, NAP, and four Provincial Adaptation Plans — to expand them from only focusing on using grey infrastructure to address the threats of floods and droughts to prioritising EbA and green infrastructure interventions. A National Working Group will be established to lead the development and adoption of these procedures into the MoWR's (including FFC's) and MoCC's mandates for flood and water resources management and climate change planning, respectively. Integrating the implementation of EbA and green infrastructure interventions into these government mandates will enable existing/planned public funds that would likely otherwise have been allocated to grey infrastructure interventions to instead be channelled towards the ongoing and future implementation and O&M of EbA and green infrastructure interventions during and beyond the project period.

Operations and maintenance

224. The GoP will undertake O&M of the project's interventions for the project lifespan (30 years) through its existing resources and infrastructure to ensure the interventions function optimally and achieve the desired adaptation impact. For the EbA interventions, MoCC and the Provincial Departments of Forestry will be primarily responsible for O&M and ensuring the optimal functioning of the interventions for the project lifespan. For the green infrastructure interventions, MoWR (including FFC) and the Provincial Departments of Irrigation will be primarily responsible for O&M and ensuring the optimal functioning of the interventions for the project lifespan. The required O&M arrangements for the project's EbA and green infrastructure interventions, including the type and frequency of O&M activities required as well as the relevant responsible entity(ies), are presented below.

Table 14. Required O&M arrangements for the project's EbA and green infrastructure interventions.

Intervention		O&M activity required	Frequency and importance of O&M activity	Responsibility
EbA interventions	Sub-activity 1.1.1.1: Restore 14,215 ha of degraded riverine ecosystems in D.I. Khan.	- Mowing - Brush cutting - Minor replanting/reseeding - Weed control - Inlet/outlet cleaning - Animal nuisance control	Routine (Recommended)	Provincial Forest Department of Khyber Pakhtunkhwa
		- Ditch stabilization - Ditch reshaping/regrading	Periodic (Recommended)	
		Major replanting/reseeding	Annual (Important)	
		Watch and Ward Systems under CBOs to protect the intervention	Annual (Important)	
	Sub-Activity 1.1.2.1: Excavate 264 m of flow paths in Badri Village, Ramak, to restore the natural hydrology of connected wetlands.	Manual clearance of debris from the channel	Routine (Recommended)	Provincial Irrigation Department of DI Khan
		Clearance of debris from the culvert/pond inlet	Periodic (Recommended)	
	Sub-Activity 1.1.2.2: De-silt the 2 ha Badri Pond in the Ramak wetlands to enhance its water-holding capacity.	Silt removal	Periodic (Recommended)	Provincial Irrigation Department of DI Khan
	Sub-Activity 1.1.2.3: Enhance and reinforce 410 m of embankments in Badri Village to reduce erosion.	Compacting embankments	Periodic (Recommended)	Provincial Irrigation Department of DI Khan
		- Mowing - Brush cutting - Minor replanting/reseeding - Weed control - Inlet/outlet cleaning - Animal nuisance control	Routine (Recommended)	
		Major replanting/reseeding	Periodic (Recommended)	
		Watch and Ward Systems under CBOs to protect the intervention	Annual (Important)	



Green infrastructure interventions	Sub-Activity 1.1.2.4: Excavate 4,000 m of flow paths in Paniala Village to restore natural hydrology of Karez Systems (surface and subsurface natural channels).	• Clearance of debris from the channel • Clearance of debris from the culvert/pond inlets.	Routine (Recommended) Periodic (Recommended)	Provincial Irrigation Department of DI Khan
	Sub-Activity 1.1.2.5: Excavate 30,000 m of flow paths in the existing water channels of Manchar Lake to restore the natural hydrology of the 25,000 ha Manchar Wetland.	• Clearance of debris from the channel • Clearance of debris from the culvert/pond inlets.	Routine (Recommended) Periodic (Recommended)	Provincial Irrigation Department of Sindh
	Sub-Activity 1.2.1.2: Build 9 flood protection embankments.	• Clearance of debris from the channel	Routine (Recommended)	Provincial Irrigation Departments of DI Khan and Khyber Pakhtunkhwa
		• Compacting embankments	Annual (Recommended)	
		• Readjustment of stone pitching on the riverside slope of the embankment	Annual (Recommended)	
	Sub-Activity 1.2.1.3: Build 7 flood dispersal embankments.	• Clearance of debris	Routine (Recommended)	Provincial Irrigation Departments of Sindh, Khyber Pakhtunkhwa and Balochistan
		• Compacting embankments	Annual (Recommended)	
		• Readjustment of stone pitching on the riverside slope of the embankment	Annual (Recommended)	
	Sub-Activity 1.2.1.4: Build 45 gabion bunds.	• Inspecting the fence to identify bulging of the fence¹¹² • Reinforcing/repairing any damage to the fence caused by localised bulging	Routine (Recommended)	Provincial Irrigation Departments of Sindh, Khyber Pakhtunkhwa and Balochistan
		• Fixing damage to the fencing of gabion bunds	Routine (Recommended)	
	Sub-Activity 1.2.1.5: Build 42 small retention areas.	• Removal of blockages to the inlets and outlets of the retention areas	Annual (Recommended)	Provincial Irrigation Departments of Sindh, Khyber Pakhtunkhwa, and Balochistan
		• Clearance of debris/sedimentation	Routine (Recommended)	
		• Protect the shoreline from erosion, slow runoff into the pond, and capture sediment and other pollutants through establishing wide planted buffers using a variety of native perennial	Annual (Recommended)	

¹¹² Gabion baskets are made of steel or aluminium. Bulging of these baskets occasionally occurs when the rock filling in the gabion basket has caused it to change shape from its original design.

	Sub-Activity 1.2.1.6: Build 9 large retention areas.	shrubs with deep root systems			Provincial Irrigation Departments of Sindh, Khyber Pakhtunkhwa and Balochistan
		Removal of blockages to the inlets and outlets of the retention areas	Annual (Recommended)		
		Clearance of debris/sedimentation	Routine (Recommended)		
	Sub-Activity 1.2.1.7: Build 15 recharge basins.	Protect the shoreline from erosion, slow runoff into the pond, and capture sediment and other pollutants through establishing wide planted buffers using a variety of native perennial shrubs with deep root systems	Annual (Recommended)		Provincial Irrigation Departments of Sindh, Khyber Pakhtunkhwa, and Balochistan
		Removal of blockages to the inlets and outlets of the retention areas	Annual (Recommended)		
		Clearance of debris/sedimentation	Routine (Recommended)		
		Protect the shoreline from erosion, slow runoff into the pond, and capture sediment and other pollutants through establishing wide planted buffers using a variety of native perennial shrubs with deep root systems	Annual (Recommended)		

225. During implementation, the project's Exit Strategy will be further developed by the PMU through: i) signing agreements with the relevant government stakeholders for O&M of the project's interventions; ii) strengthening the capacity of national and sub-national staff, as well as community leaders/representatives to implement, operate and maintain the project's EbA and green infrastructure interventions for flood and water resources management; iii) developing detailed long-term O&M plans for the project's EbA and green infrastructure interventions (in Year 3 of the project); iv) developing a scaling up and replication strategy for the project's interventions (in Year 6 of the project); and v) confirming with the GoP the financing arrangements for long-term O&M of the project's interventions and establishing an appropriate annual O&M budget in consultation with the relevant government stakeholders, including but not limited to MoCC, MoWR (and FFC) and the relevant Provincial Departments of Irrigation, Agriculture and Forestry.

Project sustainability at the local level

226. The GoP's O&M commitments will be supported at the community level by establishing community agreements between the relevant provincial line departments and CBOs. These agreements will outline arrangements for active community participation in O&M of the project's EbA and green infrastructure interventions, ecosystem protection and restoration, and impact monitoring. For example, under the CBOs, Watch and Ward Systems will be established to monitor and protect the project's EbA and green infrastructure from degradation, while Water User Groups will be created to manage community access to and use of the additional water that will be retained by the green infrastructure interventions (e.g., water stored in the small and large retention areas). These agreements will secure local-level ownership and future replication of the project's EbA and green infrastructure interventions at scale after project closure.

227. Additionally, WWF-Pakistan has learned from its more than 50 years of experience working in Pakistan that successful conservation objectives cannot be met unless local communities' livelihood needs are appropriately addressed. As a result, local communities that will benefit from the project's interventions have been extensively engaged throughout the project development process. These engagements include consultations with tribal leaders, who traditionally play an important role in leading their tribal communities. WWF-Pakistan understands the key dynamics of engaging these tribal leaders and took a proactive approach during project development to garner their support and incorporate their indigenous knowledge into the design of the project. These engagements will continue during the implementation phase to ensure sustainable delivery and adoption of the project's interventions by the target beneficiary communities in Pakistan¹¹³.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	66,000,005			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter</u> years	<u>Enter</u> years	<u>Enter</u> %		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter</u> years	<u>Enter</u> years	<u>Enter</u> %		
(iii) Equity	<u>Enter amount</u>	<u>Enter</u> years		<u>Enter</u> % Equity return		
(iv) Guarantees	<u>Enter amount</u>					
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	66,000,005					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	11,848,391			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
WWF-Pakistan	<u>In-kind</u>	1,848,391	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
United States Agency for International Development (USAID)	<u>Grant</u>	5,000,000	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
The Coca-Cola Foundation (TCCF)	<u>Grant</u>	5,000,000	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
Total financing (c) = (a)+(b)	Amount			Currency		
	77,848,397			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions	In addition to the grant resources identified above, USAID and The Coca-Cola Foundation will provide i) expert guidance on the project strategy and implementation and advice on several key project workstreams, including those focused on water resource management, climate-smart agriculture, and bankable project opportunities for climate-resilient businesses in the agriculture, forestry, water and sanitation sectors; ii) provide strategic and policy guidance inclusive of global					

¹¹³ Further details on stakeholder engagement are provided in Annex 7.



(max. 250 words,
approximately
0.5 page)

learnings and best practices; iii) support the PMU with identifying and managing project-related risks; iv) support communication and dissemination of project outcomes to key stakeholders; v) Seek to identify complementary and aligned efforts to bring Recharge Pakistan together in coordination with others in powerful ways, inclusive of connecting to existing efforts and bringing new partners to support the work. USAID and The Coca-Cola Foundation will receive regular updates on the status of the project and on the significant project developments from the PMU and participate in site visits to monitor project activity progress and engage with local communities and stakeholders.

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan.	Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.	12,448,479	7,960,595	Grant	4,487,884	Grant	TCCF
	Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.	51,697,224	51,633,958	Grant	63,266	Grant	TCCF
	Output 1.3: Strengthened community- based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.	2,004,253	1,877,721	Grant	126,532	Grant	TCCF
Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan.	Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.	1,154,849	380,537	Grant	774,312	Grant	USAID



	Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.	563,425	152,268	Grant	411,156	Grant	USAID
	Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.	977,425	340,268	Grant	637,156	Grant	USAID
Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin.	Output 3.1: Pipeline of feasible climate-resilient businesses.	1,687,550	487,794	Grant	1,038,597	Grant	USAID
					161,159	Grant	TCCF
	Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.	1,528,910	302,466	Grant	1,065,285	Grant	USAID
					161,159	Grant	TCCF
Monitoring and Evaluation		2,154,279	1,080,786	Grant	1,073,493	Grant	USAID
Project Management Costs		3,632,003	1,783,612	Grant	1,848,391	Grant	WWF-Pakistan
Indicative total cost (USD)		77,848,397	66,000,005		11,848,391		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

Technology transfer

228. Under Outputs 1.1, 1.2, 3.1, and 3.2, technology transfer will be facilitated through: i) implementing EbA interventions (including floodplain and watershed restoration and rehabilitation) in the four project sites in the Indus Basin (Activities 1.1.1 and 1.1.2); ii) establishing green infrastructure interventions at specific locations within the project target areas to reduce flood impacts and provide temporary storage and infiltration of water outside of flooding events (Activity 1.2.1); iii) developing a pipeline of sustainable climate-resilient businesses (Activity 3.1.1); and iv) introducing drought-resilient crop varieties, improved irrigation infrastructure, and sustainable agricultural and land use practices to farmers, as well as providing them with water storage tanks (Activity 3.2.1). The total project investment supporting technology transfer is US\$67,362,163.

Capacity-building interventions

229. Technical and institutional capacity of government and non-government actors involved in flood and water resources management, as well as communities in all four project sites — namely DI Khan Watershed, Ramak Watershed, Chakar Lehri Watershed, and Manchar Lake and Watershed — will be strengthened.
230. Under Output 1.3, training will include: i) strengthening CBOs to implement community-based natural resource management, including on sustainable forest management practices to ensure sustainability of interventions and ecosystems such as wetlands, floodplains, flood bypasses and washlands, watersheds and plantations; ii) developing formal WUGs to resolve and prioritize water allocation for agriculture, domestic and disaster risk reduction measures; iii) establishing community-based watch and ward systems under the CBOs to facilitate the implementation and protection of the project's EbA and green infrastructure interventions, including the required operation and maintenance, as well as protect and preserve natural forests from illegal cutting/forest fires. The establishment of these systems will be accompanied by Watch and Ward Plans, which will be developed with respective CBOs and the Provincial Forest Departments. The total project investment supporting capacity-building under Output 1.3 is US\$2,004,253.
231. Under Outputs 2.1, 2.2 and 2.3, national technical and institutional capacity, as well as the regulatory framework for flood and water resources management will be strengthened. This will be achieved through: i) undertaking an economic valuation of the project's EbA and green infrastructure interventions to analyse their cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan; ii) establishing a National Working Group to lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management on the proper execution of EbA and green infrastructure for flood and water resources management; and iii) strengthening institutional and technical capacity at the national and sub-national levels for implementing EbA and green infrastructure interventions. Local communities will also be engaged and equipped to implement, operate and maintain EbA and green infrastructure interventions. The total project investment supporting capacity-building under Outputs 2.1, 2.2 and 2.3 is US\$2,695,698.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

232. The proposed project will contribute to the following GCF Adaptation Result Areas (ARAs): ARA1.0 — Increased resilience and enhanced livelihoods of the most vulnerable people, communities, and regions; and ARA4.0 — Improved resilience of ecosystems and ecosystem services, as described below¹¹⁴.

Outcome¹¹⁵ level: Increased resilience

ARA1.0 — Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions.

233. The project will strengthen the resilience of the most vulnerable people living in Pakistan's Indus Basin — most of whom depend on agriculture for their livelihoods — to increasingly severe floods and droughts caused by climate change through implementing a combination of EbA, green infrastructure and community-based adaptation interventions in the four project sites. EbA and green infrastructure interventions implemented under the project will protect a total of 889,900 people (621,500 direct beneficiaries and 268,400 indirect beneficiaries) from climate change induced flooding. Over the 30-year modelled period for which benefits were evaluated (see Annex 3), EbA and green infrastructure interventions will avoid US\$20.2 million in losses and damages to crops, livestock, housing, and infrastructure, as well as indirect damages resulting from disruption of economic activities (e.g., because of transport severance) on or adjacent to the floodplain.
- **EbA interventions** will reduce flood impacts on populations close to the intervention sites and for people living downstream. Populations close to the EbA interventions will benefit from local ecosystem services including but not limited to slowing of flood waters delays in flood peak onset, and increased access to water during drought periods, while downstream communities will benefit from the slowing of upstream velocity and delayed peak flood. The downstream benefits will slowly decrease with increasing distance from the EbA intervention sites as their impacts and benefits are reduced.

¹¹⁴ Detailed information on the assumptions and estimates of the impact potential of the proposed project are provided in the project's results framework and economic analysis (Section E of this document and Annex 3, respectively).

¹¹⁵ A detailed description of the project activities that will be implemented under each output to achieve the GCF Outcomes is presented in Section E.6 of the Funding Proposal.

- **Green infrastructure interventions** will reduce flood impacts on nearby communities and ecosystems by reducing soil erosion and river sedimentation, diverting and reducing the velocity of intense flows in the hill torrents away from nearby villages located in the floodplain, and increasing infiltration and groundwater recharge. Populations near the green infrastructure interventions, particularly the retention and recharge areas, will benefit from localised slowing of flood waters and the subsequent temporary storage and infiltration of water outside of flooding events. The gabion bunds and flood protection embankments will primarily protect people close to these interventions as the purpose of these structures is to reduce immediate damages from floodwaters rather than reduce large-scale flood magnitude.
- **Community-based adaptation interventions** will increase the climate resilience of 50,750 of the most vulnerable smallholder farmers in the project sites. As a result of the project, these farmers will: i) adopt climate-resilient agricultural inputs (e.g., drought-resilient crop varieties); ii) have capacity (and regular access to improved extension services) to implement resilient and sustainable irrigation and soil management techniques; and iii) have access to 20,250 m³ of additional water made available through new water storage tanks. This will optimise on-farm water efficiency and support agricultural production under increasingly severe drought conditions.

ARA4.0 — Improved resilience of ecosystems and ecosystem services.

234. EbA interventions implemented under the project will enhance ecosystem functioning and the delivery of goods and services to vulnerable communities living in the Indus Basin and secure their livelihoods by restoring a total of 14,215 ha of degraded watersheds in DI Khan, restoring 34.2 km of flow paths, as well as desilting and restoring channels in Ramak Watershed and Manchar Lake. This will result in among other benefits: i) improved attenuation of flood flows through increased land cover; ii) increased infiltration and groundwater recharge through reduced runoff; and iii) reduced soil erosion from flooding through improved flow regulation in rivers, streams and channels at these sites. Moreover, the EbA and green infrastructure interventions implemented at the project sites will reduce flood extent by 50,800 ha, capture an additional 20 million m³ of water, and increase soil infiltration by 1.6 million litres.

GCF Outcome¹¹⁶ level: Enabling environment

Core indicator 5 — Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner.

235. Using the evidence of the cost-effectiveness and efficiency of the project's EbA and green infrastructure interventions (described under Core indicator 8 below), a National Working Group will be established chaired by the MoCC and comprised of regulators within the MoWR (including FFC) and the relevant Provincial Departments. This working group will lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the NAP, and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.

236. Improved policymaking and planning for integrated flood and water resources management at the national level will be supported at the local level by: i) formalising CBOs' organization and governance structure and integrating flood and drought prevention into their mandate; ii) working closely with CBOs to develop formal WUGs to resolve and prioritize water allocation for agriculture, domestic and disaster risk reduction measures; iii) establishing community-based watch and ward systems under the CBOs to secure the adoption and sustainability of EbA and green infrastructure interventions at the local level, including the required operation and maintenance; and iv) developing Watch and Ward Plans with CBOs and the Provincial Forestry Departments to protect and preserve natural forests from climate change impacts, illegal cutting/forest fires.

Core indicator 7 — Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level.

237. The project will build on WWF-Pakistan's experience and foundational work developing pipeline businesses for the DFCD to: i) identify additional businesses with that can provide positive climate change benefits to communities and ecosystems; and ii) provide these businesses — which are currently not eligible for blended financing — with technical assistance and grant funding of ~US\$700,000 required to enhance their operations. As a result of the project's sequencing of DFCD and GCF investments in pipeline development, these businesses will have the opportunity to access debt or equity finance from DFCD or other investors. The portfolio of businesses established under the project will serve as a successful case study that demonstrates the financial and economic benefits new sustainable industries and markets can provide to both investors and businesses which, in turn, can incentivize future investment in the sector.

¹¹⁶ A detailed description of the project activities that will be implemented under each output to achieve the GCF Outcomes is presented in Section E.6 of the Funding Proposal.

Core indicator 8 — Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.

238. Part of the GCF investment for the project will be used to generate an evidence base to make the case for policymakers and planners to adopt EbA and green infrastructure interventions for flood and water resources management in Pakistan. The evidence base will be generated through an economic valuation of the project's EbA and green infrastructure interventions which will include analysing the biophysical benefits of EbA and green infrastructure on ecosystems as well as their flood attenuation and water resource-related services. Results from the analysis capturing the benefits of EbA and green infrastructure, as well as lessons learned and effective adaptive management strategies used under the project, will be collated into an evidence report and delivered to national- and provincial-level policymakers and planners involved in flood and water resources management through the National Working Group and PSC. This report will be used for the development of updated procedures for the Implementation Framework for the National Water Policy, the NAP, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions.

Adaptation impact

239. Project interventions will directly benefit a total of 687,336 people, including 357,414 men (52% of the total population) and 329,922 women (48% of the total population), as well as indirectly benefit a total of 7,024,361 people, including 3,652,668 men (52% of total population) and 3,371,693 women (48% of total population) in Pakistan's Indus Basin. The project lifespan is 30 years because this is the period over which the adaptation benefits below, particularly the EbA interventions, are expected to mature and continue delivering adaptation benefits to the project beneficiaries.

The table below defines and captures the project's direct and indirect beneficiaries per output. Annex 23 presents further details on the methodology used to calculate these project beneficiaries.

Table 15. Number of direct and indirect beneficiaries.

Outputs	Definition of direct and indirect beneficiaries per output	Direct beneficiaries (sex-disaggregated)	Indirect beneficiaries (sex-disaggregated)
Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.	Direct beneficiaries: These are calculated as the number of people living within 3 km of the EbA interventions. This includes those people near the interventions and downstream who will experience a direct change in flood/drought character because of improved floodwater management and watershed ecosystems. Indirect beneficiaries: These are people living between 3 and 10 km downstream from the EbA intervention locations who will benefit from reduced knock-on effects of floods and droughts on the direct beneficiaries described above (e.g., this would be the people who would buy food at local markets from farmers who are a subset of the direct beneficiaries described above).	310,200 people (161,304 men and 148,896 women)	117,100 people (60,892 men and 56,208 women)
Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.	Direct beneficiaries: These are calculated as the number of people living within 3 km of the green infrastructure intervention locations who will directly experience a change in flood character from the interventions. Indirect beneficiaries: These are people living between 3 and 10 km from the green infrastructure interventions that will benefit from reduced disruption to the direct beneficiaries described above, as well as the co-benefit of increased water retention and infiltration.	311,300 people (161,876 men and 149,424 women)	151,300 people (78,676 men and 72,624 women)
Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions	Direct beneficiaries: This is calculated as the number of community leaders and CBO representatives trained on community-based natural resource management, as well as the people living near natural forests brought under	14,520 people (7,550 men and 6,970 women)	3,300 people (1,716 men and 1,584 women)

implemented in floodplains and watersheds.	restoration and improved management through community-based watch and ward systems. Indirect beneficiaries: This is the number of people living in the project sites who will benefit from improved knowledge related to climate-resilient livelihoods/community-based natural resources management transferred by the now trained CBO leaders.			
Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.	Direct beneficiaries: The evidence base will be disseminated to national-level stakeholders, provincial line departments, and academic and/or research organizations. These recipients who will directly use the information were calculated as direct beneficiaries. Indirect beneficiaries: Beneficiary calculations for Output 2.2 considers the benefits of Output 2.1, which will generate the economic case for EbA and green infrastructure to inform current and future planning and capacity development. As a result, indirect beneficiary numbers have not been included for Output 2.1 to avoid double counting.	120 people (62 men and 58 women)	-	
Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.	Indirect beneficiaries: The total population of the four project target sites will indirectly benefit from this project output, which involves enhancing the regulatory environment for integrated flood and water resources management, and climate change adaptation in Pakistan. The indirect beneficiaries for Output 2.2 include the indirect beneficiaries identified for the other project outputs. The total number of indirect beneficiaries for this output will therefore also be the total number of indirect beneficiaries. Site-specific indirect beneficiary numbers are provided below. • 2,729,391 people in DI Khan Watershed • 461,828 people in Chakar Lehri Watershed • 3,833,142 people in Manchar Lake and Manchar Watershed	-	7,024,361 people (3,652,668 men and 3,371,693 women)	
Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.	Direct beneficiaries: These are the total number of national and sub-national decision-makers and planners trained on EbA and responsible for integrating EbA into provincial- and district-level flood risk, water resource management and adaptation plans. The national/subnational technical staff and community members working on/executing flood risk and drought prevention work using EbA and/or green infrastructure measures will also be trained and counted as direct beneficiaries. Indirect beneficiaries: The indirect beneficiaries under this output will overlap with the indirect beneficiaries of Output 2.2 above. Therefore, to avoid double-counting, indirect beneficiaries for Output 2.3 have not been included.	320 people (166 men and 154 women)	-	
Output 3.1: Pipeline of feasible climate-resilient businesses.	Direct beneficiaries. These are the business owners/people involved in the management of the 7 bankable projects, as well as the downstream partners directly involved in implementation of these projects.	126 people (66 men and 60 women)	1,008 people (524 men and 484 women)	

	Indirect beneficiaries: These are the consumers of bankable projects and the household members ¹¹⁷ of the direct beneficiaries mentioned above.			
Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.	Direct beneficiaries: These are the local people and their families¹¹⁸ who would be direct recipients of climate-resilient agricultural interventions and alternative livelihood support. A breakdown of these benefits is provided below. • 24,786 people will benefit from training on climate-resilient agricultural practices (Sub-activity 3.2.1.1) • 6,752 people will benefit from improved irrigation (Sub-activity 3.2.1.2) • 13,277 people will benefit from climate-resilient crop varieties (Sub-activity 3.2.1.3) • 4,424 people will benefit from improved soil management (Sub-activity 3.2.1.4) • 1,512 people will benefit from increased water security from water storage tanks (Sub-activity 3.2.1.5) Indirect beneficiaries: These are the people who will benefit from increased knowledge and capacity for implementing climate-resilient, sustainable agricultural practices transferred by the direct beneficiaries named above.	50,750 people (26,390 men and 24,360 women)	16,000 people (8,320 men and 7,680 women)	
Total		687,336 people (357,415 men and 329,921 women)	7,024,361 people (3,652,668 men and 3,371,693 women)	

Mitigation co-benefits

240. While the proposed project is an adaptation-only project, the EbA interventions will provide mitigation co-benefits. These co-benefits have been calculated as follows: A total of 14,215 hectares (ha) of degraded forests, deforested areas, and rangelands in DI Khan watershed will be placed under afforestation, reforestation and Assisted Natural Regeneration (ANR). Annual CO₂ removals from these activities are estimated at 26,450 tons of CO₂ per year while the total lifetime removals are estimated at 1,037,928 tons of CO₂. The total CO₂ removals will be achieved in 10-85 years, depending on the forest type and stand age.

241. Methods: To estimate the annual CO₂ removal from afforestation, reforestation and assisted natural regeneration, the activity area was multiplied by a removal factor. To estimate the total lifetime CO₂ removals, the annual removals were multiplied by the number of years the activity is expected to continue to sequester carbon. See Table 14 for calculations. The mean stand age of each forest type where the project's activities will be undertaken was used to estimate the removal factors and number of years for carbon sequestration.

242. The Removal Factors for different forest types/ ecological zones were taken from the Sub-National Forest Monitoring Systems and Greenhouse Gas Inventory Reports for KP and Punjab Provinces developed by WWF-Pakistan for the Ministry of Climate Change (MoCC) under the National REDD+ Project, 2022. The development of removal factors followed the IPCC recommended approach and methodologies. A brief summary of the removal factor methodology is outlined in bullets below.

- Carbon stock densities were determined for each of the different forest types using the forest inventory data at sample plot level.
- The carbon stock density values included aboveground biomass of trees and shrubs, belowground biomass, dead wood, and litter.
- Both national and IPCC default allometric equations were used for estimating aboveground tree biomass.
- The default IPCC fraction (0.47) was applied to convert biomass to carbon.

¹¹⁷ Pakistan's average household size of seven was used to calculate the number of beneficiaries.

¹¹⁸ Ibid.

- The carbon stock density value (C ton/ha) of each forest type was converted to CO₂ using the conversion factor of 3.67.
- The EE/RF were originally developed for the mean age of the forest. Annual EF/RF values were determined by dividing these factors by the mean age of each forest type.
- EE/RF for forest degradation were developed by determining the carbon density values (C t/ha) of different forest strata and the difference between these values when one forest stratum is degraded into a lower stratum (open forest-canopy cover 11-30%, sparse forest-canopy cover 31-50%, medium-canopy cover 51-70% and dense-canopy cover >70%).

243. The relevant calculations are provided in the table below.

Table 16. Total annual and total lifetime CO₂ removal estimates for the proposed project.

Expected ecological zone/ area in the proposed project sites	Type of activity	Area (ha)	Removal Factor (Tons CO ₂ /ha/year) ¹¹⁹	Total annual removals (Tons CO ₂ /year) [area*removal factor]	Expected number of years for carbon sequestration	Total lifetime removals (Tons CO ₂) [Total annual removal* years]
Dry Temperate (DI Khan)	Assisted Natural Regeneration in degraded forest	2,100	2.09	4,389	85	373,023
	Block plantation in rangelands/ grasslands	1,500	2.12	3,182	85	270,450
	Farmland plantations/ Woodlots (irrigated)	40	7.59	304	10	3,036
Subtropical Broadleaved Scrub (DI Khan)	Assisted Natural Regeneration in degraded forest	2,200	3.57	7,851	17	133,471
	Block plantation in rangelands/ grasslands	3,000	1.29	3,865	17	65,700
	Farmland plantations/ Woodlots (irrigated)	105	7.59	797	10	7,970
Tropical Thorn Forest (DI Khan)	Assisted Natural Regeneration in degraded forest	2,250	1.82	4,103	34	139,500
	Block plantation in rangelands/ grasslands	2,900	0.36	1,049	34	35,670
	Farmland plantations/ Woodlots (irrigated)	120	7.59	911	10	9,108
Total		14,215		26,450		1,037,928

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

244. Current flood and drought events in Pakistan are surpassing the country's existing water infrastructure's capacity to prevent large scale economic damages and loss of human life. As recently as August 2022, as this Funding Proposal is being finalized, the project areas have been hit by disastrous floods causing the displacement of an estimated 33 million people and loss of more than 1,000 human lives¹²⁰. Flood management efforts in Pakistan

¹¹⁹ Sub-National Forest Monitoring Systems and Greenhouse Gas Inventory Reports for KP and Punjab Provinces developed by WWF-Pakistan for the Ministry of Climate Change (MoCC) under the National REDD+ Project, 2022

¹²⁰ <https://www.bbc.com/news/world-asia-62704004>

to date have been largely reactive following extreme events, relying on grey infrastructure solutions that require continuous repairs to flood embankments and irrigation infrastructure. These business-as-usual approaches fail to consider the critical role of healthy freshwater ecosystems in sustaining community livelihoods and providing natural buffers for these communities during flood and drought events.

245. Communities in Pakistan have suffered a series of external shocks that affected their living standards, including the damages inflicted by catastrophic floods in 2010 and now in 2022. Pakistan's high dependence on the agricultural sector, including smallholder farming and cattle raising to guarantee incomes for households, has left these communities highly vulnerable to these impacts. For example, in Sindh province alone, one of the four target provinces of the project, agriculture accounts for approximately 22% of the GDP and employs 44% of the labour force; almost 80% of the population relies on agriculture for their incomes through secondary and service markets¹²¹. Exacerbated by the COVID pandemic, almost 44% of the population has been identified as economically vulnerable with 24% living under the poverty line, leaving communities in the Indus Basin with very little income security or financial reserves to withstand the economic disruptions of flooding and drought brought on by climate change¹²².

Accelerating Climate Innovation

246. The project creates a paradigm shift in Pakistan's flood and water resources management efforts by establishing proofs of concept for EbA and green infrastructure interventions in the Indus Basin that validate the cost efficiency, benefits, and sustainability of these adaptation and resilience measures. As the largest investment in EbA and green infrastructure in the freshwater sector in Pakistan and one of the largest for the GCF, the project will invest US\$64 million on the ground to reduce the impacts of increasingly severe floods and droughts on vulnerable communities and ecosystems. The project's proposed EbA and Green Infrastructure interventions will reduce flood extent by 50,833 ha, capture an additional 20 million cubic meters of water, and increase soil infiltration by 1.6 million litres. This will be achieved by: i) restoring 14,215 ha of degraded forests in watersheds in DI Khan to reduce flooding intensity and restore water regulating functions such as the absorption of rainfall and maintaining soil health (Activity 1.1.1); and ii) designing and building 127 green infrastructure interventions such as flood embankments and retention ponds to reduce flood damages to community assets including households, crops, and infrastructure, as well as increase water security through additional water retention and access for households, farmers, and herders across 12 communities (Activity 1.2.1). Together these interventions will increase the resilience of the most vulnerable people and communities in Pakistan (GCF ARA1) and restore and enhance ecosystems and ecosystem services in the Indus Basin on which communities depend (GCF ARA 4).

247. Given the devastation caused by the August 2022 floods, GoP co-financing has been reallocated to relief efforts. While precise timing for restitution of GoP co-financing, or a substitute co-financier, will be reassessed in six months, the Government of Pakistan has maintained and reconfirmed its commitment to shift a significant portion of their public water management budget into green infrastructure and EbA solutions through its multi-agency GCF board.

Enabling Environment for Climate Action

248. The integration of EbA and green infrastructure interventions to address flooding and drought in the Indus Basin and across Pakistan represents a significant innovation in the country's climate change response strategy. The paradigm shift for an enabling environment for climate action is comprised of four elements that address key barriers identified in Section B.2 (a):
- Evidence for Decision Making:** The project will generate new evidence for the climate change adaptation benefits of these interventions, including setting baselines and indicators to measure the benefits of EbA in attenuating floods, conducting economic valuations of EbA and green infrastructure interventions to determine their cost effectiveness against historical loss and damages from past floods, and assessing the viability of additional EbA and green infrastructure interventions beyond the GCF project in the Haleji and Hadero Watershed, Nara-Deh Akro Watershed, and the Indus Dolphin Reserve in Sindh Province, the Isa Khel Watershed, in Khyber Pakhtunkhwa, and Lala Creek, in Punjab Province (Activity 2.1.1).
 - National and Provincial Policy and Planning Reform:** Using the evidence generated under Activity 2.1.1, the project will develop updated procedures for the integration and execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. The project will create a National Working Group chaired by the MoCC and comprised of regulators within the MoWR, and the relevant Provincial

¹²¹ Agriculture Policy Khyber Pakhtunkhwa. A Ten-Year Perspective (2015-2025). Available from <http://extwprlegs1.fao.org/docs/pdf/pak173417.pdf>

¹²² Impact of COVID-19 on Socioeconomic Situation of Pakistan. Available from https://finance.gov.pk/survey/chapters_21/Annex%20IV%20Covid.pdf

Departments to lead the development and adoption of updated procedures. These updated procedures will be co-created with the MoCC, MoWR, the Ministry of Planning, Development and Special Initiatives (MoPSI), and the Provincial Departments of Irrigation, Forestry, and Agriculture, as well as academic and research institutions including the Pakistan Council for Research in Water Resources (PCRWR) and the Pakistan Agriculture Research Council (PARC) (Activity 2.1.2). Together, these government agencies will be responsible for drafting, validating, and ultimately approving the updated procedures for integration into three of Pakistan's major climate and water management policies – the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces (Activity 2.2.1) – and used in future flood and water resources planning and management.

- c. **National and Provincial Capacity:** Following the passage of updated procedures for the Implementation Framework of the National Water Policy, the NAP, and the Provincial Adaptation Plans, project planners and technical staff within the MoCC, MoWR, and Provincial Departments of Irrigation, Agriculture, and Forestry will be trained by project team members on the requirements of the new regulatory procedures and how to apply these procedures in the design and construction of EbA and green infrastructure interventions in future country-driven projects (Activity 2.3.1).
- d. **Community Capacity:** As community ownership is essential for long-term sustainability of the project's green infrastructure and EbA investments, operational agreements will be established between 8 CBOs and the Provincial Departments of Forestry for the provinces of Sindh, Khyber Pakhtunkhwa and Balochistan. These agreements will codify at the local level the updated procedures for the Implementation Framework of the National Water Policy, the NAP, and the four Provincial Adaptation Plans, as well as assign specific operation, maintenance, and stewardship roles among communities and Provincial Departments of Forestry.

Potential for Scaling Up and Replication

249. The project approach is strongly aligned with the GCF's goal of funding initiatives that catalyse climate impact beyond a one-off investment. It will serve as the key proof point in Pakistan that EbA and green infrastructure interventions are cost-effective and impactful adaptation options. The GoP has already committed to expanding EbA and green infrastructure beyond the GCF project's 4 sites to 5 additional sites in Haleji, Hadero and Nara-Deh Akro Watersheds in Sindh Province, Indus Dolphin Reserve in Sindh Province, Isa Khel Watershed in Khyber Pakhtunkhwa, and Lala Creek in Punjab Province.

Potential for Knowledge Sharing and Lessons Learned

250. The project will generate an economic valuation on the benefits of EbA and green infrastructure, providing new evidence and lessons learned on the value of these interventions in response to floods and droughts (Activity 2.1.1). Using this evidence, the project will create updated procedures for ecosystem management and flood management for regulators and planners (Activity 2.2.1). The evidence and knowledge products will be used in Pakistan to improve flood and water resources management and also be shared through the WWF International Network to climate change and freshwater practitioners to be replicated in other flood and drought prone areas. A WWF Freshwater Adaptation Specialist will be financed under the project to facilitate this evidence and knowledge sharing through their existing networks within the scientific community. The project will also enable knowledge sharing within the target provinces by providing training to provincial governments and farmers on climate-smart agriculture practices (Activity 3.2.1) as well as to CBOs (Activities 1.3.1 and 1.3.2).

D.3. Sustainable development (max. 500 words, approximately 1 page)

251. The project's interventions will contribute to the achievement of a number of the Sustainable Development Goals (SDGs). Improving the resilience of ecosystems and communities currently vulnerable to increased instances of drought and flooding brought on by climate change (SDG 13 Climate Action), directly helping to enhance the resilience of 687,336 people, will also achieve co-benefits in the areas of GHG reduction, biodiversity conservation, poverty reduction, food insecurity attenuation, enhanced health and well-being, and gender equality promotion. Co-benefits are presented below and linked to relevant SDGs contributed to by the project. More detail on co-benefits related to EbA interventions is presented in Table 16 below and in Annex 2: Feasibility Study.

- **SDG 1 - No poverty.** The project will help alleviate poverty for 50,750 farmers vulnerable to income loss from flooding and drought by training in improved irrigation practices, the use of alternate crop and seed varieties better suited to withstand water insecurity, and improved soil and farm management practices. These new skills along with the provision of equipment and materials will allow farmers to better adapt their farming and livelihood practices to future climate conditions. The project will improve farming practices across 128 ha of farmland on which communities are directly dependent for income generation. This will help avoid income losses from flooding damaging to crops and agricultural equipment and improve water efficiency on farms reducing crop

failure and subsequent income loss for households (economic co-benefit). The impacts of climate change on poverty levels in Pakistan are evident when looking at the economic outcomes of major flooding events in 2010 and now in 2022 on Pakistan's household and overall economy. Early estimates of the ongoing damages from the floods in Pakistan are expected to surpass US\$10 billion¹²³. The project will help reduce the impacts of largescale floods and drought in Pakistan such as those in 2010 and 2022 helping to reduce economic losses.

- SDG 2 - Zero hunger. The poverty reduction benefits of the project outlined above are also expected to help reduce food insecurity in the Indus Basin. Close to 90% of agricultural production is dependent on irrigation due to Pakistan's arid climate and variable rainfall leaving food production and household access in the Indus highly vulnerable to flooding events which can disrupt irrigation networks. The activities outlined above will help reduce water dependency and enhance the resilience of agricultural systems facing increased instances of flooding and drought. The restoration of ecosystems' provisioning of services such as habitat for fisheries, freshwater availability for food cultivation and livestock raising, along with the improvement of agricultural resilience will together help enhance food security for communities in the Indus and Pakistan more broadly (social co-benefit).
- SDG 5 - Gender equality. The challenges of climate change impacts, building resilience and gender inequalities are inextricably linked. By exacerbating inequality, climate change slows progress toward gender equality and thus impedes efforts to achieve wider goals like poverty reduction and sustainable development. Conversely, gender inequality can worsen the impacts of climate change and further propel the 'feminization' of violence, poverty, marginalization and exclusion. Therefore, taking steps to narrow the gender gap and empower women can help counter the impacts of climate change. Rural women in Pakistan are considered more vulnerable to climate change as they have more limited access than men to resources for building resilience, including income, land, loans and access to off-farm employment. Women also have fewer opportunities than men to obtain trainings and knowledge. They are often excluded from local decision-making processes, in conservative areas. Yet they are responsible for ensuring their households' energy, water and subsistence farming needs are met. The Project will implement a Gender Action Plan (Annex 8) tied to the activities of the project designed to help bridge development and decision-making gaps for women by providing training to women in improved farm management and climate smart agricultural methods, providing financing and technical assistance to women owned and/or operated enterprises, and empowering women in community decision making within CBOs. The project will expand women's social and economic freedoms by focusing on their role as change agents, decision-makers and experts as well as increasing their access to resources required to meet their practical needs. It is expected that their participation in the project will serve to empower them, change societal perceptions and lead to their greater participation in other spheres of public and professional life (gender co-benefits).
- SDG 14 - Life below water and SDG 15 - Life on land. The project will restore approximately 14,215 ha of forests and place this 14,215 ha and an additional 4,000 hectares of freshwater and riverine ecosystems under improved community management. These interventions will provide additional suitable habitat and increasing landscape and freshwater ecosystem connectivity (through the creation of corridors) for vulnerable species such as the Indus River dolphin. Only about 1,100 of this unique species exist today in the lower parts of the Indus River in Pakistan. Improved water availability will also support healthy riverine forests and wetlands, and associated ecosystem services such as fish stocks, clean water, and fodder sources (environmental and social co-benefits). The restoration and increased water retention capacity of targeted wetland sites, expanding and rehabilitating riverine forest areas, and improving watershed management will also capture an estimated 26,450 M-Tons of GHG emissions annually over the project lifespan. These benefits are categorized as mitigation co-benefits under the project.

Table 17. Co-benefits associated with selected project adaptation intervention types.

Adaptation intervention type	Co-benefits		
	Environmental	Social	Economic
Restoration of riverine catchment vegetation	• Revegetation provides long-term stabilization of riverbanks, prevents erosion, and enhances protection of the bank toe from continual scour while reducing fine sediment input.	• Improved drinking water quality and supply.	• Increased lifespan and capacity of dams.

¹²³ <https://www.reuters.com/world/asia-pacific/initial-economic-losses-pakistan-floods-least-10-bln-planning-minister-2022-08-29/>

	- Improved forest cover. - Improved carbon sequestration. - Enhanced habitats for local wildlife species.		
Restoration of flow paths	- Improved deposition processes on the floodplain and the reduced movement of gravel. - Improved nutrient and pollution retention in upstream areas.	Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems.	Increase in irrigated land and hence food production.
De-siltation of wetlands	Improved health of native wetland vegetation.	Improved drinking water quality and quantity.	NA
Enhancing and reinforcing embankments	Enhanced habitats for local wildlife species.	- Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems. - Improved drinking water quality and quantity.	Lower infrastructure maintenance costs and strengthened local economies through diversification of livelihoods based on natural resources.
Flood protection embankments	- Improved hydrological connectivity across flood plains. - Improved carbon sequestration. - Erosion control and nonpoint source pollution prevention. - Increase of vegetation and biomass with extended growing seasons and enhanced soil moisture. - Sediment retention.	Improved drinking water quality and quantity.	NA
Improved efficiency of irrigation systems	Improved soil health and water content.	Reduced conflict over water resources.	- Cost savings as a result of reduced water and labour inputs. - Intensification of agriculture. - Income diversification and market access for smallholder farmers.
Drought-resistant crop varieties	Reduced degradation of natural areas through expansion of croplands.	NA	Improved food security.
Water harvesting	NA	NA	- Agricultural expansion and diversification of crops. - Additional income

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Pakistan's Vulnerability

252. Pakistan is a middle-income developing country however; it is also among the most vulnerable countries to extreme weather events due to the high recurrence and magnitude of disasters associated with changing climate conditions and is ranked 18 out of 191 countries by the 2019 [Inform Risk Index](#). Pakistan is highly exposed to flooding and drought and is ranked 8th most affected country in the world on GermanWatch's Long-Term Climate Risk Index (2000–2019).

253. As this proposal is being developed, Pakistan is experiencing unprecedented floods killing over 1,300 peoples and displacing over 33 million. The latest reports show that over 1,600 have been injured, 325,000 homes have been destroyed, 735,000 heads of livestock have been lost, and 2 million acres of crops damaged. As the floods are still on-going, these figures are expected to increase. Pakistan's Planning Ministry estimates to total cost of the

damages to exceed \$10 billion equivalent to 4% of the country's GDP¹²⁴. The floods have destroyed huge swaths of inundated farmland which will be unusable for some time. Pakistan's economy is heavily dependent on its agricultural sector both for national food production and as part of its export revenue, these most recent floods will cause significant crop revenue losses and risk for severe food shortages.

254. The floods in 2022 are Pakistan's most recent experience with climate disaster but has experienced significant events in the past decade only exacerbating its compounding vulnerability. During the catastrophic 2010 Indus flood, which affected more than 20 million people, approximately 2,000 lives were lost, 3,000 people were injured, 1.6 million houses and millions of acres of crops were also damaged, with financial damages estimated at \$40 billion. Since then, Pakistan has experienced large floods for six consecutive years¹²⁵.

Economic and financial constraints for climate change adaptation in Pakistan

255. Given Pakistan's significant vulnerability to climate change, there is urgent need of adaptation finance to implement projects which can help reduce Pakistan's vulnerability to extreme events. The country's climate adaptation needs range between US\$7 billion and US\$14 billion a year, according to an ADB report of 2017¹²⁶ which will require contributions from international public financing sources, such as the GCF. Furthermore, there is a key gap in financing for Ecosystem-based solutions (EbS) to climate change.

256. As described above, Pakistan already suffers immense losses from climate change impacts and evidence suggests that such losses will continue to have a significant and increasing effect on the GDP if no adaptation action is taken. Between 1995 and 2014, climate change cost Pakistan US\$ 3.9 billion (PPP) in average economic losses annually. Addressing these effects requires a swift increase in investment in climate change adaptation. A comparison of the global cost estimates with the current level of adaptation funding indicates that projected global adaptation needs are to be significantly greater than current investment levels, particularly in vulnerable developing countries like Pakistan.

257. Pakistan remains a resource-constrained economy, dependent on foreign financing in the shape of loans, foreign investment, and grant aid for, amongst others, infrastructure projects and assistance during post disaster situations. There is significant investment required to meet the requested budgets for National Flood Protection Plans (NFPPs) ~US\$ 431 million dependent on exchange rate fluctuations and due to insufficient government funds, foreign financing is relied on significantly for tackling the impacts of floods and droughts, and/or investing in flood and drought management initiatives.

258. National adaptation budgets are also constrained by the urgent need for investments in other development initiatives. A recent example of such a need is related to rising food and energy which is diminishing the real purchasing power of households, disproportionately affecting poor and vulnerable households that spend a larger share of their budget in these areas. In response, the Government introduced a targeted food subsidy program (Ehsaas Rashan Riyaat) in February 2022¹²⁷. Despite the country's financial constraints, estimated federal climate-related expenditure was between 5.8 and 7.6% of the total expenditures in the federal budget¹²⁸. This expenditure is expected to increase further, primarily due to the increase in frequency and intensity of climate induced disasters. The budget, however, will still fall short of that required to finance Pakistan's adaptation needs. GCF support in the form of grant financing is consequently being requested to support climate-resilient sustainable development in Pakistan.

259. While ~US\$67 million in government co-finance from the NDRMF and MoWR was committed to the project, the recent floods have required that these funds be repurposed for immediate disaster relief. Pakistan is one of the first countries in South Asia to create a Ministry of Climate Change and has had a National Climate Change Policy since 2012, demonstrating the country's commitment to counter the adversities of climate change. Despite these commitments, Pakistan's economic development challenges, large external public debt and subsequent exchange rate vulnerability, and compounding national emergencies brought on by climate change such as the floods of 2010 and now in 2022 make it very difficult for the government of Pakistan to commit significant adaptation finance at this time¹²⁹.

¹²⁴ <https://www.bbc.com/news/world-asia-62704004>

¹²⁵ Government of Pakistan, 2017.

¹²⁶ <https://pide.org.pk/research/pakistans-options-for-climate-finance/>

¹²⁷ Source: <https://www.worldbank.org/en/country/pakistan/overview#1>

¹²⁸ UNDP, 2015.

¹²⁹ <file:///C:/Users/jaudel/Downloads/Advocacy%20Brief%20Financing%20Climate%20Action%20in%20Pakistan.pdf>

260. Pakistan's large geographical area makes it the 34th largest country in the world. Its population is over 207 million with an average population density of 267 km², making it the fifth most populous country¹³⁰. Its GDP is ~US\$304 billion and GDP per capita is ~US\$1,629. Long-standing structural weaknesses of the economy and low productivity growth continue to pose risks to a sustained recovery. Strong aggregate demand pressures, in part due to previously fiscal and monetary policies, paired with the continued less conducive external environment for exports have contributed to a record-high trade deficit, putting pressure on the Rupee and the country's limited external buffers. The current account deficit (CAD) in early 2022 widened to US\$9 billion, from a surplus of US\$1.2 billion 2021, as imports values surged by 54%, doubling the 27% growth in export values.

261. Pakistan's Human Development Index (HDI) score is 0.550, resulting in its placement of 147 out of 188 countries in terms of level of human development. With the economic recovery and improved labour market conditions, poverty measured at the lower middle-income class poverty line (US\$3.20 Purchasing Power Parity (PPP) per day in 2011)¹³¹ is estimated to have declined from 37% in 2020 to 34% in 2021. According to the country's new poverty index, 38% of population lives with multi-dimensional poverty, with the highest rates in KP and Balochistan. Most of these people living in poverty reside in rural areas (54.6% rural poverty) and are disproportionately impacted by climate change¹³². Furthermore, Pakistan scores 0.546 on the Gender Inequality Index (GII), with inequality between female and male achievements in the following areas: reproductive health, empowerment, and economic activity. Pakistan's value of 0.742 for its Gender Development Index (GDI) is the highest global score in gender inequality.

262. The proposed project's contribution to supporting equitable wealth distribution includes targeting rural agriculturalists as direct and indirect beneficiaries including indigenous communities, women, and other marginalized groups. Rural agricultural livelihoods are reliant on ecosystem services, such as water provision, resulting in large portions of society being indirectly affected by the impacts of climate change on ecosystems, in addition to direct climate impacts. Agricultural and subsistence-based livelihoods in poor and marginalized communities with often limited access to modern farming technology and basic infrastructure and services are highly dependent on ecosystem services. The multi-dimensional poverty and poverty status in the project's targeted Provinces are presented in Table 18.

Table 18. Multi-dimensional poverty and poverty status of the project's target sites.

Province	Site	District	Population (2017)	Multi-dimensional poverty	Poverty status
Khyber Pakhtunkhwa	Dera Ismail Khan	Tank	391,885	0.385	Very poor
	Dera Ismail Khan & Darya Khan – Ramak	D.I. Khan	1,627,132	0.362	Very poor
Sindh	Manchar Lake Catchment	Dadu	1,550,266	0.247	Very poor
Balochistan	Chakar Lehri Sub Basin	Jhal Magsi	149,225	0.528	Extremely poor

263. Pakistan is one of the most water-stressed countries of the world, with an average per capita water availability of less than 1,000 m³ per year¹³³. Water is crucial for the country's predominantly agriculture-based economy, which accounts for 19% of GDP, 42% of jobs¹³⁴, and 91% of the annual freshwater usage¹³⁵, making it extremely vulnerable to an increasing risk of drought. Many areas of Pakistan experienced temperatures 5–8°C higher than averages, posing a risk to water supplies and agriculture, as well as human and animal health. During the country's 2015 heatwave, over 65,000 people were hospitalized with heatstroke.

Local-level (community and livelihood) vulnerability

264. Agriculture employs ~42% of the population and contributes 21% of GDP. Farmers engage in a combination of activities encompassing crop production, livestock management and non-timber forestry products, all of which are climate sensitive. As 90% of agriculture depends on irrigation from the Indus River and its tributaries, the agriculture sector is highly vulnerable to changes in the availability (too little and too much) and timing of water resources. For

¹³⁰ PBS (2019) Labour Force Statistics 2017-18 [online] accessed at <https://www.pbs.gov.pk/content/labour-force-statistics>

¹³¹ More information on Global Poverty Lines is available here: <https://www.worldbank.org/en/news/factsheet/2022/05/02/fact-sheet-an-adjustment-to-global-poverty-lines>

¹³² UNDP, 2016.

¹³³ National Water Policy, 2018.

¹³⁴ Government of Pakistan, 2017.

¹³⁵ UNDP, 2016.

example, in 2010–2014, five flood events destroyed 10.6 million acres of crops, negatively affecting farmers' incomes and land fertility. Although Pakistan has the most extensive continuous irrigation network in the world and a high level of flows (outside the drought regions), the country is still considered water stressed. This is mainly due to high demand for irrigation because of increasing evaporation rates and poor water storage capacity. Furthermore, over one-quarter of cultivable land is already degraded because of water logging, salinization, flooding and erosion. Changes in rainfall patterns and seasonal shifts in weather have resulted in climate irregularities that are making agricultural livelihoods challenging and unpredictable, thus adversely impacting farmers. All these factors compound the climate change vulnerability of agricultural communities¹³⁶.

265. Women, children, people with disabilities and smallholder farmers are among those that are the most at risk to the impacts of climate change in Pakistan. Climatic shocks particularly affect women's livelihoods. Whilst women take care of farming work when men migrate for better employment opportunities, they very seldom own or have decision-making rights over the productive assets. Only 44% of women reported being able to make decisions about major household purchases, thereby limiting their ability to respond to crisis situations. Women rarely own property (either houses or land), which limits their ability to rely on their asset base in times of emergencies or receive Government compensation if there are damages. Additionally, women also have limited access to information compared to men¹³⁷.

266. The impacts of climate change on food productivity pose a very serious challenge in a country where 60% of the population is food insecure (especially in Khyber Pakhtunkhwa) and almost half of the women and children are malnourished. Rising average temperatures are projected to reduce cereal production by up to 20% in the warmer southern parts of Pakistan, though there will be minor improvements in yield in the colder northern areas. Punjab and Sindh provinces — which together account for over 90% of agricultural produce — are the most vulnerable to climate change impacts. The production of income-generating cash crops like cotton, maize, rice, sugarcane and wheat is expected to decline due to climate change. Livestock production is also projected to decline by ~30%¹³⁸.

267. Livestock rearing constitutes 53% of Pakistan's agricultural GDP. It is dependent on climate-sensitive grasslands and rangelands, which are increasingly under pressure from longer and more frequent droughts, flash floods and a rise in temperature. Livestock rearing is an important agriculture-allied activity for 30–35 million rural farmers who earn about 40% of their income through livestock. There are regional differences with the highest proportion (87%) of people rearing livestock being in the arid province of Balochistan. The direct effects of climate change on livestock include heat stress resulting in lower milk and meat production and reduced reproduction. Rising temperatures and water-related disasters increase the risk of zoonotic epidemics as well as the loss of livestock. Indirect climate-related pressures such as lower fodder production, soil erosion and land degradation will also adversely affect the livestock sector¹³⁹.

268. Vulnerable communities in the Indus Basin depend on the water supply and recharge, food and products, flood protection, sediment control, carbon sequestration, climate regulation, and other benefits provided by ecosystems for their survival. For example, ecosystem services from Keenjhar Lake, in the lower Indus Basin, have an estimated total economic value of over USD 11,500/ha¹⁴⁰. Degradation of wetlands, forests, and pastures due to climate change is causing habitat and land productivity loss and decreasing grazing areas and fodder¹⁴¹. These impacts are compounded by the existing constraints on the Indus Basin such as irrigation infrastructure and other stresses (e.g., pollution of ecosystems). The combined impacts considerably increase the vulnerability of communities and their livelihoods.

Need for strengthening institutions and implementation capacity

269. Historically, development initiatives in Pakistan's water sector have lacked the integration of adaptation, leaving the sector vulnerable to the increasingly severe impacts of climate change. Examples of this approach include the construction of two major dams (Tarbela and Mangla) and linking canals for storage and irrigation via the Indus Waters Treaty (1960) using financing from the World Bank. At the time, the adaptation-related needs were not well understood and hence not considered in such initiatives, leaving them vulnerable to climate change impacts such

¹³⁶ ADB, 2017.

¹³⁷ IFRC, 2021. Climate Change Impacts on Health and Livelihoods: Pakistan Assessment. Available at: https://www.climatecentre.org/wp-content/uploads/RCRC_IFRC-Country-assessments-PAKISTAN-3.pdf

¹³⁸ IFRC, 2021. Climate Change Impacts on Health and Livelihoods: Pakistan Assessment. Available at: https://www.climatecentre.org/wp-content/uploads/RCRC_IFRC-Country-assessments-PAKISTAN-3.pdf

¹³⁹ IFRC, 2021. Climate Change Impacts on Health and Livelihoods: Pakistan Assessment. Available at: https://www.climatecentre.org/wp-content/uploads/RCRC_IFRC-Country-assessments-PAKISTAN-3.pdf

¹⁴⁰ Dehlavi & Nawaz, 2012.

¹⁴¹ ADB, 2017.

as increasing evaporation and siltation. While access to information and knowledge on climate change and adaptation responses (such as EbA) have since improved in Pakistan, the water sector's regulatory and planning framework does not yet reflect these important learning experiences. As a result, the sector remains vulnerable to climate change impacts. There is also a disconnect between plans, regulations and mandates for water management at the federal, provincial and district levels. The roles and responsibilities of relevant institutions such as the Departments of Water, Irrigation and Agriculture are not clearly defined, coupled with a non-existent knowledge transfer mechanism that contributes to limited coherence between sectoral plans and policies.

270. The situation is further exacerbated by severely limited public sector budgets for adaptation; only 5–7% of the GoP's budget is designated for climate change adaptation, the majority of which is used for costly and high-maintenance hard infrastructure¹⁴². This is further compounded by limited technical expertise in Pakistan for planning and implementing adaptation, especially in the areas of EbA and green infrastructure interventions. Consequently, alternatives to hard infrastructure for adaptation (that also considers future climate risks) remain poorly understood and underutilized, with an ad-hoc, reactive and uncoordinated approach to floodwater and flood risk management, leaving communities and vulnerable ecosystems at risk. The proposed project will address the institutional and implementation needs described above through mainstreaming EbA and green infrastructure as key approaches to integrated flood and water resources management in Pakistan. This will include: i) establishing an evidence base of EbA and green infrastructure interventions' effectiveness and efficiency at reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan to expand Pakistan's regulations and policies for flood and water resources management from only focusing on using grey infrastructure to address the threats of floods and droughts to prioritising EbA and green infrastructure interventions; and ii) strengthening technical and institutional capacity for scaling up and replicating EbA and green infrastructure interventions at the national and sub-national levels.

D.5. Country ownership (max. 500 words, approximately 1 page)

271. Improving resilience to climate change hazards, particularly floods and droughts, is among the top priorities of GoP. Following a Ramsar recommended exposure visit (Ramsar Advisory Mission (RAM)) to China's Yangtze Basin project (which integrates ecological solutions with 'hard' infrastructure to optimize water resource management), the GoP increasingly recognizes that "a mono-disciplinary approach based on engineering solutions alone — as in the case of the Indus Basin — cannot fully handle hydrological cycles, ecosystems, and the security of people prone to flood risks"¹⁴³. RAM's recommendations and the subsequent exchanges with flood management experts in the Yangtze Basin successfully convinced the GoP of the advantages of EbA for flood management in the Indus Basin. Specifically, a 'Diversified Portfolio Approach' is needed, combining: i) EbA interventions (e.g., restoration, reconnection and/or rehabilitation of watersheds, floodplains and wetlands¹⁴⁴); ii) green infrastructure interventions (e.g., retention areas and recharge basins); and iii) climate-resilient land and water resources management strategies (e.g., expanding flood and water resources regulations and policies to prioritise EbA and green infrastructure interventions).

272. The proposed project has been developed at the request of the GoP specifically to act on these recommendations. A participatory approach has been taken for the design of the proposed project, including close consultations with Pakistan's main water resources, flood management, and climate change adaptation stakeholders, particularly the MoWR, MoCC, and FFC, respectively. In particular, the AE and project partners have worked closely with the NDA (MoCC), who has contributed considerably to driving the project development process at a national level, as well as providing timely inputs into the development of the proposal. Twenty letters of support from federal ministries and provincial departments¹⁴⁵ — including the Letter of No Objection (Annex 1) from the GCF NDA, MoCC — have been received in support of the project, which emphasises the country's ownership and commitment. While not presented in Pakistan's GCF Country Programme (2017), the proposed project is aligned with the key climate needs and agenda presented therein, further reinforcing the ownership of the proposal by the country as presented in this section. Moreover, the GoP has highlighted the proposed project's important role of

¹⁴² Pakistan - Climate Public Expenditure and Institutional Review, UNDP, 2015.

¹⁴³ Ali, 2013.

¹⁴⁴ Opperman, et al., 2017.

¹⁴⁵ Government of Pakistan: Ministry of Water Resources, Pakistan Council of Research in Water Resources, Federal Flood Commission; Government of Balochistan: Environment Department, Provincial Disaster Management Authority (PDMA) Irrigation Department, Planning and Development Department; Forest and Wildlife Department; Government of Khyber Pakhtunkhwa: Forest Environment and Wildlife Department, Irrigation Department, Provincial Disaster Management Authority; Government of Sindh: Forest and Wildlife Department, Irrigation Department, Provincial Disaster Management Authority, Environmental Protection Agency; Government of Punjab: South Punjab Forest Company, Environment Protection Agency, Forestry, Wildlife and Fisheries Department, Irrigation Department, Provincial Disaster Management Authority.

contributing to national adaptation objectives by including it as a priority initiative under the country's updated NDCs (2021).

273. The project has been presented and approved at the highest levels of GoP, including the Standing Committees of Senate and National Assembly in 2017 and to the Advisor to the Prime Minister on Climate Change at the Ministry of Climate Change in 2018. Both presentations led to strong agreements by officials that EbA integration was necessary to manage the increasingly severe threat of floods in the Indus Basin, and harvest surplus floodwater to support water security. The Ministry of Climate Change further presented this project to the former Prime Minister of Pakistan in September 2018 — during this meeting this project was approved and prioritized for immediate funding under the “Recharge Pakistan” initiative (of which the proposed project is a major part). The Office of the Advisor to the Prime Minister on Climate Change subsequently directed relevant GoP authorities to provide necessary support for development, co-financing, and implementation of the proposed project. The Advisor to the Prime Minister on Climate Change has promoted the project as a GoP priority at various national and international environment and climate change platforms, including the UNFCCC's COPs — the proposed project (Recharge Pakistan) is one of the five initiatives on Pakistan's five-point Green Agenda.

Coherence with national climate strategy and existing policies

274. The proposed project will contribute towards achieving priorities under relevant GoP policies, including the [National Climate Change Policy](#) and updated [Pakistan's Nationally Determined Contributions](#) (NDCs). The project is a key adaptation priority within the updated NDCs and will contribute towards the short (2020–2025), medium (2030) objectives of the NDCs by: “enhancing integrated water resource management” and “harvesting rainwater and floodwater”; strengthening sub-national adaptation planning, building capacity, and investing in climate-resilient infrastructure; contributing to “additional water reservoir capacity on the river system to regulate water discharges during high floods”; and “promoting local rainwater harvesting and development of small storages on run of the rivers during peak flows”. Under the revised NDCs, Recharge Pakistan (the proposed project) is a priority initiative for which the GoP has already allocated 6 billion rupees (~US\$29.5 million). The importance of creating an enabling environment for ecological solutions to be undertaken, i.e., having relevant policies and legislation in place, a coordinated basin level approach by establishing a single coordinating body (e.g., Wetlands Management Authority under [National Forest Policy](#), and Water Council under the [National Water Policy](#)), and enhancing institutional capacity (prioritized under the [National Flood Protection Plan](#) — NFPP IV), was also stressed in the country's NDCs and is incorporated in the proposed project. In the near term it will: contribute to sub-national adaptation planning by building the capacity of relevant line departments and the development of adaptation strategies, such as the ‘National Floodplain Management Policy, and Hill Torrent Management Policy’ (as described in the NFPP IV), and the ‘Indus River Basin Management Plan’; and ensure an enabling environment for floodplain management at the national level. At the international level, the project will also contribute towards all four priorities under the [Sendai Framework](#) for Disaster Risk Reduction (2015–2030): i) understanding disaster risk; ii) strengthening disaster risk governance to manage disaster risk; iii) investing in disaster risk reduction for resilience; and iv) enhancing disaster preparedness for effective response and to “Build Back Better” in recovery, rehabilitation and reconstruction.

275. While remaining coherent with Pakistan's climate change, natural resource and disaster risk reduction frameworks, the project will also contribute to policy and regulatory change. This will be achieved through: i) establishing a National Working Group to lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan; and ii) strengthening institutional and technical capacity at the national and sub-national levels to apply the updated procedures, as well as implement, scale up and replicate EbA and green infrastructure interventions across Pakistan. Coherence with other existing policies is presented in Table 19 below.

Table 19. Coherence of the proposed project with existing policies.

Policy, Plan or Strategy	Priorities contributed to by the proposed project
The National Climate Change Policy (2012) provides a framework for addressing the issues that Pakistan faces or will face in the future due to the changing climate. The goal of the policy is to ensure that climate change is mainstreamed in the economically and socially vulnerable sectors of the economy and steer Pakistan towards climate resilient development.	Relevant objectives: 1. To pursue sustained economic growth by appropriately addressing the challenges of climate change — <i>Entire project</i> 2. To integrate climate policy with other inter-related national policies — <i>Outputs 2.1 and 2.2</i> 3. To focus on pro-poor gender sensitive adaptation will also promoting mitigation to the extent possible in a cost-effective manner — <i>Outputs 1.1, 1.2, 1.3, 3.1 and 3.2</i> 4. To ensure water, food and energy security of the country in the face of challenges posed by climate change — <i>Outputs 1.1, 1.2, 1.3 and 3.2</i>

	5. To minimize the risks arising from the expected increase in frequency and intensity of extreme weather events such as floods, droughts and tropical storms — <i>Outputs 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.2</i> 6. To strengthen inter-ministerial decision making and coordination mechanisms on climate change — <i>Outputs 2.1, 2.2 and 2.3</i> 7. To facilitate effective use of the opportunities, particularly financial, available both nationally and internationally — <i>Output 3.1</i> 8. To foster the development of appropriate economic incentives to encourage public and private sector investment in adaptation measures — <i>Outputs 2.1, 2.2 and 3.1</i> 9. To enhance awareness, skill and institutional capacity of relevant stakeholders — <i>Outputs 1.3, 2.1, 2.2, and 2.3</i> 10. To promote conservation of natural resources and long-term sustainability — <i>Entire project</i>
The aim of the National Forest Policy (2015) is to expand the national coverage of forests, protected areas, natural habitats and green areas for restoration of ecological functions and maximizing economic benefits while meeting Pakistan's obligations to international agreements related to forests. In view of Pakistan's high vulnerability to adverse impacts of climate change, in particular to extreme events, mitigation and adaptation measures are the focus of the Policy.	<u>Relevant objectives:</u> 1. Enhancing public awareness on economic, social, ecological and cultural values of forests — <i>Output 3.1</i> 2. Implementing a national level mass afforestation programme to expand and maintain forest coverage to meet international standards — <i>Output 1.1</i> 6. Facilitating implementation of international conventions and agreements related to forestry, biodiversity and climate change — <i>Entire project</i> 7. Promoting standardized and harmonized scientific planning of forests, research and education — <i>Outputs 1.3, 2.1, 2.2, and 2.3</i>
The National Water Policy provides the guiding principles and broad policy direction for the water sector in Pakistan and deals with some related issues such as climate change threats. The Policy seeks to respond to Pakistan's water crisis. The objective of the policy is to take cognizance of the emerging water crisis and provide an overall policy framework and guidelines for a comprehensive plan of action. The Policy is a national framework within which the provinces can develop their master plans for sustainable development and management of water resources.	<u>Relevant objectives:</u> 2.1 Promoting sustainable consumption and production patterns throughout the water sector from exploitation to utilization — <i>Outputs 1.3 and 3.2</i> 2.7 Providing food security and expanding water availability to help adapt to climate change, population and other large-scale stresses — <i>Outputs 1.1, 1.2 and 3.2</i> 2.10 Improving watershed management through extensive soil conservation, catchment area treatment, preservation of forests and increasing forest cover — <i>Outputs 1.1, 1.2, 1.3, and 3.2</i> 2.11 Restoring and maintaining the health of the environment and water related ecosystems — <i>Outputs 1.1, 1.2, 1.3, and 3.2</i> 2.12 Flood management to mitigate floods and minimize their damages — <i>Outputs 1.1, 1.2, 1.3, and 2.2</i> 2.13 Drought management with emphasis on long term vulnerability reduction — <i>Outputs 1.1, 1.2, 1.3, and 2.2</i> 2.15 Promoting appropriate technologies for rainwater harvesting in rural as well as urban areas — <i>Output 3.2</i> 2.19 Encouraging beneficiary participation and public private partnerships — <i>Outputs 1.3, 2.3, 3.1 and 3.2</i> 2.20 Strengthening and capacity building of water sector institutions — <i>Outputs 2.2 and 2.3</i> 2.22 Protection of wetlands and Ramsar Sites — <i>Outputs 1.1, 1.2, 1.3, 2.2, 2.3 and 3.2</i>
Pakistan's National Wetlands Policy (draft) recognises the importance of Pakistan's wetlands, which include valuable ecosystem services, such as: water regulation, wetlands and climate, biodiversity importance, human health and livelihoods.	<u>Relevant objectives:</u> 1. Addressing primary threats to Pakistan's wetlands — <i>Outputs 1.1, 1.2 and 1.3</i> 3. Greater co-ordination and collaboration between agencies and sectors on wetland issues encouraged from local to international levels — <i>Output 2.2</i> 4. Promoting wetland research, education and data management — <i>Outputs 2.1, 2.2 and 2.3</i> 5. Building Pakistan's capacity for sustainable wetland management — <i>Outputs 1.1, 1.3 and 2.3</i> 6. Promoting improved understanding, perceptions and attitudes towards wetlands conservation and wise use — <i>Outputs 1.3, 2.2, and 2.3</i>
The National Disaster Risk Management Framework (2007) has been formulated to guide the work of entire system in the area of disaster risk management. The Framework envisions, "To achieve sustainable social, economic and environmental	<u>Relevant priority areas</u> Training, education and awareness: Technical skills and knowledge of district and municipal officials in hazard-prone areas enhanced on disaster risk production and preparedness — <i>Outputs 1.3, 2.2, 2.3 and 3.2</i> Promoting disaster risk management planning:

development in Pakistan through reducing risks and vulnerabilities, particularly those of the poor and marginalized groups, and by effectively responding to and recovering from disaster impact”.	• Formulation of disaster risk management plans at the federal, provincial, district and municipality levels — <i>Outputs 1.3 and 2.2</i> Community and local level risk reduction programming: • Technical capacity of community organizations, masons, schoolteachers enhanced to deal with disaster risk reduction and preparedness issues — <i>Outputs 1.3 and 2.3</i> • Awareness of vulnerable communities and local stakeholders increased about hazards, risks, vulnerabilities, risk reduction and disaster preparedness — <i>Outputs 1.3 and 3.2</i> Mainstreaming DRR into development • Technical capacity of selected federal line ministries increased on integrating risk reduction into development plans and programmes — <i>Outputs 2.2 and 2.3</i>
The National Disaster Risk Reduction Policy (2013) provides an overall guiding framework for addressing the high levels of disaster risk permeating Pakistani Society. It covers both natural and man-made hazards. The policy seeks to promote priority measures to ameliorate already existing vulnerability to hazards, and equally important measures to ensure future development processes and programs strengthen resilience. The policy serves as a guiding framework both for DRR and relevant development plans and programs to focus attention upon priority issues.	<u>Relevant principles:</u> 1. Multi-hazard approach — <i>Outputs 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.2</i> 2. Strengthening community participation and resilience — <i>Outputs 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.2</i> 3. Strengthening the resilience of vulnerable groups — <i>Outputs 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.2</i> 4. Compatibility with local customs and norms — <i>Outputs 1.3, 2.2, 2.3 and 3.2</i> 5. Clearly defined division of roles and responsibilities between different layers of government — <i>Outputs 1.3, 2.1, 2.2, and 2.3</i> 6. Promoting inter-organizational partnerships — <i>Output 2.2</i>

Stakeholder engagement

276. The proposed project was designed following an extensive consultative process with MoCC — which is National Designated Authority (NDA) for Pakistan and is responsible for the overall management of the country’s GCF portfolio — as well as all other relevant project stakeholders. These stakeholders included representatives from:
- national and sub-national governments, such as the MoWR and the FFC, National Disaster Management Authority (NDMA), and Ministry of Planning Development and Special Initiatives (MoPDSI);
 - provincial government (line) departments, such as the Provincial Irrigation, Agriculture and Forestry Departments, Forestry, Provincial Planning and Development Departments;
 - Community-based Organisations (CBOs);
 - communities and tribal leaders;
 - Academic Institutions;
 - environmental experts;
 - social activists; and
 - private-sector actors.
277. During project development, WWF-Pakistan held numerous consultations with these national- and local-level stakeholders to: i) collect baseline information; ii) collect necessary climate data to strengthen the climate rationale of the project; iii) identify and validate potential project interventions — including appropriate EbA and green infrastructure interventions — in collaboration with relevant stakeholders; iv) engage with DFCD — as the link to the private sector— to discuss how the private sector could contribute to and benefit from the project; v) engage national and local governments to determine the most efficient and effective process for integrating climate change adaptation into Pakistan’s existing regulations for flood risk and water resources management; vi) discuss project financing — including co-financing; vii) determine roles and responsibilities of the relevant implementing entities for the project; and viii) define the implementation strategy. Details of these consultations along with stakeholder lists are presented in Annex 7.
278. These consultations included focus group discussions with women at the community level for better understanding of how they are impacted by climate-related disasters (specifically floods and droughts) to ensure the project interventions directly address their adaptation needs and promote gender-responsiveness. During this process, indigenous knowledge and best practices were also captured and incorporated in the design of the project interventions. To ensure the ongoing sensitization and consultation of stakeholders during project implementation, a project-specific Stakeholder Engagement Plan (Annex 7) has been developed in accordance with GCF and WWF ESS and stakeholder consultation guidelines.

279. The ESMF & Gender consultants held consultations and focus group discussions with over 300 stakeholders across six priority sites from which four were selected for the project. During field visit, the consultants also engaged with the community representatives in all the sites. Stakeholders involved during these sessions included villagers, tribal elders, farmers, livestock farmers, fishers, CBOs, local focal personnel and women leaders. Additionally, various government officials were also contacted at the District/Tehsil¹⁴⁶ level, officers from different departments (Agriculture, Livestock, Natural Resources, Forest, Environment, Community Development). The overarching objective of these consultations was to: i) inform communities about project objectives and activities; ii) discuss and assess possible adverse impacts and collect their views to avoid or mitigate them; and iii) discuss and assess potential project benefits and how these can be enhanced. The detailed outcome of these consultations is provided in Annex 6: ESMF and Annex 8: GAAP.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

280. The GoP is requesting US\$66 million in grant finance from the GCF to enhance the resilience of the most vulnerable people and ecosystems to the impacts of flooding and drought caused by climate change. The proposed project was originally designed to have a total of ~US\$67 million in public funds in direct co-finance committed to the project. This included ~US\$37 million from the NDRMF to finance the execution of EbA measures in the Indus Basin and ~US\$29 million from the MoWR's FFC for designing and building green infrastructure interventions, as described in Section B.3. However, the recent 2022 floods in Pakistan have required that these funds be repurposed for immediate disaster relief. As a result, the GoP is unable to commit co-financing to this project at this time¹⁴⁷. Therefore, the total project size is US\$77.8 million (including a US\$66 million GCF grant, as well as co-financing of US\$5 million from USAID, US\$5 million from TCCF and US\$1.8 million from WWF-Pakistan). Annex 3: Economic analysis has been prepared to reflect this project size.

281. The GoP has highlighted the proposed project's important role of contributing to national adaptation objectives by codifying the proposed project and its interventions as a priority initiative under the country's updated NDCs (2021). Overall, the proposed project will serve as a key case study for Pakistan, providing critical lessons and validation of EbA and green infrastructure interventions as cost-effective and impactful adaptation options.

282. GCF financing will be used to overcome the barriers to adopting EbA and green infrastructure interventions to enhance flood and water resources management in Pakistan. With GCF grant funding, the proposed project will deliver several adaptation benefits that will contribute to achieving a paradigm shift in the country towards climate-resilient sustainable development. The combined amount of US\$77.8 million will be used efficiently and effectively to: i) implement EbA and green infrastructure interventions in the four project sites most at risk to climate change in the Indus Basin to reduce the impacts of climate change-induced floods and droughts on communities and key ecosystems; ii) support the national and provincial government to integrate EbA and green infrastructure interventions into Pakistan's existing regulations and policies for flood and water resources management, namely the Implementation Framework for the National Water Policy, the NAP, and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab; iii) enhance the climate resilience and livelihoods of the most vulnerable target communities in the Indus Basin to withstand current and future climate scenarios, including working with provincial governments to adapt their existing extension services to better support farmers impacted by floods and droughts caused by climate change; iv) develop a pipeline of sustainable businesses and help MSMEs access debt and equity finance to scale these businesses across Pakistan; v) assist CBOs to integrate flooding and drought planning within their operations; and vi) create WUGs to manage community water access and disputes, as well as community Watch and Watch (W&W) systems to manage the project's EbA and green infrastructure interventions natural ecosystems. The cost-effectiveness of proposed project interventions as well as a summary of the applicable international best practices and lessons learned are presented below.

Cost-effectiveness of project interventions

283. Pakistan's Indus Basin is particularly at risk to the increasing impacts of floods and droughts caused by climate change. The intensity and frequency of these events are projected to increase under future climate scenarios. Decision-making on the type of climate change adaptation measures that the GoP should employ to respond to increased flood and drought risks and impacts, should be informed by an understanding of the cost of such measures (e.g., EbA and green infrastructure), as well as the associated potential avoided costs. To inform decision-making on adaptation interventions, an economic analysis for the project has been undertaken using a cost-benefits analysis (CBA) approach. This economic analysis, assuming a 30-year lifetime for the project and a

¹⁴⁶ second-lowest tier of local government.

¹⁴⁷ <file:///C:/Users/jaudel/Downloads/Advocacy%20Brief%20Financing%20Climate%20Action%20in%20Pakistan.pdf>

social discount rate of 9.12%, is presented in Annex 3. Included in the economic analysis is an analysis of the costs, benefits and economic rates of return associated with the proposed project's interventions in each of the four project sites in the Indus Basin, namely DI Khan, Ramak, Chakar Lehri, and Manchar Lake. All assumptions underlying the analysis are clearly set out in CBA model in Annex 3a and reported on in Annex 3b.

284. The following five key steps were used to undertake the CBA for the project:
- **Definition of an appropriate baseline or 'without intervention' scenario** against which the impacts of the proposed project's interventions were compared over their lifetime. Two scenarios were assessed. The first considers the costs of damages and losses that would occur assuming no future climate change. The second examines the impacts (costs and damages) and associated costs assuming climate change over the period 2021-2050 but without the proposed project interventions. The baseline was also used to derive a measure of the opportunity costs associated with the proposed interventions.
 - **Estimation of the costs (including externality effects)** associated with each intervention type. The analysis considered costs across the lifetime of each intervention.
 - **Estimation of the economic, social and environmental benefits** of the proposed interventions. Adaptation benefits and co-benefits were quantified and valued in monetary terms as far as possible and where it is proportionate to do so. In cases where it was not possible to quantify and value certain material benefits, these benefits (and their likely significance) were described in either quantitative (where possible) or qualitative terms.
 - **Discounting and application of the appropriate decision rules.** The costs and benefits were estimated over a 30-year period, starting in 2024. Net benefits were calculated as the difference between the 'with project' and 'without project' scenarios converted to measures of economic net present value (NPV) and economic internal rate of return (EIRR) using a social discount rate of 9.12%¹⁴⁸.
 - **Sensitivity analysis.** Sensitivity analysis was performed to test the sensitivity of the outcomes to changes in the underlying assumptions, focusing on key input variables with large uncertainties.
285. The results of the CBA at the project level are presented in Table 20. These suggest that, under the current set of assumptions, the total package of interventions is cost-beneficial, with the total project investment generating a NPV of US\$30 million, with a benefit to cost ratio (BCR) of 1:1 and an EIRR of around 13.6%. On this basis, there is a clear economic case for implementing the proposed project, as well as for the use of GCF grant funding to finance it. The EIRR for the project is 13.6%. This is higher than the EIRR hurdle rate of most international donor agencies. For example, the Millennium Challenge Corporation (MCC) of the USA employs a hurdle rate of 10% EIRR for projects¹⁴⁹. This indicates the economic value of the proposed project in the context of climate change adaptation and flood and drought risk management in Pakistan. The results of the CBA also show that, for all project sites, the costs of flooding are expected to increase in future as a result of climate change in the absence of adaptation interventions. However, the results also show that under a 'with project' scenario, the benefits of adaptation interventions under a future climate are higher than the costs for all the types of intervention to be implemented in all project target areas.

Table 20. Cost-benefit metrics for the proposed project (base case).

Base case	
Total costs of interventions (US\$, millions, undiscounted)	82
NPV (US\$, millions)	30
BCR	1:1
EIRR (%)	13.6%

286. Significant benefits are associated with carbon sequestration, and biodiversity and cultural services provided by EbA and green infrastructure interventions which, together, are valued at around US\$282.3 million. According to the sensitivity analysis presented in the Annex 3, the economic summary statistics given above are robust to changes in input values. The results of the CBA are particularly sensitive to changes in the discount rate, the shadow price of carbon, and the biodiversity and cultural ecosystem services value of wetlands. These were, therefore, used to define a 'low scenario' and a 'high scenario' which are intended to reflect the lower and upper endpoints of the range within which the actual value of benefits is expected to lie and correspond to a 'worst case'

¹⁴⁸ The discount rate computed as the average of the deposit interest rate in Pakistan in 2020 (7.4745 percent) and the discount rate at which it lends to commercial banks (10.758 percent). This follows the approach used by the FAO in their GCF Funding Proposal for Climate Resilient Agriculture and Water Management in the Indus Basin (FP108) and is also consistent with the discount rate used by Pakistan's Federal Flood Commission (FFC) in its economic valuation of schemes proposed in the Flood Protection Sector Project (FPSP) under the National Flood Protection Plan-IV (NFPP-IV).

¹⁴⁹ Millennium Challenge Corporation. 2021. Guidelines for economic and beneficiary analysis. Available at: <https://www.mcc.gov/resources/story/story-cdg-guidelines-for-economic-and-beneficiary-analysis>

and ‘best case’ scenario respectively. Under the “high scenario”, the value for wetlands (taken from the TEEB Wetland Study) appears implausibly high when compared to the estimates derived from wetland valuation studies elsewhere and therefore represent an ‘extreme’ case which is considered highly unlikely. The results are shown in the table below.

Table 21. Sensitivity analysis results.

CBA inputs and outputs	Low (worst case)	High (best case)
Total costs of interventions (US\$, millions, undiscounted)	66	66
NPV (US\$, millions)	7.1	6,324
BCR	0.92:1	95:1
EIRR (%)	14.2%	570%

Application of best practices in Pakistan’s water sector

287. Best practices from similar projects as outlined in Section B.1 were used to inform the design of EbA and green infrastructure interventions to be implemented under the proposed project. Lessons learned from key planned and ongoing investments into climate change adaptation in Pakistan’s Indus Basin, including but not limited to the NDRMF-funded project and the two GCF-funded projects in the Indus Basin, FP108 *Transforming the Indus Basin with Climate Resilient Agriculture and Water Management (led by FAO)* and FP018 *Scaling-up of Glacial Lake Outburst Flood (GLOF) risk reduction in Northern Pakistan (led by UNDP)*, respectively, were also used to guide the design of all project interventions. Such lessons include appropriate mechanisms for ensuring that project activities are implemented in a participatory, gender-inclusive and sustainable way. The specific international best practices and lessons learned that were applied to the design of the proposed project and that will contribute to the efficiency and effectiveness of project interventions are summarised below.

288. International best practices and lessons learned identified for the project include:

- ensuring that landscape planning, implementation of policies, strategies and monitoring processes are integrated and iterative to account for the dynamic nature of landscape processes;
- acknowledging the need to engage tribal leaders, Indigenous Peoples, and all relevant stakeholders at the community level in an equitable and responsive manner to fairly distribute benefits and incentives of landscape management;
- incorporating traditional knowledge into training for smallholder farmers and local communities on climate-resilient agricultural practices, community-based natural resources management and the adoption of climate-resilient livelihoods;
- drawing on regional experiences to increase institutional capacity to respond to the impacts of climate change by using EbA and green infrastructure. This is evident in the GoP’s exposure visit (Ramsar Advisory Mission (RAM)) to China’s Yangtze Basin project, which integrates ecological solutions with ‘hard’ infrastructure to optimize water resource management. During this visit, the GoP recognised that “a mono-disciplinary approach based on engineering solutions alone — as in the case of the Indus Basin — cannot fully handle hydrological cycles, ecosystems, and the security of people prone to flood risks”¹⁵⁰.
- acknowledging the need to account for the most vulnerable people who do not have sufficient collateral when developing strategies to access finance for adaptation;
- undertaking regular monitoring and evaluation of EbA, green infrastructure, climate-resilient agriculture and other climate-resilient interventions to ensure that these interventions continue to directly respond to the adaptation needs of the target communities;
- engaging and collaborating extensively and in a gender- and culturally-responsive way with relevant national and sub-national governments, as well as local-level stakeholders, to maintain buy-in to the project over the long term; and
- implementing effective financial and project management strategies to ensure the efficient use of financial resources and avoid delays during the implementation phase.

¹⁵⁰ Ali, 2013.

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Current flood management efforts in Pakistan are largely reactive to increasing extreme climate events, relying on grey infrastructure interventions. These hard interventions are often unable to manage flood waters (which are increasingly surpassing anticipated flows and frequency because of climate change), have inconsistent levels of effectiveness, and are costly to repair and operate.	<u>Low</u>	The potential target scenario is a shift away from Pakistan's business-as-usual model for flood and water resources management that is reactive and reliant on inefficient and costly grey infrastructure interventions towards a climate-resilient sustainable pathway underpinned by scalable ecosystem-based interventions (particularly EbA and green infrastructure) that are implementable at national and sub-national levels across Pakistan's flood- and drought-prone landscapes, cost-effective, cross-sectoral and integrated into flood and water resources management and regulatory frameworks.	In addition to demonstrating the case for EbA and green infrastructure interventions in Pakistan and expanding the country's current policy and regulatory instruments for flood and water resources management to prioritise these interventions (see 'Replicability' below), the project will contribute to scaling up EbA and green infrastructure interventions in Pakistan by equipping policymakers, planners and communities with the necessary skills and capacity to implement, operate and maintain EbA and green infrastructure interventions at scale and in response to increasingly severe floods and droughts caused by climate change in Pakistan.

	Additionally, Pakistan's current policy and regulatory instruments for flood and water resources management do not prioritise EbA and green infrastructure interventions as effective and efficient flood and drought risk reduction interventions, and do not consider the critical role of healthy freshwater ecosystems in increasing affected communities' resilience to climate change.			
Replicability	EbA and green infrastructure interventions have not yet been demonstrated as cost-effective flood and drought risk reduction options in Pakistan. Additionally, these interventions are not yet prioritised in the country's relevant regulatory frameworks for flood and water resources management (namely the Implementation Framework for the National Water Policy, National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces). This makes replicating these interventions difficult.	<u>Low</u>	The potential target scenario is having an evidence base demonstrating EbA and green infrastructure as cost-effective interventions for reducing the impacts of floods and droughts, as well as well documented procedures backed by a rigorous scientific base that enables the GoP to channel future investments into EbA and green infrastructure interventions instead of grey infrastructure to respond to increasingly severe floods and droughts caused by climate change.	The project will contribute to replicating EbA and green infrastructure interventions for flood and water resources management in Pakistan by: i) implementing EbA and green infrastructure interventions at target sites in the Indus Basin; ii) generating an evidence base of these interventions' adaptation benefits; iii) using this evidence to expand Pakistan's regulations and policies for flood and water resources management through updated procedures that prioritise EbA and green infrastructure interventions; and iv) strengthening the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated procedures. This will enable the GoP to channel future investments into EbA and green infrastructure interventions and replicate similar adaptation strategies in other flood- and drought prone areas in Pakistan to those demonstrated under the proposed project.

Sustainability	Because Pakistan's relevant regulatory frameworks for flood and water resources management do not prioritise EbA and green infrastructure interventions, they are not prioritised in government budgets and there is a lack of evidence, knowledge and awareness and capacity to implement and sustainably manage these interventions over the long term.	<u>Low</u>	The potential target scenario is to secure government commitment to implementing EbA and green infrastructure interventions instead of grey infrastructure in response to climate change-induced floods and droughts in Pakistan by expanding the country's regulations and policies for flood and water resources management to include EbA and green infrastructure in government mandates. This will enable existing/planned public funds that would likely otherwise have been allocated to grey infrastructure interventions to instead be channelled towards the implementation, operations and maintenance of EbA and green infrastructure interventions to address floods and droughts in Pakistan.	Sustainability of the project's interventions will be enhanced by expanding Pakistan's Implementation Framework for the National Water Policy, National Adaptation Plan (NAP), and four Provincial Adaptation Plans to prioritise EbA and green infrastructure interventions. The MoWR (through FFC) will require the application of the updated procedures for the Implementation Framework of the National Water Policy, while MoCC in collaboration with each province's Environment Protection Authority (EPA) will require the application of the updated procedures for the NAP and four Provincial Adaptation Plans. Integrating the implementation of EbA and green infrastructure interventions into these government mandates will enable future investments and country-driven flood and drought risk reduction efforts to prioritise these interventions instead of grey infrastructure. Finally, the GoP is committed to O&M of the project's interventions during and beyond the project period (see Section B.6 for further details).
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ¹⁵¹	
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Household survey; interventions' completion reports	0	Direct:	Direct:	<u>Assumptions</u> EbA and green infrastructure interventions are successful in

¹⁵¹ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

				206,200 people (107,224 men, 98,976 women) Indirect: 2,107,308 people (1,095,800 men, 1,011,508 women)	687,336 people (357,414 men, 329,922 women) Indirect: 7,024,361 people (3,652,668 men, 3,371,693 women)	attenuating the impacts of floods and droughts on communities within and around the project sites. EbA and community-based natural resource management interventions will improve the resilience of ecosystems, ecosystem service supplies and people depending on those systems for their lives and livelihoods. <u>Notes</u> The methodologies used calculate targets for direct and indirect beneficiaries are presented in Section D.1.
	Supplementary 3.1: Change in expected losses of economic assets due to the impact of extreme climate- related disasters in the geographic area of the GCF intervention	Government records ¹⁵² ; Post disaster needs assessments will be used to supplement the government records mentioned above, as necessary.	USD15.39 million in DI Khan project site USD2 million in Chakar Lehri and Manchar project sites	USD2.5 million in avoided losses of economic assets in DI Khan project site USD480,000 in avoided losses of economic assets in Chakar Lehri and Manchar project sites	USD8.6 million in avoided losses of economic assets in DI Khan project site USD1.6 million in avoided losses of economic assets in Chakar Lehri and Manchar project sites	<u>Assumptions</u> EbA and green infrastructure interventions are successful in attenuating the impacts of floods and droughts on communities, ecosystems, and infrastructure within and around the project sites. The baseline and targets are based on modelled data presented in Annex 3. <u>Notes</u> The avoided losses captured by this target include: avoided crop damages, livestock losses, damages to housing, damages to infrastructure, direct losses to crops, housing, infrastructure (road and railways), as well as

¹⁵² Government records include reports from Federal Flood Commission and disaster management authorities at district, provincial or national level. These reports are generated at least twice per year.

						to irrigation networks, water supply, electricity distribution systems, telecoms network, etc., as well as indirect damages resulting from disruption of economic activities (e.g., because of transport severance) on or adjacent to the floodplain. The mid-term target represents the expected avoided losses in US\$ for the abovementioned categories as a result of the EbA and green infrastructure interventions by Year 4. The final target represents the expected avoided losses in US\$ for the abovementioned categories as a result of the EbA and green infrastructure interventions by Year 7. The total expected avoided losses in US\$ for the abovementioned categories is US\$20.2 million over the 30-year modelled period. See Annexes 3a and b for further details.
	GCF Supplementary Indicator 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Farmers' list¹⁵³ Key informant interviews; Site surveys	0	3,705 people (1,389 women, 2,316 men) adopt climate-resilient agricultural practices 1,980 people (743 women and	7,410 people (2,779 women, 4,631 men) adopt climate-resilient agricultural practices 3,960 people (1,485 women	<u>Notes</u> The targets for GCF Supplementary Indicator 2.1 were determined using an assumed ~30% uptake rate for those people who will adopt new climate-resilient agricultural practices (Sub-activity 3.2.1.1) out of a total of 24,700 people.

¹⁵³ This is housed within the Department of Agriculture and WWF-Pakistan.

				1,238 men) adopting climate- resilient crop varieties	and 2,475 men) adopting climate- resilient crop varieties	The targets for GCF Supplementary Indicator 2.1 were determined using an assumed ~30% uptake rate for those people who will adopt new climate-resilient crop varieties (Sub-activity 3.2.1.3) out of a total of 13,200 people. The % of male beneficiaries is higher due to the agricultural sector being predominantly men.
	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience	Farmers' list ¹⁵⁴ Key informant interviews; Site surveys	0	3,350 people adopt improved irrigation (1,250 women and 2,090 men)	6,700 people adopt improved irrigation (2,513 women and 4,187 men)	<u>Notes</u> The % of male beneficiaries is higher due to the agricultural sector being predominantly men.
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	GIS maps, design documents, site surveys and photographs	0	9,600 ha of natural resources under improved climate-resilient management	18,215 ha of natural resources under improved climate-resilient management	<u>Assumptions</u> Natural resource management is owned and driven by communities, ensuring its long- term success, including behaviour change at the community level that supports the implementation of climate- resilient management practices. <u>Notes</u> The targets include: i) area under improved planning/decision making/management; and ii) area directly restored through afforestation activities under the project.
	Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal	GIS maps; site surveys; key informant interviews	0	1,600 ha of terrestrial forest brought under restoration	4,000 ha of terrestrial forest brought under restoration	<u>Assumptions</u> Ecosystem restoration (EbA) interventions are successful and supported by local

¹⁵⁴ This is housed within the Department of Agriculture and WWF-Pakistan.

	marine areas brought under resoration and/or improved ecosystems			and/or improved ecosystems 6,820 ha of terrestrial non-forest brought under restoration and/or improved ecosystems	and/or improved ecosystems 14,215 ha of terrestrial non-forest brought under restoration and/or improved ecosystems	stakeholders through the reduction of anthropomorphic drivers of ecosystem degradation (supported by community natural resource-based management interventions across project sites). <u>Notes</u> 14,215 ha in DI Khan through active restoration, while an additional 4,000 ha will be brought under community-based natural resource management and watch and ward systems.
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Pakistan's current policy and regulatory instruments for water resource management — namely the National Water Policy, National Adaptation Plan (NAP), and Provincial Adaptation Plans — do not consider EbA and green infrastructure interventions as effective and efficient flood and drought risk reduction interventions. They also do not consider the critical role of healthy freshwater ecosystems in	low	Pakistan's institutional and regulatory frameworks for flood and water resources management are strengthened through the development and adoption of updated procedures for the Implementation Framework for the National Water Policy, National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces on the cost-effective use of EbA and	Under Component 2, Pakistan's regulations and policies for flood and water resources management will be expanded from only focusing on using grey infrastructure to address the threats of floods and droughts to prioritising EbA and green infrastructure interventions. This will be achieved through a step-wise approach. An economic valuation of the project's EbA and green infrastructure interventions	National level (one country)

	increasing affected communities' resilience to climate change. Instead, the current policies and regulatory instruments promote costly grey infrastructure interventions that do not optimise the potential adaptation and other environmental and social benefits.		green infrastructure interventions. This will lead to the creation of well documented procedures backed by a rigorous scientific base that enables the GoP to channel future investments into EbA and green infrastructure interventions instead of grey infrastructure to respond to increasingly severe floods and droughts caused by climate change.	will be undertaken which will include monitoring and analysis of these interventions' cost-effectiveness and efficiency in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan, and developing an evidence report of their adaptation benefits to initiate public sector reform in flood and water resources management at the national and provincial levels in Pakistan. Using this evidence, a National Working Group will be established to lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan.	
Core indicator 7: Degree to which GCF Investments contribute to market	There is a need for business development and incubation assistance to	low	7 sustainable businesses in the Indus Basin access debt and/or equity finance	Under Component 3, due diligence will be conducted for businesses identified	Multiple sub-national areas within a country

development/transformation at the sectoral, local, or national level	incentivize businesses to deliver products and services with positive climate-resilient environmental impacts. Investors are not willing to finance the necessary origination and incubation work needed to identify such investments. Important foundational work in business sector and pipeline identification has been done by WWF-Pakistan through the forementioned DFCD Project.		to help these scale and access larger regional and national markets. This would provide an important signal that businesses which provide climate change and sustainability benefits can be viable helping to shift incentives away for environmentally destructive practices.	under the former DFCD project to identify the final pipeline of businesses to be supported by the Project. The businesses will receive technical assistance and incubation through the Project to strengthen their business practices and systems and access non grant investor finance.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	The communities, public and private sector organizations in Pakistan have limited knowledge on the impacts of climate change, (such as increasingly severe and frequent floods and droughts) and how EbA and green infrastructure can be used as appropriate adaptation responses that are flexible and cost-effective alternatives to traditional grey-infrastructure interventions. As a result, the GoP has limited technical expertise required for designing, implementing, operating and maintaining EbA and green infrastructure interventions to respond to the impacts of floods and droughts. This is	low	Enhanced knowledge and capacity of communities, public and private sector organizations at national, provincial and district levels for planning, implementing and managing EbA and green infrastructure for flood and water resources management. Strengthened evidence base on the adaptation benefits of EbA and green infrastructure, including best practices and lessons learned, required not only for the development and adoption of the updated procedures described in Core Indicator 5 above, but also to foster investment into scaling up and replicating these interventions at national and provincial levels, and	New evidence and lessons learned on the flood and drought risk reduction benefits of EbA and green infrastructure will be generated under Component 2 of the project. This evidence will not only be used to expand Pakistan's existing flood and water resources management policies and regulations (described in Core Indicator 5 above), but also: i) used to inform the selection of additional opportunities and sites across Pakistan where the project's EbA and green infrastructure interventions could be scaled up and replicated; and ii) shared through the WWF International Network to practitioners in the climate change, freshwater,	National level (one country)

	compounded by the lack of a strong evidence base that demonstrates the benefits of these interventions. Consequently, EbA and green infrastructure are not prioritised in the GoP's current policy and regulatory instruments for water resources management, and adequate knowledge (including best practices and lessons learned) required to inform the implementation of effective EbA and green infrastructure interventions is not available.		secure uptake of these interventions by communities across Pakistan.	ecosystem planning, and development sectors to be replicated in other flood- and drought-prone areas. Finally, to secure uptake and sustainability of the EbA and green infrastructure interventions by national and provincial government, as well as communities, on the implementation, operation and maintenance of these interventions. Additionally, Component 3 will enable knowledge transfer and sharing for CBOs to improve their capacity to use EbA and green infrastructure interventions to respond to the impacts of increasingly severe and frequent floods and droughts in Pakistan.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "Project/programme co-benefit indicators" section in table below. Add rows as needed.

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Outcome 1. EbA and green infrastructure interventions with improved community-led management reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin.	Area (ha) of flood extent reduced	Government records¹⁵⁵; Watershed delineation and flood map analysis; and interventions completion reports	A baseline study will be undertaken in Year 1 of the project to determine a site-specific baseline that relates to this indicator.	15,250 ha of reduced flood extent (after completion of 30% EbA and GI Work) (This total area includes: <u>EbA</u>: 2,930 ha in DI Khan from EbA interventions; and <u>Green infrastructure</u>: 23 ha in Balochistan, 6,258 ha in Manchar, and 130 ha in DI Khan from green infrastructure interventions)	50,833 ha of reduced flood extent (This total area includes: <u>EbA</u>: 9,767 ha in DI Khan from EbA interventions; and <u>Green infrastructure</u>: 75 ha in Balochistan, 20,863 ha in Manchar, and 432 ha in DI Khan from green infrastructure interventions)	<u>Assumptions</u> EbA and green infrastructure interventions are successful in attenuating the impacts of floods, reducing the area affected by flooding and the related risks on communities, ecosystems, and infrastructure within and around the project sites. <u>Notes</u> Only a final target is available for this indicator as it was only possible to model the impact of complete, fully functional EbA and green infrastructure interventions. Progress for installing green infrastructure interventions can be tracked using the indicator for Output 1.2 below (<i>Number of green infrastructure solutions installed</i>). Inundation was calculated from simulating 30 years of rainfall over the catchments. The extreme flood scenario was selected as the indicative future (2020-2050 RCP4.5) flood event. The peak area inundated, and the depth of flood were calculated from this extreme event. Depths less than 30 cm were excluded as they were less likely to cause long-term significant
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¹⁵⁵ Government records include reports from Federal Flood Commission and disaster management authorities at district, provincial or national level. These reports are generated at least twice per year.

						damages. This filter is then more indicative of the areas being damaged by the floods rather than just the area being covered with very shallow water. The difference in areas was calculated between the future flood event with no interventions and the future flood events with the GI interventions and with the EbA interventions to show the changes in the areas (hectares) inundated as a result of the interventions in the future scenario. Green infrastructure refers to any vegetative or natural material infrastructure system which enhances the natural environment through direct or indirect means . It is the range of measures that use plant or soil systems, permeable surfaces or substrates, stormwater harvest and reuse, or landscaping to store, infiltrate, or evapotranspiration stormwater and reduce flows to sewer systems or to surface waters. Examples of green infrastructure interventions that will be implemented under the project include flood protection embankments, flood dispersal embankments,
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						gabion bunds, small and large retention areas, and recharge basins. (See Annex 2 for further details.) The number of ha represents the extent of flooding that will be reduced by the green infrastructure interventions.
	Amount (litres) of soil infiltration potential/water replenished through watershed management interventions	Water replenishment calculation reports	0	320 million litres	1,600 million litres	<u>Assumptions</u> EbA and green infrastructure interventions are successful in capturing and storing floodwater and increasing groundwater recharge in the project sites. <u>Notes</u> A WWF-Pakistan report ¹⁵⁶ on project impact on groundwater recharge, water availability & sediment load was used as a reference to set the targets for this indicator.
	Amount of water captured/retained by the EbA and green infrastructure interventions	Watershed delineation analysis reports, design and specification documents	0	4 million cubic meters	20 million cubic meters	<u>Assumptions</u> EbA interventions are successful in increasing the water retention capacity of landscapes in the project sites. <u>Notes</u> This is the total water retention capacity of the EbA and green infrastructure interventions combined (including small retention areas, large retention areas, recharge basins, etc.)

¹⁵⁶ WWF-Pakistan. April 2019. Project impact on groundwater recharge, water availability & sediment load; Integrated watershed management and livelihood improvement in selected sub-catchments of Khanpur Dam.

Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.	Length (km) of natural water flow paths restored disaggregated by site (Badri, Paniala, Sehwan)	Design documents, GIS maps, photographs	0	24 km (including 4 km in Paniala and 20 km in Sehwan)	34.3 km (including 0.3 km in Badri, 4 km in Paniala, and 30 km in Sehwan)	<u>Assumptions</u> The restoration of degraded natural water flow paths is successful and leads to improved water supplies. <u>Notes</u> Mid-term and final targets were calculated by adding the lengths the areas restored together.
Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.	Number of people (disaggregated by women and men) protected from flooding	Watershed delineation and flood map analysis, GIS maps, household surveys	0	At least 138,780 people (93,390 directly and 45,390 indirectly) will be protected from flooding because of the green infrastructure interventions	At least 462,600 people (311,300 directly and 151,300 indirectly) will be protected from flooding because of the green infrastructure interventions	<u>Assumptions</u> Green infrastructure solutions are successful installed and protect the surrounding communities from flooding. <u>Notes</u> Designs and scale of green infrastructure solutions are presented in Annex 2.
	Number of green infrastructure interventions installed	Engineering design documents; engineering completion reports; photographs; government monitoring reports ¹⁵⁷	0	4 flood protection embankments (dimensions: Length: 150 m Width: 150 m Height: 10 m) 2 dispersal embankments (dimensions: Length: 150 m Width: 150 m Height: 10 m) 18 gabion bunds (dimensions: Length: 150 m Height: 6 m)	9 flood protection embankments (dimensions: Length: 150 m Width: 150 m Height: 10 m) 7 dispersal embankments (dimensions: Length: 150 m Width: 150 m Height: 10 m) 45 gabion bunds (dimensions: Length: 150 m Height: 6 m)	Methodologies for calculating project direct and indirect beneficiaries are provided in Section D.1 of the Funding Proposal.

¹⁵⁷ WWF-Pakistan will work with relevant government entities such as MoWR (FFC) and the Provincial Irrigation Departments to prepare these reports on an annual basis.

				14 small retention areas (Dimensions for small retention areas in DI Khan Watershed: Length: 275 m Width: 275 m Height: 1 m Dimensions for small retention areas in Manchar Watershed: Length: 245 m Width: 245 m Height: 1 m Dimensions for small retention areas in Chakar Lehri Watershed: Length: 180 m Width: 180 m Height: 1 m) 4 large retention areas (Dimensions for large retention areas in DI Khan Watershed: Length: 874 m Width: 874 m Height: 1 m Dimensions for large retention areas in Manchar	42 small retention areas (Dimensions for small retention areas in DI Khan Watershed: Length: 275 m Width: 275 m Height: 1 m Dimensions for small retention areas in Manchar Watershed: Length: 245 m Width: 245 m Height: 1 m Dimensions for small retention areas in Chakar Lehri Watershed: Length: 180 m Width: 180 m Height: 1 m) 9 large retention areas (Dimensions for large retention areas in DI Khan Watershed: Length: 874 m Width: 874 m Height: 1 m Dimensions for large retention areas in Manchar	
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				Watershed and Chakar Lehri Watershed: Length: 1,000 m Width: 1,000 m Height: 1 m) 6 recharge basins (Dimensions for recharge basins in DI Khan Watershed: Length: 150 m Width: 14.5 m Height: 2 m Dimensions for recharge basins in Manchar Watershed and Chakar Lehri Watershed: Length: 150 m Width: 14.5 m Height: 1 m)	Watershed and Chakar Lehri Watershed: Length: 1,000 m Width: 1,000 m Height: 1 m) 15 recharge basins (Dimensions for recharge basins in DI Khan Watershed: Length: 150 m Width: 14.5 m Height: 2 m Dimensions for recharge basins in Manchar Watershed and Chakar Lehri Watershed: Length: 150 m Width: 14.5 m Height: 1 m)	
Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.	# Community-based management bodies (disaggregated by gender-responsiveness) supported through technical/institutional capacity building	Copy of community based NRM plans, minutes of CBOs meetings, List of WUGs, TORs of WUGs, membership details, minutes of meetings	0	6 CBOs 6 Water User Groups 3 Watch and Ward Systems	8 CBOs 8 Water User Groups 4 Watch and Ward Systems	<u>Assumptions</u> Support for the maintained implementation of community-based natural resource management is continually built amongst local communities.
	# Community-based natural resource management agreements signed between the Provincial Departments (Forestry, Irrigation, and Agriculture) and communities	Signed agreements	0	At least 3 agreements	4 agreements	

Outcome 2: Enabling environment for climate action that has new procedures for implementing EbA and green infrastructure interventions for flood and water resources management in Pakistan.	Level of integration of EbA and green infrastructure considerations into the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh	Scorecard ¹⁵⁸	Level 0	Level 1	Level 3	<u>Assumptions</u> Support from government for the integration of EbA and green infrastructure considerations into the relevant implementation frameworks remains strong.
Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.	Evidence of economic benefits of EbA and green infrastructure	Evidence Report	TBD ¹⁵⁹ : Evidence base limited	Raw data collection ongoing	Evidence base for the economic benefits of EbA and green infrastructure results in plans and is used to leverage funding for replication and upscaling of project interventions	<u>Assumptions</u> EbA and green infrastructure interventions implemented by the project produce cost-effective benefits that demonstrate an economic case for the approaches.
Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.	# of plans integrating EbA and green infrastructure interventions flood and drought risk management	Revised provincial adaptation plans; List of recommendations; Copies of provincial level climate-resilient Integrated Water Resource Management Plans; Supporting documents of implementation	0	Recommendations submitted for the integration of EbA into the National Water Policy focusing on flood risk and water resource management	Recommendations submitted via PSC to the National Working Group for EbA and green infrastructure to be integrated into the	<u>Assumptions</u> The National Working Group, other government departments and relevant ministries maintain support and interest in adopting the updated procedures for implementing EbA and green infrastructure in Pakistan.

¹⁵⁸ Scorecard showing the level of integration of EbA and green infrastructure interventions into the implementation framework of the National Water Policy, national and provincial adaptation plans. Level 0: The implementation framework of the National Water Policy, national and provincial adaptation plans do not include EbA and green infrastructure interventions. Level 1: Recommendations for integrating EbA and green infrastructure interventions into the implementation framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh are produced and presented to the National Working Group for endorsement. Level 2: Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture review and endorse the recommendations for revising the implementation framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans. Level 3: Representatives from the National Working Group, including MoCC, MoWR (including FFC) and the Provincial Departments of Irrigation, Forestry and Agriculture receive final recommendations for revising and implementing the updated Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans.

¹⁵⁹ Baseline survey to be conducted in Year 1.

				National Adaptation Plan does not consider EbA and green infrastructure 0 Provincial Adaptation Plans integrate EbA and green infrastructure	implementation framework of the National Water Policy and adopted by decision-makers National Adaptation Plan integrates EbA and green infrastructure 4 Provincial Adaptation Plans for Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh integrate EbA and green infrastructure	
Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.	Number of national and sub-national decision-makers and planners trained on the application of the updated Implementation Framework of the National Water Policy, National Adaptation Plan, and three Provincial Adaptation Plans	Training reports	0	64	64	<u>Assumptions</u> A National Working Group will be established and collaborate with federal and provincial level advisory committees to create an enabling environment for integration of EbA and green infrastructure in the Implementation Framework of the National Water Policy, National Adaptation Plan, and three Provincial Adaptation Plans.
	% Change in knowledge and technical capacity of staff in the GoP on how to implement, operate and maintain EbA and green infrastructure interventions	Pre- and post-assessment questionnaires	Baseline study will be conducted in Year 1	Baseline + 25%	Baseline + 60%	

	Number of community representatives/leaders trained on how to implement, operate and maintain EbA and green infrastructure interventions	Training reports	0	32	96	
Outcome 3: Enhanced resilience of community livelihoods in the Indus Basin.	Number of people benefiting directly from the adoption of climate-resilient livelihoods and sustainable business	Key informant interviews; Site visits/surveys; GIS maps	0	25,375 people (13,195 men, 12,180 women) benefiting directly from the adoption of climate-resilient livelihood practices 63 people (33 men, 30 women) benefitting directly from sustainable business	50,750 people (26,390 men, 24,360 women) benefiting directly from the adoption of climate-resilient livelihood practices 126 people (66 men, 60 women) benefitting directly from sustainable business	<u>Assumptions</u> Local farmers and communities support and actively adopt the improved and/or new climate-resilient livelihood options introduced by the project. An adequate number of projects presented include a strong-enough business case to secure financing for businesses from FIs. Support from FIs to finance climate-resilient businesses in maintained.
Output 3.1: Pipeline of feasible climate-resilient businesses.	Number of bankable projects (disaggregated by gender ownership/inclusion) financially supported under the project	Signed copy of agreements	0	3	7	<u>Assumptions</u> An adequate number of projects presented include a strong-enough business case to secure financing from FIs. Support from FIs to finance climate-resilient businesses in maintained.
Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.	Area (ha) of land under improved climate-resilient agricultural management	Farmers' list ¹⁶⁰ ; Key informant interviews, Site surveys	0	64 ha with drought-resistant crop varieties 20 ha with improved soil management	96 ha with drought-resistant crop varieties 32 ha with improved soil management	<u>Assumptions</u> Local farmers and communities support and actively adopt the improved and/or new climate-resilient livelihood options introduced by the project.

¹⁶⁰ Ibid.

Project/programme co-benefit indicators						
Co-benefit 1: GHG emission reduction from ecosystem restoration and improved ecosystem management	Estimated GHG emissions reduced/avoided	GHG assessment: National Forest Monitoring Systems and Greenhouse Gas Inventory Reports	0	0 (The project does not anticipate significant GHG emission reductions until Year 5)	52,900 tons over the project period (7 years) 1,037,928 tons over 40 years	<u>Assumptions</u> The project's ecosystem restoration and community-based natural resource management interventions are successful. <u>Notes</u> The Removal Factors for different forest types/ ecological zones have been taken from the Sub-National Forest Monitoring Systems and Greenhouse Gas Inventory Reports for KP and Punjab Provinces developed by WWF-Pakistan for the Ministry of Climate Change (MoCC) under the National REDD+ Project, 2022. The CO2 mitigation potential calculations are based on IPCC methodology. See Section D.1 of the Funding Proposal for further details.
Co-benefit 2: Improved biodiversity as a result of the project's EbA and green infrastructure interventions.	Percentage increase in migratory birds' population at Manchar Lake	Midwinter waterfowl census report ¹⁶¹	22,176 migratory birds were recorded at Manchar Lake during the 2022 midwinter waterfowl census ¹⁶²	10% increase from the baseline for the Manchar Lake site	25% increase from the baseline for the Manchar Lake site	<u>Assumptions</u> Improved habitat will provide favourable roosting and feeding grounds for migratory birds. Targets will consider the population of species within and adjacent to the project boundaries. <u>Notes</u>

¹⁶¹ These reports will be produced annually by the Zoological Survey of Pakistan (ZSP) which is a department under the Ministry of Climate Change (MoCC).

¹⁶² Zoological Survey of Pakistan (January 2022).

						WWF-Pakistan will verify the targets for the Manchar Lake site by joining the Zoological Survey of Pakistan and the Sindh Wildlife Department team during annual waterfowl censuses.
	Percentage increase in migratory birds' population at Indus Game Reserve (D.I Khan)	Midwinter waterfowl census report ¹⁶³	9,757 migratory birds were recorded at Indus Game Reserve (D.I Khan) during the 2022 waterfowl census ¹⁶⁴	10% increase from the baseline for the Indus Game Reserve (D.I Khan)	25% increase from the baseline for the Indus Game Reserve (D.I Khan)	<u>Assumptions</u> Improved habitat will provide favourable roosting and feeding grounds for migratory birds. Targets will consider the population of species within and adjacent to the project boundaries. <u>Notes</u> WWF-Pakistan will verify the targets for the D.I Khan site by joining the Zoological Survey of Pakistan and the KP Wildlife Department team during annual waterfowl censuses.

¹⁶³ These reports will be produced annually by the Zoological Survey of Pakistan (ZSP) which is a department under the Ministry of Climate Change (MoCC).

¹⁶⁴ KP Wildlife Department (January 2022).

	Percentage increase in fish species richness and abundance /Macro-invertebrates species richness	Annual fish survey and stock assessment studies ¹⁶⁵ Annual macroinvertebrate species richness and abundance estimations ¹⁶⁶	32 species of fish ¹⁶⁷ Manchar Lake; catches 500 metric ton/year ¹⁶⁸	2% increase in the fish stocks in comparison to the baseline	10% increase in the fish stocks in comparison to the baseline	<u>Assumptions</u> Improved water quality would result in the improvement of the health of the ecosystem which would lead towards the increased abundance in the fish population and/or species diversity. Targets will consider the population of species within and adjacent to the project boundaries. <u>Notes</u> Biodiversity assessments would validate diversity and richness of candidate wildlife species at candidate sites Annual fish stock assessment would be carried by the Sindh Fisheries Department, WWF-Pakistan, Pakistan Museum of Natural History (PMNH), and academic partners.
	Percentage increase in population abundance of the Indus River Dolphin (EN) ¹⁶⁹	Five yearly population abundance estimation ¹⁷⁰	160-170 individuals Between Chashma-Taunsa Barrages 2022	3% increase	8% increase in the population abundance	<u>Assumptions</u> Improved habitat conditions and recharge capacity would ultimately lead towards the improved health of the Indus River and associated biodiversity. The population of a few indicator species may be useful to determine the visible change. Targets will consider the population of species within and

¹⁶⁵ Annual fish stock assessment would be carried by the Sindh Fisheries Department, WWF-Pakistan, Pakistan Museum of Natural History (PMNH), and academic partners.

¹⁶⁶ Annual fish stock assessment would be carried by WWF-Pakistan, Pakistan Museum of Natural History (PMNH), and academic partners.

¹⁶⁷ Mahar et al 2020

¹⁶⁸ Mahar et al 2020

¹⁶⁹ <https://www.iucnredlist.org/> IUCN RED LIST STATUS

¹⁷⁰ Five yearly population abundance assessments would be carried out by the Punjab and KP Wildlife Departments and academic partners.

						adjacent to the project boundaries. <u>Notes</u> Biodiversity assessments would validate diversity and richness of candidate wildlife species at candidate sites. Five yearly population abundance assessments would be carried out by the Punjab and KP Wildlife Departments and academic partners.
	Designation and nomination of restoration sites as sites of global ecological important such as Man and Biosphere Reserve and Ramsar Sites.	Ramsar Information Sheet (RIS) for Man Ramsar Site, Indus Flyways	Currently not designated as Ramsar site	RIS drafted in collaboration with Provincial Wildlife Department and WWF-Pakistan	RIS submitted and endorsed by Ministry of Climate Change (MoCC)	<u>Assumptions</u> Provincial wildlife department and MoCC have a priority to protect and effective management of the lake and its biodiversity.
		Key Biodiversity Areas (KBAs) proposal	Currently not designated as a KBA	Key Biodiversity Areas (KBAs) proposal drafted in collaboration with Provincial Wildlife Department and WWF-Pakistan	Key Biodiversity Areas (KBAs) proposal submitted and endorsed by MoCC	<u>Notes</u> Biodiversity assessments would validate diversity and richness of candidate sites on the bases of which RIS would be developed in consultation with the Sindh Wildlife Department who are willing to enhance protected areas coverage within the wetland habitats.
	Percentage increase in freshwater turtles (hard-shelled and soft-shelled turtles)	Population abundance estimation surveys and turtle confiscation /seizure records ^{171,172}	<u>Manchar</u>	3% increase in the population abundance	10% increase in the population abundance	<u>Assumptions</u> Improved water quality would resultant in the improvement of health of ecosystem which would lead towards the

¹⁷¹ These records will include: i) official seizures reports maintained by Provincial Wildlife Departments; and ii) WWF-Pakistan's independent data collected from multiple sources such as trade monitoring portals, social media channels and UNODC.

¹⁷² Zoological Survey of Pakistan.

			Brown roofed turtle <i>Pangshura smithii</i> (NT)¹⁷³ : 57¹⁷⁴ <u>D.I Khan</u> (Reference point: Chashma barrage) Brown roofed turtle <i>Pangshura smithii</i> (NT)¹⁷⁵ : 38¹⁷⁶ Indian softshell turtle <i>Nilssonia</i> <i>gangetica</i> (EN)¹⁷⁷: 07¹⁷⁸ Black pond turtle <i>Geoclemys</i> <i>hamiltonii</i>¹⁷⁹(EN): 45¹⁸⁰ Indian flapshell turtle <i>Lissemys</i> <i>punctata</i>¹⁸¹(VU)¹⁸² : 07¹⁸³			increases in abundance in the turtle population and/or species diversity <u>Notes</u> Turtle population assessments following standard protocols engaging wildlife departments, zoological survey of Pakistan.
Co-benefit 3: New jobs created through the project's EbA and green infrastructure interventions	Number of people benefiting from new jobs created and related new skills developed	Employment contracts/agreements; household surveys	0	4,000 people	8,000 people	<u>Assumptions</u> Local community members will be involved as permanent, part-time or temporary (e.g., daily) labours for the installation,

¹⁷³ <https://www.iucnredlist.org/>Near Threatened

¹⁷⁴ Zoological Survey of Pakistan 2012

¹⁷⁵ <https://www.iucnredlist.org/> /Near Threatened

¹⁷⁶ Zoological Survey of Pakistan 2012

¹⁷⁷ <https://www.iucnredlist.org/>Endangered

¹⁷⁸ Zoological Survey of Pakistan 2012

¹⁷⁹ <https://www.iucnredlist.org/>

¹⁸⁰ Zoological Survey of Pakistan 2012

¹⁸¹ <https://www.iucnredlist.org/>

¹⁸² <https://www.iucnredlist.org/>Vulnerable

¹⁸³ Zoological Survey of Pakistan 2012

						operations and maintenance of the project's EbA and green infrastructure interventions. <u>Notes</u> Jobs created could be permanent, part-time, or temporary.
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E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.			
Activity 1.1.1: Restore 14,215 ha of degraded agro-ecological landscapes in watersheds.	Block and linear plantations will be established along identified hill torrent and catchment areas (D.I. Khan) in collaboration with provincial Forest and Irrigation Departments. Through this activity vegetation cover will be restored (natural regeneration and afforestation) across 14,215 ha of degraded riverine forests in DI Khan project site. Revegetation of native species at this scale will help reduce flooding, erosion, flow velocities, and increase infiltration through enhancing soil porosity.	Sub-activity 1.1.1.1: Restore 14,215 ha of degraded riverine ecosystems in D.I. Khan.	1.1.1.1a: 14,215 ha of riverine ecosystems restored
Activity 1.1.2: Rehabilitate degraded wetlands, flow paths, and channels.	The project will restore ecosystems capable of holding and channelling water flows. This will improve ecosystem functioning and reduce flooding of nearby community agricultural lands and assets and increase water access especially during periods of drought for agriculture and domestic uses. 34,674 m of pathways/channels will be restored, and	Sub-Activity 1.1.2.1: Excavate 264 m of flow paths in Badri Village, Ramak, to restore the natural hydrology of connected wetlands. Sub-Activity 1.1.2.2: De-silt the 2 ha Badri Pond in the Ramak wetlands to enhance its water-holding capacity.	1.1.2.1a: 264 m of natural flow paths rehabilitated 1.1.2.2a: 2 ha (1 pond, i.e., Badri pond) de-silted 1.1.2.3a: 410 m of embankment enhanced and reinforced

	50,000 m ² of wetlands will be rehabilitated in and around the Ramak watershed and Manchar Lake.	Sub-Activity 1.1.2.3: Enhance and reinforce 410 m of embankments in Badri Village to reduce erosion. Sub-Activity 1.1.2.4: Excavate 4 km of flow paths in Paniala Village to restore natural hydrology of Karez Systems (surface and subsurface natural channels). Sub-Activity 1.1.2.5: Excavate 30 km of flow paths in the existing water channels of Manchar Lake to restore the natural hydrology of the 25,000 ha Manchar Wetland.	1.1.2.4a: 4 km of natural flow path restored 1.1.2.5a: 30 km of flow path restored
Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.			
Activity 1.2.1: Design and implement 127 green infrastructure interventions for flood risk reduction.	Under this activity, green infrastructure interventions will be designed and implemented to reduce the impacts of flooding on nearby communities and provide additional water for these communities during droughts. Green infrastructure interventions will include: i) 9 flood protection embankments spread across the DI Khan and Ramak Watersheds; ii) 7 dispersal embankments in the D.I. Khan and Ramak Watersheds; iii) 45 gabion bunds across the DI Khan, Ramak, Manchar, and Chakar Lehri Watersheds; iv) 42 small and 9 large retention areas, and 15 recharge basins across the DI Khan, Manchar, and Chakar Lehri Watersheds. These interventions will be implemented in close consultation with communities and the Provincial Departments of Irrigation.	Sub-Activity 1.2.1.1: Design the green infrastructure measures. Sub-Activity 1.2.1.2: Build 9 flood protection embankments Sub-Activity 1.2.1.3: Build 7 flood dispersal embankments. Sub-Activity 1.2.1.4: Build 45 gabion bunds. Sub-Activity 1.2.1.5: Build 42 small retention areas. Sub-Activity 1.2.1.6: Build 9 large retention areas. Sub-Activity 1.2.1.7: Build 15 recharge basins.	1.2.1.1a: 1 green infrastructure design report with the designs for all green infrastructure interventions 1.2.1.2a: 9 flood protection embankments 1.2.1.3a: 7 dispersal embankments 1.2.1.4a: 45 gabion bunds 1.2.1.5a: 42 small retention areas 1.2.1.6a: 9 large retention areas 1.2.1.7a: 15 recharge basins
Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.			
Activity 1.3.1: Strengthen the capacity of CBOs to adopt EbA and green infrastructure interventions and undertake climate-resilient community-based natural resource management.	CBOs will be strengthened to implement community-based natural resource management. They will be trained and equipped to implement sustainable forest management practices to ensure	Sub-Activity 1.3.1.1: Strengthen CBOs to adopt EbA and green infrastructure interventions and implement community-based natural resource management.	1.3.1.1a: Training workshops to strengthen the capacity of 8 CBOs to protect, monitor and maintain the project's EbA and green infrastructure interventions, as well as

	sustainability of the project's EbA and green infrastructure interventions and health of ecosystems. The project team will work closely with the CBOs and local government departments to establish formal WUGs to standardize water use allocations for agriculture, domestic and disaster risk reduction measures. In addition, community-based Watch and Ward Systems will be established to facilitate the implementation, monitoring and protection of the project's EbA and green infrastructure interventions, including the required operation and maintenance.	Sub-Activity 1.3.1.2: Organize WUGs under CBOs and build their capacity for improved water resources management. Sub-Activity 1.3.1.3: Establish community-based watch and ward systems under CBOs to protect the project's EbA and green infrastructure interventions from degradation.	implement community-based natural resource management 1.3.1.2a: Training workshops for 8 established WUGs on efficient and effective water resources management under future climate change conditions. 1.3.1.3a: Training workshops for the 4 established community-based Watch and Ward Systems to protect the project's EbA and green infrastructure interventions from degradation.
Activity 1.3.2: Strengthen communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.	Under Activity 1.3.2, communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions will be strengthened through a tailored communications strategy. The audience will include members of vulnerable communities, particularly women and children, students and academia, policymakers, and the general public. The objectives of the communication strategy include: i) educating communities, students, tourists and other project stakeholders on the impacts of climate change, particularly floods and droughts, on people and ecosystems, as well as on how to respond to these impacts through implementing EbA, green infrastructure and efficient community-based natural resources management; ii) informing relevant stakeholders about the project's EbA and green infrastructure interventions, their benefits and the progress made; and iii) communicating the successes and impacts of the project interventions to build a case for scaling up EbA and green infrastructure interventions.	Sub-Activity 1.3.2.1: Consult with tribal elders to support the sensitization of the project's adaptation interventions in beneficiary communities. Sub-Activity 1.3.2.2: Develop and implement awareness-raising campaigns for target communities to increase their knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.	1.3.2.1a: Annual consultation reports 1.3.2.2a: Annual awareness-raising campaigns communicated through, among other methods, digital and print, workshops, videos, pictures, and farmers' field days.

Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.

Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan.	The evidence will be generated under the project by monitoring and analysing the implementation and cost-effectiveness of EbA and green infrastructure interventions (described in detail in Component 1) in reducing the impacts of climate change-induced floods and droughts on vulnerable communities and ecosystems in Pakistan. The monitoring and analysis will consider: i) avoided costs and losses — impacts or losses that would otherwise occur in the absence of ecosystem services, such as increased flood extent, damages and/or losses to agricultural land and the subsequent reduced productivity, and reduced water resource quantity and quality; ii) alternative costs — the implication of implementing ‘grey’ artificial systems (as opposed to ‘green’ ecosystem-based adaptation solutions) that would be required to deliver the same adaptation benefits as ecosystems; and iii) a wide variety of sustainable development co-benefits, including but not limited to enhanced biodiversity, decreased pollution of water resources and improved access to clean water. The evidence generated under this activity will be used under Activity 2.2.1 for the development of updated procedures under Output 2.2 for Pakistan’s key regulations and policies for flood and water resources management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions	Sub-Activity 2.1.1.1: Prepare baselines and indicators for the biophysical benefits of EbA on ecosystems and their flood attenuation and water resource-related services. Sub-Activity 2.1.1.2: Monitor the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services using the baselines and indicators prepared under Sub-activity 2.1.1.1. Sub-Activity 2.1.1.3: Conduct an economic valuation of the benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services. Sub-Activity 2.1.1.4: Identify additional opportunities and sites at the national level across Pakistan for scaling up and replicating the project’s EbA and green infrastructure interventions for flood risk and water resource management. Sub-Activity 2.1.1.5: Collate and store the data and information generated under 2.1.1.1–2.1.1.4	2.1.1.1a: 1 report on baselines and indicators 2.1.1.2a: 1 Monitoring reports of the biophysical benefits of EbA and green infrastructure on ecosystems and their flood attenuation and water resource-related services using the baselines and indicators prepared under Sub-activity 2.1.1.1. 2.1.1.3a: 1 economic evaluation report 2.1.1.4a: 1 report identifying sites for scaling up and replicating EbA interventions 2.1.1.5a: 1 centralized data and knowledge management system is in place at the organization level
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	to reduce the impacts of floods and droughts in Pakistan.		
Activity 2.1.2: Develop and disseminate an evidence report to national and provincial government for implementing EbA and green infrastructure interventions in response to flooding and droughts in Pakistan.	The data and information generated under Activity 2.1.1 will be collated and synthesised into an evidence report, highlighting EbA and green infrastructure interventions' flood and drought risk reduction potential, economic value and environmental and social benefits. This report will: i) be used under Activity 2.2.1 to develop updated procedures for Pakistan's key regulations and policies for flood and water resources management on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan; and ii) stand alone as an important knowledge product and decision-making tool for policymakers and planners in Pakistan to use in other decision making around climate change planning and flood and water resources management. WWF-Pakistan will work with FFC to disseminate this evidence report to the MoCC, the Provincial Irrigation Departments, Provincial Agriculture Departments, Provincial Forestry Departments, academic and research institutions, as well as other relevant national and provincial stakeholders.	Sub-activity 2.1.2.1: Prepare an evidence report on the use and benefits of EbA and green infrastructure interventions in response to flooding and droughts in Pakistan. Sub-activity 2.1.2.2: Disseminate the evidence report developed under Sub-activity 2.1.2.1 to national and provincial planners and regulators involved in flood and water resources management, as well as academic and research institutions.	2.1.2.1a: 1 evidence report 2.1.2.2a: Evidence report disseminated to 41 institutions — Federal (3) and Provincial line departments (5 x 3 = 15), Academia (12) and Research Organizations (6).
Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.			
Activity 2.2.1: Develop updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan and four Provincial Adaptation Plans for implementing EbA and green infrastructure interventions in Pakistan.	Using the evidence from Activity 2.1.1, a National Working Group will be established under the project, chaired by the MoCC and comprised of regulators within the MoWR and its Executive Department, the FFC, as well as the relevant Provincial Departments. This working group will lead the development and adoption of updated procedures for Pakistan's key regulations and policies for flood and water resources	Sub-Activity 2.2.1.1: Establish a National Working Group to develop and adopt updated procedures for using EbA and green infrastructure interventions in flood and water resources management in Pakistan. Sub-Activity 2.2.1.2: Review the Implementation Framework of the National Water Policy, National Adaptation Plan and	2.2.1.1a: 1 National Working Group established 2.2.1.2a: Report including identified gaps and opportunities for the consideration of EbA and green infrastructure as cost-effective adaptation interventions

	management — namely the Implementation Framework for the National Water Policy, the National Adaptation Plan (NAP), and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. This will lead to the creation of well documented procedures backed by a rigorous scientific base that enables the GoP to channel future investments into EbA and green infrastructure interventions instead of grey infrastructure to respond to increasingly severe floods and droughts caused by climate change.	four Provincial Adaptation Plans for the four provinces of Khyber Pakhtunkhwa, Punjab, Balochistan and Sindh to identify entry points for EbA and green infrastructure as cost-effective adaptation interventions for addressing floods and droughts. Sub-Activity 2.2.1.3: Draft updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. Sub-Activity 2.2.1.4: Validate the updated procedures within the National Working Group for adoption into the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans.	2.2.1.3a: Report with a draft of the updated procedures for the Implementation Framework of the National Water Policy 2.2.1.3b: Report with a draft of the updated procedures for the National Adaptation Plan, and four Provincial Adaptation Plans 2.2.1.4a: Final updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans drafted and submitted to the National Working Group for endorsement and adoption. 2.2.1.4b: Validation consultations report 2.2.1.4c: Validated set of final updated procedures for the Implementation Framework of the National Water Policy, National Adaptation Plan, and four Provincial Adaptation Plans drafted and submitted to the National Working Group for endorsement and adoption.
Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.			
Activity 2.3.1: Strengthen the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated procedures under Activity 2.2.1 and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management.	Capacity-building interventions under this activity will include: i) developing a national pool of experts for scaling up EbA through investments in curriculum development, academic programmes, training of trainers approaches, and exposure visits; ii) training national and sub-national technical staff involved in flood risk and water resource management to implement EbA and green infrastructure interventions, including operations and maintenance; and iii) training local communities on the impacts of climate change, benefits of EbA and green infrastructure interventions and how to operate and maintain these interventions.	Sub-Activity 2.3.1.1: Train and raise awareness of national and sub-national policymakers and planners on the contents and application of the updated procedures developed under Activity 2.2.1. Sub-Activity 2.3.1.2: Train national and sub-national technical staff on the implementation, monitoring, operations, and maintenance of EbA and green infrastructure interventions. Sub-Activity 2.3.1.3: Train local community leaders/representatives on EbA and green infrastructure interventions and their benefits, as well as on the implementation,	2.3.1.1a: 80 national and sub-national policymakers and planners trained 2.3.1.1b: 1 training report 2.3.1.2a: 120 national and sub-national technical staff trained 2.3.1.2b: 1 training report 2.3.1.3a: 120 local community leaders/representatives trained 2.3.1.3b: 1 training report

		monitoring, operations, and maintenance of these interventions.	
Output 3.1: Pipeline of feasible climate-resilient businesses.			
Activity 3.1.1: Develop a pipeline of sustainable climate-resilient businesses.	Under this activity, due diligence and detailed needs assessments will be conducted over the existing DFCD Project identified businesses, and business development plans will be prepared for each business. Additionally, technical assistance, incubation, business support will be provided to each business based on their individual needs, with further support provided to these businesses in closing financial agreements with investors, including with the DFCD investment window. This activity will contribute to the project's efforts to enhance community resilience by providing important case studies for business models which create positive environmental impacts while creating economic resilience for vulnerable communities and business owners.	Sub-Activity 3.1.1.1: Build on the initial study completed by WWF-Pakistan identifying bankable project opportunities in the Indus Basin in the agriculture, forestry, and water and sanitation sectors. Sub-Activity 3.1.1.2: Formulate technical assistance and business development plans for shortlisted climate-resilient DFCD businesses. Sub-Activity 3.1.1.3: Assist Bankable enterprises with financial close within the DFCD funding window (loaning application, investment pitch preparation, etc.).	3.1.1.1a: Report identifying community, micro and small business opportunities, including due diligence results 3.1.1.2a: Technical assistance reports 3.1.1.2b Business development plans 3.1.1.2c: Readiness reports for DFCD Bankable Enterprises 3.1.1.3a: Financial close documents and reports
Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.			
Activity 3.2.1: Improve the climate resilience of vulnerable agricultural livelihoods.	Pakistan's agricultural sector is one of the most vulnerable to the impacts of floods and droughts in Pakistan. This activity will respond to this vulnerability by strengthening the climate resilience of small-sale farmers and vulnerable people in the Indus Basin that depend on agriculture for their livelihoods. Specifically, WWF-Pakistan will assist the Provincial Departments of Agriculture to deliver climate-responsive extension services on measures such as: i) improved water-efficient irrigation practices; ii) farming of drought-resistant crop varieties; iii) soil management practices responsive to changing climate conditions; and iv) installation and use of water storage tanks	Sub-Activity 3.2.1.1: Strengthen the capacity of small-scale farmers and people dependent on agriculture on climate-resilient agriculture and sustainable practices. Sub-Activity 3.2.1.2: Improve the water efficiency of farming through improved farm management and irrigation systems. Sub-Activity 3.2.1.3: Introduce drought-resistant crop varieties. Sub-Activity 3.2.1.4: Improve the climate resilience of soil management practices.	3.2.1.1a: Annual training workshops for farmers, extension agents and staff from the Provincial Departments of Agriculture. 3.2.1.2a: 8 efficient irrigation systems (2 per site in the 4 project sites) 3.2.1.3a: Drought resistant crop varieties introduced on 40 ha of agricultural land 3.2.1.4a: 14 ha of agricultural land under improved climate-resilient soil management 3.2.1.5a: 15 water storage tanks installed

	for vulnerable small-scale farming households.	Sub-Activity 3.2.1.5: Provide water storage tanks to vulnerable agricultural communities.	
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

289. In addition to the AE's obligations set out in the Accreditation Master Agreement (AMA), project-specific monitoring and evaluation (M&E) will consist of the following arrangements:

M&E Plan

290. The project includes monitoring and evaluation systems to track progress towards the planned outcomes over the seven-year term of the project. A Monitoring and Evaluation Plan (M&E Plan) has been prepared for the project and is included as Annex 11 to the Funding Proposal. Fund-level monitoring and evaluation of the project will be based on the Fund-level Indicators identified in Section E.3. above.

291. The AE will require adherence with the M&E Plan and the relevant monitoring and evaluation systems described in this Funding Proposal in its Subsidiary Agreements with the EEs. As such, WWF-Pakistan will be required to comply with their obligations to the AE, including for reporting on project indicators, implementation challenges, and financial status. See Section B.4, Implementing Arrangements, for an illustration of those agreements within the institutional arrangements for the project.

Primary Responsibility for Monitoring & Evaluation

292. Primary responsibility for project M&E will rest with WWF-Pakistan, through a full-time **M&E Manager** within the PMU. WWF-Pakistan will be required to deliver reports to the AE on project indicators, implementation challenges, and financial status to allow the AE to monitor and evaluate the project, and to report to the GCF. The PMU will apply standard management tools such as work plans to monitor progress and financial reporting, as well as action plans for gender, stakeholder engagement, and environmental and social safeguards.

293. The M&E Manager will be the focal point for project-level M&E information as and when required by the AE, PSC, Project Manager (PM) and WWF-Pakistan (EE). The M&E Manager will conduct their work in accordance with GCF standards and WWF-Pakistan's M&E Manual¹⁸⁴, which describes the standard operating procedures and protocols for M&E required in all WWF-Pakistan's projects and programmes. Technical support will be provided to the M&E Manager by WWF-Pakistan's M&E Unit, which forms part of its Programme Development Department, to ensure compliance with WWF Network standards, including linking the project with the Global Goals and Outcomes of the WWF Network.

294. The M&E Manager will be the custodian of all project-level M&E data, including the development and maintenance of a centralized database for data collation, consolidation, analysis and reporting. This manager will be supported by the following personnel:

- **Three full-time Provincial M&E Officers** (project staff) will report directly to and receive technical guidance from the M&E Manager. They will develop data collection tools for the project's interventions and train the project field teams on their use. The M&E Officers will monitor the project's progress against the targets specified in the Logical Framework (see Section E1–6) and be custodians of provincial-level M&E data.
- **Full-time Environment and Social Safeguards (ESS) and Gender Managers** (PMU staff) will collect M&E data related to the implementation of the ESMF (Annex 6) and Gender Action Plan (Annex 8), respectively, under the guidance of the M&E Manager.
- **Part-time independent consultants** will conduct interim and final evaluations for the project in accordance with AE and GCF requirements.

295. During the implementation phase, specific means of verification (MoVs) will be used to assess the: i) progress of project interventions towards achieving GCF Result Areas and Outcomes; and ii) performance of interventions against the project performance indicators presented in Section E.5 of the Funding Proposal. WWFUS The M&E costs, as well as the salaries of the project management staff are presented in Annex 4: Detailed budget. The M&E process for the proposed project is described below.

296. **First level:** The project field teams will collect M&E data on site-specific activities (based on the indicators, targets and means of verification presented in Sections E.1–6 and Annex 11). The collected information will be quality assured and validated by the Provincial M&E Officers, who will conduct quarterly monitoring visits to the

¹⁸⁴ Monitoring and Evaluation Manual. 2012. Monitoring and Evaluation Unit, Programme Development Department, WWF-Pakistan.

project sites and develop site-specific quarterly internal monitoring reports. M&E data as well as their sources will be validated by the M&E Manager who will conduct biannual field monitoring visits to the project sites.

297. **Second level:** The M&E Manager will consolidate the findings of site-specific quarterly monitoring reports (delivered by the Provincial M&E Officers) and present them in quarterly planning and progress review meetings. The M&E Manager will facilitate these meetings, which will be attended by the PM, ESS and Gender Managers, and Project Coordinators (based in the SIUs). During these meetings, the ESS and Gender Managers will present updates on the implementation of the ESMF (Annex 6) and Gender Action Plan (Annex 8), respectively. The M&E Manager will coordinate with provincial M&E Officers and ensure updated information is shared timeously with relevant project partners. The M&E Manager will also organize at least one reflection session per year with the PM, EEs and relevant technical project staff to assess the project monitoring data against the project's Theory of Change and Logical Framework to adaptively manage the project.

298. **Third level:** In accordance with GCF requirements, the M&E Manager will compile Annual Progress Reports (APRs) supported by project M&E staff, and ESS and Gender Managers. First, these reports will be submitted to WWF-Pakistan's M&E Unit (based at the Head Office) for clearance, with the findings presented at quarterly WWF-Pakistan's internal Programme Support Unit (PSU) meetings (WWF-Pakistan's PSUs will meet quarterly for project progress reviews and technical oversight.) After these internal presentations, the APRs will be submitted to WWF-Pakistan's Project Review Committee (PRC) for review and clearance. Following this review, the APRs will be shared with the AE as per agreed frequencies mentioned in the signed project agreements. The APRs will explain the progress of the project against result indicators, implementation challenges/delays, achievements against targets and financial status.

Periodic Reports

299. The PMU will submit an Annual Performance Report (APR) to the AE by January 31 each year of implementation, consolidating information from all EEs, subgrants, and subcontracts, including a narrative report on implementation progress based on the logical framework submitted in this Funding Proposal and considerations on the ongoing performance of the project against the GCF's investment framework criteria. This narrative report will include updates on the indicators described in the GCF's Results Management Framework, and a report on safeguards, that includes updates on the GRM, including any SEAH-specific grievances; gender; and co-benefits indicators. The PMU's Project Manager and M&E Manager will be responsible for preparing the APRs, for controlling quality, and for submitting them to the AE. The PMU will share these APRs with the PSC and other relevant stakeholders.

300. The PMU will submit semi-annual technical reports to the AE, one at the mid-year and one at the end of the calendar year, to allow the AE to make its reports to the GCF. Additionally, the PMU will submit to the AE quarterly financial reports that consolidate all financial information from the other EEs, to allow the AE to make its financial reports to the GCF. WWF-Pakistan will be responsible for preparing the financial reports, controlling their quality, and submitting them to the AE. M&E reports to be produced, including details of their frequency, authors and recipients are presented in Table 22 below.

Table 22. Project-specific M&E reporting summary.

M&E Report	Frequency	Responsibility	Recipient/s
Site-specific quarterly internal monitoring reports	Quarterly	Provincial M&E Officers	M&E Manager
Mid-year technical report	Mid-year	PM and M&E Manager (reviewed by WWF-Pakistan's PRC)	AE
Annual Progress Reports, including quarterly financial reports	Annually	PM and M&E Manager (reviewed by WWF-Pakistan's PRC)	AE, NDA. The AE will submit these reports to the GCF.
Interim evaluation report	At project mid-point	Independent consultant	AE, GCF, NDA, WWF-Pakistan's M&E Unit, PMU

Final evaluation report	At project end point	Independent consultant	AE, GCF, NDA, WWF-Pakistan's M&E Unit, PMU
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AE Monitoring and Evaluation

301. The AE will carry out the following project-level monitoring and evaluation activities:

- a. The AE will maintain and comply with an adequate system to monitor the performance of the EEs and contractually cause regular reporting from them in the Subsidiary Agreements in accordance with the AMA.
- b. The AE will carry out an inception workshop to: (a) inform project stakeholders of the project strategy and discuss any changes in the overall context that influence project implementation; (b) discuss the roles and responsibilities of the project team, including reporting and communication lines; (c) review the results framework, discuss reporting, monitoring and evaluation roles and responsibilities, and finalize the M&E plan; (d) review financial reporting requirements and procedures, and agree on the arrangements for the annual audit; (e) agree on templates and the timelines for technical and financial reporting with the PMU and executing partners; and (f) finalize the first year's work plan.
- c. The AE will carry out annual supervision missions during which, among other things, it will review the previous Annual Performance Report with stakeholders including the EEs and the NDA. During these missions, the AE will conduct workshops with the EEs, the NDA, and other stakeholders to (a) review whether the project strategies are having the expected results according to the project theory of change; (b) analyse risks and assumptions that hinder project success, to discuss modifications to make the project more efficient and effective; (c) discuss lessons from the past year(s) of project implementation; and (d) evaluate the project's gender responsiveness and application of social and environmental safeguards.
- d. The AE will submit a copy of the APR to the NDA by the end of February each year.
- e. During the project, based on the PMU's reporting to the AE described above, the AE will submit to the GCF Annual Performance Reports, including financial management reports, which will include, among other things, the dates and amounts disbursed for each funded activity and compliance with financial covenants.
- f. The AE will also provide to the GCF on an annual basis (a) a self-assessment of compliance with the GCF's Fiduciary Principles and Standards, Environmental and Social Safeguards, and Gender Policy; and (b) a report on its actions carried out or planned to be carried out to strengthen the capacities of, or otherwise support, potential subnational, national and regional entities.
- g. The AE will arrange and contract for independent interim and final evaluations that will contain the information described in Section 15.02(b) of the AMA and will apply the relevant GCF and AE policies identified in the AMA. The AE will, in collaboration with the PMU, prepare a formal management response to the findings of the independent evaluations, and will provide the evaluation reports and the management response to the GCF.
- h. The interim evaluation will take place at the mid-point of project implementation. This mid-term evaluation will evaluate progress towards the achievement of outcomes and will suggest corrective actions if needed. The findings of the interim evaluation and responses in the management response will be incorporated as recommendations for enhanced implementation during the second half of the project.
- i. The final evaluation will be submitted six months after the project's completion date. It will aim at identifying project outcomes, their sustainability, and future actions needed to assure continuity of these outcomes.
- j. Within three months of the project's completion date, the PMU will prepare the Project Completion Report for the AE's review and revision, and the AE will submit the final report to the GCF. This comprehensive report will also be made available to the public. It will summarize the results achieved (objectives, outcomes, outputs), lessons learned, problems met, and areas where results may not have been achieved. It will also provide recommendations for any further steps that may need to be taken to ensure sustainability and replication of the project's results.

E. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/program objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistle-blowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>

Description

The project includes a new mechanism for coordination among national agencies, provincial agencies, and non-government members through the National Working Group, created under Sub-Activity 2.2.1.1, to operationalize EbA and green infrastructure in Pakistan's Water Policy Implementation Framework, NAP, and Provincial Adaptation Plans.

Mitigation Measure(s)

The NWG will be established to lead the development and adoption of updated procedures for Pakistan's Implementation Framework for the National Water Policy, the NAP, and Provincial Adaptation Plans of Sindh, Khyber, Balochistan, and Punjab provinces — on the proper execution of EbA and green infrastructure interventions to reduce the impacts of floods and droughts in Pakistan. This working group will also ensure these procedures align with current national and provincial climate change planning frameworks. The NWG will include representation from the MoCC, MoWR (including FFC), the Provincial Departments of Irrigation, Forestry and Agriculture local NGOs, CSOs, and academic and research organisations¹⁸⁵. The NWG members will be selected based on: (a) their subject matter expertise and capacity to influence government regulation and procedures, and/or (b) position within government and decision-making authority or influence to change government regulation and procedures.

The project Steering Committee (PSC) and Provincial Oversight Committees (POCs) will be critical in facilitating participation within the NWG, securing government and community buy-in, and driving approval/decisions at the national and provincial levels related to adopting an integrated approach to flood and water resources management under the project. The Project Manager will coordinate with the PSC and the NWG closely to ensure the timely formation of the NWG, selection of its members, routine scheduling of NWG meetings, progress on the review of recommended amendments to the Implementation Framework for the National Water Policy, the NAP, and Provincial Adaptation Plans following the implementation timetable in Annex 5.

Selected Risk Factor 2

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Extreme climate events, such as floods, may disrupt project implementation. Physical accessibility to the sites could be constrained for up to six months. The government's reprioritization on disaster recovery could reduce their attention and resources for implementation of project activities.

Mitigation Measure(s)

The project implementation plan (Annex 5) has been designed to accommodate potential delays in implementing project activities due to climate events. The Project Manager will be responsible for coordinating adaptive management planning and response with the PSC and POC.

Selected Risk Factor 3

Category	Probability	Impact
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¹⁸⁵ Such as the Pakistan Council for Research in Water Resources (PCRWR), Pakistan Agriculture Research Council (PARC), Lahore University of Management Sciences (LUMS), National University of Science and Technology (NUST), and University of Engineering and Technology (UET).

<u>Technical and operational</u>	<u>Medium</u>	<u>High</u>
Description		
The communities in DI Khan and Chakkar Lehri have high levels of autonomy from the national and provincial governments in Pakistan. Due to their special status, the communities' Tribal Elders and leadership may not maintain their commitment to participate in the project if engagement wanes		
Mitigation Measure(s)		
WWF Pakistan has engaged with the Tribal Elders in Di Khan and the Chakkar Lehri and secured letters of commitment to participate in the project, included in Annex 6. The ESMF (Annex 6) and SEP (Annex 8) have established both safeguard and stakeholder engagement procedures which take into account local governance status. During implementation, activities in DI Khan and the Chakkar Lehri will be executed through local Site Implementation Units which will work with communities on the project activities included in Section B.3 following the agreed upon engagement mechanisms defined in Annexes 6 and 7.		
Selected Risk Factor 4		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>High</u>
Description		
The reforestation and revegetation activities under Output 1.1 may require limiting access to certain areas during planting and early maturation of saplings. Limiting access for local peoples to community areas which they traditionally access for food, fodder, and other uses could lead to wavering support for the project and ultimately disrupt activity delivery.		
Mitigation Measure(s)		
During development of the project ESMF, SEP, and GAP communities have been fully briefed on the project activities including the need to limit access to certain areas during activity execution. Community leaders have provided their approval of the project and will be essential partners for communicating the project's long-term benefits to their communities, and the need for temporary limitations to achieve these benefits. The ESMF and SEP outline measures to accommodate communities during execution of the project for any restrictions which may arise. During execution of the project and before planting begins, WWF Pakistan will use the Safeguard Eligibility and Impacts Screening, which can be found in Annex 6 to determine if any restrictions are necessary. The screening will assess community dependency on the project areas, and work with communities to create Alternative Livelihood Plans that are mutually agreed upon.		
Selected Risk Factor 5		
Category	Probability	Impact
<u>Forex</u>	<u>Medium</u>	<u>Medium</u>
Description		
Foreign exchange risk could affect the financial performance of the project. Significant fluctuation of exchange rate between the PKR to USD has been observed. In March 2022 1 USD = 178 PKR, whereas during July 2022 it went up to 1 USD = 239 PKR. Furthermore, the PKR gained 10% against the USD on Sept 1, 2022, from PKR = 239 to PKR = 219. Depreciation or appreciation of currencies could lead to under or overspending of the budget.		
Mitigation Measure(s)		
A clause in the agreement should be added that 'In case of significant exchange gain/loss, the implementer should immediately inform the donor and the donor will take appropriate action by mutual agreement with the implementer within the approved financing amount . The expected change in foreign exchange rates can also be analysed and incorporated into the project budget.		
Selected Risk Factor 6		
Category	Probability	Impact
<u>ML/FT</u>	<u>Low</u>	<u>Medium</u>
Description		
Money laundering or terrorist financing leads to improper use of property, assets, or GCF funding. Project actors fail to comply with all applicable anti-money laundering and countering financing of terrorism laws.		
Mitigation Measure(s)		
- As part of its financial and operational due diligence, the AE has assessed WWF-Pakistan in the areas of financial management, internal controls, accounting, human resources, procurement and procurement systems, equipment management, and the ability to comply with donor requirements, and will assess whether their overall policies and procedures, experience, and level of supervision proposed for this project are sufficient to mitigate any potential risks or vulnerabilities related to money laundering, terrorist financing, or prohibited practices. - The AE's Subsidiary Agreements with the EE will require compliance with the AE's and the GCF's anti-money laundering and anti-terrorist financing policies described in the Funded Activity Agreement and AMA and will flow		

down remedies for non-compliance. The Subsidiary Agreement will require that the EE in turn flow down those provisions to all project subrecipients.

- Before submission of the project to the GCF Board, the AE will compare the name of the EE and the known executing partners against UN and US sanctions lists, including sanctions established by the United Nations Security Council; sanctions administered by the Office of Foreign Assets Control within the U.S. Department of Treasury pursuant to the Global Terrorism Sanctions Regulations, the Foreign Terrorist Organizations Sanctions Regulations, and Counter Narcotics Trafficking Sanctions. At the time of submission of this proposal the AE is not aware of any individual or entity expected to receive project funds or material support or resources that is included on those lists.
- The following mechanisms will be available for the reporting of complaints and allegations of impropriety, wrongdoing or other related issues in the project and its activities (i.e., whistle-blower programs): [WWF's whistleblower system](#) is open to staff, partners, communities, and other stakeholders to report suspected illegal or inappropriate activity, or concerns about the implementation of WWF projects. This system is an online and phone mechanism hosted by a third-party provider, [WhistleB](#), which can receive reports online or by phone in multiple languages. All complaints submitted through the mechanism are investigated.

Selected Risk Factor 7

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>High</u>
Description		
In Pakistan, there are both external and internal threats from terrorist activity related to Taliban activities in neighbouring Afghanistan spilling across the border, as well Taliban groups within Pakistan and separatist movements in the Balochistan region. Although no instances of terrorism have been reported in the project areas, possible future terrorism activities, especially in DI Khan, may lead to disruption of project activities.		
Mitigation Measure(s)		
• A conflict Sensitivity analysis has been prepared for the project, included under Annex 6, which outlines security measures for the project. These include engagement with security forces in the target provinces and detailed measures in the case of any security threats, including immediate withdrawal from project areas where security threats are identified. • Additionally, the Stakeholder Engagement Plan in Annex 7 outlines how the project will work with Tribal elders in the DI Khan area to build trust and cooperation. While the project will not be working with those engaged in separatist activities, close collaboration with Tribal elders in the region can act as an additional mitigation measure by providing legitimacy to the work and offering advice to the Project team on how to avoid becoming unintentionally part of any conflict.		

Selected Risk Factor 8

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Interference or delays with procurement or transportation of inputs/equipment and challenges with supply chains and labour could delay the development of green infrastructure interventions and the execution of agricultural activities requiring specific materials.		
Mitigation Measure(s)		
Local supply chains and locally available materials and labour will be used wherever possible. Similarly for other items and inputs, reputable suppliers will be engaged through a competitive procurement to ensure timely supply and transportation of inputs.		

Selected Risk Factor 9

Category	Probability	Impact
<u>Prohibited practices</u>	<u>Medium</u>	<u>Medium</u>
Description		
Corruption, fraud, or abuse leads to theft, misappropriation, waste, or improper use of property, assets, or GCF funding.		
Mitigation Measure(s)		
As part of its financial and operational due diligence, the AE has assessed WWF-Pakistan in the areas of financial management, internal controls, accounting, human resources, procurement and procurement systems, equipment management, the ability to comply with donor requirements, and whether their overall policies and procedures, experience, and level of supervision proposed for this project are sufficient to mitigate any potential risks or vulnerabilities related to prohibited practices.		
The following controls are designed to assure that any materials or technology procured under this project are used only for the purposes intended and are not diverted or misused for unauthorized, improper or illicit purposes: The		

covenants and warranties in the AE's Subsidiary Agreements with the EE require, among other things, (a) that GCF funds are not used by the EE, or any entity to whom the funds are disbursed, for any illegal or improper purposes, including by incorporating in subrecipient and subcontractor agreements provisions corresponding to the EE's own rules, policies, and procedures to comply with the GCF Policy of Prohibited Practices, (b) compliance with the GCF Policy of Prohibited Practices; and (c) compliance with the Recharge Pakistan Operating Manual.

In the case of disbursement of cash, vouchers, commodities, or other items of value directly or indirectly to individual beneficiaries, agreements will be signed with the relevant beneficiary or service provider that includes guidelines for distribution and reporting including proof of expenses and receipts. If that is not possible in some rural areas, it may be replaced by a declaration on the use of the funds that includes expense details. Administrative monitoring, including physical verification as appropriate, will be included in the relevant EE's roles and responsibilities.

Selected Risk Factor 10

Category	Probability	Impact
<u>Prohibited practices</u>	<u>Low</u>	<u>Medium</u>

Description

Conflict arises between communities related to water management and planning and land use.

Mitigation Measure(s)

The mitigation measures include improved governance over water resources through the strengthening of CBOs and the formation of water user groups charged with the allocation of water among communities for household and agricultural use and resolving conflicts related to water access and use rights. As part of this process, Rod Kohi, the traditional method for managing water use and rights will be strengthened among project stakeholders. Additionally, as part of the ESMF requirements, the project-level screening in Appendix 1 of Annex 6 will be used to screen for community-based conflict potential, and any additional mitigation measures needed will be decided in consultation with the relevant communities. The screening will be supported by WWF-Pakistan's FPIC process, which will be operationalised during project implementation to obtain the required consent and participation from tribes, particularly for project activities focused on water resources management. A conflict sensitivity assessment has also been prepared as Appendix 5 of Annex 6.

Selected Risk Factor 11

Category	Probability	Impact
<u>Prohibited practices</u>	<u>Low</u>	<u>Medium</u>

Description

WWF Pakistan will now also be responsible for executing Activity 1.2.1 following the removal of MoWR as an EE on the project. This could increase the risk of project management and execution challenges for WWF Pakistan, such as delays in activity delivery.

Mitigation Measure(s)

The AE conducted technical, financial, and operational due diligence over the EE, WWF Pakistan, to evaluate its capacity and to determine whether it has the financial ability, controls and policies and procedures to ensure the adequate financial management of GCF funds. Based on the findings of this assessment (Annex 9), the AE is assured that WWF Pakistan has the capacity, policies and procedures, systems and experience to execute the Recharge Pakistan project and provide adequate stewardship of GCF funding.

WWF Pakistan has included finance and operational teams to manage the service providers under the project within the PMU and the site implementation units including: an Operations and Finance Manager, a Contracts Manager, a Contracts Officer, and three Senior Finance Officers, one for each of the site implementation units.

F. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

302. The project will comply with GCF's Environmental and Social Policy (ESP), including the Policy on the Prevention and Protection from Sexual Exploitation, Sexual Abuse, and Sexual Harassment (SEAH) and Indigenous Peoples Policy (IPP) through application of WWF's Environmental and Social Safeguards Framework, as detailed in the Safeguards Integrated Policies and Procedures (SIPP) and Guidance Note on GBV & SEAH. The project has been screened as Category "B" given that this is a climate change mitigation and adaptation initiative, designed to create integrated approaches to flood and drought risk management in the Indus Basin while enhancing the resilience of vulnerable rural communities to climate change and enabling a paradigm shift towards Ecosystem-Based Adaptation in Pakistan. It is expected to generate significant positive and durable social,

economic, and environmental benefits. Any adverse environmental and social impacts are expected to be minor and site specific and can be mitigated.

303. An Environmental and Social Management Framework (ESMF) (Annex 6), including an Indigenous Peoples Planning Framework (IPPF), a Process Framework (PF), and a Grievance Redress Mechanism (GRM) has been prepared to define procedures for managing the project activities' potential environmental and social risks and impacts. Additionally, the project will have an ESS Manager within the PMU, 3 ESS/ M&E Sr. Officers at the landscapes levels and a male and female Community Engagement specialist at each landscape to manage ESS risks and ensure the project complies with all policies and mitigation measures. This is a complex project in a complex country and the level of staffing for ESS reflects that.
304. The proposal formulation process has involved consultations with a variety of stakeholders including communities, national and local governments, public entities and universities using a gender-responsive approach, to ensure equitable and meaningful participation from women and men.
305. A summary of the ESS risks and mitigation measures is given below, divided by the Standards found in WWF's SIPP. For full details on the environmental and social risks and mitigation measures, please refer to Section 4 in Annex 6 of the FP.

Standard on the Protection of Natural Habitats

306. The project-related activities associated with this standard include re-vegetation of riverside and wetland habitat, excavation of flow paths and water channels, creating water retention areas in river systems and de-silting of riverbeds. Mitigation measures include ensuring the use of native species adapted to the ecosystems being re-vegetated, preparing a proper plan for the excavation and dumping of excavated material, and where possible / mentioned, using the debris from de-silting as the building block for retention areas to avoid leaving it elsewhere in the landscape.

Standard on Restriction of Access and Resettlement

307. There will be no land acquisition or involuntary resettlement under the project. This standard is triggered because there is the possibility of access restriction to fishing or use changes in water access if unmitigated. A Process Framework has been prepared as part of the ESMF and a stakeholder engagement plan (Annex 7) has been prepared. Any restrictions on use or access will only come with the full and effective participation of communities and their leaders. In the case of a few project areas, an FPIC Process as outlined in the IPPF will be followed as well.

Standard on Indigenous People

308. This standard is triggered because there are indigenous and tribal peoples in several project landscapes- the Indigenous Mohana peoples of Lake Manchar, and Kehal, Mor, and Pakhtoon tribes in the DI Khan area. An Indigenous Peoples Planning Framework has been prepared as part of the ESMF. Once project implementation begins and landscape-level ESS Screenings have been conducted, IPPs will be created through an FPIC process in partnership with the relevant communities. Stakeholder engagement has already begun in these areas during the FP stage, and tribal elders have expressed their interest in participating in the project. More details about the consultation process to date and stakeholder engagement during implementation can be found in Annex 7: Stakeholder Engagement Plan.

Standard on Community Health and Safety

309. This Standard is triggered due to activities involving small-scale construction works involving pond retention areas, gabion bunds used to stabilize riverbeds and banks, de-silting of riverbeds and wetland restoration. These will take specialized knowledge and labour and the Project will use a competitive process to hire environmental engineers with a strong safety record. The selected firm will be required to comply with WWF's guidance note on Labour and Working Conditions.

Standard on Cultural Heritage

310. This Standard has been triggered on a precautionary principle only, as there are a few shrines of cultural significance in the Kaha region, although they are not anticipated to be within the range of project activities. The ESS landscape screening during implementation will verify this, and any project activity that may impact those shrines will be relocated.

Grievance Mechanism

311. The gender and SEAH-responsive project-level grievance mechanism as detailed in Section 5.8 of Annex 6 will be implemented within the first 3 months of implementation. WWF-Pakistan's (EE) grievance mechanism and WWF US's (AE) grievance mechanism will be available throughout the project lifecycle, and accessible to stakeholders and project-affected peoples. The GCF's IRM will also be socialized.

Country-level Conflict

312. Country-level threats including ongoing conflict along the Afghanistan border and internal threats within some tribal areas in DI Khan and Balochistan have been addressed in F.1. However, there are also inter-tribal threats that will require extensive stakeholder engagement with tribal elders, and close collaboration with security officials to maintain the safety of project staff and ensure project success. More details can be found in Section 4.2 and Appendix 5 of Annex 6.
313. The final ESMF (including IPPF and PF) and the SEP will be translated into Urdu, Pashto, Balochi, and Sindhi before GCF Board Submission allowing for the 30-day public disclosure required by GCF's Information Disclosure Policy and final documentation will be disclosed in country in a locally accessible manner for 45 days per WWF policy before AE approval.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

314. **Gender Assessment:** A detailed Gender Assessment was conducted on gender-related issues in the project sites to understand the gender situation in Pakistan in general and more specifically in the project sites as it pertains to the proposed project's objectives. To ensure gender is given due consideration and is integrated into the design and implementation phases of the project, a Gender Action Plan (GAP) has been developed (see the Gender Assessment and Action Plan, Annex 8).
315. The framework for the Gender Assessment was based on an "Assessment Grid" containing 26 questions, outlining the secondary and primary data requirements. A mixed method approach (qualitative and quantitative research techniques) was adopted to conduct the gender assessment and to collect gender-specific data to identify gaps, opportunities, challenges, and to provide an analysis of the prevalence of Gender-based Violence (GBV) and Sexual Exploitation and Abuse and Sexual Harassment (SEAH's) risks. This was to understand the roles women and men play at various levels and the decision-making processes among them at the community level; access and control over the management of natural resources and the types of resources; and access to land by women and men in the project sites. The following methods were used for the collection of primary data: 311 household surveys, 14 focus group discussions (FGDs) with male and female respondents of the local communities, and 10 key informant interviews (KII) with government officials were conducted to assess the gender situation of each project site. To collect the secondary data, a literature review / desk study was conducted of the academic literature and development sector reports covering legal frameworks, laws, policies, institutional practices on gender equality, and women's empowerment in the wider socio-cultural context. The main documents used for the Gender Analysis were the National Gender Policy Framework 2022; the Climate Change Policy 2012; the proposed GCF Funding Proposal; the GCF Gender policy; and the WWF Gender Policy and Annual Reports 2019, 2020, and 2021.
316. The following is a summary of the gender assessment:
- Gender equality is provided in various forms in legal documents, including the Constitution and international treaties signed by Pakistan. However, the position and status of women in Pakistan have historically been inconsistent. Over the last three decades, the status of women has progressed, remained static, regressed, and is constantly changing depending on many factors that are beyond women's control. Numerous laws have been promulgated to ensure women's rights and protection, but their implementation and lack of political will render them ineffective. Therefore, women have generally not benefited from these declarations. As Pakistan is a patriarchal society, men are consistently given favour where women are given an inferior status before or at birth, which leads to their lower human development and more unfavourable socioeconomic circumstances, especially in rural areas.
 - Communities in the project sites have been facing the brunt of climate change with their livelihoods being adversely affected. Pakistan is among those countries that are most vulnerable to the risks associated with climate change. In the focus group discussions, the impact of climate change was explored, and it was reported that there are changes in weather patterns and that different diseases including heart diseases, skin and stomach diseases are more prevalent now in the communities especially among women and children. The rapid changes in weather and the feared migrations associated with these changes, poverty and loss of

livelihoods are a reality to different degrees in all four project sites. Therefore, it can be clearly summed up that climate change has already negatively affected the equation and balance between people and nature.

- Women lack exposure, confidence, knowledge, skills, information about opportunities, and play a limited role in politics, policies and programs regarding their aspirations and lives. Formal institutions fall short of facilitating women's access to human, financial and physical capital i.e. education, skills, health facilities, loans and physical assets. Informal institutions (family, community, religion, segregation, marriage) define and affirm norms that give precedence to marriage/child marriage over education and employment, prescribe gendered roles, and condone domestic violence. Laws in place to support gender equality and women's empowerment have therefore not had the expected impact.
- In the project areas, women literacy level is low as compared to men. Women are fully responsible for livestock including goats and chickens and are reported to fulfil all tasks in the management of livestock. Women grow food for their families and have a role in some part of the tasks involved in field crops. However, women generally do not earn wages or cash incomes from these or other tasks that they perform. They also have limited decision-making power over natural and management resources including crops, livestock and associated household income.
- No report or data sets under the title SEAH (sexual exploitation, abuse and harassment) were found for Pakistan. Therefore, it was determined that in Pakistan Sexual Exploitation, Abuse and Harassment (SEAH) are subsumed under the laws and terms of Gender Based Violence (GBV), 'honour killing' and 'domestic abuse'. The assessment showed that the situation in Pakistan regarding GBV and SEAH is alarming, and it exists at different levels and at varying degrees in all provinces. The focus group discussions revealed the existence of various forms of GBV in the project. It can also be assumed that within the ambit of GBV, sexual exploitation, abuse, and harassment also exist. Even though direct questions were not asked about SEAH and GBV (investigating issues related to GBV and SEAH is difficult, as it is a sensitive issue and, in most cases, men usually deny the existence of any aspects of GBV, whereas women recognize it and subtly acknowledge it as part of traditional family norms) physical violence was reported by female respondents.

317. **Gender Action Plan:** The results obtained through the gender assessment and the stakeholder consultations during project design and development have been used to produce a detailed GAP for the project (see Gender Assessment and Action Plan, Annex 8) that covers the following:

- Equitably and inclusively strengthening and building the capacity and resilience of the vulnerable communities towards climate change. It also focuses on promoting gender inclusiveness in the design and implementation of project activities.
- Builds on the differences in tasks and interests, implementing trainings for women at separate locations, as appropriate for each site, that focus on subjects that are specifically tailored to women's interests, roles and responsibilities. These include raising awareness, providing capacity building and various skills such as leadership and entrepreneurship, of the vulnerable groups, addressing constraints and issues and aiming to improve livelihood activities at the project sites.
- Ensures gender mainstreaming processes actively involve men and women at different levels of management, who are trained in the subject, including how to recognize and address GBV and SEAH through the GRM reporting mechanism.
- Strengthening institutional capacity of local Government departments and extension services for gender orientation, sensitization and mainstreaming for planning and implementing project interventions.

318. The GAP's implementation and monitoring will be the responsibility of the Gender Manager hired in the PMU, and supported by the three Coordinator, Gender Empowerment and Livelihood Development, one in each landscape. This staff will work closely with the community mobilizers (male and female), ESS Manager and M&E Manager to provide the necessary support and capacity building to staff, partners and stakeholders to ensure the gender mainstreaming goals of the project are achieved. Ultimately, the Project Manager will be responsible for leading his/her whole team for proper implementation of the GAP as a cross-cutting theme in the project.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

319. WWF grants management and subrecipient monitoring is supported by a grants management system that is integrated across its donor management, accounting, and budgeting systems. This system provides notifications for due dates of deliverables (from the grant recipient and to the donor), tracks disbursements, project expenses, milestones, audit findings (if applicable) and identified risks so that project supervision as a whole is informed and documented.

320. During project implementation, WWF will provide oversight and quality assurance in accordance with its policies and procedures, and any additional specific requirements contained in the subsidiary agreement (in the form of a grant agreement). This may include, but is not limited to, monitoring missions, spot checks, facilitation, and participation in project steering committee meetings, quarterly progress and annual implementation reviews, and audits at project level on the resources received from WWF.
321. WWF will advance funds to WWF-Pakistan, (for the implementation of agreed and approved project activities), through quarterly disbursements based on spending projections included with the quarterly financial reports, in accordance with WWF standard grants management policies. Consolidated project expenses will be reported semi-annually to the GCF (at the mid-year financial report and the Annual Performance Report (APR)). A dedicated project account will be setup by WWF-Pakistan to receive these disbursements, and any interest accrued during the project will be reflowed to the GCF, as will any unused funds at the time of the project's financial close. A statement of Investment Income earned on GCF Proceeds, as well as the amount of such Investment Income paid to the Fund for each calendar year of the project will be submitted annually by March 30 along with an unaudited financial statement per WWF's AMA. GCF.
322. All projects are audited annually following the WWF project audit guidelines. The project will have two annual audits: (1) WWF engages Grant Thornton (an internationally recognized audit firm, who was identified from an RFP) to conduct the annual audit of GCF-funded activity contained within its accounts; and (2) WWF will require that WWF-Pakistan engage an internationally recognized audit firm (approved by the AE) to complete an annual audit on all activities funded by GCF. A scheduled audit is used to determine whether the funds transferred to the Executing Entity were used for the appropriate purpose and in accordance with the approved project work plan and budget and the EE's assessed policies and procedures. The annual project audit, based on the calendar year will be submitted to the GCF by June 30 for each year of execution.
323. *For this project, the PMU will submit reports, annual work plans, budgets and annual procurement plans to the AE. WWF-Pakistan will submit requests for disbursements. WWF will then submit Annual Performance Reports (APRs) and financial reports to the GCF as defined within the AMA and FAA. Procurement will be executed according to the approved annual procurement plans. The Accredited Entity has assessed and confirmed that the procurement rules, policies and procedures of the Executing Entity are equivalent with the procurement rules, policies and procedures of the Accredited Entity to the extent and scope of its Accreditation. Furthermore, WWF-Pakistan has requirements in their procurement policies that exceed the requirements of the WWF AE. Because of this, all procurement, whether by the Executing Entity or by an Involved Third Party with the Executing Entity, will be done in accordance with the rules, policies and procedures of the Executing Entity (including by those governmental entities involved with the Executing Entity, which will comply with the applicable laws and regulations on public procurement, subject to the Executing Entity's review and final approval over such procurement and selection processes, in accordance with clauses 10.02 and 8.01 of the AMA. For the avoidance of any doubt, the Accredited Entity shall ensure that overall management, implementation and supervision of such procurement is in accordance with clause 8.01 of the AMA.) The project will also follow WWF's financial reporting templates and formats which are in line with internationally recognized reporting standards, as well as the GCF's required reporting templates.*
324. *Procurement will be executed according to the approved annual procurement plan and the EE's policies and procedures.*

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

G. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) ([template provided](#))
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan ([template provided](#))
- ☒ Annex 5 Implementation timetable including key project/programme milestones ([template provided](#))
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan ([template provided](#))
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan ([template provided](#))
- ☒ Annex 11 Monitoring and evaluation plan ([template provided](#))
- ☒ Annex 12 AE fee request ([template provided](#))
- ☒ Annex 13 Co-financing commitment letter, if applicable ([template provided](#))
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval ([template provided](#))
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information ([template provided](#))
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project

- | | | |
|-------------------------------------|----------|---|
| ☐ | Annex 19 | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity |
| ☒ | Annex 20 | First level AML/CFT (KYC) assessment |
| ☐ | Annex 21 | Operations manual (Operations and maintenance) |
| ☐ | Annex 22 | Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects) ¹⁸⁶ |
| ☒ | Annex 23 | Methodology for calculating project beneficiaries |

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

¹⁸⁶ Annex 22 is mandatory for mitigation and cross-cutting projects.



F.No. CFU/GCF/002/2015
Government of Pakistan
Ministry of Climate Change
(LG&RD Complex, 3rd Floor, Sector G-5/2, Islamabad)
(Ph: +92-51-9245585)

Islamabad, the 30th March, 2023

To: The Green Climate Fund ("GCF") Secretariat
Song Do,
South Korea.

Re: Funding proposal for the GCF by World Wildlife Fund (WWF) regarding Recharge Pakistan: Building Pakistan Resilience to Climate Change through Ecosystem based Adaption (EbA) and Green Infrastructure for Integrated Flood Risk Management

Dear Madam/ Sir,

We refer to the project titled *Recharge Pakistan: Building Pakistan Resilience to Climate Change through Ecosystem based Adaption (EbA) and Green Infrastructure for Integrated Flood Risk Management* in Pakistan as included in the funding proposal submitted by World Wildlife Fund to us on 9th March, 2023.

2. The undersigned is the duly authorized representative of Ministry of Climate Change, the National Designated Authority of Pakistan.
3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.
4. By communicating our no-objection, it is implied that:
 - (a) The government of Pakistan has no-objection to the project as included in the funding proposal;
 - (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Pakistan;
 - (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.
5. We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.
6. We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Syed Mujtaba Hussain
Additional Secretary /GCF Focal Point
Ministry of Climate Change
Pakistan

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	World Wildlife Fund, Inc. (WWF)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Pakistan
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, June 7, 2023
Language(s) of disclosure	English and Urdu
Explanation on language	Urdu and English are the official languages of Pakistan.
Link to disclosure	English: https://files.worldwildlife.org/wwfmsprod/files/Publication/file/45hryi4itz_Recharge_Pakistan_Annex_6_ESMF_Final.pdf Urdu*: https://files.worldwildlife.org/wwfmsprod/files/Publication/file/69dwt3to47_Annex_6_ESMF_Urdu_NH_16062023_Cleaned.pdf?ga=2.252200576.826075779.1686873744-360649369.1634576132
Other link(s)	WWF GCF Agency: https://www.worldwildlife.org/pages/recharge-pakistan WWF Pakistan: https://www.wwfpak.org/our_work/_recharge_pakistan/_safeguards_documents/
Remarks	An ESIA consistent with the requirements for a Category B project is contained in "Annex 6: Environmental and Social Management Framework, Indigenous Peoples Planning Framework & Process Framework".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, June 7, 2023
Language(s) of disclosure	English and Urdu
Explanation on language	Urdu and English are the official languages of Pakistan.
Link to disclosure	English: https://files.worldwildlife.org/wwfmsprod/files/Publication/file/45hryi4itz_Recharge_Pakistan_Annex_6_ESMF_Final.pdf

	Urdu*: https://files.worldwildlife.org/wwfcmprod/files/Publication/file/69dwt3to47_Annex_6_ESMF_Urdu_NH_16062023_Cleaned.pdf?_ga=2.252200576.826075779.1686873744-360649369.1634576132
Other link(s)	WWF GCF Agency: https://www.worldwildlife.org/pages/recharge-pakistan WWF Pakistan: https://www.wwfpak.org/our_work/recharge_pakistan/safeguards_documents/
Remarks	An ESMP consistent with the requirements for a Category B project is contained in "Annex 6: Environmental and Social Management Framework, Indigenous Peoples Planning Framework & Process Framework".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous People Planning Framework (IPPF), Process Framework (PF)*, Stakeholder Engagement Plan / Wednesday, June 7, 2023
Language(s) of disclosure	English and Urdu
Explanation on language	Urdu and English are the official languages of Pakistan
Link to disclosure	Indigenous Peoples Planning Framework and Process Framework: English: https://files.worldwildlife.org/wwfcmprod/files/Publication/file/45hryi4itz_Recharge_Pakistan_Annex_6_ESMF_Final.pdf Urdu*: https://files.worldwildlife.org/wwfcmprod/files/Publication/file/69dwt3to47_Annex_6_ESMF_Urdu_NH_16062023_Cleaned.pdf?_ga=2.252200576.826075779.1686873744-360649369.1634576132 Stakeholder Engagement Plan: English: https://files.worldwildlife.org/wwfcmprod/files/Publication/file/2lxmp18pfw_Recharge_Pakistan_Annex_7_SEP_Final.pdf Urdu*: https://files.worldwildlife.org/wwfcmprod/files/Publication/file/8wackds5kp_Recharge_Pakistan_Annex_7_SEP_Fin

	al Urdu.pdf? ga=2.172510890.826075779.1686873744-360649369.1634576132
Other link(s)	WWF GCF Agency: https://www.worldwildlife.org/pages/recharge-pakistan WWF Pakistan: https://www.wwfpak.org/our_work/recharge_pakistan/safeguards_documents/
Remarks	An Indigenous People Planning Framework (IPPF) and a Process Framework (PF)* consistent with the requirements for a Category B project is contained in “Annex 6: Environmental and Social Management Framework, Indigenous Peoples Planning Framework & Process Framework”. The Stakeholder Engagement Plan will also be disclosed in the same manner and means as “Annex 6: Environmental and Social Management Framework, Indigenous Peoples Planning Framework & Process Framework”. *a PF is the WWF equivalent to a RAP
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Please see below. Disclosure of the documents in locations convenient to affected peoples happened between May 18 and May 26, 2023 per GCF requirements. In-person disclosures meetings happened in May 2023 in each of the landscape at a gathering of large groups of stakeholders per WWF US requirements. See below for locations of document disclosure as well as dates of the in-person meetings.
Place	All ESS documents were disclosed to local stakeholders between May 18-26, 2023 at the following locations: • DI Khan and Ramak Watershed - Provincial Irrigation Department, Flood Division – Opposite Degree No. 1 College, Dera Ismail Khan, Khyber Pakhtunkhwa • Manchar Watershed – Provincial Irrigation Department – Barrage Colony, Hyderabad, Sindh • Chakar Lehri Watershed – Local Government Office, Circuit House, Dera Murad Jamali, Balochistan ESS Documents are also available in Urdu and English in printed form at the offices of WWF Pakistan: • Islamabad Office – Ground Floor, 3 Constitution Avenue, Pakistan Academy of Science Building, Sector G-5/2, Islamabad • Karachi Office – Bungalow no. 35-D, P.E.C.H.S, Block 6, Shahrah e Faisal, Karachi • Quetta Office – Killi Gishkori, Sabzal Road, Quetta Dates of in-person consultations to further aid in disclosure of ESS documents per WWF requirements:

	24th May 2023: Chakar Lehri Watershed – Local Government Office, Circuit House, Dera Murad Jamali, Balochistan 22nd May 2023: Manchar Watershed: Manchar Lake, Sehwan Sharif, Sindh 18th May 2023: DI Khan & Ramak: Provincial Irrigation Department, Flood Division – Opposite Degree No. 1 College, Dera Ismail Khan, Khyber Pakhtunkhwa
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 10, 2023

* Subsequent to the disclosure of this form to the Board and active observers on 9 June 2023, the following updates have been made: The “Annex 6: The Environmental and Social Management Framework, Indigenous Peoples Planning Framework & Process Framework” has been disclosed in Urdu on 17 June 2023.

** Subsequent to the disclosure of this form to the Board and active observers on 9 June 2023, the following updates have been made: The Stakeholder Engagement Plan has been disclosed in Urdu on 17 June 2023.

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP207

Proposal name:	Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management
Accredited entity:	World Wildlife Fund, Inc. (WWF)
Country:	Pakistan
Project/programme size:	Medium

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project has a high degree of country ownership. The project is one of the two high priority goals listed in Pakistan's nationally determined contribution. The project also supports the national Indus Basin Initiative.	Critical to the success of the project will be the formulation of an effective exit strategy that includes an expansion plan and the maintenance of extant activities. Strengthening the commitment to devising a scaling-up and replication strategy for the project's interventions would be advantageous to the project. The AE must affirm to the Government of Pakistan the financing arrangements for long-term operation and maintenance of the project's interventions during implementation.
Through the use of retention areas and the protection of embankments and other flood risk areas, the project uses ecosystem-based approaches to address the challenges of drought and flood. The project is pertinent to the country's plans to address long-term drought and resilience to floods.	
The initiative is founded on in-depth scientific studies, inclusive planning, hydrological assessments and climate analysis. This is a paradigm that could be replicated in other locations throughout the country. The determination of proofs of concept will be important for the development of the sustainability strategy and scaling up to and adoption in other areas in the country.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled “List of proposed conditions and recommendations”, respectively.

II. Summary of the Secretariat’s assessment

2.1 Project background

3. The objective of the project is to transform Pakistan’s approach to flood and water resources management. The focus of the project is on implementing ecosystems-based adaptation (EbA) and green infrastructure interventions on the ground in four watershed sites in the Indus Basin: Dera Ismail Khan Ramak, Chakar Lehri and Lake Manchar.
4. According to Pakistan's Global Change Impact Studies Centre, the mean annual temperature has risen in recent years, with significant increases in the Sindh and Balochistan provinces. The average temperature in Pakistan has risen by 0.6 °C over the last century, roughly in line with the global temperature rise. Beyond a rise in temperature, climate change is already having a significant impact, including changes to the magnitude and frequency of extreme flooding events, changes in mean rainfall and decreased predictability, which will lead to an increase in the frequency and magnitude of impacts such as droughts and low flows by the end of the century. Simultaneously, groundwater depletion and shifting water use patterns (primarily for agricultural and industrial purposes) will limit access for people and vulnerable communities. As climate variability affects water quantity and quality, water temperature, freshwater ecosystems, and the magnitude and occurrence of extreme weather events such as floods and droughts, building climate resilience becomes severely challenging.
5. According to future climate change projections based on all four Representative Concentration Pathway scenarios of the Fifth Assessment Report of the Intergovernmental Panel on Climate Change,¹ Pakistan's average temperature changes are projected to be approximately 1°C higher than the global average changes by 2100. Such an increase in temperature is likely to cause (1) major shifts in rainfall patterns, including an increase in the intensity and frequency of extreme precipitation events; (2) an increase in the number and severity of extreme events (i.e. floods, droughts, low river flows, heatwaves, cyclonic activity, etc.); and (3) steady retreat of glaciers, which are major contributors to bulk water supply.²
6. The accredited entity (AE), WWF or WWF-US, has undertaken hydrological analysis as well as meteorological data analysis.
7. The absence of a shift in Pakistan’s business as usual flooding and water resources management approach will continue to exacerbate communities’ vulnerability to extreme events caused by climate change such as the 2022 floods.
8. The project aims to address the following barriers:
 - (a) Lack of proofs of concept for EbA and green infrastructure as viable adaptation interventions in Pakistan;
 - (b) EbA and green infrastructure interventions are not included in Pakistan’s National Water Policy, national adaptation plan and provincial adaptation plans;
 - (c) Inadequate technical capacity across government and communities for implementing EbA and green infrastructure interventions in Pakistan;

¹ Available at <https://www.ipcc.ch/assessment-report/ar5/>.

² Asian Development Bank. 2017. *Climate Change Profile of Pakistan*. Available at <https://www.adb.org/sites/default/files/publication/357876/climate-change-profile-pakistan.pdf>.

- (d) Limited access to finance among the most vulnerable communities and micro, small and medium-sized enterprises for climate-resilient, sustainable businesses; and
 - (e) Current land use practices and livelihoods do not consider climate risk.
9. The total project funding amount is USD 77.8 million, of which GCF funding is requested for USD 66 million in grants. The United States Agency for International Development and the Coca-Cola Foundation will provide USD 5 million each in the form of grants to the project. The executing entity (EE), WWF-Pakistan, is also providing co-financing of USD 1.8 million in grants for project management.
10. The project's environmental and social safeguards risk is categorized as B.
11. In terms of implementation arrangements, the WWF (or WWF-US) is the AE and the EE is WWF-Pakistan. WWF-Pakistan will work with procured parties, including engineering firms, community-based organizations (CBOs), community-managed nurseries and provincial government.

2.2 Component-by-component analysis

Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan (total cost: USD 66.1 million; GCF cost: USD 61.5 million)

12. Component 1 will implement EbA and green infrastructure interventions with improved community-led management to reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin. This will be achieved by (i) implementing EbA interventions to reduce flood impacts on populations close to the intervention sites and people living downstream through enhanced ecosystem services such as slowing of flood waters, delays in flood peak onset and increased access to water during drought periods, while downstream communities will benefit from the slowing of upstream velocity and delayed peak flood; (ii) implementing green infrastructure interventions to reduce flood impacts on nearby communities and ecosystems by reducing soil erosion and river sedimentation, diverting and reducing the velocity of intense flows in the hill torrents away from nearby villages located in the floodplain, and increasing infiltration and groundwater recharge; and (iii) providing organization and capacity development to CBOs and creating water user groups and community 'watch and ward' systems to support community-based natural resource management and enhance the water supply and flood attenuation services provided by the EbA interventions to restore floodplains and watersheds and empower communities to responsibly manage the ecosystems on which they rely.
13. Component 1 is structured in three main outputs:
- (a) Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems;
 - (b) Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin; and
 - (c) Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented floodplains and watersheds.

Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan (total cost: USD 2.7 million; GCF cost: USD 0.9 million)

14. Component 2 will improve the capacity and enabling environment for climate action by using Outcome 1's evidence of the effectiveness and efficiency of EbA and green infrastructure

interventions in improving flood and water resources management to expand Pakistan's regulations and policies. National, subnational and community leaders and representatives will be trained to use the modified procedures. This work will allow the Government of Pakistan to invest in EbA and green infrastructure solutions and replicate similar adaptation measures in other flood- and drought-prone areas of Pakistan.

15. Component 2 comprises the following outputs:
 - (a) Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan;
 - (b) Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan; and
 - (c) Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.

Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin (total cost: USD 3.2 million; GCF cost: USD 2.1 million)

16. Component 3 will increase the resilience of community livelihoods in the Indus Basin by training communities and smallholder farmers on how to improve the existing agricultural practices that are prevalent throughout the Indus Basin to better withstand current and future climate change scenarios through improved irrigation practices to conserve water, improved soil management to increase yields and reduce agricultural expansion, and the introduction of new crop varieties that require less water. In addition, new opportunities for sustainable enterprises will be supported to provide alternative revenue-generating opportunities for communities in the Indus Basin and to reduce ecosystem degradation caused by traditional livelihoods that are directly dependent on natural resources. The procurement of project pipelines, the incubation of environmentally sustainable businesses and the provision of technical assistance will help to reduce ecosystem degradation and increase revenue generation for businesses and families.

17. Component 3 will be achieved through the two key outputs:
 - (a) Output 3.1: Pipeline of feasible climate-resilient businesses; and
 - (b) Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.

Monitoring and evaluation costs (total cost: USD 2.2 million; GCF cost: USD 1 million)

18. This cost will cover the relevant activities required to monitor and evaluate the project's progress towards the targets set in the project's logical framework, particularly at the GCF outcome and project level, and with respect to co-benefits. The cost is about 2.9 per cent of the total project cost, which is in alignment with the GCF Evaluation Policy.

Project management costs (total cost: USD 3.6 million; GCF cost: USD 1.8 million)

19. This cost will support the management, coordination and implementation of the project activities, in compliance with the contractual obligations that are included or referred to in the legal agreements entered into with the AE. The GCF portion of the project management cost is less than 5 per cent of the grant portion of requested funding from GCF and is compliant with the GCF policy on fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

20. The project will directly benefit 687,336 people, of which 48 per cent are female. The number of indirect beneficiaries is estimated to reach 7,024,361 people, which accounts for 3.5 per cent of the total population of the country.

21. The direct benefits will be as follows:

- (a) EbA interventions: reduction of flood impacts on populations close to the intervention sites (within 3 km) and for people living downstream;
- (b) Green infrastructure interventions: localized slowing of flood waters and the subsequent temporary storage and infiltration of water outside of flooding events; and
- (c) Community-based adaptation interventions: increased climate resilience of 50,750 of the most vulnerable smallholder farmers in the project sites.

22. The EbA interventions will provide mitigation co-benefits from afforestation, reforestation and assisted natural regeneration of a reduction in emissions of 26,450 tonnes of carbon dioxide equivalent (t CO₂ eq) per year, with the total lifetime removals estimated at 1,037,928 t CO₂ eq.

3.2 Paradigm shift potential

Scale: Medium to high

23. The project creates a paradigm shift in Pakistan's flood and water resources management efforts by establishing proofs of concept for EbA and green infrastructure interventions in the Indus Basin that validate the cost-efficiency, benefits and sustainability of these adaptation and resilience measures. The project will implement watershed basin-wide EbA interventions to manage flood risk, as a complement to grey infrastructure, and mainstreaming EbA into water, agriculture, natural resources management and disaster reduction management plans.

24. The AE is working with partners to seek financing to support the project to scale up and replicate across the country to ensure the expansion of the Recharge Pakistan initiative.

3.3 Sustainable development potential

Scale: High

25. The proposal will contribute to achieving the Sustainable Development Goals (SDGs), including SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 5 (Gender equality), SDG 14 (Life below water) and SDG 15 (Life on land).

26. The proposal aims to deliver environmental, social and economic co-benefits as shown below.

Adaptation intervention type	Co-benefits		
	Environmental	Social	Economic
Restoration of riverine catchment vegetation	- Revegetation provides long-term stabilization of riverbanks, prevents erosion and enhances protection of the riverbank toe from continual scouring while reducing fine sediment input. - Improved forest cover.	Improved drinking water quality and supply.	Increased lifespan and capacity of dams.



Adaptation intervention type	Co-benefits		
	Environmental	Social	Economic
	- Improved carbon sequestration. - Enhanced habitats for local wildlife species.		
Restoration of flow paths	- Improved deposition processes on the floodplain and reduced movement of gravel. - Improved nutrient and pollution retention in upstream areas.	Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems.	Increase in irrigated land and hence food production.
Desiltation of wetlands	Improved health of native wetland vegetation.	Improved drinking water quality and quantity.	Not applicable
Enhancing and reinforcing embankments	Enhanced habitats for local wildlife species.	- Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems. - Improved drinking water quality and quantity.	Lower infrastructure maintenance costs and strengthened local economies through diversification of livelihoods based on natural resources.
Flood protection embankments	- Improved hydrological connectivity across floodplains. - Improved carbon sequestration. - Erosion control and nonpoint source pollution prevention. - Increase of vegetation and biomass with extended growing seasons and enhanced soil moisture. - Sediment retention.	Improved drinking water quality and quantity.	Not applicable
Improved efficiency of irrigation systems	Improved soil health and water content.	Reduced conflict over water resources.	- Cost savings as a result of reduced water and labour inputs. - Intensification of agriculture.

Adaptation intervention type	Co-benefits		
	Environmental	Social	Economic
			Income diversification and market access for smallholder farmers.
Drought-resistant crop varieties	Reduced degradation of natural areas through expansion of croplands.	Not applicable	Improved food security.
Water harvesting	Not applicable	Not applicable	- Agricultural expansion and diversification of crops. - Additional income.

27. The proposal is expected to create 8,000 jobs and support livelihoods development.

28. The proposal recognizes the important roles that women play, and has put strong emphasis on women's roles and responsibilities. The proposal aims to remove the barriers that women face in decision-making, training and resource management and support women's growing role in production.

3.4 Needs of the recipient

Scale: High

29. Pakistan is highly exposed to flooding and drought. In 2022 floods devastated the country; over 1,300 lives were lost and over 33 million people displaced.

30. Pakistan is facing financial constraints for climate change adaptation.

31. The majority of the targeted communities are living below the poverty line in the Indus floodplains, which are highly susceptible to flooding.

3.5 Country ownership

Scale: High

32. The project is supported at the highest levels of government and is aligned with the National Climate Change Policy and its implementation framework. It supports both of the objectives of Pakistan's Green Stimulus developed in response to the coronavirus disease 2019 crisis, namely job creation and ecosystem restoration

33. The project will continue to seek additional financing opportunities from private and public sector funding during its implementation. The AE will develop a scaling up and replication strategy during implementation to continue expanding the Recharge Pakistan initiative to the rest of the country. The AE will be requested to report on the potential leveraged private finance.

3.6 Efficiency and effectiveness

Scale: Medium

34. The funding proposal package presents a high-quality and robust economic analysis with clear and well-referenced assumptions as well as a clear definition of scope and the counterfactual scenarios. The economic analysis contains an estimation of all the relevant economic values from the project activities, which are consistent with the information provided

in the funding proposal. The model uses a social discount rate of 9.12 per cent, which has been computed as the average of the deposit interest rate in Pakistan in 2020 (7.5 per cent) and the rate of the country's commercial banks (10.6 per cent). The rate used also follows the approved FAO project (FP108) in the GCF portfolio. In summary, the economic model results show the net present value as USD 30 million and the economic internal rate of return as 13.6 per cent, indicating that the proposed project is economically viable.

35. Financial adequacy and the appropriateness of concessionality is assessed to be high. From the supply side, banks and financial institutions in the target areas of Pakistan are currently not providing loans to smallholder farmers owing to the risks associated with uncertainty in profitability. From the demand side, smallholder farmers (around 63–85 per cent of farmers in Pakistan) have limited financial literacy, collateral and credit history, and thus very limited access to finance in general. To address this issue, a strong business case is needed to ease financial risks, and the proposal is requesting grant funding from GCF to showcase such a business case. The justification is deemed satisfactory.

36. The need for grants in this project is also justified by the public nature of project activities. In summary, 95.9 per cent of the project costs are for public goods (e.g. rehabilitation of degraded lands for flood risk reduction). For output 3.1 (pipeline development for climate-resilient businesses), GCF resources are to be used for technical assistance and capacity-building for potential businesses. For output 3.2 (climate-resilient agriculture), 84 per cent of funding is for technical assistance and training. The remaining USD 222,600 is for climate-smart agriculture inputs and water saving schemes (irrigation and water storage tanks) but the size of the funding was argued to be too small to conduct a formal financial model.

37. The cost-effectiveness of the project is assessed to be medium. The project's adaptation cost is USD 111/direct beneficiary and USD 10/total beneficiary. Cost per direct beneficiary is in the high range of the GCF portfolio for climate resilience in agriculture, compared with the other GCF-approved projects in the portfolio in Pakistan. Although the project is targeting adaptation results areas only, the AE estimated mitigation co-benefits from carbon sequestration through forest afforestation, reforestation and assisted natural regeneration. If the total GCF funding amount is regarded as mitigation cost, the cost per t CO₂ eq would then be USD 73.22/t CO₂ eq over the 40-year lifetime, which is within the price range of USD 40–80/t CO₂ eq to meet the Paris Agreement target (World Bank, 2021).³

38. The co-financing ratio is 1:0.18, which is lower than the target under the updated Strategic Plan for the GCF 2020–2023. The AE has provided justification for the low co-financing, particularly as substantial government budget, once planned as cofinance, has been redirected towards emergency relief because of the climate-related disasters and subsequent economic impacts in the country. The proposal does not describe the potential to leverage or mobilize additional financial resources, but the AE has indicated that it will implement mobilization strategies to ensure the sustainability of the project and the possible scale-up of the project to different areas, if successful.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

39. **Project background.** The project aims to create integrated approaches to flood and drought risk management in the Indus Basin in Pakistan to strengthen the climate resilience of the most vulnerable rural communities while enabling a paradigm shift towards Ecosystem-

³ "World Bank. 2021. State and Trends of Carbon Pricing 2021. © Washington, DC: World Bank. <http://hdl.handle.net/10986/35620> License: CC BY 3.0 IGO."

Based Adaptation (EbA) in Pakistan. The project will restore 14,215 ha of degraded watersheds and 34 km of flow paths and channels as EbA interventions. Furthermore, it will deploy 127 green infrastructure interventions tailored for specific locations in four project sites in the Indus Basin. The project is expected to have positive impacts on people and the environment. It will improve both the quality and quantity of water for irrigation and drinking purposes, augment water governance system, enhance flora and fauna, and facilitate the adoption of smart agriculture techniques. Ultimately the enduring effects of the project will lead to reinforced climate resilience and overall food and water security in the arid and semi-arid regions of Pakistan.

40. **Environmental and social risk category and safeguards instrument.** The AE has classified the project medium environmental and social (E&S) risk, which is equivalent to Category B as per GCF's environmental and social risk categorization and within the AE's E&S risk accreditation level. Given that the project itself is an environmental and social enhancement project, the Secretariat agrees with this categorization while noting the challenges faced by the project related to conflict sensitivity in certain project sites and the changing security context in the region. The project design and built-in safeguards have brought the overall risk level to moderate, including to mitigate the project's security risk through operationalizing a Security Plan. As the detailed activities and their exact locations are yet to be defined, the AE has prepared an environmental and social management framework (ESMF), which also includes a process framework (PF) to address restrictions on land use and access to resources, and an indigenous peoples planning framework (IPPF) to address requirements for activities that are located near or within indigenous peoples' communities and territories. A stakeholder engagement plan (SEP) was prepared separately, which contains consultations conducted to date for the project design and outlines a detail process for further stakeholder engagement during project implementation.

41. **Compliance with GCF environmental and social safeguards (ESS) standards.** The paragraphs below provide a brief description of the project's compliance with ESS standards.

42. **ESS1: Assessment and management of environmental and social risks and impacts.** The project is categorized as medium risk against the AE's safeguards integrated policies and procedures. The ESMF provides general information about the profiles of the project areas, the baseline information about the project socio-economic conditions, the archetypes of project interventions and a related methodological framework for their further screening and planning at the site level. While the current baseline information is derived from socio-economic, ecological, and biophysical data collected for the project's initial scoping, the project will conduct environmental scoping studies and develop thorough environmental baselines, including indicators for assessing the biophysical impacts of the proposed interventions, during the project implementation. The ESMF identifies the project's E&S risks assessed against the project's safeguards screening criteria per each project component, with the general mitigation measures and roles and responsibilities to bring the mitigation measures into practice specified. The ESMF provides the relevant tools and processes for further assessing the site-specific E&S risks and impacts, including the potential cumulative effects. If the screening reveals any adverse environmental or social impacts that may arise from the planned activity, an ESMP will be required before the initiation of the project activity and following the guidelines provided in the ESMF.

43. The ESMF includes a conflict and security assessment to delineate the project's conflict and security related risks and potential impacts. Given the complex security situation in the region, the ESMF identifies that two project sites (Chakkar Lehri and DI Khan) overlay with the conflict areas, which are considered by the AE as medium security risk zones. The project site in Chakkar Lehri is situated in Balochistan, the largest and most underdeveloped region, which has undergone inter-tribal conflicts and conflicts between law enforcement agencies and the separatist movement. The assessment notes sporadic incidents of violence and attacks in DI

Khan and that the changing security situation in Afghanistan may embolden terrorist factions in Pakistan. The AE has devised a Security Plan based on its institutional experience of working in conflict-prone settings and with local institutions and communities to establish credibility and install confidence. The Security Plan, as part of the ESMF, includes a set of mitigation measures, risk reduction techniques, and security protocols to be followed by the project to manage such conflict sensitivity, such as through the engagement with the security forces and law enforcement agencies, close collaboration with tribal elders in the targeted provinces to ensure the project's legitimacy and seek their advice avoid becoming unintentionally part of any conflict. In the case of security threats, the project will pursue measures including immediate withdrawal from project areas where security threats are identified. In addition to the screening of conflict sensitivity risk that will take place at activity- or site-level during implementation, the AE is suggested to conduct the site-specific conflict sensitivity assessments to inform the detailed design, siting, and stakeholder engagements. The AE is also suggested to continuously monitor the effectiveness of the Security Plan to ensure robust safety and security arrangements are put in place. Where high security or other risks of the project arise, the implementation of the ESMF and the project should be responsive to the changing conditions and the results of monitoring throughout the project's lifecycle.

44. **ESS2: Labour and working conditions.** While the ground activities are yet to be identified and specified during the project implementation, the project will require hiring and engagement of project staff, field workers and unskilled labour. The ESMF has identified potential risks of unsafe labour practices, inadequate occupational health and safety equipment, child labour, and workplace discrimination, pertaining to the construction works, such as building flood dispersal embankments, excavating flow paths, and desilting ponds to enhance the water-holding capacity, to be conducted as part of the EbA and green infrastructure interventions. Following the safeguard eligibility and impacts screening list to be applied at site-level, project activities involving child labour, forced labour, sexual exploitation, or other forms of exploitation, will be screened out as ineligible activities. The ESMF requires that the project implement all reasonable precautions to protect the health and safety of workers through the introduction of preventive and protective measures, and that all labour hired for the excavation work be provided with proper safety gear, temporary housing, first aid training and the relevant health and safety related insurance cover. The project will hire labour from locals of the area and ensure the labour rights of project-employed workers will be observed. Further screening on labour and working conditions will be conducted and detail mitigation measures will be addressed in the site-specific ESMPs.

45. **ESS3: Resource efficiency and pollution prevention.** The ESMF identifies that the works involved in the EbA and green instruction solutions, such as excavation of silt, debris, and other obstructions and pest use in afforestation can cause pollution. The misplacement and improper disposal of excavated material could lead to polluting the landscape and degrading the natural habitat. The ESMF requires a waste management plan to be formulated as part of the site-specific ESMP before the commencement of the relevant activities. The project will use the silt from the excavation work to rehabilitate and reinforce the sides of the flow paths as part of the EbA solutions. In terms of potential pollution caused by pest use, the project will exclude activities involving the procurement and/or use of pesticides and other chemicals specified as persistent organic pollutants or hazardous chemicals. Although the project will not directly promote the procurement and use of pesticides, there are chances that the project interventions, such as riverine afforestation, re-vegetation, nursery raising, and the change in agricultural practices, could lead to the procurement and use of pesticides. The ESMF includes a pest management plan that includes the guiding principles and a five-step approach on integrated pest management to be followed during project implementation. The site-specific E&S screening and ESMPs will further identify and address the project's potential environmental impacts.

46. **ESS4: Community health, safety and security.** The project is expected to benefit the health and safety of the local communities by reducing the risk of water directly flooding their agricultural lands, assets, and infrastructure. However, local communities may be exposed to risks related to the construction works, which may affect water turbidity and water table levels and lead to potential reduction in fish catch. Some project activities may reduce the access to water for irrigation purposes, especially for communities living downstream of the Indus River. There is also the potential risk that the project may inadvertently exacerbate community conflicts due to the restriction of access to land or resources and due to the selection of project beneficiaries and the access to the project benefits. The project will mitigate such risks by preventing adverse impact on water quality and supply to affected communities; ensuring the proper design and safety of the project infrastructure; putting in place protective mechanisms for the use of hazardous materials; following disease prevention procedures; and implementing the project's security plan to ensure emergency preparedness and response. To avoid escalating or creating community conflicts, the project will further screen the risks and potential impacts on community health and safety at the site-level during implementation. Additionally, following the process framework (PF) provided in the ESMF, affected communities will participate in the identification, design, implementation and monitoring of project activities and the related mitigation measures to prevent and minimize the project's potential adverse impacts on communities. As the exact impacts of the project on community health, safety and security will be further identified at the site-level through the safeguards screening and during project implementation, the AE is suggested to ensure site-specific assessment and mitigation measures and the operation of the project's PF and security plan will mutually reinforce one another.

47. **ESS5: Land acquisition and involuntary resettlement.** The project will exclude activities that lead to private land acquisition, physical displacement, and voluntary or involuntary relocation of people, including non-titled and migrant people. The ESMF identifies that the project activities may involve novel or rigorous restrictions on the use of or access to natural resources, and that there may be some small areas of land acquisition, and that certain interventions, such as those pertaining to cleaning up the lake, could lead to changes in tenure arrangements, community-based property rights or customary rights to land, territories, and/or resources. The AE has triggered its own standard on restriction of access and resettlement and developed a process framework (PF) to guarantee transparency and equity in the planning and implementation of project activities and to avoid and minimize the project's adverse impacts on the resource-dependent local communities and their livelihoods. The PF outlines the processes of engaging the local communities in the design, implementation and monitoring of the project; delineating the prospective negative impacts of the restriction on the encompassing communities, including any gender differences; defining the eligibility criteria of the economically displaced communities to receive compensation benefits and development assistance; identifying mitigation measures to assist any economically displaced people to restore and improve livelihoods; and setting out the participatory monitoring arrangements and a grievance redress procedure to resolve possible disputes on restrictions on land use and access to resources pertaining to the project. The PF will work with the safeguards screening to identify the exact social impacts of the project and address such impacts through site-specific ESMPs or developing specific livelihood restoration plans (LRPs). The PF details the processes and mitigation measures required, including on how to calculate and allocate the compensation to the affected communities. In the case of insufficient budget for an LRP as agreed with local communities, the project will redesign the specific activity/activities so that a LRP will no longer be needed, or the impact is lessened in a manner that a new LRP can be agreed upon.

48. **ESS6: Biodiversity conservation and sustainable management of living natural resources.** The project is expected to produce significant conservation benefits while harnessing biodiversity and ecosystem services to build climate resilience. It will exclude activities that could negatively affect areas of critical natural habitats or any breeding ground of known rare/endangered species, and activities that result in the loss of biodiversity, alteration

of the functioning of ecosystems, and introduction of new invasive alien species. The ESMF identifies four types of activities considered by the project that can adversely affect biodiversity and the integrity of terrestrial or aquatic ecosystems, if not managed properly. That is, revegetation and reforestation may lead to decrease in the water availability downstream and exotic or non-native tree species being planted; the construction of retention areas or rain ponds may affect local biodiversity; while dredging can improve water quality in lakes and wetlands, to clean the sediment may lead to large disturbance to the lake/ wetland and that temporarily increased sediment concentrations can disturb aquatic life; and terracing may lead to habitat conversion and soil degradation, and can have impact on local hydrology and limiting water availability downstream. Further screening will be conducted at the site-level to screen out activities that may use genetically modified organisms or modern biotechnologies or their products; involve the procurement and/or use of pesticides and other chemicals specified as persistent organic pollutants or hazardous substances; or develop forest plantations. The AE is suggested to identify the potential impacts on biodiversity and natural resources per each archetype of the potential interventions and ensure that caution must be exercised in the design, selection, and implementation of site-specific interventions. The analysis of the project intervention should also be informed by the environmental baseline to be developed at each site and be continuously monitored.

49. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** Through screening undertaken in the context of the GCF Indigenous Peoples Policy, the AE has identified that there are indigenous peoples within the project sites. These include Mohana or Mallah communities in Machar Lake Bagri in Chaker Lehri and adjacent areas and Kehal and Mores of Taunsa and DI Khan areas. The AE has prepared an indigenous peoples planning framework as part of the environmental and social management framework with the intention of preparing indigenous peoples plans through a free, prior and informed consent process in partnership with the relevant communities. The AE noted that engagement has already been undertaken in these areas during the development of the funding proposal and that at this stage there is broad support from tribal elders for participation in the project. The stakeholder engagement plan acknowledges the influence of tribal groups in the project areas and the need to engage with and seek the approval of tribal elders.

50. While planned activities could restore livelihoods, including for the Mohana, descendants of the first inhabitants of the Indus Valley who live in one of the last remaining floating villages on Lake Manchar and are dependent on the lake for fishing, there is also the risk that project activities could result in restricted access of Mohana people to fishing. This will be determined once the project activity locations are finalized, which is expected to be in the first six months of project implementation. The indigenous peoples planning framework notes that as many as 450 Mohana families may be affected by the project activities. Tribal groups exert influence in the project areas and there have also been historical tribal conflicts, including over water resources, in DI Khan and Balochistan, although the funding proposal documentation notes that this is less prevalent now and that the tribal elders have assured the AE of their cooperation, including through letters of commitment. To mitigate these risks, especially to Mohana communities and overall, the AE will implement free, prior and informed consent processes, taking into consideration the participation of women and men, and not implement in areas that will present adverse impacts to the local communities. The approval of tribal elders will be sought through their *jirgah* system before the start of all project activities.

51. In relation to local and indigenous knowledge systems, EbA should provide opportunities for the inclusion and promotion of local and indigenous knowledge. The bulk of the activities are currently dedicated to green infrastructure as opposed to promoting bottom-up knowledge exchange, although traditional knowledge of water management, cultural landscape systems such as rain-fed ponds, wells and the *karez* system, are acknowledged in addition to other traditional forms of water-sharing. The *karez* system in particular is an ancient form of water management that allows communal water-sharing. One of the project activities

involves restoring the natural hydrology of a *karez* system. The bedrock of the *karez* lies in the sociocultural institutions behind it and the implementing bodies would need to also ensure the engagement of local stakeholders – in this system also known as shareholders – and the strengthening of the accompanying knowledge and value systems to ensure the sustainability of the *karez*.

52. In line with its roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and EE, who are also encouraged to share emerging good practices and success stories with the group.

53. **ESS8: Cultural heritage.** The project will exclude activities that can negatively impact areas with cultural, historical, or transcendent values for individuals and communities. While no significant impact on the cultural heritage is expected from the project, the pre-feasibility study identifies possible location of historic sites in project areas therefore the AE has triggered its own standard on cultural resources to safeguard the project's potential impacts applicable to both physical and intangible cultural resources. The pre-feasibility study recommended the need for further information for the project site in DI Khan, where historical structures built on the Luni and Gomol River diversion point near to Kot Azam were completely damaged during the super floods in 2010. More information is needed as to the extent of the damage and whether they continue to hold cultural significance regardless of the damage. The project site in the Manchar-Hamal Watershed, Lake Manchar, is also an important historical area. Home to several archaeological sites, there are likely intangible cultural resources for the Mohana people who have innumerable tangible and intangible cultural ties to the lake. An initial baseline information has been collected based on community surveys and field visits. Given the specific activities and locations of the project's activities are yet to be determined, the ESMF provides a set of potential mitigation measures on cultural heritage, including to: conduct further baseline study and impact assessment on cultural heritage, as part of the landscape-level E&S screening and the efforts of identifying the indigenous peoples' cultural heritage and complying with the regulatory requirements; prevent the negative impacts of the projects at the design stage to ensure that the planned activities do not harm the physical or cultural heritage; and develop a cultural heritage management plan prior to the implementation of project activities to preserve both tangible and intangible cultural heritage.

54. **Sexual exploitation, abuse and harassment (SEAH).** The GCF Revised Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from SEAH in GCF-financed activities. From a policy perspective, SEAH is grouped in the laws and terms of gender-based violence (GBV), honour killings and domestic abuse. During gender assessment consultations, the AE found that GBV in the country is alarming and noted that it has included some GBV/SEAH mitigation measures in the gender action plan. The AE has a zero-tolerance approach to GBV and acknowledges that some of its project interventions could potentially have a high risk of GBV/SEAH owing to power imbalances. It recognizes that the achievement of biodiversity conservation is vital, while promoting gender equality, and makes every effort to ensure that project activities implemented by WWF respect integrity and human rights and mitigate any risk that gives rise to discriminatory and exploitative gender inequalities. The guidance note on GBV and SEAH will assist project teams in identifying any risks of GBV and SEAH that may emerge in conservation projects. Furthermore, this note will support decision-making and inform planning and implementation of possible mitigation measures to address any such risks identified during project preparation and execution. Where applicable, the project team may be required to prepare a detailed GBV/SEAH action plan and include any identified GBV/SEAH mitigation measures in project monitoring and reporting. The project management unit (PMU) will also facilitate the identification of any potential SEAH risks; SEAH complaints handling; and the provision of necessary training/capacity-building to project staff and implementation partners. GCF recommends that the SEAH grievance and reporting mechanisms be survivor-centred and gender-responsive and that modalities be put in place that will provide timely services and redress to survivors, including, as appropriate, medical care,

psychosocial support, legal support, community-driven protection measures and reintegration where applicable.

55. **Implementation arrangements.** The ESMF describes the management structure and the roles and responsibilities related to project implementation and safeguards implementation. While the AE is responsible for ensuring both the GCF and AE's own policy requirements and standards are applied and met, WWF Pakistan, as the EE, will be responsible for the implementation of all project components and activities, including ensuring the implementation of and compliance with the ESMF. A Project Management Unit (PMU), comprised of safeguards-related senior management including an E&S safeguards manager and a gender manager, will supervise the implementation of the ESMF and other safeguard plans at the site/landscape levels. It will also be responsible for the implementation. The PMU will be accountable to a Project Steering Committee (PSC), which will be established to provide technical guidance to the PMU and ensure alignment among the PMU with national, provincial and project-site level entities. The PSC will oversee the project's compliance with the safeguards commitments and provide recommendations on specific safeguards issues where needed. In addition to the E&S safeguards manager, the project will also deploy three safeguards staff at the site-level responsible for the on-the-ground implementation of the ESMF and related plans once they have been finalized.

56. **Stakeholder engagement and information disclosure.** The project underwent extensive consultations with key stakeholders respectively during the project design phase and during the preparation of the ESMF. Workshops, stakeholder meetings, and field-level consultations with local stakeholders and community groups were organized, and complemented by site visits, field inspections, and focus group discussions. The consultations during the ESMF preparation focused on identifying the potential adverse impacts of the project with mitigation options as well as assessing the potential project benefits and how these can be enhanced in favour of the local communities. It is concluded by the AE that the project affected communities were supportive of the planned project activities. A SEP has been prepared which summarizes the process and results of the consultations conducted so far and provides a detailed plan for further stakeholder engagement during the project's implementation phase, covering the project's executing partners and stakeholders, including non-government stakeholders, to be further engaged at the national, provincial, and local levels. The ESMF requires that all affected communities and relevant stakeholders be informed about the ESMF requirements and commitments, and that the project's safeguards documentation, including site-specific ESMPs, will be disclosed to all affected communities and the public following a disclosure framework specified in the ESMF.

57. **Grievance redress mechanism (GRM).** The ESMF provides the principles for the development of an activity-level GRM that will be finalized with training provided to the project's safeguards personnels within the first six months of the project implementation. The project-level GRM will be operationalized, in addition to and independently from the EE's own internal GRM system, following a separate structure, process flow, and hierarchy as outlined in the ESMF, which will enable all complaints going to straight to the PMU and with a whistle-blower mechanism to allow project communities to raise complaints anonymously. In addition to the project specific GRM, a complainant can submit a grievance to the AE's independent, third-party GRM as well as to the project complaints officer, who is fully independent from the project team, hosted by the AE. The project will provide information of the GRMs available to the project, including GCF's Independent Redress Mechanism as an integral, into the community consultations. All the consultation sessions will be conducted in the local language of the area, and the local communities will also receive training on how to file and lodge complaints. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

58. The AE provided a gender assessment and action plan with the funding proposal, thereby complying with the requirements of the GCF Gender Policy.

59. The status of women in Pakistan over the last three decades has been fluctuating and has seen many structural factors/changes over time. The existing numerous laws put in place provisions that will ensure women's rights and protection, but the implementation of the same and lack of political will is still a challenge. Women's participation in decision-making processes remains a challenge, and formal institutions, including private and public institutions, fall short of facilitating women's access to financial resources education, skills, health facilities, etc. Informal/community institutions, including religious spaces, define and reinforce child marriages and gender roles at the household and/or community level. The Global Database on Violence against Women shows that 24.5 per cent of women experienced lifetime physical and/or sexual intimate partner violence between 2017 and 2018. The impact patterns are also different for women and men. A shrinking natural resource base has led to migration and shifts in livelihoods and roles, and there is a need for alternative livelihoods.

60. The gender action plan for the Recharge Pakistan project was developed on the basis of the results and recommendations from the gender assessment, outlining gender-specific actions against each activity that aim to ensure that key gender issues are adequately incorporated in the overall project design and that stand-alone gender activities are carried out. The gender specialist/manager based at the PMU will lead the implementation of the gender action plan and facilitate various capacity needs assessments and training at the onset of the project implementation phase. The AE will also coordinate (on other outputs) with the National Commission on the Status of Women, the Women Development Department, the Council on Gender Equality, vocational training institutions and government agencies and line departments at the national and subnational level working on gender equality and women's empowerment. The gender-specific activities that will be implemented under outcome 1 include (a) integrating gender dimensions into the integrated floods and drought management Tool; (b) procurement of plant saplings from nurseries raised by local women; (c) formation of women's groups on natural resources, water, climate change and environment; (d) capacity-building of local women and men to understand what climate change is, building their leadership skills so that they will be able to actively engage in the CBO meetings; and (e) holding consultative meetings with CBO women's groups on operation and maintenance (O&M) of green infrastructure. Under outcome 2 the AE will (a) ensure that gender considerations are included in the resource management regulatory frameworks during the development of policy briefs on EbA interventions; (b) sensitize relevant government line departments on gender and women's empowerment in EbA across their planning and policy processes; and (c) The AE will set up a Project Steering Committee and provincial oversight committees, which will facilitate mainstreaming gender in initiatives and policies that focus on improving natural resource management through EbA and similar programmes. Finally, under outcome 3, the AE will (a) ensure that women and men have access to capacity-building workshops focused on bankable projects/green financing; (b) ensure that women and men are trained on climate-resilient agriculture practices covering both drought and flooding scenarios; and (c) ensure that gender issues are mainstreamed in the implementation of climate-resilient agriculture interventions, including through highly efficient irrigation systems, kitchen gardening, fruit orchards, fodder cultivation and water storage tanks; (d) build the technical capacity of local women on alternative livelihoods and entrepreneurship; and (e) implement gender-specific diversified income-generating activities including, but not limited to, fruit and vegetable processing units (packaging, labelling and post-harvest processing along with value addition), handicrafts, poultry farming and apiculture.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

61. GCF is requested to provide up to USD 66 million by way of a grant, which will finance green infrastructure interventions using an EbA approach at the four project sites for flood risk management. The AE is providing co-financing of USD 1.8 million in kind and the United States Agency for International Development and the Coca-Cola Foundation are providing USD 5 million each in the form of a grant. The total funding amount is USD 77.8 million. The Government of Pakistan is unable to provide any co-financing for the project owing to the diversion of the initial co-financing commitment to the national recovery following the floods in 2022.

62. The AE is requesting a full grant support because EbA and green infrastructure are often not considered to be economically viable in view of the high transaction cost, and the market is skewed towards hard infrastructure without the consideration of wider benefits from EbA. The analysis of the external service provider (Moody's) assessed that the recent floods have exacerbated Pakistan's liquidity and external credit weaknesses.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

63. WWF-US will serve as AE for this project. It will provide the EE with the GCF proceeds and be responsible for providing guidance to and supporting the EE.

64. WWF-Pakistan, the EE, has been operating in the country for over 50 years. The EE has a track record of working with the Government of Pakistan in policy and regulatory design and review as well as implementing EbA projects in Pakistan. The AE assessed the EE's capacity and concluded that the EE is capable of applying WWF-US and GCF standards and policies in the execution of the project.

4.3.3. Programme-specific execution risks (medium risk)

65. O&M risk: the project stated that the O&M risk beyond the project implementation can be addressed at the government and community level. The funding proposal sets out the responsibilities of the relevant government departments for the required O&M activity. This will be further elaborated, including financing arrangements and budgetary allocation, during the implementation period. Given the current budgetary constraints, the government's capacity to finance O&M cost beyond the project implementation is uncertain and the current commitment assumed that there would still be interest from the relevant ministries/governments. At the community level, the relevant provincial departments and CBOs will establish agreements on continuous monitoring and management of the green infrastructure and ecosystem protection. The AE mentioned that the CBOs will work with a water user group to explore potential options for fee structures for irrigation and drinking water which may support the O&M. Another GCF-financed project (FP108) in Pakistan is working on water tariff matters, which could serve as a basis for the CBOs/water user groups to develop fee structures.

66. Foreign exchange rate risk: the AE identified foreign exchange rate risk as one of the risks which could affect the financial performance of the project such as overspending or underspending of the budget. The country has experienced significant local currency fluctuation (from -30 to +10 per cent) in the past three years and its foreign exchange reserves have fallen to very low levels according to the information provided by the external service provider in March 2023. In the case of significant exchange gain or loss, the AE should immediately inform GCF, which will take appropriate action by mutual agreement with the AE within the Board-approved financing amount. Moreover, the project is financed by full grant without government budgetary support, which will be less influenced by any repayment obligation and fiscal situation of the government.

67. The Dutch Fund for Climate and Development (DFCD) support projects (pre-identified): GCF proceeds under component 3 will be used to support projects pre-identified by DFCD. Since the preparatory funding from DFCD is over, the GCF proceeds will fill up the gap to allow these projects to be bankable and find investors. Mobilizing funds to cover the capital expenditure will be the key for the actual investment to materialize on the ground. The funding proposal includes an indicator of the number of signed agreements to ensure this.

68. Economic analysis and project viability: the AE carried out an economic analysis, which resulted in an economic internal rate of return of 13 per cent over 30 years. The analysis considers the benefits from the avoided damage, cost savings in agriculture and carbon sequestration. Carbon sequestration is the greatest benefit derived from EbA and green infrastructure. The analysis includes a sensitivity testing with changes of inputs in underlying assumptions such as the low and high shadow price of carbon. The assumptions and referenced data used for analysis are deemed to be robust and justify the request for the grant. As the analysis assumed that the cost of O&M will continue beyond the project implementation, the long-term sustainability and viability of the project will depend on the capacity of the government and local communities to mobilize the resources to cover the O&M cost.

4.3.4. Compliance risk (medium risk)

69. Pakistan is not currently subject to any United Nations Security Council (UNSC) resolutions (i.e. sanctions).

70. WWF-US, as the AE for the project, confirmed that no persons or entities listed on any UNSC sanctions list, including the UNSC Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.

71. The AE confirmed that it has conducted a risk assessment concerning prohibited practices – that is, corruption, fraud or abuse leading to theft, misappropriation, waste or improper use of property, assets or GCF funding. The probability of risk occurrence was assessed as low, and impact severity has been assessed as medium.

72. The AE confirmed that it has also conducted a risk assessment concerning money-laundering and terrorist financing – i.e. improper use of property, assets or GCF funding, as well as project actors that fail to comply with all applicable laws on anti-money-laundering and countering the financing of terrorism (AML/CFT). The probability of risk occurrence was assessed as low, and impact severity has been assessed as medium. The risk assessment:

- (a) Verified the identity of EE;
- (b) Checked EE against a series of watch lists using LexisNexis Bridger Insight software;
- (c) Assessed the purpose, economic rationale and overall risks associated with the recipient on the basis of all relevant information, including, for example, the type of third parties involved and the financial instrument proposed by the third parties; and
- (d) Includes the legal and financial due diligence assessments completed on EE.

73. The AE confirmed that GCF funds will be administered by WWF-Pakistan. WWF-Pakistan will be required to agree, in writing, not to engage in transactions with, or provide resources or support to, individuals and organizations associated with terrorism, including those individuals or entities that appear on the Specially Designated Nationals and Blocked Persons List maintained by the United States Department of the Treasury⁴ or the UNSC terrorist designation list.⁵

⁴ See <https://www.treasury.gov/resource-center/sanctions/SDNList/Pages/default.aspx>.

⁵ See https://www.un.org/sc/suborg/en/sanctions/1267/aq_sanctions_list.

74. WWF-Pakistan is designated as a non-profit organization and does not have beneficial owners.

75. As part of its financial and operational due diligence, WWF has assessed the WWF-Pakistan office in the areas of financial management, internal controls, accounting, human resources, procurement and procurement systems, equipment management, the ability to comply with donor requirements, and whether their overall policies and procedures, experience and level of supervision proposed for this project are sufficient to mitigate any potential risks or vulnerabilities related to prohibited practices.

76. The AE's subsidiary agreement with the EE will require compliance with WWF and GCF AML/CFT policies described in the funded activity agreement and accreditation master agreement (AMA) and will flow down remedies for non-compliance. The subsidiary agreement will require that the EE in turn flow down those provisions to all project subrecipients.

77. Before submission of the project to the Board, the AE confirmed that it will compare the names of the EE and the known executing partners against United Nations and United States sanctions lists, including sanctions established by UNSC, sanctions administered by the Office of Foreign Assets Control within the United States Department of the Treasury pursuant to the Global Terrorism Sanctions Regulations, the Foreign Terrorist Organizations Sanctions Regulations and the Counter Narcotics Trafficking Sanctions. The AE confirmed that, at the time of submission of the proposal, it was not aware of any individual or entity expected to receive project funds or material support or resources that are included on those lists.

78. The AE confirmed that the EE, WWF-Pakistan, maintains a rigorous practice of conducting due diligence on beneficiaries and project partners. Under the terms of the subsidiary agreement between the AE and the EE, it will be responsible for ensuring that funds are only issued to procured parties once their due diligence process has been completed satisfactorily and a report has been submitted. In addition, WWF-Pakistan will maintain regular communications and tracking of funds disbursed to procured parties to verify activity delivery and spending.

79. The AE indicated that the due diligence process typically involves the collection and analysis of information about beneficiaries and downstream partners. This information includes, but is not limited to, their identity, background, financial status, accounting standards, cash handling procedures, safeguarding of assets and legal history. Such information is gathered through various means, including the examination of publicly available records, conducting interviews and making site visits. The purpose of due diligence is to identify any potential red flags or risks associated with the beneficiaries and downstream partners, such as involvement in illegal activities, financial instability, conflict of interest or reputational issues.

80. The AE confirmed that, other than those entities included in the implementation arrangements which will be subject to due diligence processes, there will be no intention to distribute or disburse, directly or indirectly, cash vouchers, commodities or other items of value to any other beneficiaries.

81. The AE confirmed that several mechanisms will be available for the reporting of complaints and allegations of impropriety, wrong-doing or other related issues in the project and its activities (i.e. whistle-blower programmes). The WWF whistle-blower system is open to staff, partners, communities and other stakeholders to report suspected illegal or inappropriate activity, or concerns about the implementation of WWF projects. This system is in the form of an online and telephone mechanism hosted by a third-party provider, WhistleB, which can receive reports online or by telephone in multiple languages. All complaints submitted through the mechanism are investigated.

82. **Recommended risk rating:** the Office of Risk Management and Compliance (ORMC)/Compliance Team has conducted a review of the project in accordance with relevant

Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team have determined a risk rating of “medium” and have no objection to this request proceeding to the next steps.

83. ORMC/Compliance would like to remind WWF-US, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money-laundering, terrorist financing or prohibited practices among the intended counterparties, EE, beneficiaries, persons involved or any of the proposed activities.

4.3.5. GCF portfolio concentration risk (low risk)

84. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

85. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity/executing entity capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

86. WWF-US will serve as the AE for the project and will provide GCF proceeds to the EE. The AE will provide project oversight, including technical, financial and administrative monitoring and supervision (through reporting, audits and annual site visits) and review and approval of the EE's yearly workplans and budgets.

87. WWF-Pakistan, a private independent organization under the international WWF network and a separate legal entity from WWF-US, will serve as an EE for the project. The EE will be responsible for project execution, including the management of grantees, recipients of goods or services, procured parties and their activities, and reporting to the AE.

88. WWF-US will also provide support, guidance and backstopping to the EE, monitoring the achievement of project results and outputs, reporting to GCF, and project closure and evaluation. WWF-US will conduct these responsibilities and disburse GCF proceeds to the EE, in line with the GCF AMA and funded activity agreement. The AE assessed WWF-Pakistan's capacity to execute the project and concluded that the EE is capable of applying WWF-US and GCF standards and policies in the execution of the project.

89. WWF-Pakistan will establish and host the PMU. The PMU will be led by the Project Director and operate under the guidance of the Project Steering Committee described below. The PMU, through the Project Manager, will report to the AE. It will also be responsible for overall project management and planning, providing support to the execution of day-to-day activities, coordinating with the national government and project partners, coordination with the AE, the direct supervision of contracted project activities and coordinating project execution across the four provinces.

90. The AE will advance funds to the EE through quarterly disbursements based on spending projections included in the quarterly financial reports in accordance with WWF standard grants management policies. WWF-Pakistan will set up a dedicated project account to

receive these disbursements. In addition, the project will be audited annually following the WWF project audit guidelines.

4.5 Results monitoring and reporting

91. The funding proposal is an adaptation project focusing on two results areas of the Integrated Results Management Framework: (i) most vulnerable people and communities and (ii) ecosystems and ecosystem services. Through key interventions in EbA, green infrastructure and community-based adaptation in selected watershed areas of Pakistan, the project aims to benefit a total of 7.7 million individuals, with 0.7 million direct and 7.0 million indirect beneficiaries' adaptation interventions.

92. The FP has been found to use the templates and provide contents in line with the Integrated Results Management Framework requirements. The sections on Theory of Change and logical framework have been aligned with the suggested structure and format. A continuous feedback mechanism has enabled refinement of the elements of the logical framework in terms of identification of results areas, indicators, means of verification, targets, assumptions, etc. They are found to be adequate and are expected to allow smooth monitoring and reporting of project results. The information provided in different sections of the logical framework are clearly articulated; their proper implementation will help to yield adaptation mechanisms, tools and practices for the communities in the country.

93. **The assessment of adaptation benefits and beneficiaries is a strength of the funding proposal.** Following the Secretariat's guidance, a separate annex on a methodology for the estimation of adaptation beneficiaries have been prepared, which provides a description of vulnerability context, identification of adaptation benefits and related beneficiaries. The direct and indirect beneficiaries are distinguished under a sound assumption. The methodology has allowed a systematic estimation of adaptation beneficiaries, which will help in devising approaches to reach them during implementation, as well as monitoring and reporting on adaptation beneficiaries.

94. **Implementation timetable:** the implementation timetable has been provided in the required structure and format. The milestones and deliverables that are envisaged at different stages of project implementation are adequately planned. Adherence to the proposed implementation timetable will facilitate smooth implementation of the project as well as timely delivery of the results at planned intervals.

4.6 Legal assessment

95. The Accreditation Master Agreement was signed with the AE on 16 November 2017, and was amended and restated by the first amendment and restatement agreement on 12 December 2022 which became effective on 13 December 2022 (as so amended and restated, the "AMA").

96. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and has the capacity and authority to implement the project. It is recommended that, prior to submission of the funding proposal to the Board (a) the AE has obtained all its internal approvals and (b) the Fund has received a certificate or legal opinion from the AE in form and substance satisfactory to the Fund confirming that all final internal approvals by the AE have been obtained and that it has the authority and capacity to implement the project.

97. The proposed project will be implemented in Pakistan, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further

assessed. GCF provided a draft agreement on privileges and immunities, along with a background note, to Pakistan on 15 September 2015; however, Pakistan has yet not indicated that it is ready to start the negotiations on the agreement.

98. The Heads of the Independent Redress Mechanism and the Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

99. Pursuant to clause 4.05(b) of the AMA, the AE is required to perform, among other things, legal due diligence on the proposed project. This legal due diligence is to be carried out by the AE for the purposes of developing the funding proposal and, when reviewing the funding proposal, GCF will rely on such due diligence (see decision B.12/31, annex XXVII, chapter II). The Secretariat notes that the AE's legal due diligence for the project (see annex 9 to the funding proposal), has revealed that the agreement between the Executing Entity and the Government of Pakistan, pursuant to which the Executing Entity was authorized to work in Pakistan, has expired on 5 August 2021, and that it has not been renewed. The AE's due diligence further revealed that the application to renew was still underway and the Executing Entity was in continuous contact with relevant authorities. To address implementation risks associated with the foregoing, a condition precedent to the execution of the funded activity agreement has been inserted into the term sheet according to which the AE shall deliver to GCF Secretariat a certificate or a legal opinion confirming that the Executing Entity is authorized to implement the project in Pakistan.

4.7 List of proposed conditions (including legal)

100. In order to mitigate the risks set out in this assessment, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed project;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later;
- (c) Delivery by the Accredited Entity to the GCF of a certificate or a legal opinion, in form and substance satisfactory to the GCF Secretariat, issued by the Executing Entity and addressed to the Accredited Entity and the GCF, confirming that the Executing Entity is authorized to implement the Project in the Host Country; and
- (d) Completion of the GCF Secretariat's legal due diligence on the Funding Proposal.

Independent Technical Advisory Panel's assessment of FP207

Proposal name:	Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management
Accredited entity:	World Wildlife Fund, Inc. (WWF)
Country/(ies):	Pakistan
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. The proposed project will take place in Pakistan, a country characterized by a diversity of ecosystems, climate zones and landscapes (approximately 60 per cent rough mountainous terrain and plateaus, and 40 per cent level plain), and a wide variety of natural resources. With an economy long based on agriculture, Pakistan is now considered a semi-industrialized country facing considerable challenges in balancing the competing objectives of socioeconomic development and environmental protection. The numbers of people living below the poverty line (2013 data) and malnourished (2018 data) are estimated at 29.5 per cent and 12.3 per cent, respectively. Most of Pakistan's population, estimated at 225.1 million, live in the Indus River basin, an area highly exposed to severe floods in July and August.
2. The proposed project targets two catchments of the Indus River basin, with interventions at four sites, namely Dera Ismail (DI) Khan and Ramak located in the DI Khan catchment, and Chakar Lehri and Manchar located in the Chakar Lehri catchment. Situated respectively in the Khyber Pakhtunkhwa province, and the Balochistan and Sindh provinces, the DI Khan and Chakar Lehri catchments have been selected because their ecosystems and communities are highly vulnerable to the impacts of climate change, and because there are opportunities for adaptation benefits through ecosystem-based adaptation (EbA) and green infrastructure interventions. Agriculture is the main resource for the subsistence of local people in the selected sites.
3. The total budget required to implement the project over its 7-year duration is USD 77,848,396, including USD 66,000,005 in the form of a grant from the GCF. The remainder will consist of USD 10 million from United States Agency for International Development (USAID) and the Coca-Cola Foundation, as well as an in-kind contribution from WWF-Pakistan estimated at USD 1,848,391.
4. Pakistan's vulnerability to climate change is largely due to its geographical location and socioeconomic conditions:
 - (a) Pakistan is located in a semi-arid to mostly arid region prone to recurrent large-scale floods, earthquakes, droughts and landslides. Its economy, notably the agricultural and hydroelectric sectors, is based on the development of water resources from rivers and glacial melt, which are highly dependent on climatic conditions. In recent years, the country's climate has changed, with rising temperatures and more variable rainfall, compromising not only the functioning of the above-mentioned economic sectors, but

also causing considerable loss and damage to Pakistan society as a whole. The major floods of 2010 and 2022 and their impacts are the most recent examples. With the climate conditions anticipated for the country – particularly an average temperature increase of up to 3.7 °C by 2060 and up to 6 °C by 2090, a decrease in the volume of glaciers and snow cover, and especially an increase in the frequency and intensity of extreme and slow-onset events – the situation is set to worsen if nothing is done to sustainably address these adverse climatic conditions, including major floods and droughts; and

- (b) The adaptive capacity of Pakistani stakeholders to adverse climatic conditions attributable to climate change is greatly compromised by a number of unfavourable socioeconomic situations. Indeed, the country faces relatively high levels of malnutrition and of people living below the poverty line, and contains communities and groups marginalized by economic status, location and political circumstances. In addition, complex institutional context of Pakistan makes it difficult to adopt sustainable solutions to the challenges facing the country.

5. The proposal's theory of change states that “**If** the interventions envisaged by the funding proposal (EbA and green infrastructure) are integrated into the country's planning systems and into future adaptation actions, **then** the targeted communities will be less vulnerable to floods and droughts, and governments at different levels will have the planning tools in hand to support and operationalize the interventions **because** the main barriers, notably the lack of proofs of concepts for EbA and green infrastructure as viable adaptation intervention in Pakistan, the inadequate technical capacity across government and communities for implementing EbA and green infrastructure intervention in Pakistan, and the limited access to finance among the most vulnerable communities and micro-, small- and medium size enterprises for climate-resilient, sustainable businesses, will have been removed.”

6. The main objective of the proposed project is to address climate change-induced floods and droughts in a sustainable way, using an anticipatory approach that combines EbA, green and green-grey infrastructures. To achieve this objective, the proposed project is built around three main components: component 1: proofs of concept for EbA and green and green-grey infrastructure interventions as efficient and effective solutions to flood and drought risk reduction in Pakistan; component 2: enabling a paradigm shift towards EbA and green and green-grey infrastructure in Pakistan; and component 3: enhanced community resilience and adoption of EbA and green and green-grey infrastructure interventions in Pakistan's Indus basin. These components are expected to deliver the following results: outcome 1: EbA and green and green-grey infrastructure interventions with improved community-led management will reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus basin; outcome 2: the development of an enabling environment for climate action that has new procedures for implementing EbA and green and green-grey infrastructure interventions for flood and water resources management in Pakistan; and outcome 3: increased resilience of community livelihoods in the Indus basin.

7. The World Wildlife Fund, Inc. (WWF-US) will serve as the accredited entity (AE) for the project and WWF-Pakistan will be the executing entity, which will establish and host the project management unit within WWF-Pakistan's office in the Federal Capital, Islamabad. Furthermore, there will be four provincial implementation units hosted within WWF-Pakistan's regional offices located in the provincial capitals (Khyber Pakhtunkhwa, Sindh and Balochistan), and site implementation units will be based at field offices in the following locations: (i) DI Khan Watershed, covering the DI Khan and Ramak Watershed project areas; (ii) Chakar Lehri Watershed; and (iii) Manchar Lake Watershed.

1.2 Impact potential

Scale: *Medium to high*

8. The project interventions will directly benefit an estimated total of 687,336 people, including 357,414 men (52 per cent of the total population of the region) and 329,922 women (48 per cent of the total population), as well as indirectly benefiting 7,024,361 people, including 3,652,668 men (52 per cent of total population) and 3,371,693 women (48 per cent of total population) in Pakistan's Indus basin.
9. By strengthening the adaptive capacity of institutions and stakeholders at different administrative levels, the interventions promoted by the project are likely to be implemented and consequently have positive impacts in the management of large-scale floods and droughts in the country. This will be achieved through interventions on the ground in the four project sites to redirect and contain the flow of water, in order to minimize downstream flash floods and to facilitate infiltration to replenish groundwater resources and help address droughts. These interventions include EbA interventions such as restoration, reconnection and/or rehabilitation of watersheds, floodplains and wetlands; and green (and green-grey) infrastructure interventions such as retention areas and recharge basins. Community-based adaptation interventions will support vulnerable smallholder farmers to adopt drought-resilient crop varieties, to increase their capacity for resilient and sustainable irrigation and soil management, and to enhance irrigation through new water storage tanks.
10. The proposed project should also contribute to changes in the regulatory and procedural arsenal for flood and drought management in Pakistan, to include climate-resilient land and water resources management strategies (e.g. expanding flood and water resources regulations and policies to prioritize EbA and green and green-grey infrastructure interventions). The implementation of the regulations and procedures developed under the project should contribute to ensuring that the proposed interventions are adopted and consequently contribute to positive impacts on the socio-ecological systems facing the socioeconomic development challenges posed by floods and droughts in the targeted regions.
11. Adaptation impact potential is rated as Medium to High.

1.3 Paradigm shift potential

Scale: Medium to high

12. The proposed project aims to bring about a change in approach to planning for the management of major floods and droughts in Pakistan. Building on the positive results already achieved through the implementation of the EbA and green infrastructure interventions in other regions of the world, it intends to foster an anticipatory and integrated approach to the management of the above-mentioned climate risks, combining both EbA and green and green-grey infrastructure interventions. In addition, to facilitate this change in the approach of the Government of Pakistan and various actors at the departmental level, the proposed project envisages setting up a new coordination structure, the National Working Group. Chaired by the Ministry of Climate Change and composed of regulators within the Ministry of Water Resources and its executive department, the Federal Flood Commission, as well as the relevant provincial departments, this National Working Group will lead the development and adoption of the procedures integrating the interventions promoted by the project.
13. The proposed project will create an enabling environment for the long-term adoption of flood and drought management interventions in Pakistan. Indeed, it will be able to provide evidence on the efficiency and effectiveness of EbA and green infrastructure interventions, and promote reforms in the planning process and national and provincial policies for large-scale flood and drought management in Pakistan. In addition, the project will contribute to building community capacity on the issue. All this is likely to encourage the adoption of the proposed interventions beyond the project's duration.
14. The proposed project has a high potential for scaling up and replication. It will serve as the key proof point in Pakistan that EbA and green infrastructure interventions are cost-

effective and impactful adaptation options. The Government of Pakistan has already committed to expanding EbA and green infrastructure beyond the GCF project's four sites to five additional sites in Haleji, Hadero and Nara-Deh Akro Watersheds in Sindh Province, Indus Dolphin Reserve in Sindh Province, Isa Khel Watershed in Khyber Pakhtunkhwa, and Lala Creek in Punjab Province.

15. It is important to note that the potential for paradigm shift is linked to the proposed project's ability to establish robust evidence of the efficiency and effectiveness of its interventions.

16. Paradigm shift potential of the proposed project is rated as Medium to High.

1.4 Sustainable development potential

Scale: Medium to high

17. The proposed project will contribute to several Sustainable Development Goals (SDG), namely SDG 1 on poverty, SDG 2 on zero hunger; SDG 5 on gender equality, SDG 14 on life below water and SDG 15 on life on land.

18. Furthermore, most of the proposed interventions will have environmental, social and economic co-benefits. Table 17 of the funding proposal (reproduced as a figure below) lists the co-benefits associated with some adaptation interventions.

Co-benefits associated with selected project adaptation intervention types

Adaptation intervention type	Co-benefits		
	Environmental	Social	Economic
Restoration of riverine catchment vegetation	• Revegetation provides long-term stabilization of riverbanks, prevents erosion, and enhances protection of the bank toe from continual scour while reducing fine sediment input. • Improved forest cover. • Improved carbon sequestration. • Enhanced habitats for local wildlife species.	• Improved drinking water quality and supply.	• Increased lifespan and capacity of dams.
Restoration of flow paths	• Improved deposition processes on the floodplain and the reduced movement of gravel. • Improved nutrient and pollution retention in upstream areas.	• Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems.	• Increase in irrigated land and hence food production.
De-siltation of wetlands	• Improved health of native wetland vegetation.	• Improved drinking water quality and quantity.	• NA
Enhancing and reinforcing embankments	• Enhanced habitats for local wildlife species.	• Continuation of cultural, recreational and aesthetic value associated with biodiversity of ecosystems. • Improved drinking water quality and quantity.	• Lower infrastructure maintenance costs and strengthened local

			economies through diversification of livelihoods based on natural resources.
Flood protection embankments	- Improved hydrological connectivity across flood plains. - Improved carbon sequestration. - Erosion control and non-point-source pollution prevention. - Increase of vegetation and biomass with extended growing seasons and enhanced soil moisture. - Sediment retention.	Improved drinking water quality and quantity.	NA
Improved efficiency of irrigation systems	Improved soil health and water content.	Reduced conflict over water resources.	- Cost savings as a result of reduced water and labour inputs. - Intensification of agriculture. - Income diversification and market access for smallholder farmers.
Drought-resistant crop varieties	Reduced degradation of natural areas through expansion of croplands.	NA	Improved food security.
Water harvesting	NA	NA	- Agricultural expansion and diversification of crops. - Additional income

Note: NA: Non-applicable

19. It is important to highlight that the sustainable development potential of the proposed project is also closely linked to the efficiency and effectiveness of the selected interventions.

20. Sustainable development potential is rated as Medium to High.

1.5 Needs of the recipient

Scale: High

21. Pakistan is very exposed and vulnerable to floods and droughts, and it is ranked as the eighth most affected country in the world on the German Watch Long-Term Climate Risk Index (2000–2019). The exceptional floods of 2010 and 2022 clearly demonstrated the country's

great vulnerability to these extreme, slow-onset climatic events, which are set to intensify with climate change. The aforementioned floods caused considerable material and human damage, forcing the country to devote its meagre resources to repairing the damage caused. For example, the Government of Pakistan is no longer in a position to provide the co-financing originally envisaged for the proposed project, as resources have been diverted to emergencies linked to the major floods of 2022.

22. Pakistan's adaptation needs, as well as those related to loss and damage caused by adverse weather conditions induced by climate change, are enormous. They are estimated at several billion United States dollars. Although Pakistan, one of the first countries in Asia to create a Ministry of Climate Change, has funds in the national budget for adaptation actions, it is unable to meet the immense needs in this field because of the urgent actions required by other development initiatives. In addition, the country's high level of indebtedness, its economic development challenges and the national emergency resulting from the impacts of climate change mean that it can only provide limited financial support for adaptation actions. As a result, the country is obliged to turn to international public funding sources such as the Green Climate Fund.

23. Although climate change information and adaptation responses such as EbA exist, Pakistan's water resources sector is still unable to integrate them into its planning processes, making it even more vulnerable to floods and droughts. This lack of integration of climate change issues into water resources management is due in particular to a weak institutional capacity to plan and implement adaptation actions. The proposed project intends to address these shortcomings/gaps by implementing activities such as EbA, and green and green-grey infrastructure in the project sites, and by improving the enabling environment to scale up such activities elsewhere in the country.

24. Needs of the recipients is rated as High.

1.6 Country ownership

Scale: High

25. The proposed project contributes to a set of objectives and priorities defined in the National Climate Change Policy, the framework document for addressing the current and future climate change challenges facing Pakistan. Indeed, the entire proposed project contributes to the objective of pursuing sustained economic growth by appropriately addressing the challenge of climate change. More specifically, outputs 2.1 and 2.2 contribute to the objective of integrating climate policy with other interrelated national policies; outputs 1.1, 1.2, 1.3, 3.1 and 3.2 contribute to the objective of focusing on pro-poor sensitive adaptation; outputs 1.1, 1.2, 1.3 and 3.2 contribute to the objective of ensuring water, food and energy security for the country in the face of challenges posed by climate change; outputs 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.2 contribute to the objective of minimizing the risks arising from the expected increase in frequency and intensity of extreme weather events such as floods, droughts and tropical storms; outputs 2.1, 2.2 and 2.3 contribute to the objective of strengthening interministerial decision-making and coordination mechanisms on climate change; and output 3.1 contributes to the objective of facilitating effective use of the opportunities, particularly financial, available both nationally and internationally. The proposed project is also in line with the National Forest Policy, the National Wetlands Policy, the National Disaster Risk Management Framework, the National Disaster Risk Reduction Policy, and the updated nationally determined contribution.

26. An extensive and inclusive stakeholder consultation process was used to formulate the proposed project. These stakeholders included the Ministry of Climate Change, representatives from national and subnational government such as the Ministry of Water Resources (including the Federal Flood Commission), the National Disaster Management Authority and the Ministry of Planning Development and Special Initiatives; provincial government departments covering

irrigation, agriculture and forestry, and provincial planning and development departments; and community-based organizations, communities and tribal leaders, academic institutions, environmental experts, social activists, and private sector actors. Furthermore, the consultation process included focus group discussions with women at the community level to obtain a better understanding of their needs relating to climate change.

27. Even though key ministries such as water resources, climate change, forestry and agriculture committed to working with departmental programming officers to integrate the project's positive results into their operations, it is still unclear how the national government intends to mobilize the funds needed to ensure the continuity and maintenance of the proposed project's activities. In this respect, it is recommended that the project management works with the established authorities to formulate, from the very first years of the proposed project, a realistic and sustainable strategy and plan for mobilizing resources for actions integrating EbA, green and green-grey infrastructure beyond the project implementation period.

28. Country ownership is rated as High.

1.7 Efficiency and effectiveness

Scale: Medium to high

29. On the basis of the results of the cost-benefit analysis carried out in the context of the proposed project, the package of interventions envisaged is cost-beneficial. The total project investment will generate a net present value of USD 30 million with a benefit-to-cost ratio of 1:1 and an economic internal rate of return of approximately 13.6 per cent.

30. The project has no government co-finance, as a result of the diversion of funds to relief efforts after the major floods of 2022, but the AE is able to make a small contribution and to secure an additional USD 10 million in co-financing from Coca Cola and USAID, bringing the GCF to co-financing ratio to 1:0.18.

31. To inform the design of EbA and green and green-grey infrastructure interventions, the proposed project includes best practices from similar projects including an optimized set of adaptation approaches that are described as "EbA and green infrastructure" in the funding proposal, but which are not all narrowly "green", and can be considered to include soft engineered or hybrid "green-grey" measures, such as flood protection embankments, dispersal embankments, gabion bunds, retention areas and recharge basins.

32. In addition to the information gathered during the stakeholder consultation process, the results of the cost-benefit analysis and the case studies (notably from China¹ and Afghanistan²) greatly guided the design of the project's interventions. Indeed, the approach used in the case of the China project served as a model for most of the green infrastructure and EbA interventions considered by the proposed project.

33. According to the Working Group II contribution to the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report, the "understanding of EbA has improved considerably but there is medium agreement regarding its benefits (Doswald et al., 2014; Nalau et al., 2018a)"³ and "limited evidence and low agreement on its economic efficiency and long-

¹ Paragraph 272 of the FP

² Responses to iTAP questions (Question # 4)

³ IPCC, 2022: *Climate Change 2022: Impacts, Adaptation and Vulnerability*. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844. Page 1340.

Available at https://www.ipcc.ch/report/ar6/wg2/downloads/report/IPCC_AR6_WGII_FullReport.pdf

See also, Doswald, N. et al., 2014: Effectiveness of ecosystem-based approaches for adaptation: review of the evidence-base. *Climate and Development*. 6 (2), 185–201, doi:10.1080/17565529.2013.867247 and Nalau, J., S.

term effectiveness.” This issue was raised during the question-and-answer session with the AE. In response to the question the AE provided a list of peer-reviewed literature and reports to support the efficacy results of the proposed interventions. However, some of them did not directly address the issues raised.

34. Given that the interventions envisaged will be screened again, as indicated in the project document, it would be important to take into account further scientific references confirming their effectiveness when making the final choice. In particular, published literature on the groundwater resources of the Indus basin and the potential for aquifer recharge using monsoon floodwaters should be consulted.⁴

35. Furthermore, establishing evidence of the effectiveness of the EbA and green infrastructure interventions is a fundamental element of the proposed project. However, the implementation plan does not indicate when the monitoring and evaluation plan will be developed. According to the answers provided to the questions submitted by the independent Technical Advisory Panel (iTAP) and the statements made by the AE during the above-mentioned question-and-answer session, this aspect will be taken into account from the earliest stages of the proposed project. With this in mind, it is strongly recommended that the AE present a clear protocol for this complex task of assessing the effectiveness of the proposed interventions. This protocol should be validated by peers to ensure its scientific robustness and consequently guarantee the validity of the evaluation process.

36. Efficiency and effectiveness of the proposed project is rated as Medium to High.

II. Overall remarks from the independent Technical Advisory Panel

37. Based on the above explanations, iTAP endorses the funding proposal “Recharge Pakistan: Building Pakistan’s resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management”.

38. The iTAP recommends the following improvements to the funding proposal:

- (a) The formulation, from the very first years of the proposed project, of a realistic and sustainable strategy and plan for mobilizing resources for actions integrating EbA, green and green-grey infrastructures;
- (b) The formulation, at the early stage of the project implementation, of a clear protocol for assessing the efficiency and effectiveness of EbA and green and green-grey infrastructure interventions;
- (c) Making an additional effort to better take into account scientific and technical references in the design of the EbA, green and green-grey infrastructure interventions; and
- (d) Due to the risk of disease propagation associated with the establishment of retention ponds (resulting from the probability of standing water), it is important to carefully reassess these risks and to ensure that appropriate mitigation measures are put in place.

39. Considering that the proposal has largely satisfied all the GCF’s investment criteria, the above recommendations are left to the discretion of the AE.

Becken and B. Mackey, 2018: Ecosystem-based adaptation: a review of the constraints. *Environ. Sci. Policy*. 89, 357–364, [doi:10.1016/j.envsci.2018.08.014](https://doi.org/10.1016/j.envsci.2018.08.014).

⁴ Four examples are:

<https://wap.hillpublisher.com/ArticleDetails.aspx?cid=777>, https://www.researchgate.net/figure/Three-distinct-groundwater-zones-of-the-Indus-Basin-of-Pakistan_fig4_283421956, <https://hess.copernicus.org/articles/16/1063/2012/hess-16-1063-2012.pdf> and https://www.researchgate.net/publication/259174744_Indus_Basin_Floods_Mechanisms_Impacts_and_Management.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP207)

Proposal name:	Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management
Accredited entity:	World Wildlife Fund, Inc. (WWF)
Country/(ies):	Pakistan
Project/programme size:	Medium

Impact potential
We would like to thank the iTAP for this detailed assessment.
Paradigm shift potential
We would like to thank the iTAP for this detailed assessment.
Sustainable development potential
We would like to thank the iTAP for this detailed assessment.
Needs of the recipient
We would like to thank the iTAP for this detailed assessment.
Country ownership
We would like to thank the iTAP for this detailed assessment.
Efficiency and effectiveness
We would like to thank the iTAP for this detailed assessment.

Overall remarks from the independent Technical Advisory Panel:

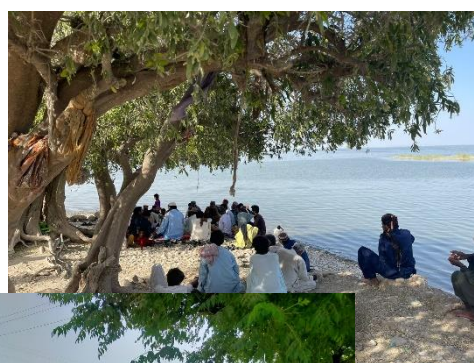
We would like to express our appreciation to the iTAP for their overall remarks on this project and for the recommendations provided.

- (a) During implementation under Sub-Activity 2.1.1.4, the Government of Pakistan will develop a proposed plan for future EbA and grey-green infrastructure projects in additional sites. This will include a proposed resource mobilization plan. The Government of Pakistan and its donor partners, such as USAID, have expressed an interest in supporting the replication of these interventions in other sites beyond the Recharge Pakistan project.
- (b) Under Sub-Activity 2.1.1.2, the project will establish protocols for assessing the efficiency and effectiveness of EbA and green and green-grey infrastructure interventions on ecosystems and their flood attenuation and water resource-related services. We will also subject the protocols to peer review (ideally through a third-party scientific journal), to ensure their credibility and make them more widely available to the adaptation community, including the GCF.
- (c) Additional efforts will be undertaken to take into account other relevant scientific and technical references in the design of the EbA, and green-grey infrastructure interventions. Under the project, a PhD level freshwater scientist will be responsible for ensuring that additional scientific references are consulted.
- (d) The project's ESMF has been updated to include the risk of waterborne diseases associated with the establishment of retention ponds. To mitigate these risks all project outputs, meaning all activities under a given output, in each landscape will be subject to the project's ESS screening, which includes a category on health and safety, to identify possible risks and create any necessary mitigation measures or plans.

World Wide Fund – Recharge Pakistan Project

Recharge Pakistan: Building Pakistan’s resilience to climate change through Ecosystem-based Adaptation (EbA) and Green Infrastructure for integrated flood risk management

***Gender
Assessment
and
Gender Action
Plan***



Integrated Research Solutions Global (Pvt) Ltd.

August 2022



List of Acronyms	4
Preface	6
Acknowledgement	7
Executive Summary	8
CHAPTER 1: Background and Introduction	15
1.1 Background	15
1.2 Introduction	15
1.3 Gender Assessment Approach	16
1.4 Research Methodology	16
1.4.1 Development of Assessment Grid	16
1.4.2 Literature Review	17
1.4.3 Development of Assessment Tools and Guidelines	17
1.4.4 Pre-Testing and Finalization of Tools / Guidelines	17
1.4.5 Sample Size	18
1.4.6 Sampling Methodology	18
CHAPTER 2: Status of Women in Pakistan	20
2.1 Constitutional Rights of Women in Pakistan	20
2.2 Pakistan as Signatory to International Conventions and Treaties	20
2.3 Women's Quotas	20
2.4 Women's Rights Policies and Legislation	21
2.5 National and Provincial Commissions on the Status of Women (NCSW)	21
2.6 National Gender Policy Framework 2022	22
2.7 Gender Gaps in Pakistan	22
2.8 Conclusions	24
CHAPTER 3: Gender Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH)	25
3.1 Global Overview	26
3.2 Green Climate Fund, WWF Gender Policy and Related GBV & SEAH Guidelines	27
3.3 GBV and SEAH in Pakistan	28
3.4 Legislation on GBV and SEAH	29
3.5 GBV and SEAH in the Provinces	30
3.6 GBV and SEAH in WWF Recharge Pakistan Project Sites	31
3.7 Conclusions	32
CHAPTER 4: Gender Considerations in Global & Pakistan Climate Change & Environment Policies	34
4.1 Global Overview	34
4.2 Women & Environment Nexus in Green Climate Fund (GCF) & WWF Gender Policies	35
4.3 Gender Considerations in Environment, Climate Change & Water Policies in Pakistan	36
4.3.1 National Climate Change Policy (NCCP) 2012 / Updated 2021	36
4.3.2 National Water Policy of Pakistan 2018	37
4.4 Conclusions	37
CHAPTER 5: Gender Roles in the Context of Local Ecosystems	39
5.1 Relationship of Women and Men to Natural Resources	39
5.2 Access and Control of Natural Resources	39

5.3	Degradation of Natural Resources	40
5.4	Relationship of Women and Men to Water	40
5.5	Distance to Environmental and Social Services	41
5.6	Impact of Climate Change	41
5.7	Protection of Natural Resources	42
5.8	Participation of Women in Government Projects	42
5.9	Roles and Responsibilities of Government Departments in the Provinces	42
5.10	Management of Water and Environmental Resources	42
5.11	Conclusions	43
CHAPTER 6: Socio-Economic Profile		45
6.1	Age-Sex Distribution and Marital Status	45
6.2	Educational Status	46
6.3	Employment and Professional Status	46
6.4	Household Ownership and Construction	47
6.5	Household Expenditure	48
6.6	Ownership of Assets	48
6.7	Status of Family Indebtedness	49
6.8	Women Involvement in Decision Making Process	49
6.9	Activities of Men and Women Inside and Outside Home	50
6.10	Conclusions	50
CHAPTER 7: Gender Mainstreaming in WWF		52
7.1	WWF Gender Policy	52
7.2	Review of WWF Annual Reports 2019-2021	52
7.3	WWF Gender Projects	53
7.4	Case Studies from the Field	53
7.5	Focus Group Discussion with WWF's Female Staff	54
7.5.1	Recruitment, Initiation and Orientation	54
7.5.2	Familiarity with WWF's Gender Policy	55
7.5.3	Gender Design and Development	55
7.5.4	Organogram and Gender Presence	56
7.5.5	Work Environment	56
7.5.6	Changes in Women's Position in Last 10 Years	57
7.6	Conclusions	57
Gender Action Plan		59
Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan		60
Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan		65
Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin		68
Bibliography		72

List of Tables

• Table 1: Field Sites of the Project	18
• Table 2: Household Survey Respondents	19
• Table 3: Number of FGDs and Number of Participants	19
• Table 4: Key Informant Interviews (KII) Conducted in 4 Provinces	19
• Table 5: Gender Development Indicators Mapped	23
• Table-6A Distance to Nearest Water Sources from Home	40
• Table-6B Distance to Agriculture Land from Home	41
• Table-6C Distance to Fuel/Firewood from home	41
• Table-6D Distance to Animal Fodder from Home	41
• Table-6E Distance to Sanitation	41
• Table-7 Age - Sex Distribution	45
• Table-8 Educational Status of Children (age 4 to 16 years)	46
• Table-9 Educational Status of Adult (age 17 years and above)	46
• Table-10A Employment / Profession of Children (age 16 years)	46
• Table-10B Employment / Profession of Adults (age 17 years and above)	47
• Table-11 Type of Family Expenditures	48
• Table-12 Ownership of Family Assets	49
• Table-13 Source of Loan	49
• Table-14A Women Involvement in Decision Making Process	50
• Table-14B Type of Decisions in which Women Participate	50

List of Annexes

- A1 WWF Assessment Grid.
- A2 WWF Recharge Pakistan Literature Review.
- A3 Informed Consent Form English.
- A4 Informed Consent Form Urdu.
- A5 Family Survey Form English.
- A6 Household Survey Tool English and Urdu.
- A7 FGD Guidelines for Community Men and Women English and Urdu.
- A8 Interview Guidelines for Government Officials.
- A9 FGD Guidelines for WWF Female Staff Members.
- A10 List of Government Official Interviewed
- A11 WWF Female Staff - Positions and Departments.
- A12 WWF Organization Structure.
- A13 Guidelines for Moderator, Note Taker & Logistic In charge.
- A14 Field Images.

List of Acronyms

AIDS	Acquired Immune Deficiency Syndrome
CAPNET	Capacity Development for Sustainable Water Management Network
CBOs	Community Based Organizations.
ccGAP	Climate Change and Gender Action Plan.
CEDAW	Convention for the Elimination of All Discrimination Against Women.
CSOs	Civil Society Organizations.
DCSW	District Committee on the Status of Women.
DPU s	Domestic Poultry Units.
EbA	Ecosystem based Adaptation.
EWT	Every Woman Treaty
FAO	Food and Agriculture Organization.
FATA	Federally Administered Tribal Areas.
FGDs	Focus Group Discussions.
GAD	Gender and Development.
GBV	Gender Based Violence.
GCF	Green Climate Fund.
GDI	Gross Domestic Income.
GEF	Global Environment Facility.
GEP	Gender Equity Program.
GGCA	Global Gender and Climate Alliance.
GGGI	Global Gender Gap Index.
GHG	Green House Gases.
GII	Gender Inequality Index.
GLOF	Glacial Lake Outburst Floods.
GWP	Global Water Partnership.
HH	Household.
HIV	Human Immunodeficiency Virus.
HRC P	Human Rights Commission of Pakistan.
IDH	The Sustainable Trade Initiative.
IECLP	Indus Eco Region Community Livelihood Project.
IKEA	Ingvar Kamprad Elmtaryd Agunnaryd (Swedish Funding Agency)
ILO	International Labour Organization.
IRS	Integrated Research Solutions Global.
IUCN	International Union for Conservation of Nature.
KPK	Khyber Pakhtunkhwa.
MFF	Mangroves for the Future.
MOCC	Ministry of Climate Change.
MOU	Memorandum of Understanding.
NCCP	National Climate Change Policy.
NCSW	National Commission on Status of Women.
NDC	Nationally Determined Contributions.
NGCCC	National Gender and Climate Change Committee.
NGOs	Non – Governmental Organizations.
NPA	National Plan of Action.
NSAP	National Strategic Action Plan.
NWP	National Water Policy.
OECD	Organization for Economic Co-operation and Development.
PCSWs	Pakistan Commission on Status of Women.
PDHS	Pakistan Demographic and Health Survey.
PIDE	Pakistan Institute of Development Economics.
PKR	Pakistan Rupee.
PPAF	Pakistan Poverty Alleviation Fund.
PRA	Participatory Rural Appraisal.
RNE	Royal Netherlands Embassy.
RSPs	Rural Support Programs.
SDGs	Sustainable Development Goals.
SEAH	Sexual Exploitation, Abuse and Harassment.
SHRC	Sindh Human Rights Commission.

SIGI	Social Institutions and Gender Index.
TOR	Terms of References.
TV / AC	Television and Air Conditioner
UN	United Nations.
UNDP	United Nations Development Program.
UNFCCC	United Nations Framework Convention on Climate Change.
UNICEF	United Nations Children's Emergency Fund.
UNODC	United Nations Office on Drugs and Crime.
USD	US Dollar
VAW	Violence Against Women.
VOs	Village Organizations.
WDD	Women Development Department.
WEDI	Women Entrepreneurial Development Initiative
WEDO	Women Environment and Development Organization.
WEF	World Economic Forum.
WGs	Women Groups.
WID	Women in Development.
WWF	World Wide Fund.

Preface

For the project team that has delivered this assessment, it was much more a labor of love than a routine consulting assignment. The core team gave much more time and effort than originally intended, to cover the expanding scope of the work, but did so without complaint because the subject was very close to our hearts. In addition to their vast combined experience in water, environment, and climate change sectors, three of the four members of the core team have been actively involved in the global and Pakistani women's movement and in women-led research organizations and have been able to bring their vast experience in these sectors to inform and enrich this assessment.

The core team that has envisaged, designed, managed, analyzed and reported on this assessment are the following:

- Ms. Simi Kamal (Team Leader)
- Ms. Meher M. Noshirwani (Gender Specialist)
- Ms. Kausar Hashmi (Field / Data Collection Expert)
- Mr. Farzand Ali (IRS Manager Operations)

The field and logistical team from IRS Global, colleagues from DevCon, government officers from the field sites and WWF staff members have been duly mentioned under the Acknowledgements that follow separately.

We believe that the material in this assessment can be used for a number of purposes. It can directly feed into the Gender Action Plan (GAP) for the proposed GCF project (as has been done in the last chapter of this report). It can also help WWF improve its own systems to enhance gender inclusion and gender leadership, and initiate measures in its programmatic interventions to propel women's voices, enhance women's empowerment and institute gender mainstreaming.

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- Aziza Burfat, Shafquat Ali, Rafia – Sindh field team.
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- Dr. Owais Ahmed Data Manager (DevCon).
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Executive Summary

The World Wildlife Fund (WWF) / Green Climate Fund (GCF) on behalf of and in collaboration with the Government of Pakistan, is preparing a proposal for the project **Recharge Pakistan: Building Pakistan's resilience to Climate Change through ecosystem-based adaptation for Integrated Flood Risk Management**.

In this context, the WWF Recharge Pakistan project has conducted a Gender Analysis and developed a Gender Action Plan to ensure that gender mainstreaming is integrated into the design phase of the project.

The project's proposed activities will be implemented under three components:

- Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan.
- Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan.
- Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin.

As a result of WWF-Pakistan's engagements during Concept Note and Funding Proposal development with national (MoCC and MoWR), local (provincial irrigation departments) and international (e.g., World Bank, ADB, KfW, GIZ, etc.), the proposed project was designed to have a total of ~US\$67 million in government co-finance from the NDRMF (~US\$37 million) and MoWR (~US\$29 million) committed to the project to complement the GCF grant. While ~US\$67 million in government co-finance from the NDRMF and MoWR was committed to the project to complement the GCF grant, the recent 2022 floods have required that these funds be repurposed for immediate disaster relief making it very difficult for the Government of Pakistan to commit significant adaptation finance at this time. Despite these constraints, the GoP remains committed to the proposed project but is unable to commit any co-financing for the project due to the economic constraints on public spending due to the high costs for recovery from the 2022 floods.

As a result of the above-mentioned co-financing no longer being available for the project, the project has been re-scaled from 6 sites [Manchar, Kaha, Taunsa, Chakar Lehri, DI Khan and Ramak] to four sites [Manchar, Chakar Lehri, DI Khan and Ramak]. This gender assessment was undertaken for all six sites and includes, in full, the gender assessment for the remaining four sites that will be targeted under the project. The Gender Action Plan has been revised to reflect the activities required only in the four project sites to ensure the project interventions are gender-responsive.

The framework for the Gender Analysis was based on an Assessment Grid with 26 questions, outlining the secondary and primary data requirements. The following steps were designed and implemented:

- Development of Assessment Grid
- Literature Review and Secondary Data covering the status of women in Pakistan, global and Pakistani GBV and SEAH (gender-based violence including sexual exploitation, abuse and harassment) situation, policies of relevant sectors, examples of relevant projects, women's participation and leadership in the environment sectors, impact of climate change, and participation and women's leadership in WWF

- Development of Assessment Tools and Guidelines (household survey, focus group discussions (FGDs) and key informant interviews (KII)
- Pre-Testing and Finalization of Tools / Guidelines
- Data Collection through 15 FGDs, 527 households and 10 key informants
- Analysis and Reporting
- Gender Action Plan

Status of Women in Pakistan

The position and status of women in Pakistan has a checkered history. Over the last three decades the status of women has progressed, remained static, regressed, and constantly changes depending on many structural factors. Numerous laws have been promulgated ensuring women rights and protection, but their implementation and lack of political will renders them largely ineffective. Very little attention is placed on the crucial nexus of culture, religion and policy development and implementation of laws. The plethora of laws, negative socio-norms and practices have become concrete in Pakistan's patriarchal society in a way that they consistently favor men and place women at disadvantage.

Due to some of the above-mentioned reasons, women in Pakistan lack (or have inadequate) capabilities to fully and meaningfully participate in decision making processes, political/governance spaces, and in policies and programs. Formal institutions including both private and public, fall short of facilitating women's access to human, financial and physical capital i.e. education, skills, health facilities, loans and physical assets. Informal institutions (family, community, religion, segregation, marriage) define and affirm norms that give precedence to marriage/child marriage over education and employment, prescribe gendered roles, and condone domestic violence/gender-based violence. Laws have therefore not had the expected impact.

The World Economic Forum Global Gender Gap Report of 2022¹ found that Pakistan is the second worst country in the world in terms of gender parity; ranking Pakistan 145 out of 146 countries worldwide.

Pakistan is among the countries that are most vulnerable to the risks associated with climate change and ranks 8th on the Global Climate Risk Index 2021.

Pakistan's women are further disadvantaged because 9.1 million women agricultural workers play a substantial role in food production and food security, but they are largely unpaid, suffer from greater time poverty, and are far more vulnerable to exploitation than men. Women's ownership of land, and control over physical assets is minimal: only 2 percentage of women own a house or agricultural land.

Unfortunately, the recent Gender Policy Framework (March 2022) of the Government of Pakistan needs to do more to close the gender gaps and propel the levels and scope of action required to counter anti-women norms and practices. A study conducted by UN highlights that the recent floods of 2022 "serve as a case in point, pushing back many of the gains made by the women in Pakistan in over a decade"² due to poor governance, weak response mechanisms and plans that are made without the gender lens.

The experiences of the 2005 earthquake, and the floods of 2010 and 2011 have shown that the patriarchal norms that shield men from their crimes against women, and victimize the women instead, become even stronger in emergencies. The floods of 2022 have once again

¹ World Economic Forum, Insight Report: Global Gender Gap Report 2022.

² UNDP, Gender Equality in Pakistan: Climatic & Politico-Economic Stressors VOLUME 9 / ISSUE 4 | DECEMBER 2022 - JANUARY 2023

exposed women to increased risk of gender-based violence and discrimination which could lead to sexual abuse, harassment, exploitation, and early and forced marriages. The United Nations Population Fund³ noted that there are almost 650,000 pregnant women in flood-affected areas, of which almost 73,000 were due to give birth in September 2022. These women are particularly vulnerable and face numerous health and safety risks. With the lack of adequate protection and safeguarding mechanisms in flood-affected districts, women and girls will be largely cut off from the support they need.

However, the presence of the National Commission on the Status of Women, Provincial Commissions, the Women's Caucus in Parliament and NGOs working on women's rights, gender mainstreaming, combatting violence against women should not be underestimated. The project should make links with these organizations and work with them proactively to develop the support actions needed to support women in the four project sites. Links with local government offices, extension services, law enforcement and justice offices will be key to helping women and men to access government services and obtain redress.

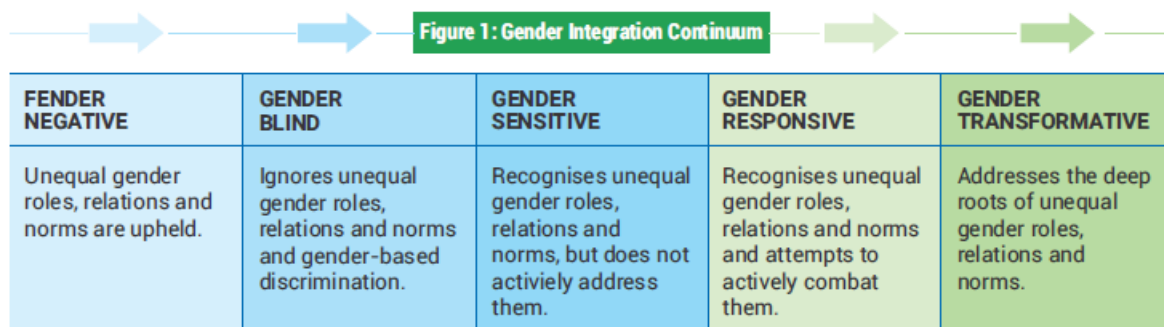
Gender Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH)

According to Global Database on violence against women, 24.5% have experienced lifetime physical and/or sexual intimate partner violence between 2017 and 2018. There is no confirmed national level comprehensive statistics on GBV and SEAH. Post 2022 floods in Pakistan, there have been multiple studies and gender needs assessments conducted by multi-lateral donors, but no specific study has been conducted that focuses purely on GBV and SEAH related incidence in the aftermath of this disastrous floods. However, it can be safely said that SEAH is subsumed under the terms Gender Based Violence (GBV), 'honor killing' and 'domestic abuse'. GBV and SEAH are complex issues and are a means of control to maintain power structures, traditional norms, customary laws, and access to resources, all resulting in the oppression of women and gender inequalities.

Another dimension of GBV is the connection with the environment - **pressures on the environment and its natural resources increase gender inequalities and power imbalances in communities and households coping with resource scarcity, climate change, and environmental degradation.** The linkages between environmental degradation, climate change and the impact of covid-19 has led to an increase in GBV and SEAH, as has been documented globally.

In terms of legislation, Pakistan is signatory to international conventions, and has many progressive and supportive laws that could provide holistic mitigation to and address GBV and SEAH. Unfortunately, other laws that remain on the books (such as the Hudood Ordinances) counter the progressive laws and thus setting back any progress already made. The net result is that women rarely get the justice and redress they need. The effectiveness of these laws in relation to gender issues can be measured according to the Gender Integration Continuum given below. The scale has been coined to measure the level of responsiveness of a particular policy or law to not only identify but also address gender issues specifically climate change, insecurity (GBV and SEAH) and gender nexus.

³ <https://pakistan.unfpa.org/en/news/women-and-girls-bearing-brunt-pakistan-monsoon-floods>



Based on the Gender Integration scale, the following policies and laws are advantageous and supportive to the project.

Supportive Gender Laws	Law	Advantage	Gender Integration
	Climate Change Gender Action Plan 2022	ccGAP addresses gender-specific experiences of climate impact in disasters, agriculture and food security, biodiversity, coastal management, water and energy.	Gender Responsive: The policy is an integral guidelines in establishing women's agency and inclusion in problem identification and interventions specific to those covered under the Recharge Pakistan Gender Action Plan
	National Gender Policy Framework 2022	The policy identifies and maps the major gender issues and lack of protections currently prevalent in the country.	Gender Sensitive: While the policies gives a detailed insight into the national gender issues, it does not provide adequate guideline to address the challenges.
	Protection Against Harassment of Women at Workplace Act 2010	The Act identifies that 80% of the female employees working in the formal and informal sectors in Pakistan are sexually harassed.	Gender Responsive: The law develops jurisprudence and legal infrastructures to address SEAH at workplace.
	Domestic Violence (Prevention & Protection Bill) 2009-10	The Domestic Violence Bill makes violence against women and children an offence, punishable by time in jail and imposition of fines.	Gender Responsive: The law is responsive to domestic violence experiences of women in Pakistan but places the onus of reporting and conviction on the victims.
	Anti-Honour Killing Laws (Criminal Amendment Bill) 2015	The law criminalizes the act of violence against and killing of women in the name of 'honor'.	Gender Responsive: The law is responsive to gender-based violence against women in the name of 'honor'.

The focus group discussions have revealed the existence of various forms of gender-based violence in the field sites. Even though direct questions were not asked about SEAH to community members participating in FGDs (since this would have required a specialized approach using trained psychologists due to the sensitive nature of these questions), physical violence has been mentioned by women respondents as issues they are faced with in their community. Disrespect for women has also been cited which indicates a deeper insidious side of violence which must be addressed if gender equality and women empowerment is to bring real change to the lives of the women in the communities, and their household and country in general.

Reducing violence and SEAH against women and girls requires a community-based, multi-pronged approach, and sustained engagement with multiple stakeholders. To be effective, these initiatives would have to address social norms on gender roles and the acceptability of violence. This is very hard to do in rural patriarchal feudal societies, of the kind where WWF is attempting its Recharge Pakistan initiatives.

Community engagement is vital to reduce GBV and SEAH and identify victims and survivors that need help. In order to do this community members (men, women, elders, youth, religious figures, influential people, local law enforcement and government officials, local magistrates and journalist, local health personnel, school teachers etc have to be sensitized to the issues, supportive laws, cultural positives and negatives and where redress may be found. For example, nearest lawyers, legal aid centres, shelters etc.

GBV and SEAH are challenging yet necessary to address. The issues are ingrained in traditionally held gender relations mixed in conflicts on property, land ownership and increasing poverty. Redress needs actual action and systems on the ground, not just training and orienting people. A woman attacked with acid or who has been raped needs immediate shelter (sometimes for month or years), medical support, legal aid, economic support, community support, rehabilitation, education for her children and psycho-social counselling over extended periods. WWF does not currently have a team or program or cross-cutting approach to cover these components. However, the organization believes that climate resilience can only be achieved when rights and privileges of vulnerable populations are protected by the state. Despite GBV and SEAHs being critical to women rights, preventive measures can only be taken within the boundaries of the project. WWF will ensure that preventive measures are in place to mitigate future GBV and SEAHs related instances in the project sites. The GAP also outlines gender specific actions to proactively recognize and address issues related to GBV and SEAHs.

GBV and SEAH will be addressed in the GCF project, through various strategies and recruiting relevant expertise in the project team, who will be dedicated to work constantly on engaging with parliamentarians (to influence better laws), especially the Women's Caucus and women parliamentarians. This team should also engage with the National Commission on the Status of Women (NCSW), the Provincial Commissions on the Status of Women (PCSWs), the Human Rights Commission (HRC) and the Provincial Commissions on Human Rights (PCHRs). Orientation is required of law enforcement officers and the judiciary to handle GBV and SEAH cases. The GCF and WWF's own policies and documents form a strong basis for training and orientation on gender-based violence issues and for mainstreaming GBV concerns in its projects. Other strategies to ensure challenges relating to GBV and SEAH the project will employ include: the development of a terms of reference to recruit a gender specialist in the PMU, who has strong and proven expertise in working on issues of GBV and SEAH in Pakistan, among other essential and important experience. The gender specialist will be the resource person in the project who will conduct capacity needs assessment, provide training (including recurrent/refresher trainings) and support capacity building of all project staff and relevant partners, to recognize and address SEAH and GBV issues identified during project implementation, which will be reported through the gender-responsive and survivor-centered grievance mechanism established for the project (See Section 5.3 in Annex 6 for detailed guidance on SEAH Risk Mitigation).

Gender Considerations in Global & Pakistani Climate Change & Environment Policies

The challenges of climate change impacts, building resilience and countering gender inequalities are all interlinked.

By exacerbating inequality, climate change slows progress toward gender equality and thus impedes efforts to achieve wider goals like poverty reduction and sustainable development. Conversely, gender inequality can worsen the impacts of climate change and further propel the 'feminization' of violence, poverty, marginalization and exclusion. Therefore, taking steps to narrow the gender gap and empower women can help counter the impacts of climate change.

Women continue to face serious challenges amid threats to climate degradation, particularly in occupations that are dependent on natural environments or geographic locations and where climate stress may cause natural hazards. Agriculture, water, and forestry are among the sectors most impacted by climate change and women are heavily engaged in these sectors as full-time labour or secondary workers in addition to their dependence on natural environments for sustenance. Declining crop yields disrupted water availability, rural out-migration, and frequent natural disasters triggered by climate stress produce a precarious survival scenario for communities, especially women.

As women are disproportionately impacted by the adverse effects of climate change, Pakistan ratified the Paris Agreement (2015) which mandates gender equity and justice in climate action by member states.

The Ministry of Climate Change, Government of Pakistan has developed the Climate Change Gender Action Plan (ccGAP) in partnership with International Union for the Conservation of Nature (IUCN) Pakistan. The ccGAP demonstrates the commitment by the Government of Pakistan to ensure the inclusion and leadership of women in climate action. While aligned with the priority sectors identified in the National Climate Change Policy (NCCP) of 2021, it is an important step towards achieving a key objective of the NCCP to focus on pro-poor gender sensitive adaptation

The National Climate Change Policy updated 2021 is the current policy which has a specific focus on gender and states that climate change is likely to affect poor and underprivileged communities, and that rural women are likely to be strongly affected by climate change. Much progress has been made in setting ambitious targets for adaptation and mitigation, there is reappraisal of women's vulnerability and gender-sensitive objectives to address women's differentiated burdens in climate stress. The recently updated National Climate Change Policy and the Nationally Determined Contribution (2021) pay due attention to women's differentiated burden in climate emergencies and how to help them.

Therefore, it can be said that the policy environment is feasible for positive reinforcement of gender mainstreaming in ecosystem and resilience-based programming in Pakistan. The GCF and WWF's gender policies are both good instruments to guide the design, and implementation of the intended WWF project in Pakistan.

Socio-economic Profile of Communities

Any gender assessment requires a socio economic and environmental context within which gender issues can be understood correctly and the most suitable and effective interventions developed and implemented. This is especially required in the case of projects and programs that are community based where, in addition to national and provincial situation, the local poverty, control, power and normative environment has to be taken into consideration.

The Socio-economic picture that emerges of the communities in the six WWF sites shows clearly that they are very poor, indebted and live from hand-to-mouth. They are not much used to accounting for money handling and are not very accurate in terms of their income and expenditures.

We also see that mostly men are involved in work that is usually paid (when they get the work), women usually look after domestic chores and are unpaid. In some instances, women and men worked together in fishing, fetching water. Except for fisheries and livestock, women did not report working in any professions or carrying any jobs. While men left their communities for labor work elsewhere, women tend to remain.

The low numbers of women respondents recording that they do not spend money of any kind reinforces the findings that women do not have much say in decision-making or household spending, and may not even be aware of how much is being spent by the men of the family on key households' expenditures.

Most of the people there reported unemployment - while men and some women have skills and professions (traditionally through the decades), **currently they do not have a means of income. We can see that the environment for their traditional professions and means of income (fisheries, agriculture, livestock and handicrafts) is dwindling and degrading – through climate change, privatization and government take-over.** Lifting the socio-economic profiles of the poorest communities in the six project sites will be a challenge, so that earning opportunities for men and women have to be an integral part of the design of the project.

Gender Roles in the Context of Local Ecosystems

Climate change has already negatively affected the equation and balance between people and nature. The rapid changes are visible and the feared migrations, poverty and loss of livelihoods are a reality to different degrees in all six project sites.

The impact of climate change in the form of changing weather patterns and extreme temperatures are changes that are a reality for the women and men in the project sites. Adaptation to these changes and interventions for addressing these changes need to be designed with the participation of the community including the women. Shrinking natural resource base has led to migration and a shift from traditional livelihoods to manual labor. The women left to tend animals and mind the home face daily struggles for survival. Alternative livelihoods are now needed.

The tendency of local landowning elites to commandeer resources and their ability to manipulate government functionaries in difficult times means that meager environmental and ecosystem resources will continue to be denied to the landless and the poor, especially women.

WWF has to be careful in the way the gender dialectic is used, because it can lead away from the plight of both poor men and women in the face of elite capture. The focus should not only be on building gender parity between community women and men, but in providing interventions that also protect communities from unilateral action by elites and well-intentioned but poorly resourced government institutions that render poor men and women bereft of their traditional livelihoods. Extra care is needed to ensure that women's environmental entitlements remain and they continue to access environmental resources for their water and food security.

As for the government, it is clear from the interviews of functionaries that they are mostly clueless on gender terminology, policies and actions and would require a lot of attention in the intended project design and implementation of the Gender Action Plan.

Gender Mainstreaming in WWF

WWF already has a strong Gender policy and follows the GCF Gender Policy Guidelines. These already contain very good guidelines for gender mainstreaming.

WWF also has several projects that focus on women and support women's empowerment, but gender data is scattered over different reports and websites. An Annual Gender Report covering all the current projects with disaggregated data, number of male and female beneficiaries and impact on women's empowerment would provide a valuable resource for gender mainstreaming in future WWF projects.

However, it was seen that women staff members were not fully aware of these resources or used them well. Gender trainings for both men and women should be developed and integrated into the orientation process for both existing and new employees. Gender concepts, WWF's gender policies, and gender laws and policies in Pakistan should be shared with staff members from time to time. Gender analysis training in designing gender projects, collection of data, tabulation, and reporting would increase the understanding and capacity of the staff for gender responsive and gender transformative programming.

A gender focal person designated to ensure that gender policies are translated through the organization at all levels and to organize and deliver the required training would be very helpful.

And finally, there are several women in WWF at different strata of the organization, but only a few in leadership and decision-making positions. Therefore, these women should be trained and provided opportunities for move towards leadership positions, and a clear career path should be offered to women as they progress within WWF.

Gender Action Plan

Based on the gender assessment results, , stakeholder consultation recommendations, and the KII, the project aims to mainstream gender using a gender-responsive approach throughout each project component. The Gender Action Plan below reflects the activities required in the four project sites to ensure the project interventions are gender-responsive.

CHAPTER 1: Background and Introduction

1.1 Background

The World Wildlife Fund (WWF) / Green Climate Fund (GCF) on behalf of and in collaboration with the Government of Pakistan, is preparing a proposal for the project **Recharge Pakistan: Building Pakistan's resilience to Climate Change through ecosystem-based adaptation for Integrated Flood Risk Management**. This project brings a paradigm shift towards managing floods through large-scale Ecosystem Based Adaptation (EbA), because the Indus River the lifeline of water-stressed Pakistan causes regular floods exacerbated by climate change, and Pakistan currently relies on expensive infrastructural flood management measures with limited efficiency⁴.

WWF GCF Accredited Entity requires that all GCF projects comply with WWF Gender Policy in the design and development of the gender analysis and action plan. WWF adopted its Gender policy in 2011, and this policy reflects WWF's ongoing commitment to equity and integrating a gender perspective in its policies, program, and projects, as well as in its own institutional structure⁵.

The WWF Global Network Policy: Gender Policy Statement 2011 states that *'the policy statement is to ensure that WWF's conservation policies, programs and activities benefit women and men equally and contribute to gender equity, as part of WWF's broader commitment to strengthen the social dimensions of its projects, programs, and policy work. It describes the rationale for gender mainstreaming in the context of WWF mission, biodiversity and footprint goals and outlines WWF commitment to integrate a gender perspective in our programmatic and operational structures and procedures.'*

The policy outlines WWF's belief in promoting gender equity and integrating a gender perspective in projects and programs. WWF is also committed to ensuring an organizational culture and work environment where management and other staff are aware of discrimination and respect diversity in work⁶.

1.2 Introduction

In this context, the WWF Recharge Pakistan project has conducted a Gender Analysis and developed a Gender Action Plan to ensure that gender mainstreaming is integrated into the design phase of the project, which reflects WWF's continuing commitment to its policy and future action.

The Recharge Pakistan Project has a three-pronged approach which includes: ecosystem-based flood management through wetland restoration, hill-torrent management, and green infrastructure at several priority sites; community-based natural resource management, climate-adapted livelihoods, and risk reduction; and a paradigm shift to scale up the Ecosystem Based Adaptation (EbA) approach. The proposed project will have multiple, long-term, wide-ranging benefits for Pakistan, including:

- a) EbA that improves resilience to flooding and drought;
- b) Increased capacity of water-stressed areas to efficiently store and use floodwater;
- c) Enhanced recharge of groundwater through wetland restoration interventions, and hill torrent management interventions, e.g. watershed restoration, check dams, and terracing.

⁴ Technical Proposal Gender Assessment Integrated Research Solutions (PVT) Ltd. 2022.

⁵ https://www.wwfpak.org/about_us/_wwf_s_policies/gender/.

⁶ WWF Policy 2011, Global Network Policy: Gender Policy Statement.

- d) Other environmental and socio-economic benefits, e.g. biodiversity and habitat conservation, carbon sequestration, wetlands-based livelihoods.

The project's proposed activities will be implemented under three components:

- Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan.
- Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan.
- Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin⁷.

1.3 Gender Assessment Approach

A Gender analysis is a methodology to examine the differences in roles and norms for women and men, and differences within social groups, the different levels of power they hold, their differing needs, constraints, and opportunities, and the impact of these differences in their lives. Gender analysis also attempts to explain the mechanisms by which gender inequality is maintained through cultural norms, and institutional structures, as well as the ways with which social groups of men and women act to transform their own existing roles, relationships, and processes in their own interest⁸.

To conduct a Gender Analysis and develop a Gender Action Plan, a gender assessment of the project sites was conducted to collect gender-specific data to identify gaps, opportunities, and challenges and increase the understanding of the roles of women and men in the Recharge Pakistan project sites. Decision-making processes among women and men at the community level, control over the management of natural resources and the types of resources, access to land by women and men in the project area, how benefits from projects and other opportunities are shared, were part of the criteria of the gender assessment. The aim was to identify options for equal participation of both women, men, in climate change adaptation and promote more equitable management of and benefits from, the use and protection of natural resources⁹.

The framework for the Gender Analysis was based on an Assessment Grid¹⁰ prepared by IRS Global. This Grid had 26 questions to be answered by the assessment. A literature review of the status of women in Pakistan, gender-based violence (including sexual exploitation, abuse and harassment - SEAH), women's participation and leadership in the environment sectors, impact of climate change, and participation and women's leadership in WWF. Primary data collection then followed.

1.4 Research Methodology

Both quantitative and qualitative data collection techniques were employed, using research tools such as a household survey, focus group discussions (FGDs) and key informant interviews (KII) with government officials. Details of the Methodology are given below:

1.4.1 Development of Assessment Grid

An Assessment Grid to conduct the Gender Analysis was developed at the beginning of the assignment, where the team leader, with the full team, developed a framework to be followed for the assignment. All the questions to be answered were identified, as were the sources of

⁷ Technical Proposal Gender Assessment Integrated Research Solutions (PVT) Ltd. 2022.

⁸ <http://www.mangrovesforthe future.org/assets/Repository/Documents/Gender-Analysis-Toolkit-for-Coastal-Management-Practitioners.pdf>.

⁹ Technical Proposal Gender Assessment Integrated Research Solutions (PVT) Ltd. 2022.

¹⁰ Gender Assessment for Recharge Pakistan Program (WWF) Assessment Grid, 15 March 2022.

primary and secondary information. Methods to access the information and data collection instruments were also developed and the Assessment Grid was shared and agreed with WWF (Annex A1).

When the Gender Assessment Grid was discussed with WWF, the question of exploring gender-based violence and sexual exploitation, abuse and harassment (SEAH) was taken up. The research team conveyed to the WWF team that to ask direct questions on these sensitive issues in small communities through household survey or in focus group discussions would spread alarm in these communities, the field teams would be viewed with suspicion and communities will close ranks. **It was suggested by the assessment team that a separate investigation with psychologists collecting the data was called for and that this would require additional budget.** WWF advised the research team to go ahead as planned with indirect questions in the FGDs and use existing reports for overview.

1.4.2 Literature Review

A literature review was conducted of the academic literature and development sector reports covering legal frameworks, laws, policies, institutional practices on gender equality, women's empowerment in the wider socio-cultural context, that will promote and support the gender mainstreaming contextualized in the project. The overarching documents used for the Gender Analysis is the National Gender Policy Framework 2022, the Climate Change Policy 2012, Recharge Pakistan: Building Pakistan's Resilience to Climate Change through Ecosystem-Based Adaptation for Integrated Flood Risk Management, WWF Annual Reports 2019, 2020, and 2021. Details of all the reports and documents consulted are in Annex A2.

1.4.3 Development of Assessment Tools and Guidelines

Based on the Assessment Grid the following tools were developed:

- Gender Assessment Family Survey Form.
- WWF Household (HH) Survey Tool.
- Interview Guidelines for Government Officials.
- FGD Guidelines for Community Men and Women (including members of community groups or community-based organizations, if existing in the area)
- FGD Guidelines for WWF Female Staff.
- Guidelines for Moderator, Note Taker and Logistic In charge.

All the research tools were developed in English and relevant tools were translated into Urdu. All survey tools are presented in Annexes A5 – A9. Consent forms in English and Urdu were also used as required (Annex A3 – A4).

1.4.4 Pre-Testing and Finalization of Tools / Guidelines

The tools and guidelines were pre-tested, and changes incorporated. The pre-testing was carried out in Halaji Lake and around Manchar Lake in Sindh. This was also an opportunity for training of field staff.

The pre-tested tools were shared and discussed with WWF team members in several online meetings and then finalized in a meeting at the WWF office in Islamabad.

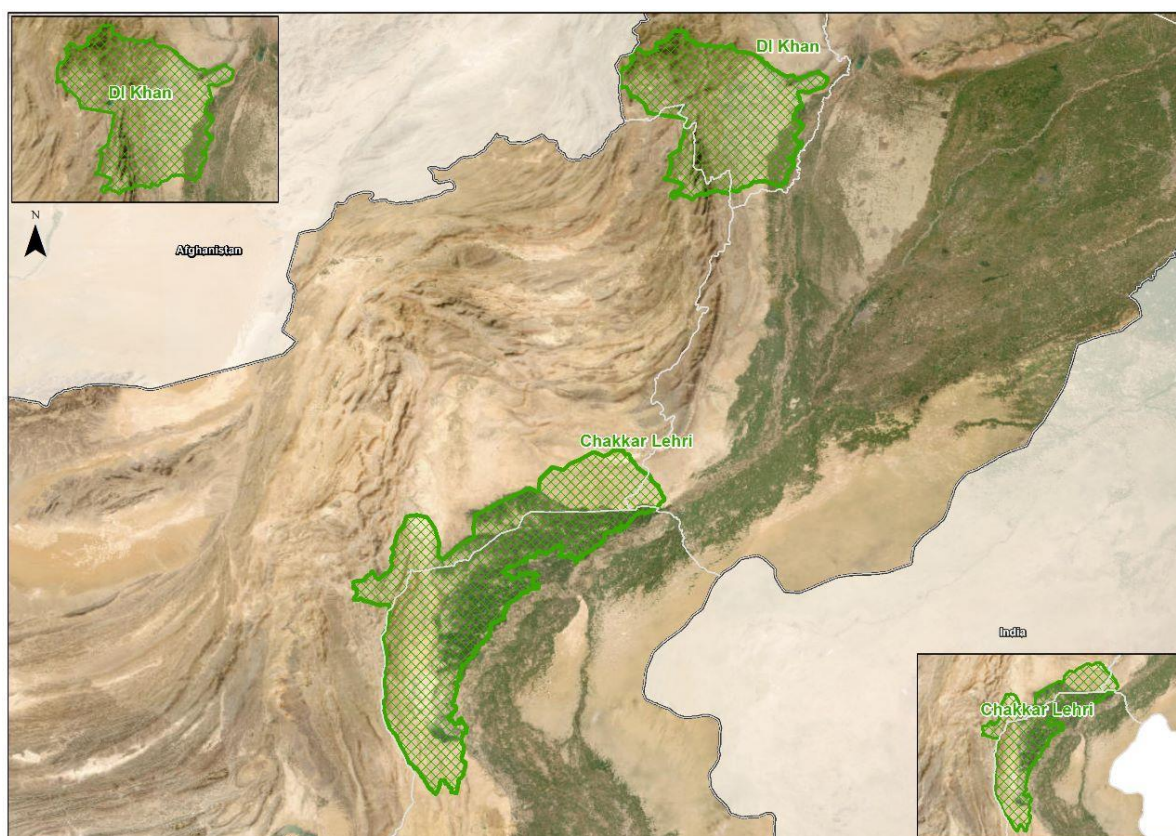
1.4.5 Sample Size

The gender assessment field work was carried out in 6 locations of all four provinces as identified by WWF. These locations were:

Table 1: Field Sites of the Project

S. No.	Location	District	Province
1.	Manchar Wetlands Complex	Dadu	Sindh
2.	Kaha Hills	Rajanpur	Punjab
3.	Taunsa Barrage Wildlife Sanctuary	Dera Ghazi Khan	Punjab
4.	Chakar Lehri Sub-Basin	Sibi	Balochistan
5.	Hill Torrents	Dera Ismail Khan	Khyber Pakhtunkhwa
6.	Ramak	Dera Ismail Khan	Khyber Pakhtunkhwa.

Map of the Field Sites



1.4.6 Sampling Methodology

The sampling methodology for quantitative and qualitative information is given below.

- Household Survey**

Quantitative data collection was the joint responsibility of IRS and DevCon, as requested by WWF. IRS prepared the tool in English and Urdu and shared with DevCon. This tool was incorporated in a wider data collection tool used by DevCon. The household survey was conducted by DevCon and covered a total of 527 households from 6 locations/sites of the total respondents 311 were men and 216 women.

Table 2: Household Survey Respondents

Household Survey Respondents		Numbers
Men		311
Women		216
Total		527

- Focus Group Discussions (FGDs)**

FGDs were conducted with community men and women separately, with 15-17 participants in each FGD. A total of 226 community members participated in a total of 14 FGDs (7 FGDs with men and 7 FGDs with women) in 6 locations.

An additional FGD with 12 women staff members of WWF was conducted online to explore participation and women's leadership in WWF (see list of WWF female staff in Annex-A11).

Table 3: Number of FGDs and Number of Participants

Target Groups	Number of FGDs	Number of Participants
Men Community Members	7	113
Women Community Members	7	113
Women Staff Members of WWF	1	12
Total	15	238

- Key Informant Interviews (KII)**

Key Informant Interviews were conducted with 10 government officials. Details by province is given below. A list of people interviewed is presented in Annex A10.

Table 4: Key Informant Interviews (KII) Conducted in 4 Provinces

Target	Numbers of KII
Sindh	2

Punjab	4
KPK	2
Balochistan	2
Total	10

CHAPTER 2: Status of Women in Pakistan

“All citizens are equal before law and are entitled to equal protection of law. There shall be no discrimination on the basis of sex”.

Article 25 Fundamental Rights and Principles of Policy, The Constitution of Pakistan 1973.

2.1 Constitutional Rights of Women in Pakistan

According to the Constitution of Pakistan all citizens are equal and there should be no discrimination on the basis of sex. Hence, women are guaranteed fundamental rights and are considered equal citizens in all respects¹¹.

Formal laws in Pakistan include the Constitution of Pakistan (1973), all current statutory laws, including Islamic laws, personal laws covering other religions, Indo-British laws, laws made by the federal and provincial governments of Pakistan, the Muslim Family Laws Ordinance of 1965 (a very progressive law), 'Shariah' laws introduced by General Zia-ul-Haq, as well as the international conventions and treaties to which Pakistan is a signatory, and the Women Protection Bill 2006¹².

The 1973 Constitution of Pakistan provides several safeguards to ensure that women are treated equally and to eliminate discrimination on the basis of sex in all spheres of life. Pakistan's Constitution establishes the principles of equality of citizens as well as protection and affirmative action for women and children and laws have been enacted to operationalize them at the federal level, and since the adoption of the 18th Constitutional Amendment in the provinces as well. Legislation is wide ranging and is guided by the Constitution and by Pakistan's international commitments (CEDAW, SDGs, Beijing Platform for Action, Child Rights Convention) and other human rights treaties that have been signed and ratified.

2.2 Pakistan as Signatory to International Conventions and Treaties

As a UN member state, Pakistan is signatory to the 2030 Agenda - the 17 Sustainable Development Goals (SDGs), and targets and indicators under each goal. Pakistan stands among one of the first countries to formally endorse the 2030 Agenda, through a unanimous parliamentary resolution. Pakistan is also committed to the Convention for the Elimination of all Discrimination against Women (CEDAW), the Beijing Platform for Action, ILO conventions, and Child Rights Conventions, all directed to ensure women's rightful place in society¹³.

The Alternative CSO report to CEDAW 2020¹⁴, states that *“Pakistan has passed numerous laws and formulated policies to promote women's rights in this period, unfortunately, many lack enabling laws, regulations, rules, and actions to be effective. And the WEF/ Global Gender Gap 2021 also indicates that in reality the status of women in Pakistan remains low.*

2.3 Women's Quotas

¹¹ Gender Differences: Understanding Perceptions National Baseline Study / Gender Equality Program (GEP), Aurat Foundation and USAID 2012.

¹² Differential Effects of the Interplay of Formal and Customary Laws on Women and Men in Pakistan, December 2006.

¹³ National Gender Policy Framework Ministry of Planning, Development and Special initiatives, Government of Pakistan March 2022.

¹⁴ Unmet Promises: Alternative CEDAW Report 2020 on Pakistan's Fifth Periodic Report, Alternative CSO Report, Submitted by Shirkat Gah-Women's Resource Centre.

Pakistan has a 17 percent quota for women in the national and provincial assemblies, and studies reflect that while women constitute about 20 percent¹⁵ (or one-fifth) of the total membership of both houses in parliament, they contribute almost 33 percent to the parliamentary business which indicates the potential of increased representation and participation of women in parliament. However, despite this quota women legislators are confronted with multiple challenges, including silencing and harassment by male colleagues, and undermining by religious parties. The study¹⁶ conducted by the Pakistan Institute of Legislative Development and Transparency (PILDAT) highlights the perceived discrimination against the women elect in the reserved seats as compared to the elected members of the parliament. However, exercise of political power to influence women-centered legislation and policy to foster political and familial loyalties towards male political actors and leaders. The political landscape has recently begun to accept women's role as policy makers with leaders now becoming more engaged in critical policy making to address climate change. This situation is still complex due to male-dominated political space at the constituency level and political patriarchy. Political parties are mandated to give five percent of party tickets to women, but there is the need to increase women's political participation and mainstreaming in active political participation. The share of women ministers in Pakistan has decreased from 12 percent in 2020 to 10.7 percent in 2021, with their acute underrepresentation in the political sphere both locally and nationally¹⁷.

2.4 Women's Rights Policies and Legislation

The history of laws affecting women's rights and empowerment in Pakistan involves a complex pattern of advances and setbacks, with the state's efforts to articulate a definition of women's rights complicated by the need to balance divergent views on the place of women in Pakistani society.

In 2002, the first National Policy for Development and Empowerment of Women was formulated, a landmark for securing women's rights, and ensuring statutory provisions for gender equality in Pakistan¹⁸. *"The policy contains a vision, goals, aims and objectives, lays down the guiding principles, gives key policy measures, specifically addressing the empowerment dimensions in the social, economic, and political fields. The policy is a statement of intent of the government of Pakistan to specify its measure for the development and empowerment of women with the policy providing the guidelines and will ensure in the overall operating framework of the government that a gender perspective is reflected in all national policies and plans"*¹⁹.

This Policy has not been updated, but it still provides a historical perspective of the recognition for women's rights, and gender mainstreaming in Pakistan.

At the policy level various other progressive laws supporting women's empowerment have also been passed. These are: The Anti-Sexual Harassment Act, Anti Women Practices Act, Acid Crimes and Control Act, National Commission on Status of Women Act, and the Domestic Violence Bill are all legal instruments for the protection of women. The enabling environment for women, as manifested in the laws, policies, and institutional practices on one hand and social norms and informal institutions on the other, do not fully support women. Constitutional provisions and some progressive legislations have been enacted e.g. anti-women practices, "honour" killing, rape, domestic violence, sexual harassment at work and public places, which

¹⁵ <https://na.gov.pk/en/content.php?id=2>

¹⁶ <https://www.pildat.org/publications/Publication/women/WomenRepresentationInPakistanParliament.pdf>

¹⁷ National Gender Policy Framework Ministry of Planning, Development and Special initiatives, Government of Pakistan March 2022.

¹⁸ National Policy for Development and Empowerment of Women, Ministry of Women's Development 2002.

¹⁹ Ibid.

all form women's protection legislation, and other landmark legislation includes the Sindh Home-based Workers Act 2018, Sindh Women Agricultural Workers Rights Act 2019, and the Electoral Act 2017²⁰.

2.5 National and Provincial Commissions on the Status of Women (NCSW)

Historically, the establishment of the National Commission on the Status of Women in 2002, was another positive measure towards women's empowerment, since the NCSW is a statutory body which emerged as an outcome of the national and international commitments of the Government of Pakistan like the Beijing Declaration and Platform for Action 1995; and National Plan of Action (NPA) for Women, 1998.

The National Commission on the Status of Women (NCSW) was established with the specific purpose to examine policies, programs and other measures taken by the Government for women's development and gender equality, to review laws, rules and regulations affecting the status of women, monitor mechanisms and institutional procedures for redress of violations of women's rights and individual grievances, encourage and sponsor research to generate information, analysis and studies relating to women and gender issues, to develop and to maintain interaction and dialogue with NGOs, experts and individuals in society at the national, regional and international level²¹.

The NCSW, a custodian of women's rights did not have a Chairperson at the Federal level, or members at the provincial level since 2019, and the current Chairperson was appointed in July 2021.

Provincial Commissions on the Status of Women re now functioning in all four provinces and in KP Province District offices have been established for some years.

2.6 National Gender Policy Framework 2022

The National Gender Policy Framework was launched in March 2022. It recognizes the importance of gender equality, and Pakistan's commitment to the SDGs and Pakistan's poor status on the WEF 2021 Global Gender Gap Index (from 151 place to 153, out of 156 countries). The framework acknowledges poor progress towards gender equality, closing the gender gaps in wages and labor force participation and the negative impacts of covid-19 pandemic on women.

The report states that increasing economic participation of 48 percent of the national population is the main priority of the Government, while addressing legal barriers for women's economic participation. It emphasizes the gender wage gap, stating that the income of an average woman in Pakistan is 16 percent less than the income of the average man.

In terms of legal frameworks, the document states that considerable progress has been made at the national and provincial level, but effort will have to be made for implementation, compliance, and the uniformity of laws across all provinces. The lack of data and monitoring has also led to weak implementation, and an absence of support mechanisms. Poor coordination with a presence of gender bias reduces the effectiveness of the legislation. The document also mentions the persistent gap between women and men in their access to resources and opportunities, which impedes the progress of women.

The report identifies six key priorities and policy actions: governance, equality and quality in education, employment and economic empowerment, agency, political participation and meaningful engagement, safety and security, and health and well-being. It proposes to

²⁰ Young Women in Pakistan Status Report 2020, UN Women.

²¹ National Commission on the Status of Women / Annual Report January 2018- June 2019.

establish a Gender Transformative Accelerator which will work closely with the federal and provincial NCSW's and the Ministry of Human Rights.

2.7 Gender Gaps in Pakistan

The World Economic Forum Global Gender Gap Report of 2021²², Pakistan is the second worst country in the world in terms of gender parity, ranking Pakistan 145 out of 146 countries worldwide. Pakistan has closed 56.4 percent of the gender gap in 2022 — the highest overall level of parity the country has posted since the report launched in 2006. The country ranked 145 on economic participation and opportunity, 143 on health and survival, 135 on educational attainment and 95 on political participation. The country ranks 178th on the Gender Development Index out of 181 ranked countries and 28th on the Gender Inequality Index²³.

Table 5: Gender Development Indicators Mapped²⁴

Gender Development Index (GDI)	Gender Inequality Index (GII)	Global Gender Gap Index (GGGI)
- Measures the gender gaps in human development by accounting for disparities between women and men in 3 human development dimensions – health, education, and quality of life. - Computed by UN In 2014, Pakistan's CDI was 0.750 vs 0.837 for South Asia.	- Reflects gender inequalities in reproductive health, empowerment, and economic activity. - Computed by UN In 2017, Pakistan GII was 0.541, ranking it 133 out of 160 countries.	- Developed in 2011 by World Economic Forum, and it measures economic participation, educational attainments, health & survival, and political empowerment. In 2020, Pakistan ranked 3rd to last (151 out of 153 countries).
Social Institutions and Gender Index (SIGI)	Women's Economic Opportunity Index	Female Opportunity Index
- SIGI ranks non-OECD countries according to social institutions like social practices and legal norms contributing to observed gender inequalities. - Computed by OECD. Pakistan ranked 0.79 on SIGI, highly discriminatory towards women.	- Piloted in 2010 by Economist Intelligence Unit, it measures indicators influencing economic opportunity of a country. Pakistan is ranked 108 out of 113 countries on WEDI.	- Computed by N26 - Measures advancement in female leadership in the government, corporate sector, STEM and entrepreneurship. Pakistan ranks 100th out of 100 countries.

The literacy level of men is 71.1 percent compared to 46.5 percent for women²⁵, indicating significant disparities in access to educational opportunities and possibilities for meaningful contribution in the public domain for women. Although an estimated 60 million women are of working age, only 22.6 percent participate in 'paid' labour (GGGI 2021). Their retention in the labour market is further challenged by competing family priorities, inflexible workplace policies and structural gendered workplace barriers. Gender disparity continues to exist between men and women in the country. Negative social values, discriminatory legislation, cultural practices, the lack of implementation of affirmative policies for women, absence of political will, combined

²² World Economic Forum, Insight Report: Global Gender Gap Report 2022.

²³ Human Development Report Country Profile UNDP 2020.

²⁴ National Gender Policy Framework Ministry of Planning, Development and Special initiatives, Government of Pakistan March 2022.

²⁵ <https://data.worldbank.org/indicator/SE.ADT.LITR.MA.ZS?locations=PK>

with women's low political, economic, and social status in society, continues to perpetuate a cycle of oppression for women.

As the fifth most populous country in the world, Pakistan stands at 229 million people of which 49 percent are women. However, only 22.6 percent of these women are employed, compared to 84.8 percent of men. Pakistan is among the seven countries with largest economic gender gaps at 31.6 percent. Additionally, only 25 percent of technical roles and 5 percent of leadership roles in Pakistan are occupied by women.

The Gender Wage Gap Income of an average woman in Pakistan is less than 16 percent of the income of an average man. Pakistan has a high gender wage gap, at 50 percent, as employed women earn a median wage of PKR 7,000 per month (USD 35) compared to male counterparts who earn a median wage of PKR 14,000 per month (USD 70). The gender wage gap alone costs Pakistan PKR 500 billion annually, and this money is even more valuable because women invest 90 percent of their incomes back into their families, vs. 35 percent by men instead²⁶.

Women's participation in social processes remains constrained due to the norms that persist though there are variations in their application determined by rural-urban and geographical location and class. Their engagement in formal political processes has steadily improved though still not commensurate with their share of the population. Almost 12 million women are not registered voters; only 85 percent of women 18 years of age and above have CNICs²⁷.

2.8 Conclusions

The position and status of women in Pakistan has a checkered history. In 1987, two women's rights pioneers wrote a seminal book entitled 'Women of Pakistan, Two Steps Forward, One Step Back?'²⁸ and 35 years later the situation remains the same. Over the last three decades the status of women has progressed, remained static, regressed, and constantly changes depending on many factors that are beyond women's control. Numerous laws have been promulgated ensuring women rights and protection, but their implementation and lack of political will renders them ineffective.

While it has been articulated since 2005, very little attention is placed on the crucial nexus of culture, religion and the making and implementation of laws. The plethora of laws, norms and practices have become concrete in Pakistan's patriarchal society in a way that they consistently favor men and place women at disadvantage.

Women lack exposure, confidence, knowledge, skills, information about opportunities, and limited role in politics, and in policies and programs regarding their aspirations and lives. Formal institutions fall short of facilitating women's access to human, financial and physical capital i.e. education, skills, health facilities, loans and physical assets. Informal institutions (family, community, religion, segregation, marriage) define and affirm norms that give precedence to marriage/child marriage over education and employment, prescribe gendered roles, and condone domestic violence/gender-based violence. Laws have therefore not had the expected impact.

The experiences of the 2005 earthquake and the floods of 2010 and 2011 have shown that the patriarchal norms that shield men from their crimes against women, and victimize the women instead, become even stronger in emergencies.

²⁶ National Gender Policy Framework Ministry of Planning, Development and Special initiatives, Government of Pakistan March 2022.

²⁷ Young Women in Pakistan Status Report 2020, UN Women

²⁸ Women of Pakistan Two Steps Forward, One Step Back? Khawar Mumtaz and Farida Shaheed, 1987.

In remote rural settings and under climate change scenarios, WWF would do well to watch out for these tendencies, as they are insidious and ingrained to the extent that women will blame themselves and echo men when they extol the virtues of women's place at home or not remunerating her for economic work.

Unfortunately the recent Gender Policy Framework (March 2022) is not comprehensive enough to close the gender gaps and propel the levels and scope of action required to counter anti-women norms and practices.

CHAPTER 3: Gender Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH)

Human rights and sustainable development are inseparable as the fundamental principle of Agenda 2030 is 'leave no one behind'. To achieve this goal, we must eliminate gender-based discrimination, violence, and oppression²⁹.
(Dr. Shireen M Mazari, recent former Minister Human Rights, Government of Pakistan).

While we searched diligently, **no report or data sets under the title SEAH (sexual exploitation, abuse and harassment) per se were found for Pakistan.**

We looked at definitions of the SEAH components and then tried to find them within other expressions and acronyms used in Pakistan.

Sexual Violence is used as an umbrella term for sexual exploitation, abuse and harassment as quoted in DIFD/UKaid report 2018³⁰. It includes acts of a sexual nature against one or more persons or that cause such person or persons to engage in an act of a sexual nature by force, or by threat of force or coercion, such as that caused by fear of violence, duress, detention, psychological oppression or abuse of power, or by taking advantage of a coercive environment or such person's or persons' incapacity to give genuine consent.

According to the UN, forms of sexual violence include rape, attempted rape, forced prostitution, sexual exploitation and abuse, trafficking for the purpose of sexual exploitation, child pornography, child prostitution, sexual slavery, forced marriage, forced pregnancy, forced public nudity, forced virginity testing, etc³¹.

Sexual Exploitation, as defined by the UN, constitutes any actual or attempted abuse of a position of vulnerability, differential power or trust for sexual purposes, including profiting monetarily, socially or politically from the sexual exploitation of another. It is a broad term, but it includes transactional sex, solicitation of transactional sex and exploitative relationships.

Sexual Abuse, also as defined by the UN, means the actual or threatened physical intrusion of a sexual nature, whether by force or under unequal or coercive conditions. All sexual activity with children (as defined under the UN Convention on the Rights of the Child as any person under the age of 18) is considered to be sexual abuse, regardless of the age of majority or consent locally.

²⁹ National Gender Policy Framework, Ministry of Planning Development and Special Initiatives Government of Pakistan, March 2022, <https://www.pc.gov.pk/web/gender>.

³⁰ Sexual Exploitation, Abuse and Harassment (SEAH) in the International Aid Sector, Victim and Survivor Voice: Main Findings from a DIFD-led Listing Exercise, DIFD/UKaid 2018.

³¹ United Nations Glossary on Sexual Exploitation and Abuse, United Nations 2017.

Sexual Harassment is defined as any form of unwanted verbal, non-verbal or physical conduct of a sexual nature with the purpose or effect of violating the dignity of a person, in particular when creating an intimidating, hostile, degrading, humiliating or offensive environment. Sexual harassment may be perpetrated against beneficiaries, community members, citizens, as well as staff, personnel, etc.

Gender Base Violence (GBV) is any harm or potential of harm perpetrated against a person's will on the basis of gender. GBV encompasses many different expressions of violence including physical, sexual and emotional abuse sexual harassment; stalking; rape, including 'corrective' rape and rape as a tactic of conflict; domestic and intimate partner violence (IPV); child marriage; human trafficking; and female genital mutilation (FGM)³².

It can be safely said that in Pakistan Sexual Exploitation, Abuse and Harassment (SEAH) are subsumed under the terms Gender Based Violence (GBV), 'honor killing' and 'domestic abuse'. According to a recent study, it is estimated that one in three women and girls globally will experience gender-based violence during her lifetime³³. Embedded in discriminatory gender norms and laws GBV and SEAH occur in all societies as a means of control, oppression and exploitation of women that further reinforces gender inequality. GBV and SEAH are complex issues and are a means of control to maintain power structures, traditional norms, customary laws, and access to resources.

Another dimension of GBV is the connection with the environment and linkages have recently been established as evidence from many countries³⁴. **Pressures on the environment and its natural resources increase gender inequalities and power imbalances in communities and households coping with resource scarcity, climate change, and environmental degradation.**

3.1 Global Overview

International frameworks, SDG Goal 5 on Gender Equality, and women's rights frameworks such as the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) all address the issues of women's empowerment in general. But GBV is a complex issue and culturally context-specific which has made it difficult to be effectively addressed to date.

Failure to address this issue also entails a significant economic, social, and environmental cost for the future. Numerous studies have shown that children growing up with violence are more likely to become survivors themselves or perpetrators of violence in the future. A World Bank³⁵ brief on GBV states that gender-based violence affects women and girls of all socio-economic backgrounds, therefore this issue needs to be addressed in both developing and developed countries.

More than one in three women worldwide experience sexual or domestic violence—more than 1.3 billion women, eclipsing the number of people affected by HIV/AIDS, measles, Ebola, and malaria combined. The idea of a binding norm on violence against women and girls was first

³² World Bank 2019 /Gender-based violence and environment linkages The violence of inequality Itzá Castañeda Camey, Laura Sabater, Cate Owren and A. Emmett Boyer, Jamie Wen, editor. IUCN/USAID 2020. <https://portals.iucn.org/library/sites/library/files/documents/2020-002-En.pdf>.

³³ World Bank 2019 /Gender-based violence and environment linkages The violence of inequality Itzá Castañeda Camey, Laura Sabater, Cate Owren and A. Emmett Boyer, Jamie Wen, editor. IUCN/USAID 2020. <https://portals.iucn.org/library/sites/library/files/documents/2020-002-En.pdf>.

³⁴ Ibid.

³⁵ The World Bank Brief, Gender-Based Violence (Violence Against Women and Girls), 25th September 2019, <https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls>.

introduced in 1996 by the first UN Special Rapporteur on violence against women. Today, there is unprecedented momentum on women's safety, spurred by horrific acts of violence around the world, spearheaded by the Every Woman Treaty.

The widespread revelations revealed by the global #MeToo movement also have activists demanding action. People across the globe are asking for systems change. Strong models for rapid institutional and cultural change are available, as are global best practices that combine proven interventions and strategies from human rights and public health. With these practices contained in a global binding norm on violence against women and girls, the current system of impunity can shift to one of safety and accountability, bringing forward a safer, more just world for women and girls everywhere³⁶.

The **#MeToo movement in Pakistan** is modeled after the international #MeToo movement and began in late 2018 in Pakistani society. It has been used as a springboard to stimulate a more inclusive, organic movement, adapted to local settings, and has aimed to reach all sectors, including the lowest rungs of society. In Pakistan when the #MeToo movement began, many issues relating to sexual harassment, misconduct, and violence against Pakistani women surfaced, generating identification and discussion on the range of behaviors and roles of many alleged offenders across the media, film industry, stage, and political realm, including Pakistan's Prime Minister. In addition, clergy, charity and social welfare institutions, the judiciary, and Pakistani security establishments such as the police and military came under the spotlight of accusations.

3.2 Green Climate Fund, WWF Gender Policy and Related GBV and SEAH Guidelines

In the context of the WWF Recharge Pakistan project, there are several policies available for reference: the Green Climate Fund Gender Policy³⁷, the WWF Gender Policy 2011³⁸, and the WWF Guidance Note on Gender-Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH) for Global Environment Facility (GEF) and Green Climate Fund (GCF) Projects³⁹. They all recognize gender relations, and roles and responsibilities which influence women's and men's access to and control over decisions, assets and resources, information, and knowledge.

The GCF Gender Policy is closely aligned with the United Nations Sustainable Development Goals (SDGs), which make explicit commitments to gender equality.

Similarly, the WWF Gender Policy is consistent with WWF's Mission, Guiding Principles and Code of Ethics. It is an integral part of WWF's social policies and position statements, including Indigenous Peoples, Human Rights, Poverty and Conservation, Population and HIV/AIDS, and supports existing commitments to gender equality for those countries which are signatories to the United Nations CEDAW and the Universal Declaration of Human Rights.

Moreover, the WWF Guidance Note on GBV and SEAH for GEF and GCF Projects, establishes basic guiding principles for effectively minimizing and mitigating any identified GBV/SEAH risks in projects, and provide suggestions, recommendations, and further resources for project teams working on GBV/SEAH related to projects⁴⁰. It outlines WWF's

³⁶ EWT, Safer Sooner: Towards a global binding norm to end violence against women & girls, EWT 2020

³⁷ Gender Policy GCF [GBV%20%20WWF/GCF%20Gender%20Policy.pdf](#).

³⁸ WWF Gender Policy 2011 [GBV%20%20WWF/WWFGenderPolicy_%20FINAL_26May11.pdf](#).

³⁹ Guidance Note on Gender-Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH) for GEF and GCF Projects. [GBV%20%20WWF/Guidance%20Note%20on%20Gender%20Based%20Violence%20-%20final%2004-05-2022.pdf](#).

⁴⁰ Guidance Note on Gender-Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH) for GEF and GCF Projects. [GBV%20%20WWF/Guidance%20Note%20on%20Gender%20Based%20Violence%20-%20final%2004-05-2022.pdf](#).

General Principles; Respect for Human Rights, Compliance with WWF's Environment and Social Safeguards Framework, Applicable Laws, and provides definitions for GBV, and SEAH, which can be recognized in project sites, and activities. The Guidance note identifies potential perpetrators, and ways to identify GBV/SEAH impacts and assess GBV/SEAH risks, and recommends a monitoring plan for all projects, which includes reporting on gender and various safeguards.

3.3 GBV and SEAH in Pakistan

In Pakistan, the GBV and SEAH issues continue to remain a challenge despite the passing of integral laws and policies. **In GBV reporting and documentation in Pakistan, sexual violence, sexual abuse, sexual attacks and sexual harassment (SEAH) against women and girls are included in the term Gender Based Violence. SEAH is also included in the term 'domestic violence' and 'honor killings'.**

However, monitoring sexual exploitation, abuse and harassment (SEAH) is a challenge as often communities will see it as a direct assault on their collective 'dignity'. Even where incidents are known to have occurred, denial is common. In police stations, SEAH is often reported under 'kidnapping', because that is how the girls' families prefer the situation to be viewed. Therefore, information on sexual harassment, abuse and exploitation (SEAH) has to be monitored and obtained over a lengthy periods of time with families and individuals in a very sensitive manner with the careful inclusion of psychologists and trained counsellors in the process. Some of the studies quoted here have used such targeted methods to obtain data.

A national study on Gender Based Violence concluded that '*gender-based violence is a living reality in present day Pakistan*'⁴¹, despite the multiple actions taken the prevalence of all forms of violence against women and girls remains high⁴². In 2018, 'honor' crimes claimed 254 female victims according to the Human Rights Commission of Pakistan (HRCP). HRCP also recorded 67 acid attacks and 856 incidents of sexual exploitation, abuse and harassment (SEAH) against women the same year⁴³.

The HRCP Annual Report 2020⁴⁴ states that human rights abuses to target women include child marriage and honor killings, which are largely centered on controlling women's agency. The report highlights other forms of violence as well, including sexual assault, SEAH and domestic violence. **Actual figures for GBV are probably higher as family members are often the perpetrators, and there is a stigma attached to reporting on sexual violence and SEAH.**

The Pakistan Demographic and Health Survey (PDHS) 2017-2018 on domestic violence revealed that 56 percent of women remained silent about violence, and if they did seek help or advice, they were more likely to approach their own or their husband's family. Reporting of gender-based crimes is low because when a woman complains she gets blamed, and the lack of proper jurisprudence resulting in the onus of reporting and conviction on the victims. For many women, prevailing attitudes, lack of awareness of support or protection, perceived police inaction, or lack of confidence in the criminal justice system inhibits them from approaching the authorities.

⁴¹ Gender Differences: Understanding Perceptions National Baseline Study Aurat Foundation / USAID 2012.

⁴² State of Human Rights in Pakistan 2018'. Annual Report. Human Rights Commission of Pakistan. Accessed at: <http://hrcp-web.org/publication/wp-content/uploads/2019/04/State-of-Human-Rights-in-2018-English.pdf>.

⁴³ State of Human Rights in Pakistan 2018'. Annual Report. Human Rights Commission of Pakistan. Accessed at: <http://hrcp-web.org/publication/wp-content/uploads/2019/04/State-of-Human-Rights-in-2018-English.pdf>.

⁴⁴ HRCP Report rings alarms over plight of women in Pakistan, ANI July 2021, <https://www.aninews.in/news/world/asia/hrcp-report-rings-alarm-over-plight-of-women-in-pakistan20210706202836/>

The National Commission on the Status of Women Annual Report 2018-19 also reports almost 1,000 honor killings every year, and forms of violence include domestic violence, SEAH, forced marriages, and acid attacks⁴⁵.

A study from the International Union for Conservation of Nature (IUCN), Mangroves for the Future Program (MFF) shows that as traditional means of livelihood from fisheries decline pressures on fisherfolk to search for alternative sources of income arises. And with the loss of livelihood from fishing and limited skills and opportunities for alternate livelihood options, the ability to be the provider for the family leads to changing the dynamics at the household level and often to domestic violence⁴⁶. **This is because women are an equal participant in the fisheries sector just like the agriculture sector.**

The National Gender Policy Framework 2022 has provided data on the status of GBV during COVID-19 and cases of GBV including sexual violence against young people adolescent boys and girls has grown.

The fact that GBV is not integrated in social protection or public health responses and education or training makes it difficult to address the root causes of this issue. The lack of comprehensive information and statistical data at national and provincial level means that implementation of existing laws and developing specific interventions to address the issue of GBV remains neglected.

Projects addressing climate impacts to build resilience, such as Recharge Pakistan, have the ability to build the capacity of women to exercise agency in identifying issues, participate in the designing and planning of contextual interventions and gain economic and social empowerment through informed and powerful decision-making. Without the integration of gender context into climate change resilient actions, the compounding risks experienced by such vulnerable groups cannot be deemed effective or efficient.

Gender based crimes, violence against women, and domestic violence, have deep social, psychological, and mental effects on women. These destroy a woman's confidence, her respect, her status, and position within the family. Gender based crimes reinforce the social structures which believe that women are subservient to men and have no human rights as an individual⁴⁷.

3.4 Legislation on GBV and SEAH

Legislation on GBV and SEAH against women and girls is reflected in the federal and provincial laws for the protection of women:

- Anti-Rape Laws (Criminal Amendment Bill) 2016.
- Child Marriage Restraint Act, 1929.
- Foreigners Act, 1946 and Foreigners Order, 1951.
- Muslim Family Law Ordinance, 1961.
- Dowry and Bridal Gifts (Restriction) Act, 1976`

⁴⁵ National Commission on Status of Women January 2018-June 2019.
[Annual Report 2018-2019.pdf - Google Drive](#)

⁴⁶ Reflecting on the Beijing+20 Platform and Action Women and environment – Policy to Action Perspective of coastal resource dependent communities across South and Southeast Asia Maeve Nightingale Manager Capacity Development and Gender Focal Person, IUCN Mangroves for the Future MFF Program.
[GBV%20%20WWF/CSO%20Forum_women%20and%20environment%20Maeve%20Nightingale.pdf](#).

⁴⁷ Gender Differences: Understanding Perceptions National Baseline Study, 2012, Gender Equality Program (GEP), Aurat Foundation and USAID.

- Anti-Honour Killing Laws (Criminal Amendment Bill) 2015.
- Anti-Terrorism Act 1997.
- Prevention and Control of Human Trafficking Ordinance, 2002.
- Criminal Law (Amendment) Act, 2004.
- The Protection Against Harassment of Women at the Workplace Act, 2010.
- Criminal Law Amendment Act, 2010.
- Criminal Law (Second Amendment) Act, 2011.
- The Prevention of Anti Women Practices - Criminal Law (Third Amendment) Act, 2011.
- Domestic Violence (Prevention and Protection) Act, 2012.
- Protection for Women (Criminal Law Amendment) Act, 2006.

The Hudood Ordinances promulgated by a military dictator in 1979, brought in several anti-women laws that became an impediment in reducing GBV and SEAH. Some of the laws listed above acted to counter these so-called Islamic laws that are still on the books. For example, the Protection for Women Act 2006 removed rape from the Hudood ordinances and put it back in the criminal code.

However, implementation of protective laws remains a persistent challenge. There are limited services such as shelters, helplines, and legal aid for women survivors of violence and SEAH, especially young women. In the past, government interventions have focused on establishing helplines, women police stations, hotlines and protection centers, that try to support women after they have experienced violence and SEAH.

In addition, GBV and SEAH specific data is inadequate since most survivors do not disclose their experience to anyone and do not seek medical or psychological support.

The problems of women who live under plethora of tribal, feudal, Islamic, and Indo-British laws are often so varied and local in nature that they cannot always be solved through solutions expounded by laws and legislation alone. The lives of women, particularly in the rural areas, remain difficult and both women's empowerment and gender equality remain a challenge⁴⁸.

3.5 GBV and SEAH in the Provinces

Sindh

The women in Sindh, like other women of Pakistan, suffer from discriminatory attitudes, laws traditions and practices in a male dominated patriarchal society. The result is reflected in the low indicators of poor health status, high maternal mortality rates, a very low literacy rates, increasing numbers of 'honor' killings (mostly for SEAH reasons) and declining legal and social status. Equally significant are the relatively less documented indices of domestic violence, sexual violence, economic and physical exploitation, restricted social mobility, and psychological slavery, where women view subordination as their fate⁴⁹.

The Sindh Human Rights Commission (SHRC) found in its initial assessment report 2021, that cases of gender-based violence and other human rights abuses increased across the province during the lockdown imposed to restrict the spread of Covid-19⁵⁰. And according to the

⁴⁸ Whose Honour? Differential Effects of the Interplay of Formal and Customary Laws on Women and Men in Pakistan, Paper Presented at the International Conference on The Other Self: Conflict, Confusion or Compromise, NCSW, December 2006, by Simi Kamal.

⁴⁹ A Study of the Interplay of Formal and Customary Laws on Women' Volume V: Sindh Province, RDC/RNE, 2002, Legal Political and Socio Economic Situation of Women in Sindh, Paper Prepared for NCSW, September 2006, by Simi Kamal.

⁵⁰ <https://www.thenews.com.pk/print/789682-gender-based-violence-increased-in-sindh-during-lockdown-shrc>.

UNODC Gender and Pandemic Advocacy Brief⁵¹ despite several women-friendly laws and policies, challenges in implementation continue to impede the prospects for women and children in Sindh. Most women in Sindh lack access to basic services in education, health, justice, and social support.

Punjab

In Punjab the Pakistan Demographic and Health Survey 2017-18 indicates that 34 percent of married women have experienced physical, sexual, or emotional violence from their spouses. The most common type of spousal violence is emotional violence (26 percent), followed by physical violence (23 percent). Five percent of women have experienced spousal sexual violence, and 26 percent of married women who have experienced spousal physical or sexual violence have sustained physical injuries. Nine percent of women have experienced violence during pregnancy, and 56 percent women never sought help or disclosed GBV due to socio-cultural and other problems regarding accessibility to health care and psycho-social support services, and in Punjab the prevalence of GBV is 32 percent⁵².

Punjab's Women's Economic and Social Well-being Survey 2018 is the only one in the country that provides a baseline for the province. The other official source is the Pakistan Demographic and Health Survey (PDHS) which covers married women and reveals that 44 percent of young married women and a similar percentage of young men think a husband is justified in beating his wife.

Khyber Pakhtunkhwa (KPK)

In KPK given the traditional society and cultural issues, few GBV survivors seek support in response to violence. Women crises centres also exist in major cities of Khyber-Pakhtunkhwa under the supervision of the Social Welfare Department. And in accordance with the Constitutional protection mandate Khyber Pakhtunkhwa has a comprehensive mechanism for GBV survivors from information to re-unification and referrals. While the Government of Pakistan has passed various laws to prevent violence and support those affected by it, the conviction rate is very low for violence against women.

Resources and services for women survivors of GBV remain scarce. When encountered by a case of GBV, relief workers are often at a loss as to how to best respond given that there are few referral and response mechanisms in place to address their immediate or on-going needs. Often the staff, who interact with a survivor of GBV are not aware of the severe and long-lasting health, emotional and psychosocial problems that survivors face. Such multifaceted challenges require specialized, expert response from other agencies or service providers, knowledge of which is often limited among the staff of the health facility where GBV survivors come for help and support⁵³.

Baluchistan

In 2014, a breakdown of provinces by various VAW categories revealed that Balochistan accounted for merely 0.28 percent of the overall cases of women's kidnappings, 2.86 percent of abductions, 0.26 percent of overall rape instances, 2.26 percent of suicides, 10.8 percent of 'honour' killing cases, 6.07 percent of domestic violence related cases, 7.69 percent of acid throwing cases and 1.11 percent of miscellaneous other cases of VAW. A simplistic analysis

⁵¹ UNODC Gender and Pandemic Advocacy Brief, Protecting Women in times of Covid-19 .

https://www.unodc.org/documents/pakistan/Advocacy_Brief_3_Gender_-COVID-19-Sindh.pdf.

⁵² Multi-Sectoral Referral Pathways for GBV Survivors in Punjab and Khyber Pakhtunkhwa, Pakistan, June 2021, <https://phkh.nhsr.ppknowledge-article/multi-sectoral-referral-pathways-gender-based-violence-survivors-punjab-and-kp>.

⁵³ Ibid. [Multi%20Sectoral%20Referral%20Pathways%20for%20GBV%20Survivors%20WHO%202021.pdf](https://phkh.nhsr.ppknowledge-article/multi-sectoral-referral-pathways-gender-based-violence-survivors-punjab-and-kp).

of these figures would lead to the conclusion that Balochistan has relatively low incidences of VAW compared to the rest of Pakistan. However, it should be noted that Balochistan accounts for merely 7.46 percent of the country's overall population. Proportionately, the number of reported cases of VAW in Balochistan could, therefore, be high relative to the rest of the country. The strong influence of tribal customs and the predominance of informal justice mechanisms such as Jirgas in Balochistan are also key factors behind the low number of reported cases of VAW in the province⁵⁴.

3.6 GBV and SEAH in WWF Recharge Pakistan Project Sites

An analysis of WWF background documents indicates that there is no data or information on GBV or SEAH in the Recharge Pakistan field sites. So, this current research will probably be the first time this kind of information has been generated. However, the situation in Pakistan regarding GBV and SEAH is alarming, and exists in different levels in all provinces, so it can be inferred that GBV and SEAH exists in varying degrees in the project sites.

Investigating issues related to GBV and SEAH is difficult, as it is a sensitive issue and, in most cases, men usually deny the existence of any aspect of GBV, whereas women recognize it and subtly acknowledge it as part of traditional family norms. During the focus group discussions indirect questions to gauge this issue were asked such as, *'what kind of dangers and issues might women face within the home? And what kind of dangers and issues might women face if they go outside their home?'* Most of the male respondents in the field sites said that women faced no 'danger' within their homes, in contrast to half the number of women who said they faced 'danger' in their homes, indicating differences in the perception of danger and women's sensitivity to possible violence.

Types of violence reported were 'scolding women', and both men and women acknowledged this, but physical violence was reported by more women than men.

In terms of facing danger outside the home, both men and women has similar responses claiming that women did not face any danger outside their homes. The possible reason for this is because women's mobility is restricted to the house and women stay at home. An interesting fact which did emerge was a majority of men said they respected women, whereas almost half the female respondents said the men did not respect them.

These contradictions reflect the national data on gender-based violence, which needs to be taken into consideration when planning interventions for the Recharge Pakistan project.

3.7 Conclusions

The data from official sources confirms the prevalence of GBV and SEAH all over Pakistan. Unreported cases are probably higher, legislation is not always effective or efficient, and dominant male attitudes perceive gender-based and sexual violence as acceptable, and women keep quiet as part of the traditional family structure. The linkages between environmental degradation, climate change and the impact of covid-19 has led to an increase in GBV and SEAH, and it is this trend that the WWF Recharge Pakistan project interventions must consider as it designs and implements its program.

In terms of legislation, Pakistan is signatory to international conventions, and has many progressive and supportive laws that could bring down GBV and SEAH. Unfortunately, other laws that remain on the books (such as the Hudood Ordinances) counter the progressive laws

⁵⁴ Sustainable Development Goal 5: A Legislative and Policy Gap Analysis for Balochistan, https://www.sdgpakistan.pk/uploads/pub/SDG_5_Policy_Gap_Analysis_Balochistan.pdf.

and reflect the mind-sets of the low enforcement personnel and people serving in the judiciary. The net result is that women rarely get the justice and redress they need.

The focus group discussions have revealed the existence of various forms of gender -based violence in the field sites, which means planned interventions should take these factors into consideration. It can also be assumed that within the ambit of gender -based violence, sexual exploitation, abuse, and harassment also exists. Even though direct questions were not asked about SEAH, physical violence has been mentioned by women respondents. Disrespect for women has also been cited which indicates a deeper insidious side of violence which must be addressed if gender empowerment, and gender mainstreaming is to bring real change to the lives of the women in the communities.

Reducing violence and SEAH against women and girls requires a community-based, multi-pronged approach, and sustained engagement with multiple stakeholders. To be effective, these initiatives would have to address social norms on gender roles and the acceptability of violence.

Community engagement is vital to reduce GBV and SEAH and identify victims and survivors that need help. In order to do this community members (men, women, elders, youth, religious figures, influential people, local law enforcement and government officials, local magistrates, local health personnel, etc have to be sensitized to the issues, supportive laws, cultural positives and negatives and where redress may be found. For example, nearest lawyers, legal aid centres, shelters, etc.

GBV and SEAH are challenging yet necessary to address. The issues are ingrained in traditionally held gender relations mixed in conflicts on property, land ownership and increasing poverty. Redress needs actual action and systems on the ground, not just training and orienting people. A woman attacked with acid or who has been raped needs immediate shelter (sometimes for month or years), medical support, legal aid, economic support, community support, rehabilitation, education for her children and psycho-social counselling over extended periods. WWF does not currently have a team or program or cross-cutting approach to cover these components. However, the organization believes that climate resilience can only be achieved when rights and privileges of vulnerable populations are protected by the state. For this purpose, the GAP includes a multi-tier staffing arrangement for a cohesive approach across all project activities and offers maximum interaction with local communities and key stakeholders at national and sub-national level for implementation of components on building gendered climate resilience.

Pakistan's women, more than 9.1 million are agricultural workers and play a substantial role in food production and food security, but are largely unpaid, disadvantaged; suffer from greater time poverty, and are far more vulnerable to exploitation than men⁵⁵. Climate change and disaster vulnerability only exacerbates these disadvantages. Recharge Pakistan is limited in scope to address these vulnerabilities in entirety, however, the program has the potential to financially and intellectually empower women in its target areas especially in rural, peri-urban and small urban zones to build small business, inclusion in planning for future climate challenges as well agency in decision-making.

The project should therefore dedicate enough resources (and capacities) within the WWF team working on constantly engaging with parliamentarians (to influence better laws), especially the Women's Caucus and women parliamentarians. This team should also engage with the National Commission on the Status of Women (NCSW), the Provincial Commissions on the Status of Women (PCSWs), the Human Rights Commission (HRC) and the Provincial

⁵⁵ UNDP and NCSW, Climate Equity: Women as Agents of Change, 2022, Accessed at: <https://www.undp.org/pakistan/publications/climate-equity-women-agents-change>

Commissions on Human Rights (PCHRs). Orientation is required of law enforcement officers and the judiciary to handle GBV and SEAH cases. The GCF and WWF's own policies and documents form a strong basis for training and orientation on gender-based violence issues and for mainstreaming GBV concerns in its projects. Other strategies to ensure challenges relating to GBV and SEAH the project will employ include: the development of a terms of reference to recruit a gender specialist in the PMU, who has strong and proven expertise in working on issues of GBV and SEAH in Pakistan, among other essential and important experience. The gender specialist will be the resource person in the project who will conduct capacity needs assessment, provide training (including recurrent/refresher trainings) and support capacity building of all project staff and relevant partners, to recognize and address SEAH and GBV issues identified during project implementation, which will be reported through the gender-responsive and survivor-centered grievance mechanism established for the project (See Section 5.3 in Annex 6 for detailed guidance on SEAH Risk Mitigation).

WWF should interact with people related to enforcement of laws, identification of victims and survivors, legal aid, shelters, rehabilitation, education, training, counseling, economic support, housing and help to re-integrate in society. The wide range of stakeholders can include;

- Law enforcement agencies
 - Police
 - Women Police
 - FIA including special branch on cyber crime
- Professionals from the Jurisprudence and legal institutions
 - Local Magistrates
 - Local Lawyers
 - Local Women Lawyers Associations/Groups
- Local non-profit organizations working on legal rights for women (they offer pro-bono consultations to victims of GBV)
- Forensic Medical staff in state hospitals that document harassment/rape cases
- Local Religious and Community Leaders

As the project will progress, new stakeholders can be identified, so if cases are seen in the course of work in the four sites, they can be referred to the right institutions for engagement as well as assistance.

It is, therefore, recommended that Aurat Foundation (a Pakistani NGO working extensively on advocacy and action, with extensive grassroots links, strong relations with women parliamentarians and very good contacts state institutions) be called on to assist. The USAID-funded Gender Equity Program of Aurat Foundation already has the materials and experience to deliver seamless service concepts and programs to assist survivors of GBV and SEAH. Other NGOs and community-based organizations may also be contacted for this purpose.

The increasing destruction of the environment is fueling a rise in violence against women and SEAH, as acknowledged worldwide. If communities are to benefit from projects and programs to counter the effects of climate change, clear national and provincial frameworks on ending GBV and countering SEAH, interaction with a functioning seamless service of support, a robust mechanism for accountability, and dedicated funding will be needed. Organizations like WWF need to push for these frameworks to allow the local law enforcement and justice systems to act when women need protection.

CHAPTER 4: Gender Considerations in Global and Pakistani Climate Change and Environment Policies

Training Manual on Gender and Climate Change. IUCN, UNDP, GGCA, 2009

“Climate change and gender inequalities are inextricably linked. By exacerbating inequality overall, climate change slows progress toward gender equality and thus impedes efforts to achieve wider goals like poverty reduction and sustainable development. Gender inequality can worsen the impacts of climate change. Meanwhile, taking steps to narrow the gender gap and empower women can help reduce these impacts.”

The nexus between women and the environment emerged on the mid-1980's when research and studies on women's roles, responsibilities and interaction with the natural resources became evident and women were recognised as managers of the natural resources. Women's position in water management, forests, and conservation led to a growing body of knowledge of the importance of women's interaction with the environment⁵⁶. Over the last few decades, the recognition that women are managers of the natural resource by virtue of being primarily responsible for the household, and that environmental degradation has a direct impact on them has provided evidence and gender has been incorporated into national policies, project planning and development issues. As the focus changed from women in development (WID) to gender and development (GAD) and more emphasis was placed on recognizing women's role in all sectors, a greater understanding of women's relationship with the natural resource, and within the debate on sustainable development has also been established.

4.1 Global Overview

Climate Change is a persisting global challenge and the nexus between Gender and Climate Change has now been acknowledged and recognized at local, national, and international levels. Within the discourse of gender and climate change over the last 20 years, the linkages between women and environment have changed considerably and simultaneously the relationships between men and women have become increasingly complex. **As global climatic conditions change, its impact on people and the way in which they adapt and cope with these changes have specific gender dimensions, which are being increasingly understood at the local level.**

Existing studies⁵⁷ and research on gender and environment have focused on specific issues such as gender roles in water management, forests etc. and the analytical frameworks to understand the nexus between gender and climate change are becoming more influential as international commitments and gender responsive policy has led to greater gender mainstreaming and Gender Action Plans for implementation.

Policy measurers at global and national level have recognized that gender equality is essential to achieving climate change goals, and since 2008, several references to gender have entered the UNFCCC negotiation text⁵⁸. It is also argued that although climate change affects everyone, it is not gender neutral, since most of the poor in the world are women, the impact

⁵⁶ Gender and Climate Change: An Introduction, Edited by Irene Dankelman, Earthscan 2010, <https://gender.cgiar.org/publications-data/gender-and-climate-change-introduction>.

⁵⁷ Gender and Climate Change: An Introduction, Edited by Irene Dankelman, Earthscan 2010.

⁵⁸ The Art of Implementation, Gender Strategies Transforming National and Regional Climate Change Decision Making, by the IUCN Global Gender Office on behalf of the Global Gender and Climate Alliance, 2012, <https://portals.iucn.org/library/sites/library/files/documents/2012-086.pdf>.

of climate change is more severe for women than men⁵⁹. Thus, the impact of climate change on men and women is different, a gender equality perspective is essential for policy, and for strategies of adaptation and mitigation. In this context, women should also be seen as “powerful agents of change” whose participation in planning processes is necessary if the impact of climate change is to be addressed⁶⁰. Moreover, women’s issues and problems at the local level must be considered in the wider context of women’s empowerment, their social status, access to resources and opportunities, control over assets, and their social vulnerability, to understand the relationship between gender and climate change.

It has also been established that when women and girls are included in climate change action strategies, there are also marked improvements in outcomes. “When we engage women in climate adaptation, we see progress in reducing poverty and improving food security”⁶¹.

4.2 Women and Environment Nexus in Green Climate Fund (GCF) and WWF Gender Policies

The Green Climate Fund Policy recognizes the linkages between gender and environment and the emphasizes the impact of climate change on women.

“It clearly states that ‘the impacts of climate change can exacerbate existing gender inequalities. The Gender Policy acknowledges that climate change initiatives are more sustainable, equitable and more likely to achieve their objectives when gender equality and women’s empowerment considerations are integrated into the design and implementation of projects. Further, it recognizes that women and vulnerable communities are also part of the solution to climate change and should, therefore, be effectively engaged in discussions and decisions that affect them’⁶².

The policy reflects the SDG’s commitments to gender equality, the United Nations Framework Convention on Climate Change (UNFCCC), it is consistent with international arrangements in particular the Universal Declaration of Human Rights, the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), the International Labour Organization’s core conventions, and the Paris Agreement.

Similarly, WWF as a conservation organisation also recognises the linkages between natural resource management, the importance of the sustainable use of natural resources, and the complex relationship of women’s and men’s roles and responsibilities, and how they use and manage natural resources, and how gender relations influence households and communities’ decisions of the use of resources⁶³.

“The WWF Gender Policy supports the existing commitments to gender equality for those countries which are signatories to the United Nations Convention for the Elimination of All Forms of Discrimination Against Women (CEDAW) and the Universal Declaration of Human Rights”⁶⁴.

⁵⁹ Report of High-level Roundtable: How a Changing Climate Impacts Women, Council of Women World Leader Women Environment and Development Organization (WEDO) 21 September 2007, <https://wedo.org/wp-content/uploads/2007/12/roundtable-final-report-6-nov.pdf>.

⁶⁰ Training Manual on Gender and Climate Change. IUCN, UNDP, GGCA, 2009, <https://portals.iucn.org/library/sites/library/files/documents/2009-012.pdf>.

⁶¹ *ibid*

⁶² GCF Gender Policy:GCF%20Gender%20Policy.pdf

⁶³ WWF Gender Policy file:///C:/Users/meher/Desktop/Chapter%203%20Docs/WWF_Gender_Policy.pdf.

⁶⁴ WWF Gender Policy file:///C:/Users/meher/Desktop/Chapter%203%20Docs/WWF_Gender_Policy.pdf.

4.3 Gender Considerations in Environment, Climate Change and Water Policies in Pakistan

Pakistan is among the countries that are most vulnerable to the risks associated with climate change and ranks 8th on the Global Climate Risk Index 2021.

Women continue to face serious challenges amid threats to climate degradation, particularly in occupations that are dependent on natural environments or geographic locations and where climate stress may cause natural hazards. Agriculture, water, and forestry are among the sectors most impacted by climate change and women are heavily engaged in these sectors as full-time labour or secondary workers in addition to their dependence on natural environments for sustenance. Declining crop yields disrupted water availability, rural out-migration, and frequent natural disasters triggered by climate stress produce a precarious survival scenario for communities, especially women.

Pakistan's women are further disadvantaged because 9.1 million women agricultural workers play a substantial role in food production and food security, but they are largely unpaid, suffer from greater time poverty, and are far more vulnerable to exploitation than men. Women's ownership of land, and control over physical assets is minimal: only 2 percentage of women report owning a house or agricultural land as compared to 72 percentage of men⁶⁵. As women are disproportionately impacted by the adverse effects of climate change, Pakistan ratified the Paris Agreement (2015) which mandates gender equity and justice in climate action by member states.

4.3.1 Climate Change Gender Action Plan (ccGAP) 2022

The Pakistan ccGAP is designed to strengthen the implementation of national climate change policies and plans. It seeks to operationalize the commitments set out in the NCCP by ensuring that women are included in all stages of planning, development, and implementation of climate action in the core priority sectors of the plan. These sectors are;

1. Disaster Risk Reduction
2. Agriculture & Food Security
3. Forests and Biodiversity
4. Integrated Coastal Management
5. Water & Sanitation, and,
6. Energy & Transport

Gender constitutes an integral element that can prohibit an individual's access to resilience by its very nature of dictation by social norms and cultural values in particular areas. In Pakistan, the situation is not deemed different as women, among the gender groups, maybe as biologically vulnerable as men to the horrific impact of climate-induced disasters but are deemed so due to the cultural norms, behaviors, and social standing they are awarded in Pakistan. This is despite the fact that gender equality is enshrined in the Constitution of Pakistan, which affirms that all citizens are equal under the law and there shall be no discrimination on the basis of sex.

The ccGAP demonstrates the commitment by the Government of Pakistan to ensure the inclusion and leadership of women in climate action. While aligned with the priority sectors identified in the National Climate Change Policy (NCCP) of 2021, it is an important step towards achieving a key objective of the NCCP to focus on pro-poor gender sensitive

⁶⁵ Climate Equity - Women as Agents of Change, NCSW Pakistan, UNDP, March 2022, Accessed at: <https://www.undp.org/pakistan/publications/climate-equity-women-agents-change>

adaptation⁶⁶. The Plan adopts key principles for gender inclusiveness for effective implementation. These principles are:

- Participation and representation of women; to elevate their voices, “promote their representation, participation and leadership at all levels of climate policy, planning and decision-making”⁶⁷ (IUCN, 2022).
- Coordination and institutionalization; creating opportunities “to strengthen coordination and institutionalise measures for gender mainstreaming”.
- Policy and programmes; to address and include the needs of women and girls by updating old and developing new policies.
- Gender disaggregated data; to ensure collection and availability of gender disaggregated data to inform policy and programme decisions.
- Context specific research; to invest in context-specific research
- Monitoring and evaluation; to develop monitoring frameworks and tools to monitor and measure progress

“The ccGAP was developed through a series of consultations with representatives from government ministries and departments, civil society organizations, gender specialists, and subject experts, as well as a diverse group of key stakeholders” (IUCN, 2022).

4.3.2 National Climate Change Policy (NCCP) 2012 / Updated 2021

“Women are often viewed as vulnerable to climate change, but they are also the solution. They are often the ones with first-hand knowledge of it. They are coming face to face with it”⁶⁸.

The main policy on climate change that includes a focus on Gender is the National Climate Change Policy (NCCP) 2012, which builds on a foundation of policy, legislative and institutional measures since the early 1990s, including the Environmental Policy 2005. The NCCP provides a framework for addressing the issues that Pakistan faces or will face in the future due to the changing climate. The policy thus provides a comprehensive framework for the development of Action Plans for national efforts on adaptation and mitigation, setting out a clear goal of achieving climate resilient development for the country through mainstreaming economically and socially vulnerable sectors. Gender is recognized in the policy emphasizing that women are powerful agents of change, and that it is vital to ensure women’s participation in all policies and decisions related to climate change⁶⁹.

The National Climate Change Policy 2021 has been updated and has a specific focus on gender and states that climate change is likely to affect poor and underprivileged regions, communities, and people disproportionately as they are marginalized, vulnerable and have the least resources to adapt. In Pakistan, women are likely to be strongly affected by climate change as most rural women are engaged in the agriculture sector, which is highly climate sensitive. Climate change is expected to increase the work of agriculture production and other subsistence activities such as collecting fuel wood and water, putting extra pressure on women. Further, women are found to be more vulnerable during extreme climate events and disasters.

⁶⁶ IUCN and MOCC, Climate Change Gender Action Plan (ccGAP), 2022, Accessed at:

<https://www.iucn.org/news/202207/iucn-and-mocc-launch-pakistans-first-ever-climate-change-gender-action-plan>

⁶⁷ *Ibid*

⁶⁸ *Ibid*

⁶⁹ NCCP 2012.

Pakistan is presently developing a National Climate Change Gender Action Plan (NCCGAP), because it fully recognizes that women are powerful agents of change. To address the gender aspects of vulnerability from climate change, the Policy has several policy measures to mainstream gender perspectives into climate change efforts at national and regional levels⁷⁰.

In response to the impact of climate change and in recognition of women's diverse roles, knowledge, leadership, and constraints in relation to the environment, the Climate Change Gender Action Plan (ccGAP) includes Water and WASH as a priority sector, aligned with the National Climate Change Policy and endorsed by the National Gender and Climate Change Committee (NGCCC).

4.3.2 National Water Policy of Pakistan 2018

The National Water Policy of Pakistan 2018 (NWP) describes the situation in the country as moving from water stressed to water scarce. According to the NWP, 'Pakistan is heading towards a situation of water shortage and by corollary, a threat of food insecurity. The situation calls for rapid development and management of the country's water resources. Water resources are inextricably linked with climate and the impending climate change scenario has serious implications for Pakistan's water resources. The objective of the National Water Policy is to take cognizance of the emerging water crisis and provide an overall policy framework and guidelines for a comprehensive plan of action in response to these risks⁷¹.

Pakistan's economy is a 'water economy' with 60 percent of the population directly engaged in agriculture and livestock and 80 percent of Pakistan's exports are based on these sectors. Approximately 95 percent of surface water and almost all fresh groundwater in Pakistan is currently used in agriculture⁷².

The constitution of Pakistan states that "no person shall be deprived of life or liberty save in accordance with law" (Article 9). Water is a basic need for living, which implies that it is the responsibility of the state to provide clean drinking water. Legal frameworks for protection of consumers are available but right to safe drinking water still needs to be recognized in clear terms in the constitution of Pakistan. In Water Act of Federally Administered Tribal Areas (FATA) (DWP -FATA, 2014), water was mentioned as a fundamental human right; however, since FATA has been merged into the province of Khyber Pakhtunkhwa the act has become secondary⁷³.

In January 2021, an amendment to The Sindh Water Management Ordinance, 2002 to include great women participation in water governance was passed. The Sindh Water Management (Amendment) Bill, 2018 now guarantees women's representation in around 45,000 water course associations, over 350 farmer organisations, and 14 area water boards in the province⁷⁴.

4.4 Conclusions

The challenges of climate change impacts, building resilience and gender inequalities are inextricably linked. By exacerbating inequality, climate change slows progress toward gender equality and thus impedes efforts to achieve wider goals like poverty reduction and sustainable development. Conversely, gender inequality can worsen the impacts of climate change and

⁷⁰ National Climate Change Policy 2021 Government of Pakistan.

⁷¹ National Water Policy, Government of Pakistan, Ministry of Water Resources, April 2018.

⁷² Think Tank on the Rational Use of Water, 1st Report 2016, Recommendations for Pakistan Water Policy Framework, Hisaar Foundation.

⁷³ [Policy Brief On Addressing the Drinking Water Challenge in Pakistan](#), CRCP, November, 2018

⁷⁴ Manoj Genani, Women in Sindh Win Historic recognition to manage water <https://www.dawn.com/news/1602970>, January 2021.

further propel the ‘feminization’ of violence, poverty, marginalization and exclusion. Therefore, taking steps to narrow the gender gap and empower women can help counter the impacts of climate change.

The National Climate Change Policy 2012/ updated 2021 is the only current policy which has a specific focus on gender and states that climate change is likely to affect poor and underprivileged communities, and that rural women are likely to be strongly affected by climate change. The other policies related to the environment do not have a strong focus on women, which leaves a gap in the legislative process.

To mitigate the risks and make communities more resilient, Pakistan needs to put women right at the Centre of the policy framework on climate change. Some progress has been made over recent years in revising the climate policy framework, setting ambitious targets for adaptation and mitigation, calling for a reappraisal of women’s vulnerability and gender-sensitive objectives to address women’s differentiated burdens in climate stress. The recently introduced Climate Change Policy (2021) and submission of the updated Nationally Determined Contribution (2021) pay due attention to women’s differentiated burden in climate emergencies and how to help them.

Therefore, it can be said that the policy environment is feasible for positive reinforcement of gender mainstreaming in ecosystem and resilience-based programming in Pakistan.

The GCF policy and WWF’s gender policy are both good instruments to guide the design, and implementation of the intended WWF project in Pakistan.

CHAPTER 5: Gender Roles in the Context of Local Ecosystems

This chapter is based on the qualitative and quantitative primary data that was obtained from the six sites in four provinces identified by WWF as intended locations for ecosystem-based project interventions. The data from FGDs, Household Survey and Key Informant Interviews I juxtaposed here to see a comprehensive picture of local gender relations in the specific context of the intended program.

5.1 Relationship of Women and Men to Natural Resources

Men and women in the focus group discussions were asked to identify the natural resources around their area. The respondents identified forests, land, grazing areas, rivers, canals, and lakes.

However, both men and women said that all resources were under the control of government or feudal landlords or rich people – tube wells, forests, agricultural land and fish – and that they felt rather helpless.

More than half of both men and women spoke of equal access to the (meagre) resources. Others spoke of little or no access since everything belonged to the *waderas*/landlords. In Sindh it is ‘government’ control with ‘contractors brought in, and local people thrown off. Few owned any land, but those that did, the men had the ownership. Access to fishing has been denied in recent years. Feudal control seems to be strengthening.

In terms of custodial rights or responsibilities, stewardship and governance roles towards natural resources, a majority did not report any custodial roles by communities or government.

In terms of indigenous rights and systems for using and sharing natural resources and environmental services, both men and women could not identify any such system. In a very few instances water sharing for agriculture was reported, and also small infrastructure put up by government.

Traditional knowledge systems and practices still in use include rain-fed ponds, *karez* and wells.

5.2 Access and Control of Natural Resources

According to the respondents, powerful political landlords have increased their control over the natural resource as they declined, and both males and females also said that since everything belonged to the landlords the respondents had little or no access to natural resources. Only a few respondents owned land, and those who did were males.

In Sindh one respondent said, *‘it is government control with contractors who are brought in to take over, and the local people are thrown off.’* Access to fishing has also been denied to the fisherfolk, and feudal control appears to be becoming stronger.

However, gender differentials are positive, and both males and females said they had equal access to their meager natural resources, indicating a level of equity within the communities.

In terms of custodial rights, stewardship or governance, the respondents claimed that all resources were controlled by the government, and the powerful landlords. Forests, land, fishing, water and even the tube wells were controlled by the landlords. Similarly, questions related to indigenous rights, and traditional knowledge systems, were also asked and the

respondents identified practices such as rain fed ponds, the *karez* system, and wells, among traditional systems. And a few mentioned sharing waters for agriculture.

5.3 Degradation of Natural Resources

Because of degradation and depletion of the natural resources, decline in environmental and ecology, livelihoods are shifting from fisheries and agriculture to manual labour in towns and cities. Livestock however has not declined in the same way, and women continue with livestock rearing. Respondents spoke of powerful political and landowning people commandeering resources as they decline.

Due to rampant ecological decline the stock of environmental goods has gone down drastically. Today, in their daily lives (as opposed to livelihoods) about half the men and women can only partially depend on rainwater, water sources, common land, firewood, fodder, fuel. Some respondents said “nothing is free here”.

5.4 Relationship of Women and Men to Water

Water was also identified as a natural resource - rainfall, ponds, hill torrents, groundwater although saline and brackish, or water from borings, wells, and tube wells, indicating the importance of water to the respondents. Both men and women in all the field sites across all the provinces had similar responses. In terms of the use of natural resources, gender differences emerged where for men water was important for irrigation, and for women water was a priority for livestock.

From household data we saw that more than fifty percent of households go their water from within half km, while one quarter had a range of one kilometer and the remaining quarter between 2 -10 kms. But not all nearest water sources necessarily have sweet water fit for drinking.

Table-6A: Distance to Nearest Water Sources from Home

Distance in km	Total	
	N	%
< 0.5 km	273	52
0.5-1 km	127	24
1- 2 km	29	6
2-5 km	73	14
5-10 km	28	5
>10 km	26	5

According to the respondents in the FGDs drinking water was largely unavailable in the villages, and had to be brought from outside their areas, bought from vendors, or collected from long distances. The situation in Manchar was the worst, where the respondents spoke of prosperity 30 years ago, compared to their present situation of poverty, a ‘poisoned’ lake, a decline in the quality of water, very few fish, and a decrease in the number of species of fish over the last three decades. As one woman said, ‘*I have never seen sweet water since my birth.*’ And older women mentioned seeing migratory birds in their youth, but not for the last 30 years.

5.5 Distance to Environmental and Social Services

As expected most of the environmental and services related to the context of this assessment were located within walking distance. For example half the households could get water from within half a kilometer and a quarter within a mile, though mostly outside their very small settlements. The distance to agricultural land and the fuel and fodder profile is also similar.

Table-6B: Distance to Agriculture Land from Home

Distance in km	Total	
	N	%
< 0.5 km	323	61
0.5-1 km	62	12
1-2 km	31	6
2-5 km	60	11
5-10 km	39	7
> 10 km	40	8

Table-6C: Distance to Fuel/ Firewood from home

Distance in km	Total	
	N	%
< 0.5 km	249	47
0.5-1 km	134	25
1-2 km	43	8
2-5 km	99	19
5-10 km	22	4
> 10 km	20	4

Table-6D: Distance to Animal Fodder from Home

Distance in km	Total	
	N	%
< 0.5 km	259	49
0.5-1 km	155	29
1-2 km	44	8
2-5 km	66	13
5-10 km	33	6
> 10 km	12	2

Table-6E: Distance to Sanitation

Distance in km	Total	
	N	%
< 0.5 km	358	68
0.5-1 km	111	21
1-2 km	25	5
2-5 km	43	8
5-10 km	5	1
> 10 km	7	1

5.6 Impact of Climate Change

The impact of climate change was explored in the focus group discussions and changes in weather patterns were reported. In the past there was more rain, more water for crops, a better yield, more food, and a secure livelihood. Now, summers are longer with extreme temperatures, and shorter winters. Rainfall patterns have changed, water sources have dried up, and the water table has gone down.

Key government respondents pointed out the changes in last 20 years - temperature rise, less water, less crops, less surface water, irrigation has been affected, reduction in forest area, people are migrating out either seasonally or permanently.

Communities have been affected in many ways: fishing has gone down, and peoples' livelihoods are severely affected. Because of denudation of the lands (trees gone, vegetation dwindling) there are more dust storms. Availability of firewood has also declined.

Rainfall patterns have changed, either more or less rains and at odd times, negatively affecting cultivation. In Manchar area it was reported that poisonous water of the lake has huge impacts on the communities.

Migrations were reported in Balochistan due to change in weather patterns, denudation of vegetation cover and decrease in groundwater.

5.7 Protection of Natural Resources

According to the respondents to protect the natural resources, the government should provide free, clean drinking water, small ponds for irrigation and water storage. Pollution into rivers and canals should be stopped, and Manchar Lake should be cleaned. Alternative livelihoods should be provided to those that have lost their fishing and agriculture-based livelihoods and all cutting of forests should be stopped.

5.8 Participation of Women in Government Projects

As part of the data collection, government representatives from the Irrigation Department and Planning and Development departments were interviewed. All the Key Informants were male, and no females were employed in any of the provincial departments.

In Baluchistan, and KPK the government representatives said that they were aware of women working on certain projects at a basic level in their departments but not in senior positions. In Sindh and Punjab, the respondents said that they were unaware of any women in their departments. In terms of women's participation, the respondents were unaware of women consulted at the design stage and said that projects were already designed and given to them to implement since these were related to infrastructure, repairs, and maintenance. And since women were not perceived as relevant to infrastructure there was no need to consult women as the work related largely to engineering and they did not use social mobilization in their jobs.

Only one respondent in Punjab said that he knew of a few women in higher positions in department in the cities who participated in decision-making. The respondents were not aware of any gender equality considerations in the projects they implemented, and all of them thought that women were not suitable for the kind of work required in their departments.

However, they did believe that women were the main beneficiaries of development and infrastructure projects, as these projects provided them with safe water and better irrigation meant more food and income for their families.

5.9 Roles and Responsibilities of Government Departments in the Provinces

Most of the government respondents found on the site were field staff with very specific responsibilities. They were not able to articulate the roles and responsibilities of their departments. They could only talk about own roles and functions. Only two respondents mentioned a particular project on which they are working.

Based on responsibilities articulated by respondents we can see that the range of activities they work on cannot be construed as nature-based solutions.

Only in the Punjab project area is there UNICEF funded project in place so we can see proper planning, tendering and implement following international/UNICEF rules. In the project sites of three provinces no such projects were reported, and only routine departmental activities seen to be in place.

No social mobilization mechanism and instrument found within any department to execute a social mobilization plan. Only in Punjab where a UNICEF project is implemented is there a community mobilization mechanism through CBOs.

5.10 Management of Water and Environmental Resources

Most of the government officials mentioned groundwater, handpumps, solar pumps as the main sources of water for domestic use, while in Balochistan government water supply system identified as the main sources of water for domestic use. Tube wells and canals/barrage are the main sources of irrigation water, in KPK there were 40 solar tube wells also in use to irrigate 1000-acre land.

Drinking water quality is mostly not so good, saline/brackish, bitter, yellow and bad in taste. Only in one area of Punjab irrigation water testing facility exists, while in Balochistan and Sindh government official mentioned that irrigation department is responsible for testing of quality of irrigation water.

Due to water pollution/contamination crop production is affected in most of the areas, whereas in Manchar Lake water is poisonous and not in use for agriculture. Sources of water pollution are mostly chemical fertilizer, pesticides and industrial wastes while mixing of sewerage water is in cities. There is less water pollution in KPK and some parts of Punjab mentioned by government officials because they do not spray their crops, not use chemicals fertilizer and no industry near the fields.

Government officials said that solutions are to reduce use of pesticides, and chemical fertilizer, separate the sewerage lines from canals/irrigation water, do not release industrial waste into canals, Reverse Osmosis (RO) and filtration plants that provide drinking water.

Government officials mentioned a list of diseases associated with water pollution: diarrhea, cholera, hepatitis, jaundice, typhoid, skin diseases and other water borne diseases. While destruction of crops, reduction in fish catch and decrease in crop production, reduction of land fertility and economic loss are mentioned as economic risks associated with water reduction and pollution.

Decline in crop production, reduce land fertility, migration of communities and livelihoods shifting from agriculture and fishing to manual labour in nearby cities, are listed as impact of water pollution in Balochistan, Sindh and parts of Punjab. Government officials talked of economic crises occurring during floods and drought.

5.11 Conclusions

It can be clearly concluded that climate change has already negatively affected the equation and balance between people and nature. The rapid changes are visible and the feared migrations, poverty and loss of livelihoods are a reality to different degrees in all six project sites.

The impact of climate change in the form of changing weather patterns and extreme temperatures are changes that are a reality for the women and men in the project sites. Adaptation to these changes and interventions for addressing these changes need to be designed with the participation of the community including the women. Alternative livelihoods are now needed. Shrinking natural resource base has led to migration and a shift from traditional livelihoods to manual labor. The women left to tend animals and mind the home face daily struggles for survival. These issues are all part of the challenges that the Recharge Pakistan project will have to address.

The tendency of elites to commandeer resources and their ability to manipulate government functionaries in difficult times means that meager environmental and ecosystem resources will continue to be denied to the landless and the poor, especially women.

WWF has to be careful in the way the gender dialectic is used, because it can lead away from the plight of both poor men and women in the face of elite capture. The focus should not only be on building gender parity between community women and men, but in providing interventions that also protect communities from unilateral action by elites and government institutions that render poor men and women bereft of their traditional livelihoods. Extra care is needed to ensure that women's environmental entitlements remain and they continue to access environmental resources for their water and food security.

As for the government, it is clear from the interviews of functionaries that they are mostly clueless on gender terminology, policies and actions and would require a lot of attention in the intended project design and implementation of the Gender Action Plan.

CHAPTER 6: Socio-Economic Profile

Any gender assessment requires a socio economic and environmental context within which gender issues can be understood correctly and the most suitable and effective interventions developed and implemented. This is especially required in the case of projects and programs that are community based where, in addition to national and provincial situation, the local poverty, control, power and normative environment has to be taken into consideration.

In the case of this gender assessment the socio economic information was collected from households through a quantitative instrument that was developed and pretested by IRS Global, and administered by our counterpart consultants DevCon as part of a larger data gathering exercise managed by their field teams in the six project sites designated by WWF. We have duly acknowledged this contribution of DevCon.

Information was collected from 527 households. Of these men respondents provided information for 311 households and women respondents for 216 households. Please note that several tables below are multiple response.

6.1 Age-Sex Distribution and Marital Status

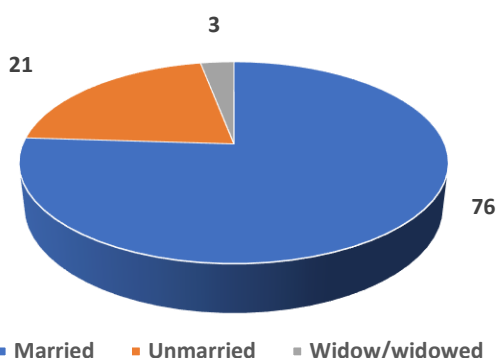
53 percent of all people recorded are 20 years or below, in keeping with national trends. While 42 percent of people are between 21-45, only 4.5 percent are over 45 years, and that too mostly women. It is quite usual for people in rural settings of Pakistan to not know birthdays and guess their age, the data still indicates a smaller proportion of people who can be bread earners.

Male female ratio is 52:48 reflecting the national figures. Marital status of people age 16 years and above is about three quarters: 83 percent of women and 70 percent of men being married.

Table-7: Age - Sex Distribution

Age in years	Men		Women		Total	
	N	%	N	%	N	%
0-20 years	999	55	852	51	1851	53
21-45 years	720	40	747	45	1467	42
> 45 years	89	5	69	4	158	4.52
No response	13	1	4	0.24	17	0.49
Total	1821	100	1672	100	3493	100

Figure-1: Marital Status (age 16 years & above)



6.2 Educational Status

In terms of educational status of children 4-16 years, primary level enrolment is 47 percent for boys, and 39 percent for girls. Middle and secondary enrolment is very low, with only 1 percent girls in secondary. **44 percent of children have had no schooling at all.** Out of school children include 54 percent girl and 36 percent of boys. In terms of adult education (17 years and above), 52 percent men and 75 percent women never went to school.

Table-8: Educational Status of Children (age 4 to 16 years)

Educational Status of Children	Boys		Girls		Total	
	N	%	N	%	N	%
Primary (class 1-5)	359	47	246	39	605	44
Middle (class 6- 8)	59	8	34	5	93	7
Secondary (class 9 -10)	68	9	6	1	74	5
No schooling	273	36	341	54	614	44
Total	759	100	627	100	1386	100

Table-9: Educational Status of Adult (age 17 years and above)

Educational Status of Adults	Men		Women		Total	
	N	%	N	%	N	%
Primary	89	10	52	6	141	8
Secondary	207	22	67	7.5	274	15
Middle	58	6	45	5	103	6
B.A/BSc	97	10	60	7	157	9
No schooling	484	52	674	75	1158	63
Total	935	100	898	100	1833	100

6.3 Employment and Professional Status

Please note that the question asked for both profession and employment as one amalgam. One can be happy that child labor is low in the designated since areas since 96 percent of girls and 88 percent of boys under 16 years do not work. For people aged 17 years and above, among men 26 percent wage labour, 16 percent small business, 12 percent agriculture, 10 percent fishing, 10 percent government. About 20 percent men and 88 percent women are unemployed. Those women who work, mostly do fishing and handicrafts.

Table-10A: Employment / Profession of Children (age 16 years)

Employment	Men		Women		Total	
	N	%	N	%	N	%
Agriculture	6	1	0	0	6	0.5
Fishing	30	4	25	4	55	4
Govt. Employee	3	0.4	0	0	3	0.2
Handicraft	1	0.14	0	0	1	0.07
Live Stock	5	1	0	0	5	0.5
Small Business	10	1.5	0	0	10	0.7
Wage Work/Labour	30	4.1	0	0	30	2.2
Unemployed	646	88.4	629	96.2	1275	92.1

*multiple responses

Table-10B: Employment / Profession of Adults (age 17 years and above)

Employment	Men		Women		Total	
	N	%	N	%	N	%
Agriculture	112	12	6	1	118	6
Fishing	97	10.4	46	5.1	143	8
Govt. Employment	94	10.1	20	2.2	114	6.2
Handicraft	0	0	25	3	25	1.4
Livestock	10	1	2	0.22	12	1
Private Employment	36	4	7	1	43	2.5
Small Business	150	16	0	0	150	8
Wage Work/Labour	247	26.4	3	0.33	250	13.64
Unemployed	189	20	789	88	978	53

*multiple responses

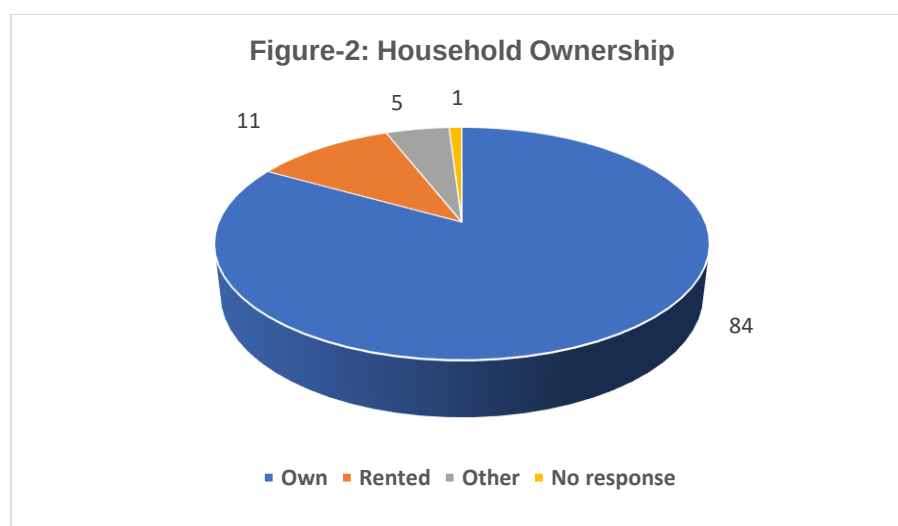
When we review this data in juxtaposition with the FGDs data we see that most of the people there reported unemployment. So except for government employees, while men and some women have skills and professions (traditionally through the decades), **currently they do not have a means of income**. When read with other findings we can see that the environment for their traditional professions and means of income (fisheries, agriculture, livestock and handicrafts) is dwindling and degrading – through climate change, privatization and government take-over.

FGDs further show that mostly men are involved in work that is usually paid (when they get the work), and women usually look after domestic chores and are unpaid. In some instances, women and men worked together in fishing, fetching water. Except for fisheries and livestock, women did not report working in any professions or carrying any jobs. While men left their communities for labor work elsewhere, women tend to remain.

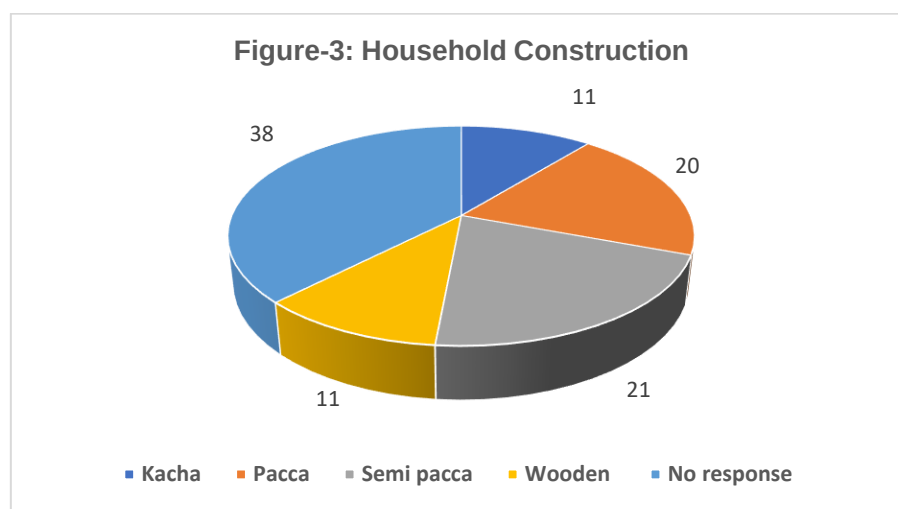
6.4 Household Ownership and Construction

A total of 84 percent of the households have their own houses, but this should be taken with a pinch of salt for typically in Sindh's rural areas, landless people (who do share cropping) are 'allowed' to build their non-permanent *kaccha* houses on the land owned by landowners. They typically do not hold any lease of ownership papers and can be displaced from these 'owned' houses at any time by the people who actually own the land.

Kaccha houses refers to structures built with mud/clay, wood parts and roofs thatched with local reeds and dried vegetation. *Pakka* houses refers to structures built with bricks, some cement or concrete and some piping. Semi *pakka* is a mix of both styles.



One can see from the ‘no response’ to the type of housing that people do not understand the nuances of *kaccha*, *pakka* and semi *pakka*, but observations from the field show that people mostly live in very rudimentary, run-down *kaccha* housing.



6.5 Household Expenditure

From field exposure we have seen that except for government servants, men and in particular women, are unable to provide a reliable estimate of their monthly or annual income because they do not get regular income. But they were able to give broad categories of what they spent on. The low numbers of women respondents recording that they do not spend money of any kind reinforces the findings from the FGDs which show that women do not have much say in decision-making or household spending, and may not even be aware of how much is being spent by the men of the family on key households expenditures.

Table-11: Type of Family Expenditures

Family Expenditure	Men		Women		Total	
	N	%	N	%	N	%
Food	274	88	13	6	287	55
Water	187	60	17	8	204	39
Fuel (including electricity, gas)	204	66	18	8	222	42
Education	142	46	10	5	152	29
Health	244	79	7	3	251	48
Transport	237	76	6	3	243	46
Leisure	135	43	0	0	135	26
Others	130	42	4	2	134	25

*multiple responses

This table shows interesting insights – while 50 percent households report they have water within half a km, but 39 percent of households are buying that water.

6.6 Ownership of Assets

Very few men reported to owned assets, mostly they owned livestock, house, motor bikes. Some others mentioned land, TV/AC/mobile, money, and jewelry. But in the FGDs livestock ownership by women was reported more emphatically.

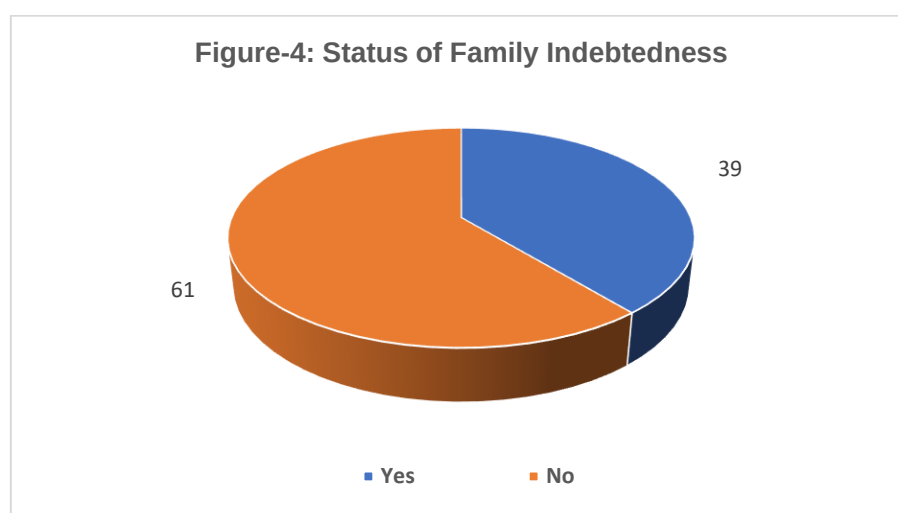
Table-12: Ownership of Family Assets

Ownership of Assets	Men		Women		Total	
	No	%	No	%	No	%
Land	16	5	1	1	17	3
Livestock	26	8	1	1	27	5
House	18	6	0	0	18	3
Car/motor bike	17	6	0	0	17	3
Bank balance/money	12	4	2	1	14	3
Jewelry	8	3	0	0	8	2
TV/AC/mobile etc	16	5	0	0	16	3
Others	3	1	0	0	3	1

*multiple responses

6.7 Status of Family Indebtedness

About 39 percent households were under debt and they mostly took loan from family/relatives and local money lenders.

**Table-13: Source of Loan**

Sources of Loan	Number	Percentage
Local money lender	44	21
Family/relatives	99	48
Other	65	31

*multiple responses

6.8 Women Involvement in Decision Making Process

72 percent men and 84 percent women reported that women are involved in decision making process mostly in family matters. Otherwise it is mostly men who rule the roost.

Table-14A: Women Involvement in Decision Making Process

Women Involvement in Decision Making	Men respondents		Women respondents		Total	
	N	%	N	%	N	%
Yes	225	72	182	84	407	77
No	72	23	29	13	101	19
No response	14	5	5	2	19	4
Total	311	100	216	100	527	100

Table-14B: Type of Decisions in which Women Participate

Type of Decisions	Men respondents		Women respondents		Total	
	N	%	N	%	N	%
Family matters	207	92	164	90	371	91
Property matter	38	17	9	5	47	12
Use of natural resource	26	12	13	7	39	10
Farming/agriculture/fishing	30	13	7	4	37	9

*multiple responses

In the FGDs more men than women seem to think that women have a role in decision-making, and that they are more aware of women's rights. However, key informants (government) did not think that women had much role in decision making, except at the higher levels in cities, and also felt that there was not much they could contribute to in field situation of infrastructure projects.

6.9 Activities of Men and Women Inside and Outside Home

Respondents were asked to state what activities the men of the family did inside the house and outside the house. In the same way respondents were asked to state what activities were undertaken by women inside and outside the house.

In terms of how the roles of men and women have changed over the last 20 years, the FGDs showed that previously men and women worked together in fisheries and agriculture. Now men have to go off to far off places for manual labor, and women stay at home. Some reported that in recent years men help in fetching water, collecting firewood and look after livestock.

6.10 Conclusions

The Socio-economic picture that emerges of the communities in the six WWF sites shows clearly that they are very poor, indebted and live from hand-to-mouth. While many of the indicators are typical of rural areas of Pakistan (education, access to public services, poor water and sanitation), many others point to abject poverty, loss of livelihoods, no opportunities and sense of hopelessness.

Except for government servants, men and in particular women, cannot provide estimate of their monthly or annual income because **they do not get regular income**. Women do not seem to spend money. They do not have much say in decision-making or household spending, and may not even be aware of how much is being spent by the men of the family on key household expenditures.

Mostly men are involved in work that is usually paid (when they get the work), and women usually look after domestic chores and are unpaid. In some instances, women and men worked together in fishing, fetching water. Except for fisheries and livestock, women did not report working in any professions or carrying any jobs. While men left their communities for labor work elsewhere, women tend to remain.

The environment for their traditional professions and means of income (fisheries, agriculture, livestock and handicrafts) is dwindling and degrading – through climate change, privatization and government take-over.

Lifting the socio-economic profiles of the poorest communities in the six project sites will be a challenge, so that earning opportunities for men and women have to be an integral part of the design of the project.

CHAPTER 7: Gender Mainstreaming in WWF

7.1 WWF Gender Policy

WWF adopted its Gender policy in 2011. This policy reflects WWF's ongoing commitment to equity and integrating a gender perspective in its policies, programs, and projects, as well as in its own institutional structure⁷⁵.

The WWF Global Network Policy: Gender Policy Statement 2011 states that *"The policy statement is to ensure that WWF's conservation policies, programs and activities benefit women and men equally and contribute to gender equity, as part of WWF's broader commitment to strengthen the social dimensions of its projects, programs, and policy work. It describes the rationale for gender mainstreaming in the context of WWF mission, biodiversity and footprint goals and outlines WWF commitment to integrate a gender perspective in our programmatic and operational structures and procedures."* The policy outlines WWF's belief in promoting gender equity and integrating a gender perspective *'in projects and programs and its commitment to creating programmatic structures and procedures that incorporate a gender perspective into program and project development processes through the application of gender awareness and analysis in the project cycle, including design, implementation, monitoring, and evaluation.'* And WWF is also committed to *'ensure an organizational culture and work environment where management and other staff are aware of what constitutes discrimination and respects diversity in work and management styles and prevent discriminatory practices including stereotyping and sexual harassment'*⁷⁶.

In this context the WWF Recharge Pakistan project has conducted this Gender Analysis and developed a Gender Action Plan to ensure that gender mainstreaming is integrated into the design phase of the project, which reflects WWF's continuing commitment to its policy and future action.

7.2 Review of WWF Annual Reports 2019-2021

A brief review of WWF Pakistan's Annual Report's over a period of three years 2019-2021 mentions gender in several places and specifically women's inclusion across its six areas of focus⁷⁷. Several examples of projects which have included women, and supported women's empowerment were described.

Examples include the Khanpur Watershed project focusing on women, support to cluster organization with trainings on value addition and marketing of *Chilghoza* nuts, and diversification of livelihoods in Balochistan⁷⁸, the Food and Markets projects where 835 farming families including working women were supported with gender trainings to develop their skill so they could explore alternate income generation opportunities⁷⁹.

The 2021 WWF installed one rainwater harvesting system, at Government College Women University Sialkot, to offset the use of groundwater during the rainy season, illustrating support to women's educational institutions. And in Sindh 30,000 mangrove saplings were planted in the Indus Delta engaging women *'sheroes'*, a community-focused plantation to enhance coastal climate resilience and provide a livelihood to the community. Similarly, 100 women

⁷⁵ https://www.wwfpak.org/about_us/_wwf_s_policies/gender/.

⁷⁶ WWF Policy 2011, Global Network Policy: Gender Policy Statement.

⁷⁷ WWF Annual Report 2021 (Food and Markets, Climate and Energy, Freshwater, Wildlife, Forests, Oceans).

⁷⁸ WWF Annual Report 2019.

⁷⁹ WWF Annual Report 2020.

fisher folks from villages of the Indus Delta were trained in home-based enterprises and traditional handicrafts to support alternative livelihoods⁸⁰.

7.3 WWF Gender Projects

Three gender project reports were received from WWF and reviewed to understand to what extent gender was addressed among WWF's overall projects. A brief analysis of these reports indicates that skill training to increase woman's income generation and potential for alternative livelihoods is integrated and considered important for women's empowerment. There are four gender specific projects in the Livelihood Improvement & Gender Empowerment Initiatives, which focus on cotton farming, biodiversity, agroforestry, and micro-enterprise development. The Indus Eco Region Community Livelihood Project 2019 integrates women by promoting livelihoods, and the report provides the number of female beneficiaries. And the Sida 2 Grow Oxfam project Annual Report monitors gender outcomes and impact as well.

However, the gender data is scattered and although it exists across reports and projects, there is the need to map the number of gender interventions, the beneficiaries, indicators of success, and impact on women to record and understand the extent of gender mainstreaming within WWF's programs.

The Implementation Completion and results report for the Indus Eco Region Community Livelihood Project 2019⁸¹ is helpful, because it has some overlap with the current GCF project.

This project integrates women into the project by promoting alternate livelihoods. This component was designed to help diversify sources of income and reduce reliance on fishing as a livelihood at Keenjhar Lake, Manchar Lake, Chotiari and the Nara Canal. This was through organizing targeted freshwater fisher folk into groups, introduction of better and sustainable fishing practices, enhancing income-generation opportunities for youth, women, and the disadvantaged through establishing community-based cottage industries and developing effective and sustainable linkages to markets and local governments.

A review of this project document indicates an overlap with the field sites of the Recharge Pakistan Project and could provide a good basis to build on the outcomes of the Indus Eco Region project, and lessons learnt and apply them to the Recharge Pakistan project process. According to the Community Livelihood Project report, *'at project closure, the project development objectives were partly achieved, and the overall efficacy of the project is rated modest. Direct female project beneficiaries were 17 percent of women HHs. The project developed three craft centers, one in each of the three project areas namely Keenjhar, Chotiari and Nara. Women were provided training in block printing, stitching and craft management. Under IECLP, Domestic Poultry Units (DPUs) were given to ten women at Nara Canal, Keenjhar Lake, Manchar Lake and Chotiari Reservoir. Gender mainstreaming was an important aspect of the project.'*

7.4 Case Studies from the Field

A review of Case Studies from the field also demonstrate how WWF has improved women's economic and social status as Case Study 1 Micro Nurseries Run by Women⁸² has led to income generation for rural women, given them a sense of empowerment and enabling women

⁸⁰ WWF Annual Report 2021.

⁸¹ IMPLEMENTATION COMPLETION AND RESULTS REPORT Indus Eco Region Community Livelihood Project (IECLP) (P146252) The World Bank 14 January 2019.

⁸²https://www.wwfpak.org/knowledge_hub/_success_stories/_micro_nurseries_run_by_rural_women/#:~:text=W WF%2DPakistan%20works%20with%20women,%2C%20shade%2C%20shelter%20and%20protection

to choose how to spend their income, and learn how to manage small nurseries. Similarly, Case Study 2, a WWF – Better Cotton Initiative⁸³ clearly shows the impact on women's livelihoods through income generation by cultivating tree nurseries and selling the saplings to generate revenue for themselves and their families.

WWF's projects, and engagement with the communities include women's participation, with gender responsive interventions, and recognise the ingrained gender bias in rural society which prevents women's participation in decision making. And as part of their program WWF also increases women's awareness of the importance of nature and the environment.

7.5 Focus Group Discussion with WWF's Women Staff

To continue a further analysis of gender mainstreaming, and opportunities for women's leadership within WWF, a focused group discussion was held with a cross section of female staff from WWF offices based in Lahore, Islamabad, Multan, and Karachi. Twelve women who had worked with WWF between one and five years (and one who had been there considerably longer) represented their programs in Climate and Energy, Environmental Education, Food and Markets, Recharge Pakistan, Audits, and Program Development. The positions of the staff were Managers, Coordinators, Senior Project Officers, Research Associates, and a Gender Specialist.

The FGD's⁸⁴ addressed recruitment procedures, opportunities for gender training, gender sensitization of men and women on gender issues in Pakistan, numbers of men and women in gender-related positions within WWF, role of women staff in projects and programs, and decision making, changes in the position of women within the organization in the last ten years, potential for an enabling environment in the workplace and challenges for staff and women's leadership.

7.5.1 Recruitment, Initiation and Orientation

The FGD showed that **recruitment and initiation of staff** into WWF is in response to advertisements in newspapers where candidates are shortlisted, interviewed, tested, and called for a final interview. It is a formal process based on TOR's developed by WWF depending on the requirements of the position. The respondents were unaware if WWF is an "equal opportunities employer" or if 'women are encouraged to apply' because it is not stated in positions advertised. From their experience the respondents stated that there is no bias in the hiring process, in fact there is a preference to hire women in agricultural projects, because travel to remote field sites is not considered a barrier for employing women.

Orientation to WWF's rules and procedures was a regular practice and an Orientation package was provided, for introductions to staff, and a short brief on harassment at the workplace and how to address it is part of the Orientation package, but gender orientation is not part of the orientation process. In projects if there are donor requirements for gender trainings, gender modules have been developed for specific projects to be used by field staff. Gender sensitization i.e., the status of women and men in Pakistan is not provided as part of orientation.

7.5.2 Familiarity with WWF's Gender Policy

⁸³ <https://stories.bettercotton.org/ruksana/>

⁸⁴ The Focus Group Discussion centred around 14 questions and was conducted through a zoom meeting ensuring the participants of complete confidentiality of their responses. The names of the participants remain anonymous. See Annex 1 for FGD questions.

*Only one woman had worked with WWF for longer than five years.

Basic **gender concepts** are not taught, and the definition of a gender specialist was perceived as *'a person who can communicate with both men and women in the agricultural sector's farming families.'* And it appeared that gender integration is in response to donor's requirements. Similarly, the legal requirement for information and mechanisms on sexual harassment or discrimination in the workplace is not formally presented at the time of orientation. And there is no locus of a gender person or gender coordinator who could link gender policy through the organization, to programs and projects. However, the respondents said that women staff are always involved in project team discussions, and women are included in the design of activities, and are part of the decision making at the project level. This was mentioned in reference to when the head of a department is a woman.

WWF's **global gender policy**, and the **GCF policy** was new information to the participants who were unaware of existing gender policies of WWF at global level. WWF Pakistan follows the WWF global gender policy, and all except one respondent had not heard of these gender policies. This lack of awareness of the strong global gender policy within WWF's female staff reflects a gap in the organization towards gender integration and mainstreaming.

7.5.3 Gender Design and Development

Project design and development takes place in individual departments, and projects are developed according to donor requirements, with technical support if needed from other departments, and if gender is a focus for the donor, then gender modules are developed for implementation in the communities.

Addressing gender in the programs, or within the organization comes from the head of the department and again in accordance with donor requirements. Gender mainstreaming is perceived as donor driven with a 'top-down approach', and male managers make the main decisions related to gender in programming, ideas also emerge from male managers, and there is no formal process for women in WWF to voice their opinions, to hear and share ideas that they can put forward in the projects.

However, in complete contrast to their experiences the respondents spoke of WWF's organizational approach which is to develop a Strategic Five-Year plan where gender is integrated, gender empowerment is a primary component, gender norms, cultural sensitivity, emphasis on gender disaggregated data which provides the basis for developing activities and women's engagement in projects is include in the strategic planning process. Gender action plans, safeguards, and specific gender interventions are also developed and included in the planning process. And the women staff also emphasised the strong involvement of women in agriculture, coastal, wildlife, and income generation projects. It appeared that the role of women was recognised and encouraged at the field, community, and project level, but not within the office workplace.

The project staff also commented that women in the communities are involved in advocacy activities, encouraged to speak at district forums, and community women's representation and leadership is supported and promoted. The respondents thought that WWF had developed its own set of informal unwritten protocols for gender and said, *'we are getting there and someday soon we will be a gender sensitive organisation'*.

7.5.4 Organogram and Gender Presence

The **WWF Organogram 2021**⁸⁵ shows that there are 31 Senior Managerial staff in WWF, of which 10 are women. The Head of WWF is a male, and in the second tier there are six Senior Management positions of which one woman is the Head / Senior Manager of Communications, in the Lahore office. In the third tier among eight Senior Managers and Directors, there is one woman in the position of Coordinator Livelihood Improvement and Gender Empowerment, in the Lahore office. In the fourth tier, among seven Managers and Coordinators there are two women in the positions of Manager Conservation and Wildlife, and a Senior Graphic Designer and Communication in the Lahore office, and one woman who is Coordinator Policy and Safeguards in the Islamabad office. In the fifth tier among five Senior Managers and Managers there is one woman who is the Head /Senior Manager in the Climate and Energy / Corporate Partnership in the Lahore office. And in the sixth tier among three Managerial positions there is one woman as Manager People Development in the Islamabad office, and one Coordinator ILES project in the Karachi office. In the seventh tier there are two positions both who are women as Coordinator Freshwater Program in Sialkot, and a Senior Office Compensation and Benefits in the Lahore office⁸⁶. An analysis of the organisation structure shows that there are only 3 percent women in WWF, and women are scattered throughout the organisation, but in the sixth and seventh tier there are only women (Annex A12)

Information received from WWF regarding the number of men to women in the organisation indicates there are 6 men and 5 women in the Governance tier. In the Management tier there are 38 men to 4 women, in Finance and Administration there are 145 men and 5 women, 1 man in the Human Resources to 3 women, and in projects there are 613 men to 67 women. The percentage of men is 91percent to 9 percent women⁸⁷.

7.5.5 Work Environment

Further discussion on an **enabling work environment** revealed a day care center at the Head Office which was not functional. Lack of a prayer room for women, and the absence of a private room for women to go to if they were feeling unwell were mentioned. Women's health during pregnancies, or reproductive health issues were not recognized, and the absence of empathy for women during menstruation was also mentioned. A subtle negative attitude towards a woman after she got married was perceived as an impediment to her work and ability 'to do the job' and pregnancies and a raising a child were considered obstacles to performance. A lack of gender sensitivity towards women travelling alone with male drivers, and examples of the lack of empathy from male colleagues for the absence of toilets for women in remote villages where they often had to spend days in the communities during their menstrual cycle were considered difficult and stressful. Requests to work from home during a pregnancy were not viewed favorably, and although an official Maternity leave policy exists, it was not always implemented with full support to the woman concerned.

The WWF Employees Rules 2022⁸⁸ contains a Maternity policy and a Paternity policy. The Maternity policy states that married female employees are permitted Maternity Leave with a salary for 3 months. The leave includes weekends and any public holidays and can only be availed twice during service. For Paternity Leave, married male employees are allowed Paternity Leave with salary equivalent to 5 workdays. The leave can only be availed twice during service.

⁸⁵ WWF Pakistan, Organization Structure 2021.

⁸⁶ WWF Female Staff - Positions and Departments.

⁸⁷ Number of men and women in WWF Pakistan, email sent by Fareeha Naseem, Gender Specialist, Recharge Pakistan Program, WWF Lahore Office, 10th May 2022, details in annex.

⁸⁸ WWF Pakistan Employees Rules 2022.

7.5.6 Changes in Women's Position in Last 10 Years

And yes, there have been **positive changes in women's position in the last ten years** within WWF. A few mentioned that they were the first women in certain projects a few years ago, but recently more women have joined the organization, and a positive change in attitude from male colleagues towards women working in the field with them has occurred and become supportive. Travel is comfortable, and hotels, and safe places to stay were appreciated by the respondents. Women have also relocated to different cities for the sake of the job, and this is accepted which indicates a shift in culture and expectations. More women are challenging stereotype roles which has led to greater acceptability. In most cases it was stated that if the head of department was supportive the workplace was comfortable, but this was not formal office protocol, and it was the responsibility of the woman to speak for herself, and for her manager to provide support.

From the FGDs it appeared that there is a clear dichotomy between women's role in projects and within the organisation. Women participate in decisions related to project design and activities in the field, the attitudes of their male colleagues are positive and supportive, and the perception that women can travel long distances and work in remote areas is not a barrier to their work. However, within the organisation the situation is different, and an enabling environment depends on the attitude of the head of department, and in most cases, women must ask for support and voice their opinions to improve their working environment. These women staff can develop a platform of their own because a critical mass already exists and if these women are supported by their senior managers, and if gender responsive mechanisms are developed gender mainstreaming and a gender sensitive environment can be created within WWF.

7.6 Conclusions

WWF has a strong Gender policy and follows the GCF Gender Policy Guidelines. It also has several projects which focus on women and support women's empowerment, but gender data is scattered over different reports and websites, and an Annual Gender Report covering all the current projects with disaggregated data, number of male and female beneficiaries and impact on women's empowerment would provide a baseline for gender mainstreaming.

A review of WWF's projects, and their interaction at the local level with the communities indicates women's inclusion and participation, with gender sensitive interventions, and addresses the deep-rooted gender bias which exists in rural society and prevents women's participation in decision making. And WWF also increases women's awareness of the importance of nature and the environment by recognising the linkages between gender and natural resources. Work in the field and projects has a large component of gender and responds to donor requirements.

However, within the organization respondents from the FGDs indicated that gender trainings for both men and women are absent and should be developed and integrated into the Orientation process for a new employee. Gender concepts, WWF's gender policies, and gender laws and policies in Pakistan should be provided in the Orientation kit and the importance of gender in WWF as an institution should be part of the initiation process.

Reproductive health issues within the office and the field are not always recognised or supported and this causes stress and has a negative impact on the working environment, these issues should be considered and become a part of formal institutional protocols. The legal requirement of the existing Protection against Harassment of Women at Workplace Act 2010, should also be understood and all staff should be informed, and the mechanisms for recourse within the law should be established in the organization.

Similarly, to develop an enabling environment a gender focal point should be designated to ensure that gender policies are translated through the organization at all levels. And this person should have a strong gender background and experience in gender issues.

In terms of increasing awareness of gender issues, gender analysis training in collection of data, tabulation, and reporting would increase the understanding and capacity of the staff to gender responsive programming.

And finally, there are several women in WWF at different strata of the organization, but only a few in leadership and decision-making positions. Therefore, these women should be trained and provided opportunities for move towards leadership positions, and a clear career path should be offered to women as they progress within WWF.

Gender Action Plan:

The Gender Action Plan (GAP) for the Recharge Pakistan project is developed based on the results and recommendations from the Gender Assessment. The GAP covers gender specific actions against each activity that aim to ensure gender issues are adequately addressed and bring noticeable benefits to women and men, especially in vulnerable groups. It aims to ensure that both men and women have equal opportunities to participate in, and benefit from, the project activities. It also ensures that both women and men can participate actively and in an informed manner in these activities and planning that can relate effectively to gender-related issues in communities. The actions under GAP have adopted the language of the overall Logical Framework developed for Recharge Pakistan.

The project activities will contribute towards gender mainstreaming in the country by:

- Promoting gender inclusiveness in EbA and all project activities.
- Establishing strategic plans, and to develop and implement community-level awareness campaigns on climate change impacts and the project's adaptation.
- Strengthening the capacity of local communities to undertake community-based natural resource management.
- Improving climate resilience of the vulnerable groups through agricultural livelihood activities and capacity building of vulnerable communities to undertake alternative income generation activities.
- Building capacity of key stakeholders including Government line departments, CBOs, and local communities to apply a gender lens into their own projects.

The Gender Specialist (Manager, Gender) based at the Project Management Unit (PMU) will lead the implementation of the GAP. In addition, the Gender Specialist will conduct a needs/capacity assessment at the onset of the project implementation phase to evaluate the level and frequency of training and support the PMU team and partner will require to efficiently implement the GAP. This shall include training and regular refresher sessions on gender concepts, gender mainstreaming, how to recognize and address GBV and SEAH issues using the preventive structures put in place in the project to respond efficiently and appropriately to these events if occur during the project life cycle (**see Activity 1.3.2 under GAP – gender specific actions c, d, e and f**). At Site Implementation Units (SIUs) a total of 12 social mobilizers have been added to the team who will have representation of 6 male and 6 female staff to facilitate implementation of the GAP. Additionally, ESSF staff at both PMU and SIU will facilitate and play a key role in implementation of gender specific actions. Dedicated roles of Coordinator, Gender Empowerment and Livelihood Development one in each landscape (DI Khan and Ramak will be covered by one Coordinator) have been budgeted to ensure a focused approach to implementation of the GAP. A multi-tier staffing arrangement ensures a cohesive approach across all project activities and offers maximum interaction with local communities and key stakeholders at national and sub-national level.

Additionally, cooperating organizations such as National Commissions on the status of Women, Women Development Department (WDD), Council on Gender Equality, Vocational Training Institutions, Government agencies and line departments at national and sub-national levels working on Gender Mainstreaming have also been listed under responsibility that can be called upon to facilitate with the implementation of these activities.

Note:

- i) To avoid duplication, beneficiaries will only be reported once irrespective of how many activities they participated in.
- ii) While setting the targets, the social and cultural norms were kept in mind to ensure a true representation of women participation across each site.
- iii) Percentage representation of women will be ensured from the very beginning of each activity, and no ramp time shall be required to attain the targets.

- iv) There is a variance in women participation for activities under Output 1.1 (40%) as compared to Output 1.2 (25%) due to the fact that participation of women in operation and maintenance trainings for green infrastructure are likely to remain low due to existing cultural norms.
- v) Under the Activity 3.2.2 Gender specific action (c), we expect to train 166 people (50% women) at each site on GBV, SEAH and GRM. In the same activity gender specific action (d) 25 people per site from local NGOs, and other relevant institutions will be trained on GBV and SEAH of which 30% will be women as the male to female ratio across target entities is around 70% male vs. 30% females.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
Component 1: Proofs of concept for EbA and green infrastructure interventions as efficient and effective solutions for flood and drought risk reduction in Pakistan.						
Outcome 1: EbA and green infrastructure interventions with improved community-led management reduce flood risk and enhance the climate resilience of the most vulnerable people in Pakistan's Indus Basin.						
Output 1.1: EbA interventions implemented to restore floodplain and watershed ecosystems.	Activity 1.1.1: Restore 14,215 ha of degraded agro-ecological landscapes in watersheds.	a) Producing Integrated Floods and Drought Management Toolbox covering ecosystem-based adaptation, including gender dimensions (not confined to gender tools, but for the whole project) b) Prior to on-ground implementation of EbA interventions, women consent is secured and are also consulted on their role for operation and maintenance during CBOs meetings. c) Procurement of plant saplings from nurseries raised by local women.	WWF Gender and project team Gender specialist and Site Implementation Units Cooperating Organizations GWP CAPNET WWF Gender and project team Gender Specialist	- Indicator a): Toolbox with policy, management, and capacity building tools, including gender dimensions, from which relevant ones can be selected for each site Target: 100% of toolbox products include gender dimensions - Indicator b): Number of women participations during free prior and informed consent (FPIC) process at the community level Target: 30 women per site - Indicator c): Number of women selling saplings from owned nurseries	Same timeline as the project activities Q3Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
				Target: 60 women		
	Activity 1.1.2: Rehabilitate degraded wetlands, flow paths, and channels.	a): Direct gender group interventions at the site for formation of women's groups around natural resources, water, climate change and environment b) EbA interventions directly contribute towards building resilience of women, through capacity building workshops and practical experiences, to adapt to flood and drought events. And ensure consultation of women over final design of interventions and their impacts c) Build capacity of local women and men on climate change is, building their leadership skills, so that they will be able to actively engage in the CBO meetings d) consultative meetings with CBO's women groups on their role for operation and maintenance of green infrastructure	WWF Gender team Social mobilizers Local NGOs Cooperating Organizations Training institutions	-Indicator a): Number of women's groups formed at each site Target: 04 women's groups formed (1 per project site) -Indicator b): Number of women at each site that are members of these groups Target: At least 30 women are members of these groups at each site - Indicator c): Number of people trained (men and women) on leadership skills Target: 100 Women and 120 men (at each site) - Indicator d): Number of women's groups involved in planning and implementation of	Q3Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
				women related interventions Target: 4 Women's groups (1 group at each site)		
		e) Development of four plans relating to the sets of vulnerabilities (addressing climate induced drought & Flood) of local women and men according to each site's context and including gender dimensions. These will cover, but not limited to locally feasible EbA and green infrastructure measures, training needs assessments, specialized trainings on O&M of EbA and GI, and ensure alignment with the climate resilient alternative livelihoods. f) Ensure that EbA interventions address climate vulnerability of local women towards drought and flooding through expert review of the interventions to	WWF Gender team and Project team (Safeguards) Cooperating Organizations GCF or similar organizations MOCC Local NGO's and CBO's Others as needed	-Indicator e): Number of plans developed to address the vulnerabilities of women and men related to drought & flood with appropriate resilience measures, taking into account gender dimensions. Target: 4 resilience plans (1 for each site) - Indicator f): Number of people with improved resilience Target: 2000 people with improved resilience (40% will be women)		Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
		ensure gender dimensions are included and that they respond to women's and men's realities and expressed needs.				
Output 1.2: Green infrastructure interventions established in Pakistan's Indus Basin.	Activity 1.2.1: Design and implement 127 green infrastructure interventions for flood risk reduction.	a): Consultative meetings with women's groups during design and implementation of green infrastructure measures b): Green infrastructure interventions address climate vulnerability of local women towards drought and flooding c): Revival and/or improvement of traditional Ecosystem-based Adaptation interventions that may help women cope with climate vulnerabilities	WWF Gender team WWF Project team Consultants Cooperating Organizations GCF or similar organizations MoCC Others as needed	-Indicator a): Number of women's groups involved in the implementation of green infrastructure measures Target: 4 women groups -Indicator b): Women's groups participate in implementation of green infrastructure measures (one in each site) Target: 4 women's groups (one at each site) - Indicator c): Number of people trained on sustainable operation and maintenance of green infrastructure measures Target: total 2000 people will be trained from which 25% will be women	Q1Y1 – Q1Y2	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
Output 1.3: Strengthened community-based natural resource management to support water supply and flood attenuation services provided by EbA interventions implemented in floodplains and watersheds.	Activity 1.3.1: Strengthen the capacity of community-based organisations (CBOs) to adopt EbA and green infrastructure interventions and undertake climate-resilient community-based natural resource management.	a): Gender mainstreaming is ensured through local women participation in capacity building sessions focussed on natural resource management b): Women representation and active participation in decision-making is ensured in Water User Groups and are consulted for defining their role in the operation and maintenance of EbA and green infrastructure interventions	WWF Gender team WWF technical teams WWF project team Consultants Cooperating Organizations Local govt. Depts.	- Indicator a): Number of people trained on community-based natural resource management Target: 800 people trained (40% will be women) - Indicator b): Number of local women resource management committees (e.g., water management groups, wetland management groups) engaged in planning and implementation of IFRM through EbA Target: 4 resources management committees (1 at each site)	Q3Y1 – Q4Y7	Covered by output budget.
	Activity 1.3.2: Strengthen communities' knowledge and awareness of climate change impacts and the benefits of the project's EbA and green infrastructure interventions.	a) Gender and youth centric awareness material is developed and disseminated among local communities, policy makers and key stakeholders to highlight their climate vulnerability	WWF Gender and project team Cooperating Organizations -Training Institutions Consultants -Social mobilizers -Local NGOs	- Indicator a): Number of women reached out through awareness raising events organized under the project Target: 1,200 people (300 per site) (40% women)	Q2Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
		b) Gender disaggregated impacts of project interventions are documented to demonstrate pathway to climate change adaptation c): Awareness of local community on GBV and SEAH and GRM Reporting Mechanism d) Establishment of grievance resolution mechanism which is sensitive to victims of GBV and SEAH to report into and connect with relevant stakeholders for domestic support and protection e): NGOs, shelters, and other local institutions trained in identifying GBV and SEAH on GRM for GBV and SEAH specific to the project.	WWF Gender team And project gender specialist; Project management unit coordinator and safeguards focal point responsible for GRM Cooperating Organizations Local police Social Welfare officers PCSW SCSW BCSW Women's Desks District Women Development officers NGOs Aurat foundation Shelters Local groups	-Indicator b): Number of case studies developed Target: 20 (50% women of case studies developed are from women) - Indicator c): Number of people attend sessions on GBV and SEAH and GRM Target: 675 people (50% women) - Indicator d): GRM formed and functional at each site Target: 4 (1 at each site) - Indicator e): Number of people from NGOs, shelters, and other local institutions trained on reporting mechanism and in identifying GBV and SEAH Target:100 people (30% women)		

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
		f) Train and continued reinforcement of capacity on Gender, GBV and SEAH for project staff, partners and key stakeholders	WWF Gender and project team	- Indicator f): Number of people trained on mainstreaming Gender, GBV and SEAH Target:200 people (30% women)		
Component 2: Enabling a paradigm shift towards EbA and green infrastructure in Pakistan						
Outcome 2: Enabling environment for climate action that has new procedures for implementing EbA and green infrastructure interventions for flood and water resources management in Pakistan.						
Output 2.1: Evidence base to support the adoption of EbA and green infrastructure interventions by the Government of Pakistan.	Activity 2.1.1: Develop an evidence-based case of the climate change adaptation benefits of EbA and green infrastructure in Pakistan.	a) Develop policy briefs that highlight gender dimensions and the role of women in adopting EbA interventions and scaling them up across Pakistan, including ensuring gender is considered in resource management regulatory frameworks.	WWF project teams Cooperating Organizations Aurat Foundation PPAF Social Welfare departments WDD PCSW SCSW NCSW BCSW DCSWs in KP RSPs Govt. agencies and Depts. at provincial levels UN Agencies Others	-Indicator a): Number of policy briefs developed focusing on gender dimensions Target: 4 policy briefs (1 at each site)	Q3Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
	Activity 2.1.2: Develop and disseminate an evidence report to national and provincial government agencies for implementing EbA and green infrastructure interventions in response to flooding and droughts in Pakistan.	a) Establishing regular liaison with local Government Departments at District level and for extension services in project sites for gender orientation, sensitization and mainstreaming. b) Sensitise relevant government line departments on the role of gender and women's empowerment in mainstreaming EbA across their planning and policy documents through promotional material, policy briefs, and regular liaison. c) Project Steering Committee (PSC) and Provincial Oversight Committees (POCs) facilitate mainstreaming gender in initiatives and policies that focus on improving natural resource management through EbA and similar programmes	WWF senior management with the project gender specialist Cooperating Organizations Provincial Government Departments District government Nearest offices of Local govt. Extension services	-Indicator a) Number of sensitization meetings conducted with different departments Target: 16 meetings (4 meetings at each site) -Indicator b) Number of people attended sensitization trainings Target: 175 (75% men) -Indicator c) Number of policies advocating for gender mainstreaming in initiatives focusing on improving NRM through EbA Target: 4 provincial policies	Q3Y1 – Q4Y6	Covered by output budget.
Output 2.2: Updated procedures for implementing EbA and green infrastructure in Pakistan.	Activity 2.2.1: Develop updated procedures for the Implementation Framework of the National Water Policy, National Adaptation	a) Development of gender responsive mitigation and/or adaptation plan for each area to ensure men and women have equal access to solutions and benefits	WWF Gender team and WWF Project team Cooperating Organizations	-Indicator a): Number of mitigation and adaptation plans with gender-specific approaches for women	Q3Y1 – Q4Y5	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
	Plan and four Provincial Adaptation Plans for implementing EbA and green infrastructure interventions in Pakistan.		GCF or similar organizations MoCC Others as needed	developed and implemented Target: 4 plans (1 mitigation plan for each site)		
Output 2.3: Strengthened capacity for implementing EbA and green infrastructure interventions in the Indus Basin for flood risk and water resources management.	Activity 2.3.1: Strengthen the capacity of national and sub-national staff, as well as community leaders/representatives, to apply the updated procedures under Activity 2.2.1 and implement, operate and maintain EbA and green infrastructure interventions for flood and water resources management.	a) Ensure women representation across capacity building trainings, so that they can play a leadership role in mainstreaming EbA and green infrastructure measures at national and sub-national level.	WWF senior management with the project gender specialist Federal Government Departments Provincial Government Departments Cooperating Organizations Academic Institutions	- Indicator a): Number of people trained on implementing EbA and green infrastructure solutions at national and sub-national level. Target: 275 people trained (40% will be women)	Q3Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
Component 3: Enhanced community resilience and adoption of EbA and green infrastructure interventions in Pakistan's Indus Basin.						
Outcome 3: Enhanced resilience of community livelihoods in the Indus Basin.						
Output 3.1: Pipeline of feasible climate-resilient businesses	Activity 3.1.1: Develop a pipeline of 7 sustainable climate-resilient businesses	a): Ensure women and men have access to capacity building workshops focused on bankable projects / green financing. Ensure that women are considered when carrying out these activities.	WWF Gender and project team Consultants Cooperating Organizations Local government departments	-Indicator a): Number of women attended capacity building workshops on bankable Projects Target: 80 women (20 women at each site) -Indicator a): Number of people adopting businesses supported under the project Target: Up to 2 women-led businesses (out of 7)	Q3Y1 – Q4Y7	Covered by output budget.

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
Output 3.2: Improved climate-resilient livelihoods among vulnerable target communities.	Activity 3.2.1: Improve the climate resilience of vulnerable agricultural livelihoods.	a): Women and men are trained on climate resilient agriculture practices covering both, drought and flooding scenarios.	WWF Gender and project team Consultants Cooperating Organizations Local government departments	-Indicator a): Number of women and men provided with agricultural livelihood support and training Target: 5,000 people (25% women and 75% men)	Q3Y1 – Q4Y7	Covered by output budget.
		b): Gender issues and Women's empowerment are mainstreamed in implementation of climate resilient agriculture interventions through implementation of high-efficient irrigation systems, kitchen gardening, fruit orchards, fodder cultivation, and water storage tanks.	WWF Project Team & Gender Specialist (in partnership with government vocational / technical training institutes. Consultants Cooperating Organizations	-Indicator b): Number of people adopting climate resilient agriculture interventions and alternative skills Target: 1500 people (25% women and 75% men)		
		c): Build awareness of the most vulnerable communities in the Indus Basin, particularly women and youth groups, on climate-resilient household business options		-Indicator c): Number of women and men who participate in awareness raising events/ campaigns Target: 800 people (40% will be women)		
		d): Build technical capacity of local women on alternative livelihoods and entrepreneurship.		-Indicator d): Number of women trained on alternative livelihoods and entrepreneurship		

Output	Activities	Gender Specific Actions	Responsible	Indicators and Targets	Timeline	Budget
		e): Implement gender specific diversified income generation activities including, but not limited to, fruit and vegetable processing units (packaging, labelling and post-harvest processing along with value addition), handicrafts, poultry farming, apiculture, etc.		Target: 800 Women (200 from each site) -Indicator e): Number of people trained on different value chains Target: 400 people (200 females and 200 males)		

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The Focus Group Discussion centred around 14 questions and was conducted through a zoom meeting ensuring the participants of complete confidentiality of their responses. The names of the participants remain anonymous. See Annex 1 for FGD questions.

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